Southeast Conservation Adaptation Strategy

Southeast Blueprint 2026 Development Process

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Contents

Background 1
Introduction to the Southeast Conservation Blueprint 1
The Need for a Blueprint 1
The History of the Blueprint 1
Southeast Blueprint 2026 Overview 3
Accessing Blueprint Data 5
Intended Uses 5
Contact Blueprint Staff 5
Indicators 6
Continental 6
Terrestrial 6
Freshwater 96
Coastal & Marine 139
Caribbean 202
Terrestrial 202
Freshwater 241
Coastal & Marine 261
Running Zonation 285
Defining Subregions 285
Continental Subregions 286
Caribbean Subregions 292
Combining Subregions 294
Defining Extent 295
Overall Zonation Approach 295
Inland Continental 296
Marine Continental 299
Caribbean 301
Combining Caribbean & Continental Results 302
Literature Cited 303
Connectivity 304
Continental 304

Continental Resistance Raster & Hubs 304
Prepare Continental Hubs & Resistance Raster for Linkage Mapper 310
Run Continental Corridor Analysis 311
Merge Outputs of Corridor Analysis 313
Make Final Continental Corridors 313
Caribbean 315
Caribbean Resistance Raster 315
Caribbean Hubs 316
Run Caribbean Corridor Analysis 317
Make Final Caribbean Corridors 319
Known Issues 320
Other Things to Keep in Mind 320
Literature Cited 321
Creating Southeast Blueprint 2026 322
Priority Categories 322
Priorities for a Connected Network of Lands & Waters 322
Combining Zonation Results with Corridors to Create the Southeast Blueprint 323
Input Data 323
Mapping Steps 323
Known Issues 325
Continental 325
Caribbean 330
Appendices 332
Appendix I: Additional information on SARP aquatic animal SGCN HUC12 summaries used in imperiled aquatic species indicator 332
Species Summaries by HUC12 Methodology* 332
Appendix II: Ancillary Blueprint Data 335
Caribbean Island Size 335
Elevation-Based Marine/Terrestrial Mask 337
Marine Mammal & Turtle Movement Areas Mask 340
Reservoirs 341

Background

Introduction to the Southeast Conservation Blueprint

The Southeast Conservation Blueprint is the primary product of the Southeast Conservation Adaptation Strategy (SECAS). It is a living, spatial plan to achieve the SECAS vision of a connected network of lands and waters across the Southeast and Caribbean. The Blueprint is regularly updated to incorporate new data, partner input, and information about on-the-ground conditions.

The Need for a Blueprint

The lands and waters of the Southeast United States and Caribbean are changing rapidly, reshaped by climate change, urban growth, and increasing human demands on resources. While these forces span the boundaries of political jurisdictions, the conservation community does not have a comprehensive, cross-boundary, and inter-organizational strategy for how to sustain shared resources in the face of future change.

The History of the Blueprint

The Southeast Conservation Blueprint is the primary product of the Southeast Conservation Adaptation Strategy, a regional conservation partnership spanning the Southeast United States and Caribbean. It identifies the most important places for collective action by the conservation community to achieve the SECAS vision of a connected network of lands and waters. The Blueprint has evolved over time through an iterative revision process, improving its accuracy, spatial resolution, regionwide consistency, and utility to conservation professionals.

In the fall of 2013, SECAS leadership set a goal of developing a first-generation Southeast Blueprint for landscape-scale conservation by the fall of 2016. Many different conservation planning efforts were already underway, but most eco-regional plans only covered parts of states, while state-specific plans stopped at the state line. The results of all this parallel planning did not yet add up to an integrated regional strategy.

Three years later, Version 1.0 of the Southeast Blueprint was released in December 2016. Development of this first Blueprint relied heavily on Landscape Conservation Cooperative (LCC) partnerships across the Southeast and Caribbean. This plan provided the first ever integration of spatial plans developed through the South Atlantic, Appalachian, Gulf Coastal Plains and Ozarks, Gulf Coast Prairie, North Atlantic, and Caribbean LCCs.

Version 2.0 of the Southeast Blueprint was released in November 2017. This plan incorporated the improved subregional Blueprints from several LCCs and established priority connections with western states through the Crucial Habitat Assessment Tool. Significant improvements over Version 1.0 included improved consistency across LCC boundaries, improved consistency in climate change response, and improved integration beyond the Southeast.

Version 3.0 of the Southeast Blueprint was first introduced at the October 2018 Southeastern Association of Fish and Wildlife Agencies (SEAFWA) annual conference and officially released in February 2019. Its development occurred during a time of transition for the LCC Network where the structure and function of some LCCs was changing. Despite these changes, the capacity and commitment to continue to support Blueprint users and improve the Southeast Blueprint remained strong, evidenced by the many examples of Blueprint implementation, as well as progress on Blueprint improvements. Blueprint 3.0 added full coverage of Texas, integrated threat layers covering the full Southeast, and a hubs and corridors layer covering part of the region.

Version 4.0 of the Southeast Blueprint was released in October 2019 at the SEAFWA annual conference. Improvements over the previous version include: corrected overprioritization in Texas, Oklahoma, and the mountains of West Virginia and Virginia; improved priorities in the Lower Mississippi Valley, Louisiana marshes, and the Southern Appalachians; updated inputs from Florida and the Middle Southeast subregion; expanded marine coverage to include state and federal waters around Florida; and expanded hubs and corridors that covered all of Florida.

Southeast Blueprint 2020 was released in October 2020 at the virtual SEAFWA annual conference. Significant improvements over the previous versions include: finer resolution and a more connected network of priorities in the inland South Atlantic subregion; updated data and a more consistent approach to cross-state prioritization in the Middle Southeast subregion; and better integration in areas of overlap between the South Atlantic Blueprint, Florida Blueprint, and Nature's Network design.

Southeast Blueprint 2021 was released in November 2021 following the virtual SEAFWA annual conference. Significant improvements over the previous versions include: updated and improved indicators, better incorporation of equity, deep-sea coral, fire, and important grasslands, and multiple connectivity improvements in the South Atlantic subregion, as well as correcting a scoring issue in the Middle Southeast subregion.

Southeast Blueprint 2022 was released in October 2022 at the SEAFWA annual conference. Rather than continuing to stitch together so many different subregional plans, the 2022 Blueprint took massive strides toward regional consistency by using the same methods and indicators across 15 states of the Southeast. To provide more complete coverage of the SECAS geography, Blueprint 2022 also incorporated two additional input plans: the latest update to the Florida Marine Blueprint for marine areas in Florida and the Caribbean Landscape Conservation Design for inland areas in Puerto Rico.

Southeast Blueprint 2023 was released in October 2023 at the SEAFWA annual conference. For the first time, this Blueprint used a consistent approach across the entire geography and did not have to stitch together any subregional inputs. The 2023 Blueprint expanded consistent methods and indicators to Puerto Rico, the U.S. Virgin Islands, and nearshore U.S. Caribbean waters, as well as to the full extent

of U.S. waters in the Atlantic Ocean and Gulf of America. It also included minor refinements to the inland continental Southeast priorities and updated hubs and corridors for the full Blueprint area.

Southeast Blueprint 2024 was released in October 2024 at the SEAFWA annual conference. Version 2024 focused on updating the priorities in the continental Southeast. It used refined prioritization and connectivity methods and incorporated a number of new and improved indicators for both inland and marine areas. The U.S. Caribbean priorities were not updated in version 2024.

Southeast Blueprint 2025 was released in November 2025 after the SEAFWA annual conference. Version 2025 included numerous new and improved indicators in the inland, freshwater, and coastal areas of the continental Southeast and U.S. Caribbean. It also used refined prioritization and connectivity methods.

Southeast Blueprint 2026 was released in October 2026 ahead of the SEAFWA annual conference. Version 2026 included several new indicators in the inland, freshwater, and coastal areas of the continental Southeast and U.S. Caribbean, including fire frequency, Gulf marine birds, and imperiled reptiles, amphibians, mammals, and aquatic species.

Southeast Blueprint 2026 Overview

The 2026 version of the Blueprint identifies priority areas based on a suite of natural and cultural resource indicators representing terrestrial, freshwater, and marine ecosystems. A connectivity analysis identifies corridors that link coastal and inland areas and span climate gradients. Figure 1 on the next page provides an overview of the Southeast Blueprint 2026 development process.

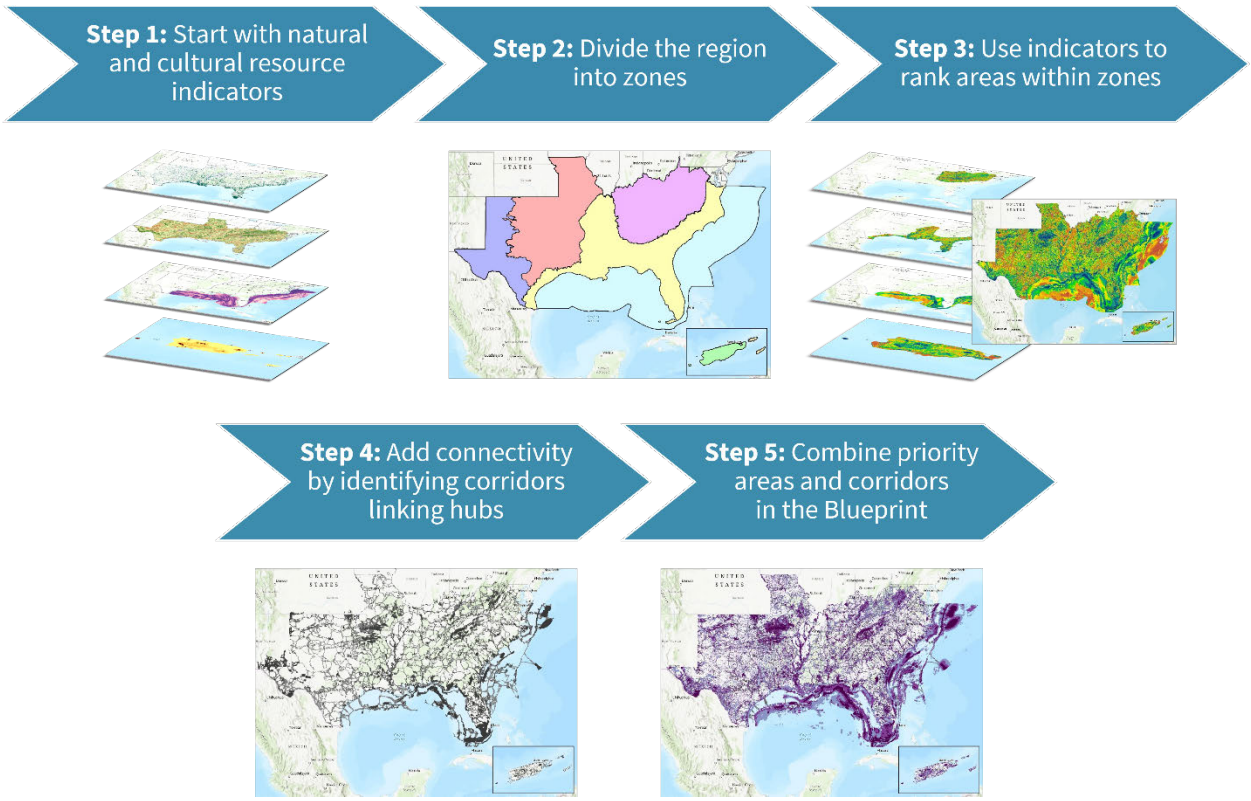


Figure 1. The overall process of developing Southeast Conservation Blueprint 2026.

Step 1: The terrestrial, freshwater, and coastal/marine ecosystems of the Southeast and U.S. Caribbean are represented by 70 natural and cultural resource indicators.

Step 2: The SECAS geography is divided into 7 zones created by grouping 22 smaller subregions.

Step 3: A program called Zonation ranks the pixels in each zone according to their indicator scores, using a modeling approach that conserves high-value representations of all indicators collectively. Pixels that rank higher in Zonation become higher priority in the Blueprint. The subregions are also used as inputs to Zonation to better balance priorities within zones.

Step 4: Using a program called Linkage Mapper, a least-cost path connectivity analysis identifies corridors that link hubs across the shortest distance possible, while also routing through as much Blueprint priority as possible. In the continental Southeast, hubs are large patches (~5,000+ acres) of highest priority Blueprint areas and/or protected lands. In the Caribbean, hubs are large patches (~125+ acres) of highest priority Blueprint areas and/or protected lands.

Step 5: Combining the areas of highest Zonation ranking with the corridors produces Southeast Blueprint 2026.

Accessing Blueprint Data

- Start simple in the Blueprint Explorer, an easy-to-use online viewer that allows you to create custom reports and discover what's driving the Blueprint priorities
- Visit the Blueprint page of the SECAS Atlas to dig deeper into the Blueprint and underlying data layers, and make your own maps
- Download the Blueprint data for direct use in your desktop GIS

Intended Uses

More than 500 people from over 200 organizations are already using the Blueprint to guide conservation action and investment across the SECAS geography. So far, it has helped bring in more than $440 million in conservation funding to protect and restore over 470,000 acres and reconnect more than 1,300 stream miles. A few things to keep in mind when using the Blueprint:

- The Blueprint identifies areas where conservation action would have the biggest impact for a connected network of lands and waters. A range of conservation activities could benefit those priority areas—not just land protection, but also management, restoration, economic incentives, outreach and communication, etc.
- The Blueprint is not intended to be used in isolation of other datasets. Instead, it provides a regional perspective that, in combination with local data and knowledge, can help inform decisions about where to focus conservation action in the face of future change.
- If you want to find your piece of the Blueprint, you can explore the underlying indicator and connectivity data. These underlying layers can help identify important areas for corridors, birds, water quality, climate resilience, prescribed burning, reforestation, and much more. They can help you tell the unique story of what makes your area of interest special. Contact a member of Blueprint user support staff if you'd like some help.
- As a living spatial plan, the Blueprint is always a work in progress. We maintain a list of the problems with the Blueprint that have been identified in the review process so far. Those "known issues" are captured in this report and on the SECAS website. The Blueprint will continue to be refined to incorporate improvements to the input data and methods, as well as feedback from Blueprint users and other partners.

Contact Blueprint Staff

Do you have a question about the Blueprint? Would you like help using the Blueprint to support a proposal or inform a decision? Staff across the Southeast are here to support you! The Blueprint is also revised based on input from people like you. So, if you have a suggestion on how to improve the priorities, let us know!

To get help or provide feedback, reach out to a member of SECAS user support staff: http://secassoutheast.org/staff.

Indicators

The indicators drive the identification of priority areas for a connected network of lands and waters within the Southeast Conservation Blueprint. One suite of indicators applies to the continental Southeast portion of the Blueprint, which spans the 15 contiguous Southeast states and surrounding marine environment out to the end of U.S. waters. Another suite of indicators applies to the U.S. Caribbean and surrounding nearshore marine environment. Spatial data for each indicator is available on the Blueprint page of the SECAS Atlas.

Continental

Terrestrial

East Coastal Plain Open Pine Birds

This indicator identifies areas within the historical longleaf pine range east of the Mississippi River where creating or maintaining open pine habitat would most benefit six focal species of birds (Bachman's sparrow, red-cockaded woodpecker, Henslow's sparrow, red-headed woodpecker, Northern bobwhite, brown-headed nuthatch). It prioritizes areas for open pine conservation based on suitability for longleaf pine, feasibility of prescribed burning, proximity to protected lands, habitat suitability for focal bird species, and proximity to bird source populations. It originates from the East Gulf Coastal Plain Joint Venture's prioritization of areas for open pine ecosystem restoration.

This indicator was last updated in October 2023.

Reason for Selection

Open longleaf pine forests once spanned nearly 90 million acres across the Southeast, supporting a rich community of wildlife and plants. Fire suppression, deforestation, and conversion to ecosystems dominated by loblolly and slash pine have dramatically reduced the extent of longleaf and caused the decline of many associated species. In addition, pine and prairie birds are experiencing significant declines and are currently off-track for meeting the SECAS 10% goal, so it is important that the Blueprint capture opportunities to conserve, restore, and manage open pine habitat. This indicator also promotes consistency with the longleaf and open pine ecosystem priorities of the East Gulf Coastal Plain Joint Venture (EGCPJV).

Input Data

- Bird priorities from Prioritization of areas for open pine ecosystem restoration in the Southeastern United States (bird_priority.tif); read the project final report; read a news article about the project; explore the data in the EGCPJV Open Pine Decision Support Tool
- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- 2019 National Land Cover Database (NLCD): Land cover
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Reproject the EGCPJV bird priority data to NAD 1983 Contiguous USA Albers (EPSG 5070).
- If an area intersects an EPA Estimated Floodplain, change the value to NoData.
- If an area is something besides Evergreen Forest or Mixed Forest in the 2019 NLCD, change the value to NoData. Because the bird priority layer was made with older landcover data, this step helps remove areas that do not currently have pine trees.
- The EGCPJV bird data provides a raster with scores of 0-100 representing the relative potential for open pine bird response to conservation action. Reclassify this raster as shown in in the legend below.
- Add zero values to represent the extent of the source data and to make it perform better in online tools. Assign a value of 0 to any pixel that had a value ≥ 0 in the original bird priorities layer but was converted to NoData above.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = High priority for open pine conservation for focal bird species (Bachman's sparrow, red-cockaded woodpecker, Henslow's sparrow, red-headed woodpecker, Northern bobwhite, and brown-headed nuthatch) (score >80-100)
4 = Medium-high priority (score >60-80)
3 = Medium priority (score >40-60)
2 = Medium-low priority (score >20-40)
1 = Low priority (score 0-20)
0 = Not a priority (not identified as upland pine)

Known Issues

- The EGCPJV's open pine bird model includes frequently inundated floodplain areas. These areas are unlikely to be potential open pine habitat. To address this in the indicator, we removed areas within the EPA estimated floodplain.
- This indicator prioritizes some areas outside of the floodplain that were not historically open pine habitat.
- The 2019 NLCD classes used to exclude non-pine areas of the open pine bird model (evergreen forest and mixed forests) do not exclusively target pine trees. For example, the evergreen class can include other evergreen tree species like Eastern red cedar or southern magnolia. In addition, the NLCD likely misclassifies other types of landcover as evergreen or mixed forests. As a result, this indicator may leave in some areas prioritized in the bird model that are not actually pine. Conversely, the NLCD also likely misclassifies some areas of pine as other landcover classes, which could cause pine areas prioritized in the model to be inadvertently excluded from the indicator.
- This indicator does not capture areas where planting new pine stands in existing agricultural areas would benefit open pine birds.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Grand, James B. and Kevin J. Kleiner. 2016. Prioritizing Landscapes for Longleaf Pine Conservation. Report provided by the Cooperative Fish and Wildlife Research Unit Program under agreement with the U.S. Fish and Wildlife Service. U.S. Department of Interior, Fish and Wildlife Service, Cooperator Science Series FWS/CSS-119-2016, National Conservation Training Center. [https://digitalmedia.fws.gov/digital/collection/document/id/2131].

Grand, J.B., and Kleiner, K.J., 2017, Prioritization of areas for open pine ecosystem restoration in the Southeastern United States: U.S. Geological Survey data release, [https://doi.org/10.5066/F7P26WN2].

Open Pine Decision Support Tool. [https://scagulf.shinyapps.io/opdst/].

U.S. Geological Survey (USGS). Published June 2021. National Land Cover Database (NLCD) 2019 Land Cover Conterminous United States. Sioux Falls, SD. [https://doi.org/10.5066/P9KZCM54].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Fire Frequency

This indicator uses remote sensing, fire treatment data, and historical landcover models to assess whether an area matches its historical fire regime now and in the recent past (1984-2024). It focuses on fire-adapted forests and grasslands that historically burned at least once every 20 years. It evaluates departure from historical conditions based on the U.S. Forest Service's fire regime condition classes. This indicator combines Landsat-based fire history metrics from the U.S. Geological Survey, fire treatment data from LANDFIRE, and historical fire regime estimates from LANDFIRE. Development of fire history metrics was supported by Southeast FireMap.

This indicator was last updated in October 2026.

Reason for Selection

Historically in the South, "fires burned as often as once a year or more in Coastal Plain pine systems or as infrequently as every 50 years or more on north-facing or cove sites in the mountains" (Waldrop and Goodrick 2012). Fires were typically started by lightning or by Indigenous Americans using fire to manage open savannas. Species adapted to the specific fire return interval, or average time between fires, that typically occurred in a given area. The term "fire regime" is often used to describe historical pattern of fire, including frequency and severity.

Ecosystems that experienced regular fire over time—such as grasslands and longleaf pine savannas—evolved into systems characterized by many species that not only tolerate fire, but require it. Fire suppression during the 20th century led to the loss and deterioration of many fire-adapted ecosystems and their associated wildlife and plant species. For example, the federally threatened red-cockaded woodpecker is considered an umbrella species for longleaf ecosystems and has suffered population declines due to losses of suitable habitat attributed in part to fire suppression (USFWS 2022). Additionally, the endangered species recovery plan for the Alabama canebrake pitcher plant, which is only found in wetland areas, explicitly identifies fire exclusion as a severe threat to this species (USFWS 1992).

When natural disturbance processes are disrupted, regeneration tends to follow a shrub and mixed hardwood trajectory in which a return to a savanna or longleaf dominated system is unlikely (Hartnett and Krofta 1989). Furthermore, prolonged fire suppression can also lead to increased fuel loads, and with it, the threat of catastrophic wildfire. Today, practices that mimic historical disturbance, such as fire, are recognized for their essential role in maintaining healthy ecosystems and reducing wildfire risk. As a result, "prescribed burning is an important tool throughout Southern forests, grasslands, and croplands" (Waldrop and Goodrick 2012).

This indicator focuses on the many Southeastern ecosystems that rely on regular, low-intensity fires to maintain habitat and encourage native plant growth. The analysis defines a fire-adapted ecosystem as one that historically burned at least once every 20 years. It assesses both whether an area currently falls within its appropriate historical fire regime (as of 2024), as well as how closely it matched its historical fire regime in the recent past (between 1984 and 2024).

The U.S. Forest Service (USFS)'s fire regime condition classes provide a framework for characterizing how closely a site matches its historical fire regime. Areas within a <33% departure from "pre-EuroAmerican settlement era" conditions fall within an estimated historical range of variation and are considered to have low departure from the historical fire regime; areas with a 33-66% departure are considered to have moderate departure; and areas with a >66% departure to have high departure (USFS 2010).

Input Data

- Southeast Blueprint 2026 extent
- 2024 Annual National Land Cover Database (NLCD): Land cover collection 1.1
- Floodplain inundation (Southeast Blueprint indicator)
- U.S. Geological Survey (USGS) Contemporary fire history metrics for the conterminous United States (1984-2024) (ver. 4.0, March 2025)
- 2020 LANDFIRE Biophysical Settings (BPS) [LF 2.2.0]; access the data
- LANDFIRE All Years Annual Disturbance (Dist) [CONUS 1999-Present]

Mapping Steps

- Start by formatting 10 years of annual disturbance rasters from LANDFIRE (2015-2024). For each annual disturbance raster, combine fire types (DIST_TYPE IN ('Fire', 'Prescribed Fire', 'Wildfire', 'Wildland Fire Use')). Then combine those rasters into a single raster so the value of each pixel is the time since last burn starting in 2024.
- Combine the integrated LANDFIRE time since last burn raster with the USGS time since last burn metric (conus_1984_2024_TSLB.tif) by taking the minimum value across both rasters. Indicator reviewers identified multiple recent fires that were missing from the USGS Landsat-based fire data. Incorporating 10 years of data from LANDFIRE, which uses many data sources that don't rely on remote sensing, helps improve coverage of recent fires.
- Now estimate current departure from the historical fire regime. For fire return intervals greater than 5 years, test whether the integrated raster of time since last burn is greater than the LANDFIRE BPS historical fire return interval (FRI_ALLFIR) * 1.33. We expand the window by an extra 33% because that is still considered within the historical range of variation (low departure) in the USFS fire regime condition classes. For fire return intervals of 5 years or less, compare the time since last burn to the historical fire return interval (FRI_ALLFIR) + 2 years. The percentage-based departures overestimate departure in these more frequent (≤5-year) fire return intervals. For example, a 33% departure from a 3-year interval is less than 1 year, which is less than the historic range of variation for many of those ecosystems in the Southeast.
- Next, estimate departure from historical fire regime from 1984-2024. Calculate the percent departure from historic fire regime using the USGS fire frequency data (conus_1984_2024_FRQ) and the expected number of fires over the 40-year period from LANDFIRE BPS historical fire return interval (40/FRI_ALLFIR). For historical fire regimes of 5 years or less, use 40/(FRI_ALLFIR+2). Convert the percent departure into categories based on the U.S. Forest Service fire regime condition classes (≤33%, 33-66%, >66%) (USFS 2010).

- Classify as zero any pixel where the historical fire return interval is >20 years. These pixels receive the lowest indicator score because regular fire is less important in the function of these ecosystems and they are therefore not the focus of this indicator.
- Combine the current departure from historical fire regime and departure from historical fire regime from 1984-2024 calculated above to generate the indicator categories as shown below.
- Rebuild the 2024 NLCD raster attribute table and calculate statistics. This fixes Esri software problems with reclassifying based on landcover types.
- Reclassify to a value of 0 the burned areas that are most likely to be false positives.
    - Assign a value of 0 to pixels classified in the 2024 NLCD as one of the following landcover types: Open water (11), all urban (21, 22, 23, 24), barren (31), and cultivated crops (82). Fire in these pixels was often either not natural or likely misclassified. Clusters of pixels in barren landcover were often industrial sites and quarries.
    - Assign a value of 0 to floodplain inundation indicator categories with likely misclassifications: "2 = inundation index of 91-100 (persistent inundation)", "3 = inundation index of 6-20 (regular inundation)", and "4 = inundation index of 21-90 (frequent inundation)". Inundated vegetation is often misclassified as burned area since they have similar spectral signatures in remote sensing.
- As a final step, clip to the spatial extent of Southeast Blueprint 2026.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Fire-adapted area, currently in historical fire regime, low (<33%) departure from historical fire regime from 1984-2024

5 = Fire-adapted area, currently in historical fire regime, medium (33-66%) departure from historical fire regime from 1984-2024

4 = Fire-adapted area, currently in historical fire regime, high (>66%) departure from historical fire regime from 1984-2024

3 = Fire-adapted area, not currently in historical fire regime, low (<33%) departure from historical fire regime from 1984-2024

2 = Fire-adapted area, not currently in historical fire regime, medium (33-66%) departure from historical fire regime from 1984-2024

1 = Fire-adapted area, not currently in historical fire regime, high (>66%) departure from historical fire regime from 1984-2024

0 = Urban, row crop, or not a fire-adapted area (>20-year fire regime)

Known Issues

- This indicator does not include fires less than 10 acres in size. 10 acres is the minimum mapping unit in the fire history metric data.
- The Landsat-based fire history metrics overestimate fire frequency in open areas with wet soils. Wet soils can be much darker than dry soils and may be misclassified as burned areas. This misclassification was improved by removing areas classified as cultivated crops. A mask built on floodplain inundation frequency was also used to reduce misclassifications.
- The Landsat-based fire history metrics underestimate fire frequency in places with major impediments to burned area detection/mapping. Impediments include rapid green-up following a burn, cloud cover and shadows obscuring burn signatures, difficulty detecting or differentiating a low intensity burn signature beneath tree canopies, and the satellite product resolution often being too coarse to capture fine-scale differences or small burns.
- The indicator slightly underestimates areas currently within historic fire regimes on private lands when compared to public lands. While adding in LANDFIRE annual disturbance data reduced underestimates on public lands, its coverage isn't as good on private lands.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Allen, Y. 2016. Landscape Scale Assessment of Floodplain Inundation Frequency Using Landsat Imagery. River Research and Applications 32:1609–1620. [https://doi.org/10.1002/rra.2987].

Landscape Fire and Resource Management Planning Tools (LANDFIRE), Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published 2022-09-28. LANDFIRE 2020 Biophysical Settings (BPS) CONUS. LF 2020. Raster digital data. Sioux Falls, SD. [https://landfire.gov/vegetation/bps].

Hartnett, D.C; Krofta, D.M. 1989. Fifty-five Years of Post-fire Succession in the Southern Mixed Hardwood Forest. *Bulletin of the Torrey Botanical,* 116(2), 107-113. [https://doi.org/10.2307/2997194].

Teske, Casey, Melanie K. Vanderhoof, Todd J. Hawbaker, Joe Noble, and John K. Hiers. 2021. Using the Landsat Burned Area Products to Derive Fire History Relevant for Fire Management and Conservation in the State of Florida, Southeastern USA, Fire, 4, no. 2: 26. [https://doi.org/10.3390/fire4020026].

U.S. Fish and Wildlife Service. 1992. Alabama Canebrake Pitcher Plant Recovery Plan. Jackson, Mississippi. 21pp. [https://ecos.fws.gov/docs/recovery_plan/921008.pdf].

U.S. Fish and Wildlife Service. 2022. Species status assessment for the red-cockaded woodpecker (*Picoides borealis*). Version 1.4, October 2022. Atlanta, GA. [https://iris.fws.gov/APPS/ServCat/DownloadFile/257450].

U.S. Forest Service. September 2010. Interagency Fire Regime Condition Class (FRCC) Guidebook (Version 3.0). National Interagency Fuels, Fire, and Vegetation Technology Transfer (NIFTT). [https://www.fs.usda.gov/rm/pubs/rmrs_gtr292/2010_barrett.pdf].

U.S. Geological Survey (USGS), 2024, Annual NLCD Collection 1 Science Products (ver. 1.1, June 2025): U.S. Geological Survey data release. [https://doi.org/10.5066/P94UXNTS].

Vanderhoof, M.K., Hawbaker, T.J., Teske, C., Menick, C., Noble, J., and Smith, J., 2025, Contemporary fire history metrics for the conterminous United States (1984-2024) (ver. 4.0, March 2025): U.S. Geological Survey data release. [https://doi.org/10.5066/P98996IH].

Waldrop, Thomas A.; Goodrick, Scott L. 2012. (Slightly revised 2018). Introduction to prescribed fires in Southern ecosystems. Science Update SRS-054. Asheville, NC: U.S. Department of Agriculture Forest Service, Southern Research Station. 80 p. [https://www.srs.fs.usda.gov/pubs/su/su_srs054.pdf].

Grassland & Savanna Extent

This indicator represents grasslands and savannas in the southeastern United States. It considers known grassland and savanna locations, predicted locations within protected areas, and surrounding pollinator buffers. It also incorporates other potential grassland and savanna locations within natural and altered landscapes, and restoration opportunities within historical locations based on past fire intervals and historical ecosystem predictions. This indicator combines data from multiple sources, including the Southeastern Grasslands Institute, the National Land Cover Database, LANDFIRE, Oklahoma and Texas ecological systems maps, and more.

This indicator was last updated in November 2025.

Reason for Selection

Native grasslands and savannas are important for many endemic species, provide critical habitat and food for pollinators, and are often hotspots for biodiversity. Once a predominant ecosystem type, grasslands and savannas have significantly declined from their historical extent. In part because of the regular disturbance (e.g., mowing, fire) typically required to maintain high-quality grasslands, they are difficult to detect through remote sensing and are not well-captured by other indicators. In addition, grassland and savanna birds are experiencing significant declines and are currently off-track for meeting the SECAS 10% goal, so it is important that the Blueprint captures known and potential habitat.

Input Data

- Texas Ecological Mapping Systems: statewide raster, accessed 12-2023
- Oklahoma Ecological Systems Map: download the raster, accessed 12-2023
- Protected Areas Database of the United States (PAD-US): PAD-US 3.0 national geodatabase - Combined Proclamation Marine Fee Designation Easement; PAD-US 4.0 national geodatabase - Combined Proclamation Marine Fee Designation Easement
- National Land Cover Database (NLCD): 2021 Land Cover, 2021 U.S. Forest Service (USFS) Tree Canopy Cover, 2013 Land Cover (from 2016 data release), and 2013 USFS Tree Canopy Cover (from 2021 data release)
- 2020 LANDFIRE Biophysical Settings (BPS) [LF 2.2.0]; access the data
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0]; access the data
- U.S. Geological Survey (USGS) Watershed Boundary Dataset (WBD), accessed 1-5-2023: HUC12s; download the data
- Southeast Blueprint 2025 extent

Known grasslands

- Known grassland prairies dataset for the Middle Southeast subregion, provided by Toby Gray with Mississippi State University in Oct 2020 (available on request by emailing rua_mordecai@fws.gov); this is an improved version of the Known Prairie Patches in the Gulf Coastal Plains and Ozarks (GCPO) layer
- Known Piedmont prairie locations in the South Atlantic subregion: We identified known prairie locations by requesting spatial data on known prairies from the 74 members of the Piedmont Prairie Partnership mailing list and other prairie managers (Wake County Open Space program and Prairie Ridge Ecostation in NC). We combined that information with known locations in Virginia aggregated by the Virginia Natural Heritage Program (available on request by emailing rua_mordecai@fws.gov).
- Grassland polygons from the Catawba Indian Nation, provided by Aaron Baumgardner, Natural Resources Director, in July 2023 (for more information email rua_mordecai@fws.gov)
- Grassland polygons from two iNaturalist projects in Texas: erwin-park-prairie-restoration-area, stella-rowan-prairie
- Confirmed prairie remnants and prairie plantings from the 2017 U.S. Geological Survey (USGS) grassland quality and pollinator habitat potential in Southwest Louisiana project
- Dixon Water Foundation Texas ranch boundaries, accessed 4-8-2025
- Lathrop Island, FL savanna boundaries, provided on 3-24-2025 by the U.S. Fish and Wildlife Service (U.S. FWS), available on request (rua_mordecai@fws.gov)
- Smiley Meadow Preserve boundaries, provided in October 2024 by The Nature Conservancy (TNC) Texas (also available on the TNC lands public layer)
- Grassland polygons for Key Cave National Wildlife Refuge in Alabama, provided by the U.S. FWS on 3-30-2023, available on request (rua_mordecai@fws.gov)

- Southeastern Grasslands Institute polygons from selected iNaturalist projects. We used only projects with polygons digitized at a fine resolution and did not include projects with more coarse polygons covering a large area. Specific projects used were:
    - allegheny-mountains-riverscour-barrens, big-south-fork-riverscour-barrens-1, big-south-fork-riverscour-barrens-2, big-south-fork-riverscour-barrens-4-us, big-south-fork-riverscour-barrens-6, biodiversity-of-piedmont-granite-glades-outcrops, bluff-mountain-fen, caney-fork-sandstone-riverscour-barrens-and-glades, clear-creek-sandstone-riverscour-barrens, clear-fork-river-riverscour-barrens, craggy-mountains-mafic-outcrops-and-barrens, cumberland-plateau-escarpment-limestone-barrens, cumberland-river-limestone-riverscour-glades, daddy-s-creek-riverscour-barrens, dunbar-cave-prairie-restoration, eastern-highland-rim-limestone-riverscour-glade, emory-river-sandstone-riverscour-barrens, falls-of-the-ohio-river-limestone-riverscour-glade, flat-rock-cedar-glades-and-barrens-state-natural-area, grasshopper-hollow-fen, gunstocker-glade, hiwassee-river-phyllite-riverscour-glade, ketona-dolomite-barrens, laurel-river-riverscour-barrens-and-glades, lime-hills-limestone-barrens, limestone-barrens-of-the-western-valley-of-the-tennessee-river, little-mountains-limestone-barrens, little-river-canyon-riverscour-barrens-and-glades, moulton-valley-limestone-glades, mulberry-fork-of-black-warrior-river-riverscour-barrens-and-glades, muldraugh-s-hill-limestone-barrens, nashville- basin-limestone-glades, new-river-riverscour-barrens, obed-river-sandstone-riverscour-barrens, outer-bluegrass-dolomite-barrens, ridge-and-valley-sandstone-outcrops, rock-creek-sandstone-riverscour-barrens, rockcastle-river-sandstone-riverscour-barrens, shawnee-hills-sandstone-glades-and-outcrops, southern-blue-ridge-mountains-grass-balds, southern-blue-ridge-mountains-serpentine-barrens, southern-blue-ridge-phyllite-outcrops, southern-ridge-and-valley-limestone-glades, southern-ridge-and-valley-shale-barrens, southern-ridge-and-valley-siltstone-barrens, tennessee-ridge-and-valley-dolomite-barrens-and-woodlands-tn-us, the-farm-prairie-and-oak-savanna, tin-top-road-savanna, western-allegheny-escarpment-limestone-barrens, western-highland-rim-limestone-glade-and-barrens, western-valley-limestone-barrens-decatur-co-north-us, western-valley-limestone-barrens-hardin-wayne-cos, western-valley-limestone-barrens-perry-co, western-valley-silurian-limestone-barrens, white-s-creek-sandstone-riverscour-barrens-and-glades, folder-six-glades

Mapping Steps

- Combine all known grasslands polygons into one layer and convert to raster. Create a mask based on 2021 NLCD classes ‘Developed, medium intensity’ (23) and ‘Developed, high intensity’ (24) to remove parts of the known grasslands that have been recently developed. Assign the remaining known grassland pixels a value of 7.
- From the 2021 and 2013 NLCD landcover, create two rasters that only include classes with possible grasslands and savannas (one for each timestep). The classes included are based on NLCD classes that overlap known grassland and savanna polygons. Any class that covered

>1% of known grasslands and savannas is included: 31 Barren Land, 41 Deciduous Forest, 42 Evergreen Forest, 43 Mixed Forest, 52 Scrub/Shrub, 71 Grassland/Herbaceous, 81 Pasture/Hay.

- Use LANDFIRE EVT group name (EVT_GP_N = 'Longleaf Pine Woodland') to add as possible grasslands wet longleaf pine savannas that can be classified as forested wetland by NLCD. Add these to both the 2013 and 2021 possible grassland and savanna timesteps.
- Use LANDFIRE BPS to identify historical grasslands in the western part of the region that wouldn't typically have trees by selecting the following values in the BPS_NAME field: 'Southern Blackland Tallgrass Prairie', 'Central Mixedgrass Prairie', 'Chihuahuan Loamy Plains Desert Grassland', 'Southeastern Great Plains Tallgrass Prairie', 'Western Great Plains Sand Prairie', 'Western Great Plains Shortgrass Prairie'. This will be used to define where canopy cover thresholds apply in the next step.
- For the two possible grassland and savanna rasters created above, use the 2013 and 2021 NLCD USFS Tree Canopy Cover to refine the depiction of potential grasslands and savannas for each year. Remove 1) forest with ≥ 60% canopy cover and 2) Western relatively treeless grasslands with >0% canopy cover (as defined by the LANDFIRE BPS earlier).
- Make a combined possible grassland and savanna raster that only includes pixels that are possible grasslands and savannas in both 2013 and 2021. This removes temporary grasslands and savannas that result from clearcuts.
- From the Texas and Oklahoma ecological systems maps, extract classes that predict areas invaded by mesquite, a non-native tree that spreads aggressively in the grasslands and savannas of the Southwest and disrupts natural ecosystems through its heavy water consumption. For Oklahoma, this is VegName = 'Ruderal Mesquite Shrubland'. For Texas, this is CommonName = 'Native Invasive: Mesquite Shrubland'. Combine these and use them to remove from the possible grasslands and savannas layer areas that are no longer grassland or savanna due to mesquite invasion. The resulting layer represents potential grasslands and savannas.
- To identify likely grasslands and savannas, overlay the potential grasslands and savannas raster with select polygons from PAD-US 4.0. To pull out types of protected lands that commonly manage grasslands and savannas, we used GAP status, designation type, manager name, and easement holder. We also identified a number of protected areas directly by name that had important areas of grassland and savanna but weren't captured by the other rules.
    - GAP status (GAP_sts) 1 or 2: Gap status 1 and 2 refer to areas managed for biodiversity that are not subject to extractive uses like logging and mining. GAP status 2 is technically intended to encompass areas where disturbance events are suppressed, but in practice, most protected areas in the Southeast that actively manage grasslands and savannas are classified as GAP status 2.
    - Designation type (Des_Tp) of 'NWR', 'MIL', 'NF', or 'NG' (i.e. National Wildlife Refuge, military installation, National Forest, or National Grassland)
    - Manager name (Mang_Name) of 'RWD' (i.e. Regional Water District)
    - Local manager name (Loc_Mang) of 'Ducks Unlimited (Wetlands America Trust)'
    - Easement holder (EsmtHldr) of 'Tall Timbers Research Station & Land Conservancy'

    - Unit name (Unit_Nm) of 'Point Washington State Forest', 'Pine Log State Forest', 'M. C. Davis - Seven Runs Creek Conservation Easement', 'Nokuse Plantation Conservation Easements', 'Tate's Hell State Forest', 'Box-R Wildlife Management Area', 'Aucilla Wildlife Management Area', 'Snipe Island Unit', 'Big Bend Wildlife Management Area', 'Goethe State Forest', 'Amelia Wildlife Management Area', 'Powhatan Wildlife Management Area', 'Cumberland State Forest', 'Appomattox-Buckingham State Forest', 'Haw River State Park', 'R. Wayne Bailey - Caswell Game Land', 'Medoc Mountain State Park', 'Embro Game Land', 'Dupont State Forest', 'Hanging Rock State Park', 'Bladen Lakes State Forest', 'Whitehall Plantation Game Land', 'Suggs Mill Pond Game Land', 'Bushy Lake State Natural Area', 'Pondberry Bay Plant Conservation Preserve', 'Green Swamp Game Land', 'Holly Shelter Game Land', 'Chowan Swamp Game Land', 'Brookgreen Gardens', 'Cary State Forest', 'Suwannee Ridge Mitigation Park Wildlife and Environmental Area', 'Adams-Alapha Ag & Conservation Easement', 'Twin Rivers State Forest', 'Chattahoochee Fall Line Wildlife Management Area', 'Enon Plantation', or 'Georgia-Alabama Land Trust Easement #214', 'Covington Wildlife Management Area', ' Magnolia Branch Wildlife Reserve', 'Little River State Forest', or 'Susan Turner Plantation', or have the local name (Loc_Nm) 'Sandhills Game Land', 'Blackwater River State Forest', 'Three Lakes Wildlife Management Area', 'Herky Huffman/Bull Creek Wildlife Management Area', 'Triple N Ranch Wildlife Management Area', 'Whaley Conservation Easement', 'Babcock Ranch Preserve', 'Withlacoochee State Forest', 'Lake Wales Ridge State Forest', 'EVERGLADES HEADWATERS NATIONAL WILDLIFE REFUGE AND CONSERVATION AREA', 'Seminole State Forest', 'Schofield Tract', 'LAKE WALES RIDGE NATIONAL WILDLIFE REFUGE', 'Platt Branch Mitigation Park Wildlife and Environmental Area', 'Fisheating Creek/Lykes Brothers Conservation Easement', 'Bright Hour Watershed', 'Fred C. Babcock-Cecil M. Webb Wildlife Management Area', or 'BICY'
- PAD-US 4.0 is missing state wildlife management area boundaries in Oklahoma. Extract those from PAD 3.0 by using a combination of state name, designation, and GAP status criteria: State_Nm = 'OK' AND Loc_Ds = 'State Wildlife Management Area' AND GAP_Sts IN ('1','2'). Add the results to the polygons from PAD-US 4.0 generated in the previous step.
- Many areas of city-owned land in PAD-US are classified as GAP status 2, but don't appear to be managed for biodiversity. These are often golf courses, ball fields, and other recreational amenities that are not beneficial for biodiversity. Further filter the merged PAD-US layer to include only GAP 2 city-owned land (Mang_Name = 'CITY') if it has the following local designations: 'Conservation Area', 'Conservation Park', 'Local Conservation Area', 'Natural Area', 'Nature Park', 'Nature Preserve', 'Nature Reserve', 'Preserve', 'Regional Nature Preserve', 'Wildlife Area', or 'Wildlife Refuge'.
- Use the ArcGIS Region Group function to remove any single isolated likely grassland and savanna pixels. Single isolated pixels are often mapping errors that do not represent ecosystems within the intended scope of this indicator. The resulting layer represents likely grasslands.
- Use Region Group to separate patches of likely grassland and savanna that are >10 acres and those that are ≤10 acres. Grassland birds like grasshopper sparrows avoid patches <10 acres

(Dechant et al. 2002, Slater 2004). Assign pixels in >10 acre patches a value of 6 and those in <10 acre patches a value of 5.

- Use Region Group to identify likely grassland and savanna patches >5 acres to use as cores for identifying pollinator buffers. While important pollinators can persist in patches smaller than those needed for grasslands birds, larger patches are more likely to support the diversity of plants and nesting habitat needed to support a diverse pollinator community.
- Combine known grasslands and savannas with likely grassland and savanna patches >5 acres into a single class to create pollinator buffers. Buffer by 300 m to align with foraging distances for solitary bees (Antoine and Forrest 2021, Zurbuchen et al. 2010).
- Use NLCD 2021 to remove any open water in the pollinator buffers. Assign the remaining pollinator buffer pixels a value of 4.
- Use the Zonal Statistics function to identify HUC12 subwatersheds that contain at least one known or likely grassland/savanna patch >5 acres. These subwatersheds will be the grassland/savanna hubs. This is intended to elevate potential grasslands that are nearby other established grassland patches. We used subwatersheds as the summary unit because they are: 1) ecologically based boundaries 2) smaller than counties, and 3) a unit often used to focus conservation actions. The output from this step is "HUC12_knownOrLikely5acPlus.tif", which is also used in the grassland and savanna restoration indicator.
- Use LANDFIRE Biophysical Settings to create a raster of places with a historical fire interval (FRI_ALLFIR) of 9 years or less. That aligns with the role of regular fire in maintaining the grasslands and savannas of the Southeast.
- Add additional historical grassland and savanna areas where the LANDFIRE BPS vegetation group (GROUPVEG) is either 'Savanna' or 'Grassland'. That adds other grasslands and savannas that are maintained by disturbance other than fire.
- Remove areas where the LANDFIRE BPS vegetation group (GROUPVEG) is 'Shrubland'. This removes historical shrublands that were introduced based on fire history.
- Use 2021 NLCD to remove any areas classified as 95 Emergent Herbaceous Wetlands from the historical grasslands and savannas layer to remove fresh and saltwater marshes. Assign the remaining historical grasslands pixels a value of 1. The output from this step is "historicGrass_notMarsh.tif", which is also used in the grassland and savanna restoration indicator.
- Use conditional statements to combine all rasters together to get the final indicator values, retaining the highest value a pixel can have.
- Create an additional output layer to be used in the grassland and savanna restoration indicator ("testHistoricNotCurrentGrass_selectedPAD_SECAS.tif"). To get this layer, first use Extract by Mask to identify historical grassland and savanna in protected lands that commonly manage grasslands and savannas. This is based on two rasters created earlier in the mapping steps. Then remove areas that are currently potential, likely, or known grasslands. That results in a layer depicting areas that are historically but not currently grassland or savanna and are in protected lands that commonly manage grasslands and savannas.
- Adding zero values helps users better understand the extent of this indicator and makes this indicator layer perform better in online tools. Any pixel within the extent of the HUC12

subwatersheds that does not receive a score from the above steps receives a value of 0 (not identified as grassland/savanna). All pixels outside the extent of the HUC12 subwatersheds (i.e., ocean) are coded as NoData.

- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Known grassland/savanna
6 = Likely grassland/savanna >10 acres
5 = Likely grassland/savanna ≤10 acres
4 = Pollinator buffer around known or likely grassland/savanna
3 = Potential grassland/savanna in grassland/savanna hub
2 = Potential grassland/savanna outside grassland/savanna hub
1 = Historic grassland/savanna
0 = Not identified as grassland/savanna

Known Issues

- Some known grassland and savanna locations are omitted due to confidentiality concerns around potential poaching and disturbance of rare plant species.
- Some important grasslands and savannas on private land and in protected areas managed for multiple use are underprioritized.
- Some small areas of non-grassland and non-savanna are overprioritized due to the coarseness of the 30 m NLCD pixels. This often happens in places where a single pixel overlaps with both a dirt road and dense forest. That overlap makes it look like a savanna. Roadside edges in general are a challenge for this indicator. Roadsides can be important refugia for many grassland and savanna plants, so the indicator does not remove them. However, this also results in many roadsides being overprioritized.
- Some small areas of savanna and sections of powerline rights-of-way are underprioritized due to the coarseness of the 30 m NLCD pixels. This often happens where a single 30 m pixel overlaps with an area where >60% of the pixel looks forested, despite there being significant savanna-like openings in the remaining 40%.
- Some grasslands that have been fully invaded by shrubs other than mesquite are overprioritized.
- This indicator may underestimate the extent of historical grasslands and savannas in the Kentucky portions of the Western Pennyroyal Plain. LANDFIRE predicts grassland and savanna in this area but also includes more extensive areas of wet forests with closed canopies than other sources. Much of this area is now converted to agriculture, making it more difficult to estimate historical conditions.

- This indicator overestimates likely grasslands in some areas in the Everglades Headwaters National Wildlife Refuge proclamation boundary. Many, but not all, pastures in that area function similarly to native grasslands.
- This indicator overestimates likely grasslands in some areas of beach without vegetation (e.g., Santa Rosa Island just west of Destin, FL). Methods to improve this issue in the future are under development.
- Some temporary grasslands in tree plantations, which are probably no longer grasslands now, are still included in the indicator. While the indicator attempts to remove temporary grasslands from tree plantations, a relatively small number of pixels do get missed by the current methods. Improved remote sensing products could fix this issue in the near future.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Antoine C. M., Forrest J. R. K. (2021). Nesting habitat of ground-nesting bees: a review. Ecol. Entomol. 46, 143–159. doi: 10.1111/een.12986. [https://resjournals.onlinelibrary.wiley.com/doi/full/10.1111/een.12986].

Baldwin, H.Q., Allain, L.K., and Hartley, S.B., 2017, Grassland quality and pollinator habitat potential in Southwest Louisiana: U.S. Geological Survey data release. [https://doi.org/10.5066/F76M359X].

Dechant, J. A., M. L. Sondreal, D. H. Johnson, L. D. Igl, C. M. Goldade, M. P. Nenneman, and B. R. Euliss. 1998 (revised 2002). Effects of management practices on grassland birds: Grasshopper Sparrow. Northern Prairie Wildlife Research Center, Jamestown, ND. 28 pages. [https://pubs.usgs.gov/unnumbered/93877/report.pdf].

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

Dewitz, J., 2019, National Land Cover Database (NLCD) 2016 Products (ver. 3.0, November 2023): U.S. Geological Survey data release. [https://doi.org/10.5066/P96HHBIE].

Landscape Fire and Resource Management Planning Tools (LANDFIRE), Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published 2022-09-28. LANDFIRE 2020 Biophysical Settings (BPS) CONUS. LF 2020. Raster digital data. Sioux Falls, SD. [https://landfire.gov/vegetation/bps].

Landscape Fire and Resource Management Planning Tools (LANDFIRE), Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published 2022-09-28. LANDFIRE 2020 Existing Vegetation Type (EVT) CONUS. LF 2020. Raster digital data. Sioux Falls, SD. [https://landfire.gov/vegetation/evt].

Slater, G.L. (2004, October 7). Grasshopper Sparrow *(Ammodramus savannarum)*: a technical conservation assessment. [Online]. USDA Forest Service, Rocky Mountain Region. Accessed February 24, 2024. [https://www.ecoinst.org/wp-content/uploads/2014/03/Slater_2004_GRSP.pdf].

Southeast Conservation Adaptation Strategy. 2021. Recent Trends in Southeastern Ecosystems (2021): Measuring Progress toward the SECAS Goal. [https://secassoutheast.org/pdf/SECAS-goal-report-2021.pdf].

U.S. Department of Agriculture, Natural Resources Conservation Service. 2010. Management considerations for grassland birds in northeastern haylands and pasturelands. Wildlife Insight. Washington, D.C. [https://www.bobolinkproject.com/docs/NRCS_Grassland_leaflet.pdf].

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2022, Protected Areas Database of the United States (PAD-US) 3.0: U.S. Geological Survey data release. [https://doi.org/10.5066/P9Q9LQ4B].

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2024, Protected Areas Database of the United States (PAD-US) 4: U.S. Geological Survey data release. [https://doi.org/10.5066/P96WBCHS].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Zurbuchen, A., Landert, L., Klaiber, J., Mueller, A., Hein, S., Dorn, S., 2010. Maximum foraging ranges in solitary bees: only few individuals have the capability to cover long foraging distances. Biol. Conserv. 143, 669–676. [https://doi.org/10.1016/j.biocon.2009.12.003].

Grassland & Savanna Restoration

This indicator represents potential restoration opportunities for grasslands and savannas within their historical range in the southeastern United States. It considers proximity to hubs of existing grasslands and savannas, protected areas with management potential, landcover classes with good restoration potential, priority areas for restoration of longleaf and shortleaf pine, and historical grassland and savanna locations. This indicator combines data from multiple sources, including the Southeast Blueprint grassland and savanna extent indicator, the Longleaf Suitability Analysis, shortleaf pine suitability, landcover from the National Land Cover Database and LANDFIRE, and more.

This indicator was last updated in November 2025.

Reason for Selection

Native grasslands and savannas are important for many endemic species, provide critical habitat and food for pollinators, and are often hotspots for biodiversity. Once a predominant ecosystem type, grasslands and savannas have significantly declined from their historical extent. In part because of the regular disturbance (e.g., mowing, fire) typically required to maintain high-quality grasslands, they are difficult to detect through remote sensing and are not well-captured by other indicators. In addition, grassland and savanna birds are experiencing significant declines and are currently off-track for meeting the SECAS 10% goal, so it is important that the Blueprint captures known and potential habitat.

Input Data

- National Land Cover Database (NLCD): 2021 Land Cover
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0]; access the data
- Grassland and savanna extent (Southeast Blueprint indicator) - intermediary code output
- Florida Natural Areas Inventory Longleaf Sustainability Analysis v1, accessed 2-23-2024 (LSA_Restoration_Priorities_EqualWeights_v1.tif)

  This dataset combines equally weighted measures of longleaf pine suitability and sustainability to identify areas that are highly suitable for reestablishing longleaf pine. The suitability metric calculates the probability of suitability for longleaf restoration based on the relationship between known longleaf pine occurrences and a suite of environmental variables, such as climate, fire regime, soils, and ecosystem characteristics. The sustainability metric examines landscape connectivity, a measure called landscape integrity that evaluates overall naturalness at multiple scales, climate resilience, and ability to burn (based on proximity to buildings and urban growth projections).

- U.S. Forest Service Shortleaf pine range (Digital Representation of Shortleaf Pine Range Map from “Atlas of United States Trees” by Elbert L. Little, Jr. and other publications), accessed 4-26-2025; access on ArcGIvS Online
- Natural Resources Conservation Service shortleaf pine suitability, accessed 4-14-2025

  This dataset calculates overall suitability for shortleaf pine by relating shortleaf pine occurrences to multiple soil and site properties from the National Soil Information System. It considers temperature, precipitation, slope, aspect, and numerous soil characteristics such as pH, clay content, depth, drainage, and more.

- Southeast Blueprint 2025 extent

Mapping Steps

- Use LANDFIRE EVT to make a layer of agricultural classes with high grassland restoration potential using the EVT Group Name field. Select pixels classified as 'Agricultural-Fallow/Idle Cropland', 'Agricultural-Pasture and Hayland', or 'Agricultural-Wheat'.
- From the Longleaf Suitability Analysis, reclassify the longleaf restoration priorities (LSA_Restoration_Priorities_EqualWeights_v1.tif) to extract the top half of the restoration classes for longleaf.
- From the shortleaf pine potential layer, reclassify any areas without site limitations to 1 and assign all other areas a value of 0. Clip this layer by the shortleaf pine range. Then create a mask using 2021 NLCD landcover to include only upland forest classes (41, 42, or 43) and use it to further limit the shortleaf pine layer to only areas currently in those upland forest types.
- Combine the agricultural classes with restoration potential, the top half of the longleaf restoration priorities, and the refined shortleaf pine layer into a single layer of landcover with good grassland/savanna restoration potential (referred to in the legend as "restorable landcover"). Then remove any areas that weren't historically grassland and savanna using an intermediate input from the Blueprint grassland and savanna extent indicator (historicGrass_notMarsh.tif) that defines historical grassland and savanna areas.
- From the resulting layer, use the ArcGIS Region Group function to identify patches >10 acres.
- Use an intermediate output from the Blueprint grassland and savanna extent indicator (HUC12_knownOrLikely5acPlus.tif) to define grassland and savanna hubs. These are intended to elevate grassland restoration opportunities that are near other established grassland patches. Any HUC12 subwatershed that contain known or likely grassland/savanna patches >5 acres is considered a grassland hub.
- To identify historical grasslands and savannas in protected areas with management potential, use an intermediate output from the grassland and savanna extent indicator (testHistoricNotCurrentGrass_selectedPAD_SECAS.tif). Use Region Group to identify patches > 10 acres.
- Create a known or likely grassland/savanna layer using all classes > 4 in the grassland and savanna extent indicator. Use this to ensure all pixels in those classes are assigned a value of 1 in the final indicator. Because they are already grassland/savanna, they are a lower priority for restoration.
- Create a developed areas mask from the NLCD using only the medium and high density developed classes (43 and 44), which are not easily restorable. Use the mask to make sure medium and high developed areas don't get included in the top 4 indicator classes (highest to medium restoration priority). We do still include low density development and developed open spaces, as these less developed classes are often targets of grassland restoration.
- Use conditional statements to combine all rasters together to get the final indicator values, retaining the highest value a pixel can have.
    - Assign the highest value to protected areas with management potential that fall within the grassland/savanna hubs. Assign a value of 6 to any undeveloped pixel within a grassland/savanna hub AND within a protected area with management potential AND within a historical grassland/savanna patch >10 acres.

    - Next look at other areas within the grassland/savanna hubs. Assign a value of 5 to any pixel in landcover with good restoration potential that is within a grassland/savanna hub AND within a historical grassland/savanna patch >10 acres, but NOT within a protected area with management potential.
    - Now look at protected areas that fall outside the grassland/savanna hubs. Assign a value of 4 to any undeveloped pixel that is within a protected area with management potential AND within a historical grassland/savanna patch >10 acres, but NOT within a grassland/savanna hub.
    - Next look at opportunities that are outside the grassland/savanna hubs and outside the protected areas with management potential. Assign a value of 3 to any pixel in landcover with good restoration potential that is within a historical grassland/savanna patch >10 acres, but NOT within a protected area with management potential or a grassland/savanna hub.
    - Assign a value of 2 to all remaining pixels of restorable grassland landcover that fall within a historical grassland/savanna patch >10 acres.
    - Assign a value of 1 to all known or likely current grasslands.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Highest restoration priority (in a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
5 = Very high restoration priority (in a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
4 = High restoration priority (outside a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
3 = Medium restoration priority (outside a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
2 = Low restoration priority (other historic grassland/savanna)
1 = Very low restoration priority (already known or likely grassland/savanna)
0 = Lowest restoration priority (not identified as historic grassland/savanna)

Known Issues

- Savanna restoration potential is underprioritized in forests without historical longleaf or shortleaf pine.
- Restoration potential is underprioritized in the Arid West region of Texas and Oklahoma.
- This indicator underprioritizes some important grassland and savanna restoration opportunities on private land and in protected areas managed for multiple use.

- This indicator may underestimate the extent of historical grasslands and savannas in the Kentucky portions of the Western Pennyroyal Plain. LANDFIRE predicts grassland and savanna in this area but also includes more extensive areas of wet forests with closed canopies than other sources. Much of this area is now converted to agriculture, making it more difficult to estimate historical conditions.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Antoine C. M., Forrest J. R. K. (2021). Nesting habitat of ground-nesting bees: a review. Ecol. Entomol. 46, 143–159. doi: 10.1111/een.12986. [https://resjournals.onlinelibrary.wiley.com/doi/full/10.1111/een.12986].

Dechant, J. A., M. L. Sondreal, D. H. Johnson, L. D. Igl, C. M. Goldade, M. P. Nenneman, and B. R. Euliss. 1998 (revised 2002). Effects of management practices on grassland birds: Grasshopper Sparrow. Northern Prairie Wildlife Research Center, Jamestown, ND. 28 pages. [https://pubs.usgs.gov/unnumbered/93877/report.pdf].

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

Florida Natural Areas Inventory and University of Florida Center for Landscape Conservation Planning. 2023. Longleaf Pine Sustainability Analysis Version 1 Technical Report. Tallahassee, FL.

Landscape Fire and Resource Management Planning Tools (LANDFIRE), Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published 2022-09-28. LANDFIRE 2020 Biophysical Settings (BPS) CONUS. LF 2020. Raster digital data. Sioux Falls, SD. [https://www.landfire.gov/bps.php].

Reid, Richard. Natural Resources Conservation Service. Shortleaf Pine Suitability. [https://tiles.arcgis.com/tiles/SXbDpmb7xQkk44JV/arcgis/rest/services/FOR_Shortleaf_Pine_Suitability_R/MapServer?cacheKey=9ec3894970528105].

Slater, G.L. (2004, October 7). Grasshopper Sparrow *(Ammodramus savannarum)*: a technical conservation assessment. [Online]. USDA Forest Service, Rocky Mountain Region. Accessed February 24, 2024. [https://www.ecoinst.org/wp-content/uploads/2014/03/Slater_2004_GRSP.pdf].

Southeast Conservation Adaptation Strategy. 2021. Recent Trends in Southeastern Ecosystems (2021): Measuring Progress toward the SECAS Goal. [https://secassoutheast.org/pdf/SECAS-goal-report-2021.pdf].

U.S. Department of Agriculture, Natural Resources Conservation Service. 2010. Management considerations for grassland birds in northeastern haylands and pasturelands. Wildlife Insight. Washington, D.C. [https://www.bobolinkproject.com/docs/NRCS_Grassland_leaflet.pdf].

U.S. Forest Service. Shortleaf pine range. Digital Representation of Shortleaf Pine Range Map from "Atlas of United States Trees" by Elbert L. Little, Jr. and other publications. Accessed 4-26-202 [https://services1.arcgis.com/gGHDlz6USftL5Pau/arcgis/rest/services/shortleaf_pine_range/FeatureServer/0].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Zurbuchen, A., Landert, L., Klaiber, J., Mueller, A., Hein, S., Dorn, S., 2010. Maximum foraging ranges in solitary bees: only few individuals have the capability to cover long foraging distances. Biol. Conserv. 143, 669–676. [https://doi.org/10.1016/j.biocon.2009.12.003].

Greenways & Trails

This cultural resource indicator measures both the natural condition and connected length of greenways and trails to characterize the quality of the recreational experience. Natural condition is based on the amount of impervious surface surrounding the path. Connected length captures how far a person can go without leaving a dedicated path, based on common distances for walking, running, and biking. This indicator originates from OpenStreetMap data and the National Land Cover Database.

This indicator was last updated in October 2023.

Reason for Selection

This indicator captures the recreational value and opportunities to connect with nature provided by greenways and trails. Greenways and trails provide many well-established social and economic benefits ranging from improving human health, reducing traffic congestion and air and noise pollution, increasing property values, and generating new jobs and business revenue (ITRE 2018). The locations of greenways and trails are regularly updated through the open-source database OpenStreetMap, while data on condition are regularly updated through the National Land Cover Database (NLCD).

Input Data

- Base Blueprint 2022 extent
- OpenStreetMap data "roads" layer, accessed 2-27-2023

A line from this dataset is considered a potential greenway/trail if the value in the "fclass" attribute is either bridleway, cycleway, footway, or path. OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- 2019 National Land Cover Database (NLCD): Percent developed imperviousness
- Southeast Blueprint 2023 extent

Mapping Steps

The greenways and trails indicator score reflects both the natural condition and connected length of the greenway/trail.

Natural condition

Natural condition is based on the amount of impervious surface surrounding the greenway/trail. Since perceptions of a greenway's "naturalness" are influenced both by the immediate surroundings adjacent to the path, and the greater viewshed, natural condition is calculated by averaging two measurements: local impervious and nearby impervious.

Local impervious is defined as the percent impervious surface of the 30 m pixel that intersects the trail. Nearby impervious is defined as the average impervious surface within a 300 m radius circle surrounding the path (note: along a 300 m stretch of trail, we only count the impervious surface within a 45 m buffer on either side of the trail, since pixels nearer the trail have a bigger impact on the greenway/trail experience). The natural classes are defined as follows:

3 = Mostly natural: average of local and nearby impervious is ≤1%
2 = Partly natural: average of local and nearby impervious is >1 and <10%
1 = Developed: average of local and nearby impervious is ≥10%

Connected length

The connected length of the path is calculated using the entire extent of the potential greenways/trails dataset. A trail is considered connected to another trail if it is within 2 m of the other trail. Length thresholds are defined by typical lengths of three common recreational greenway activities: walking, running, and biking. The 40 km threshold for biking is based on the standard triathlon biking segment of 40 km (~25 mi). Because a 5K is the most common road race distance, the running threshold is set at 5 km (~3.1 mi) (Running USA 2017). The 1.9 km (1.2 mi) walking threshold is based on the average walking trip on a summer day (U.S. DOT 2002).

- Using the statistics software R, download the OpenStreetMap data for the continental Southeast area.
- Select all lines from the OpenStreetMap data that have a highway tag of either footway, cycleway, bridleway, or path. These are all considered potential trails.
- Removed all lines marked as private.

- Identify lines from the potential trails that are tagged as sidewalks. Assign them a value of 1 in the indicator.

Final scores

If the potential greenway/trail was tagged as a sidewalk in the "other tags" field, it is given a value of 1 to separate sidewalks from what most people think of as a trail or greenway. If a pixel does not intersect a potential greenway/trail but overlaps with a value that is not NoData in the 2019 NLCD impervious surface layer, it is coded with a value of 0. Then clip to the spatial extent of Base Blueprint 2022. As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Table 1. Greenways and trails indicator scoring.

		Trail length				**Sidewalk**
		L ≥ 40 km	40 km > L ≥ 5 km	5 km > L ≥ 1.9 km	L < 1.9 km	
Average of local and nearby impervious	Imp ≤ 1%	7	6	5	4	1
	1% < Imp < 10%	6	5	4	3	
	Imp ≥ 10%	5	4	3	2	
Sidewalk		1				

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Mostly natural and connected for ≥40 km
6 = Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km
5 = Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km
4 = Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km
3 = Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km
2 = Developed and connected for <1.9 km
1 = Sidewalk
0 = Not identified as trail, sidewalk, or other path

Known Issues

- This indicator sometimes misclassifies sidewalks as greenways and trails because they are not tagged as sidewalks in the OpenStreetMap data.
- This indicator occasionally misclassifies driveways as sidewalks in places where they are not correctly tagged as private in OpenStreetMap. These typically appear as isolated pixels receiving a score of 1 on the indicator.
- OpenStreetMap does not provide a complete inventory of greenways and trails in the Southeast. Paths that are missing from the source data will be underprioritized in this indicator. For example, some trails are missing within National Wildlife Refuges.
- When calculating nearby impervious for one greenway, if there's another greenway within 300 m, impervious surface from the different but overlapping greenway buffer area is also used to compute natural condition. This is an unintended issue with the analysis methods. Investigation into potential fixes is ongoing.
- This indicator includes trails and sidewalks from OpenStreetMap, which is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the path of a greenway) or incorrect tags (e.g., mislabeling a path as a footway that is actually a road for vehicles). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new greenways and trails to improve the accuracy and coverage of this indicator in the future.
- This indicator sometimes underestimates greenway length when connections route under bridges or along abandoned dirt roads. Some of these issues have been fixed through active testing and improvement, but some likely remain.
- The indicator doesn't currently include areas where future greenways are planned.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

American Planning Association. 2018. Recommendations for Future Enhancements to the Blueprint. [https://secassoutheast.org/pdf/Recommendations-for-Future-Enhancements-to-the-Blueprint-FINAL.pdf].

Institute for Transportation Research and Education (ITRE) & Alta Planning and Design. February 2018. Evaluating the Economic Impact of Shared Use Paths in North Carolina: 2015-2017 Final Report. [https://itre.ncsu.edu/wp-content/uploads/2018/03/NCDOT-2015-44_SUP-Project_Final-Report_optimized.pdf].

OpenStreetMap. Highways. Data extracted through Geofabrik downloads. Accessed February 23, 2022. [https://wiki.openstreetmap.org/wiki/Highways].

Running USA. 23 March 2017. U.S. Road Race Trends. Road race finisher total experiences slight year-over-year decline in 2016. [https://web.archive.org/web/20170404232619/https://www.runningusa.org/2017-us-road-race-trends].

U.S. Department of Transportation. National Highway Traffic Safety Administration and the Bureau of Transportation Statistics. 2002. National Survey of Pedestrian & Bicyclist Attitudes and Behaviors: Highlights Report. [https://www.bts.gov/sites/bts.dot.gov/files/docs/browse-statistical-products-and-data/bts-publications/archive/203331/entire-1.pdf].

U.S. Geological Survey (USGS). Published June 2021. National Land Cover Database (NLCD) 2019 Land Cover Conterminous United States. Sioux Falls, SD. [https://doi.org/10.5066/P9KZCM54].

Yang, Limin, Jin, Suming, Danielson, Patrick, Homer, Collin G., Gass, L., Bender, S.M., Case, Adam, Costello, C., Dewitz, Jon A., Fry, Joyce A., Funk, M., Granneman, Brian J., Liknes, G.C., Rigge, Matthew B., Xian, George. 2018. A new generation of the United States National Land Cover Database—Requirements, research priorities, design, and implementation strategies: ISPRS Journal of Photogrammetry and Remote Sensing, v. 146, p. 108–123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Imperiled Amphibians & Reptiles

This indicator identifies potential habitat to support amphibian and reptile Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. This indicator uses 190 species and subspecies models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.

This indicator was last updated in October 2026.

Reason for Selection

The Southeast is an important region for amphibian and reptile conservation. For example, the North American Coastal Plain is considered a biodiversity hotspot (CEPF 2016) and is considered an area of exceptional endemism (Noss et al. 2013) with approximately half of North American native herpetofauna occurring in the Southeast (Tuberville et al. 2005). Additionally, many reptile and amphibian species rely on a mix of aquatic and upland habitat types and are often associated with better habitat connectivity, habitat diversity, and water quality (Blaustein and Kiesecker 2002).

The species used in this indicator come from the Southeast animal Regional Species of Greatest Conservation Need (RSGCN) list. For their Wildlife Action Plans, each state and territorial fish and wildlife agency identifies a list of species most in need of conservation action, known as Species of Greatest Conservation Need (SGCN). Using the SGCN lists developed for the 2015 Wildlife Action Plans,

the Southeast Association of Fish and Wildlife Agencies' (SEAFWA) Wildlife Diversity Committee (WDC) created the RSGCN list to identify regional priority species. The Southeast animal RSGCN list includes 960 species representing amphibians, birds, bumblebees, fish, invertebrates (crayfish and mussels), mammals, and reptiles. In addition, the SEAFWA WDC ranked each species by degree of conservation concern (very high, high, or moderate concern).

This indicator identifies potential habitat for 105 amphibian and 85 reptile species and subspecies on the RSGCN list. While all RSGCN warrant consideration, this indicator emphasizes places that support potential habitat for range-restricted species and narrow endemics that only occur in very small and often singular places, making them more vulnerable to extinction. For example, the only known habitat for the Austin blind salamander (*Eurycea waterlooensis*) is found within Barton Springs, fed by the Edwards Aquifer in Texas. In contrast, the green salamander (*Aneides aeneus*) has a much broader range and is found in nine states within the Southeast region.

To highlight the most important places for range-restricted species, we used a measure called "rarity-weighted richness" that measures the importance of each place to the species that occur there (IUCN 2025). With this approach, a place important for just one particularly rare species may score more highly than another place where multiple common species occur. Rarity-weighted richness is complementary to species richness, which measures the total number of species that occur in a given location.

Input Data

- 2024 Annual National Land Cover Database (NLCD): Land cover collection 1.1
- Southeast Blueprint 2026 extent
- Elevation-based mask used to separate the marine environment from terrestrial areas: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- U.S. Geological Survey (USGS) Southeast Gap Analysis Project (GAP) species distribution models for amphibian and reptile RSGCN, provided by Dr. Steve Williams on December 18, 2025 (available on request by emailing sgwillia@ncsu.edu)

  The Southeast Gap Analysis Project (SE-GAP) is part of the National GAP, which is sponsored by the USGS. "Gap analysis" is the science of examining how well we are protecting common plants and animals (USGS 2025). Using known species ranges and habitat relationships, SE-GAP models the distributions of terrestrial vertebrate species that breed in the Southeast or use habitat there for an important part of their life history. The original SE-GAP models used 2001 landcover, and they have recently been updated to 2023 landcover. This indicator uses the latest SE-GAP models for 105 amphibian and 85 reptile species and subspecies on the RSGCN list.

A full list of all species included appears in the Full Species List section below.

Mapping Steps

- Estimate rarity-weighted richness. This means the smaller the area of potential habitat for the species, the higher the weight. Conceptually, to determine the weight for a particular species, we look at the proportion of all the potential habitat for a species that's represented by a single 30 m pixel. However, there are species with very large ranges and some with very small ranges, which gives ArcGIS a hard time. So, we tweak the general idea of just calculating 1 pixel/(number of pixels of potential habitat). To deal with large areas of potential habitat, we use 10,000/(number of pixels of potential habitat). If we just use 1, some of the resulting decimals are so small that ArcGIS just rounds them down to zero. To deal with species with very small ranges, we set the maximum weighted value of a species to 1. As a final step, we sum the weights for all species to get the total rarity-weighted richness for a given pixel.
- Clip to clip to the spatial extent of Southeast Blueprint 2026.
- Use the elevation-based mask raster to remove marine areas from the rarity-weighted richness layer by assigning them a value of NoData.
- Reclassify the NLCD to create a developed areas mask. Assign all developed classes a value of 0, and all other classes a value of 1. Developed classes include "Developed, Open Space", "Developed, Low Intensity", "Developed, Medium Intensity", and "Developed, High Intensity".
- To remove developed areas, multiply the amphibian and reptile rarity-weighted richness layer (with marine areas removed) by the developed mask. While there may be a limited amount of potential habitat for some species in the lower intensity urban classes, GAP tends to greatly overpredict potential habitat for species in urban areas.
- Reclassify the resulting raster into the indicator classes seen below. The ranges of legend values are of the lowest value of the class (e.g., value 5 includes exactly 5%; value 4 includes exactly 10% but not exactly 5%; value 3 includes exactly 20% but not exactly 10%).
- As a final step, clip to the spatial extent of Southeast Blueprint 2026.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Highest importance for range-restricted species (top 5%)
4 = Very high importance for range-restricted species (upper 5-10%)
3 = High importance for range-restricted species (upper 10-20%)
2 = Medium importance for range-restricted species (above average)
1 = Low importance for range-restricted species (below average)
0 = Developed landcover or no potential habitat

Known Issues

- Misclassification underestimates some potential habitat in restoration areas that were previously row crop (e.g., large areas of Catahoula National Wildlife Refuge in LA).
- GAP provides the following disclaimer for their predicted habitat maps: "While the actual distributions of many species are likely to be habitat limited, suitable habitat will not always be occupied because of population dynamics and species interactions. Furthermore, these maps correspond to midscale characterizations of landscapes, but individual animals may deem areas to be unsuitable because of presence or absence of fine-scale features and characteristics that are not represented in our models (e.g. snags, vernal pools, shrubby undergrowth)."
- GAP species-habitat distribution maps predict suitable landcover and environmental conditions for individual species based on their ranges and life history requirements. This data tends to overpredict habitat and cannot always separate development from potential habitat.
- This indicator undervalues habitat for a small number of subspecies with narrow ranges where the GAP model covers the full species range and is not limited to the narrow range of that particular subspecies. We are working to limit those specific GAP models to a closer approximation of subspecies ranges for future indicator updates.
- Because this indicator is intended to represent terrestrial and freshwater ecosystems, we attempted to remove any predicted amphibian and reptile habitat in the marine environment. However, in aquatic systems, it is difficult to determine where freshwater ends and saltwater begins, as that transition is highly dynamic and influenced by many factors including topography, instream flow, tides, storms, and more. To define the extent of this indicator, we used an elevation-based approach that includes all areas greater than -1 m in elevation (i.e. all areas above 1 m below sea level), then backfilled to capture low-lying coastal areas that are disconnected from the ocean. We intentionally erred on the side of including more terrestrial and freshwater areas so as not to exclude small islands and coastal freshwater lakes (like Lake Mattamuskeet in NC). This is an imperfect methodology but produced better results than some of the other alternatives we explored.

Full Species List

Reptiles

Table 2. List of reptile RSGCN included in this indicator.

	Scientific Name	**Common Name**
1	*Apalone mutica*	Smooth softshell turtle
2	*Caretta caretta*	Atlantic loggerhead sea turtle
3	*Cemophora lineri*	Texas scarlet snake
4	*Chelonia mydas*	Atlantic green sea turtle
5	*Chrysemys dorsalis*	Southern painted turtle
6	*Clemmys guttata*	Spotted turtle
7	*Crocodylus acutus*	American crocodile

	Scientific Name	Common Name
8	*Crotalus adamanteus*	Eastern diamond-backed rattlesnake
9	*Crotalus horridus*	Timber rattlesnake
10	*Crotaphytus reticulatus*	Reticulate collared lizard
11	*Deirochelys reticularia* (including *miaria* & *reticularia)*	Chicken turtle (Including Eastern & Western)
12	*Dermochelys coriacea*	Atlantic leatherback sea turtle
13	*Diadophis punctatus acricus*	Key ringneck snake
14	*Drymarchon couperi*	Eastern indigo snake
15	*Drymarchon melanurus erebennus*	Texas indigo snake
16	*Eretmochelys imbricata imbricata*	Atlantic hawksbill sea turtle
17	*Eumeces septentrionalis obtusirostris*	Southern prairie skink
18	*Farancia abacura* (including *reinwardtii)*	Mudsnake (including Western)
19	*Farancia erytrogramma* (including *erytrogramma)*	Rainbow snake (including common)
20	*Glyptemys muhlenbergii*	Bog turtle (Southern Distinct Pop. Segment)
21	*Gopherus berlandieri*	Texas tortoise
22	*Gopherus polyphemus*	Gopher tortoise
23	*Graptemys barbouri*	Barbour's map turtle
24	*Graptemys caglei*	Cagle's map turtle
25	*Graptemys ernsti*	Escambia map turtle
26	*Graptemys flavimaculata*	Yellow-blotched map turtle
27	*Graptemys gibbonsi*	Pascagoula map turtle
28	*Graptemys nigrinoda*	Black-knobbed map turtle
29	*Graptemys oculifera*	Ringed map turtle
30	*Graptemys ouachitensis*	Ouachita map turtle
31	*Graptemys pearlensis*	Pearl River map turtle
32	*Graptemys pulchra*	Alabama map turtle
33	*Graptemys sabinensis*	Sabine map turtle
34	*Graptemys versa*	Texas map turtle
35	*Heterodon gloydi*	Dusty hog-nosed snake
36	*Heterodon simus*	Southern hognose snake
37	*Holbrookia lacerata lacerata*	Plateau earless lizard
38	*Holbrookia lacerata subcaudalis*	Southern earless lizard
39	*Kinosternon baurii*	Striped mud turtle
40	*Lampropeltis extenuata*	Short-tailed snake
41	*Lampropeltis meansi*	Apalachicola kingsnake
42	*Lepidochelys kempii*	Kemp's Atlantic ridley sea turtle
43	*Liodytes pygaea* (including *cyclas* & *paludis)*	Black swampsnake (Including Southern Florida & Carolina)

	Scientific Name	Common Name
44	*Liodytes rigida* (including *deltae* & *sinicola)*	Glossy swampsnake (Including Delta & Gulf crayfish)
45	*Macrochelys suwanniensis*	Suwannee alligator snapping turtle
46	*Macrochelys temminckii*	Alligator snapping turtle
47	*Malaclemys terrapin* (including *terrapin, centrata, littoralis, macrospilota, pileata, rhizophorarum* & *tequesta)*	Diamondback terrapin (including Carolina, Texas, Mississippi, ornate, mangrove, Northern & Eastern Florida)
48	*Micrurus fulvius*	Eastern coral snake
49	*Nerodia clarkii* (including *clarkii, compressicauda* & *taeniata)*	Saltmarsh snake (including Gulf, mangrove & Atlantic)
50	*Nerodia floridana*	Florida green watersnake
51	*Nerodia harteri*	Brazos water snake
52	*Nerodia paucimaculata*	Concho water snake
53	*Nerodia sipedon williamengelsi*	Carolina water snake
54	*Ophisaurus attenuatus* (including *longicaudus)*	Slender glass lizard (including eastern)
55	*Ophisaurus compressus*	Island glass lizard
56	*Ophisaurus mimicus*	Mimic glass lizard
57	*Ophisaurus ventralis*	Eastern glass lizard
58	*Phrynosoma cornutum*	Texas horned lizard
59	*Pituophis melanoleucus lodingi*	Black pinesnake
60	*Pituophis melanoleucus melanoleucus*	Northern pine snake
61	*Pituophis melanoleucus mugitus*	Florida pine snake
62	*Pituophis ruthveni*	Louisiana pinesnake
63	*Plestiodon anthracinus* (including *anthracinus* & *pluvialis)*	Coal skink (including Northern & Southern)
64	*Plestiodon egregius* (including *egregius, insularis, lividus* & *onocrepis)*	Florida Keys mole skink, Cedar Key mole skink, blue-tailed mole skink, peninsula mole skink
65	*Plestiodon reynoldsi*	Florida sand skink
66	*Pseudemys alabamensis*	Alabama red-bellied turtle
67	*Pseudemys gorzugi*	Rio Grande cooter
68	*Pseudemys rubriventris*	Northern red-bellied cooter
69	*Rhadinaea flavilata*	Pine woods littersnake
70	*Rhineura floridana*	Florida wormlizard
71	*Sceloporus arenicolus*	Dunes sagebrush lizard
72	*Sceloporus woodi*	Florida scrub lizard
73	*Sistrurus miliarius* (including *miliarius* & *streckeri)*	Pygmy rattlesnake (Including Carolina & Western)
74	*Sistrurus tergeminus*	Western massasauga
75	*Sternotherus carinatus*	Razor-backed musk turtle

	Scientific Name	Common Name
76	*Sternotherus depressus*	Flattened musk turtle
77	*Sternotherus minor peltifer*	Stripe-necked musk turtle
78	*Storeria victa*	Florida brown snake (Lower Keys population)
79	*Tantilla cucullata*	Big Bend blackheaded snake
80	*Tantilla oolitica*	Rim rock crowned snake
81	*Tantilla relicta pamlica*	Coastal dunes crowned snake
82	*Terrapene carolina* (including *carolina, major* & *triunguis)*	Eastern box turtle (Including Eastern, Gulf Coast & three-toed)
83	*Terrapene ornata*	Ornate box turtle
84	*Trachemys gaigeae*	Big Bend slider
85	*Trachemys scripta troostii*	Cumberland slider

Amphibians

Table 3. List of amphibian RSGCN included in this indicator.

	Scientific Name	Common Name
1	*Ambystoma annulatum*	Ringed salamander
2	*Ambystoma barbouri*	Streamside salamander
3	*Ambystoma bishopi*	Reticulated flatwoods salamander
4	*Ambystoma cingulatum*	Flatwoods salamander (frosted)
5	*Ambystoma mabeei*	Mabee's salamander
6	*Amphiuma pholeter*	One-toed amphiuma
7	*Amphiuma tridactylum*	Three-toed amphiuma
8	*Anaxyrus (Bufo) houstonensis*	Houston toad
9	*Anaxyrus quercicus*	Oak toad
10	*Aneides aeneus*	Green salamander
11	*Cryptobranchus alleganiensis* (including *alleganiensis* & *bishopi)*	Hellbender (including Eastern & Ozark)
12	*Desmognathus abditus*	Cumberland dusky salamander
13	*Desmognathus aeneus*	Seepage salamander
14	*Desmognathus apalachicolae*	Apalachicola dusky salamander
15	*Desmognathus auriculatus*	Southern dusky salamander
16	*Desmognathus brimleyorum*	Ouachita dusky salamander
17	*Desmognathus folkertsi*	Dwarf black-bellied salamander
18	*Desmognathus imitator* (including pop. 1)	Imitator salamander (Including Waterrock Knob pop.)
19	*Desmognathus marmoratus*	Shovel-nosed salamander
20	*Desmognathus orestes*	Blue Ridge dusky salamander
21	*Desmognathus organi*	Northern pygmy salamander
22	*Desmognathus santeetlah*	Santeetlah dusky salamander
23	*Desmognathus valentinei*	Valentine's southern dusky salamander

	Scientific Name	Common Name
24	*Desmognathus welteri*	Black Mountain salamander
25	*Desmognathus wrighti*	Pygmy salamander
26	*Eurycea aquatica*	Brown-backed salamander
27	*Eurycea chamberlaini*	Chamberlain's dwarf salamander
28	*Eurycea chisolmensis*	Salado Springs salamander
29	*Eurycea hillisi*	Hillis's dwarf salamander
30	*Eurycea junaluska*	Junaluska salamander
31	*Eurycea latitans*	Cascade Caverns salamander
32	*Eurycea nana*	San Marcos salamander
33	*Eurycea naufragia*	Georgetown salamander
34	*Eurycea neotenes*	Texas salamander
35	*Eurycea paludicola*	Western dwarf salamander
36	*Eurycea pterophila*	Fern bank (Blanco River springs) salamander
37	*Eurycea rathbuni*	Texas blind salamander
38	*Eurycea robusta*	Blanco blind salamander
39	*Eurycea sosorum*	Barton springs salamander
40	*Eurycea sphagnicola*	Bog dwarf salamander
41	*Eurycea ssp.* (including *braggi, nerea* & *spelaea)*	Grotto salamander (Including Eastern, Western & Northern)
42	*Eurycea subfluvicola*	Ouachita streambed salamander
43	*Eurycea tonkawae*	Jollyville Plateau salamander
44	*Eurycea tynerensis*	Oklahoma salamander
45	*Eurycea wallacei*	Georgia blind salamander
46	*Eurycea waterlooensis*	Austin blind salamander
47	*Eurycea wilderae*	Blue Ridge two-lined salamander
48	*Gastrophryne olivacea*	Western narrowmouth toad
49	*Gyrinophilus gulolineatus*	Berry cave salamander
50	*Gyrinophilus palleucus* (including *necturoides* & *palleucus)*	Tennessee cave salamander (Including Big Mouth cave & pale)
51	*Gyrinophilus subterraneus*	West Virginia spring salamander
52	*Hyla andersonii*	Pine barrens treefrog
53	*Hyla gratiosa*	Barking treefrog
54	*Lithobates areolatus* (including *areolatus* & *circulosus)*	Southern crawfish frog (Including southern & northern)
55	*Lithobates capito*	Gopher frog
56	*Lithobates heckscheri*	River frog
57	*Lithobates okaloosae*	Florida bog frog
58	*Lithobates sevosus*	Mississippi gopher frog
59	*Necturus alabamensis*	Black warrior waterdog
60	*Necturus beyeri*	Gulf Coast waterdog

	Scientific Name	Common Name
61	*Necturus lewisi*	Neuse River waterdog
62	*Necturus maculosus louisianensis*	Red River mudpuppy
63	*Necturus punctatus*	Dwarf waterdog
64	*Notophthalmus meridionalis*	Black-spotted newt
65	*Notophthalmus perstriatus*	Striped newt
66	*Phaeognathus hubrichti*	Red Hills salamander
67	*Plethodon amplus*	Blue Ridge gray-cheeked salamander
68	*Plethodon aureolus*	Tellico salamander
69	*Plethodon caddoensis*	Caddo Mountain salamander
70	*Plethodon chattahoochee*	Chattahoochee slimy salamander
71	*Plethodon cheoah*	Cheoah Bald salamander
72	*Plethodon fourchensis*	Fourche Mountain salamander
73	*Plethodon hubrichti*	Peaks of Otter salamander
74	*Plethodon jordani*	Jordan's salamander
75	*Plethodon kentucki*	Cumberland Plateau salamander
76	*Plethodon kiamichi*	Kiamichi slimy salamander
77	*Plethodon kisatchie*	Louisiana slimy salamander
78	*Plethodon meridianus*	South Mountain gray-cheeked salamander
79	*Plethodon nettingi*	Cheat Mountain salamander
80	*Plethodon ouachitae*	Rich Mountain salamander
81	*Plethodon petraeus*	Pigeon Mountain salamander
82	*Plethodon punctatus*	Cow Knob salamander
83	*Plethodon richmondi*	Ravine salamander
84	*Plethodon savannah*	Savannah slimy salamander
85	*Plethodon sequoyah*	Sequoyah slimy salamander
86	*Plethodon serratus*	Southern red-backed salamander
87	*Plethodon shenandoah*	Shenandoah salamander
88	*Plethodon shermani*	Red-legged salamander
89	*Plethodon teyahalee (=P. oconaluftee)*	Southern Appalachian salamander
90	*Plethodon ventralis*	Southern zigzag salamander
91	*Plethodon virginia*	Shenandoah Mountain salamander
92	*Plethodon websteri*	Webster's salamander
93	*Plethodon wehrlei*	Wehrle's salamander
94	*Plethodon welleri*	Weller's salamander
95	*Plethodon yonahlossee* (including pop. 1)	Yonahlossee salamander
96	*Pseudacris illinoensis*	Illinois chorus frog
97	*Pseudacris ocularis*	Little grass frog
98	*Pseudacris ornata*	Ornate chorus frog
99	*Pseudacris streckeri*	Strecker's chorus frog

	Scientific Name	Common Name
100	*Pseudobranchus striatus* (including *striatus* & *lustricolus)*	Dwarf siren (Including broad-striped & Gulf hammock)
101	*Pseudotriton montanus* (including *flavissimus* & *montanus)*	Mud salamander (Including Gulf Coast & Eastern)
102	*Scaphiopus hurterii*	Hurter's spadefoot
103	*Siren lacertina*	Greater siren
104	*Stereochilus marginatus*	Many-lined salamander
105	*Urspelerpes brucei*	Patch-nosed salamander

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Blaustein, A.R. & Kiesecker, J.M. (2002). Complexity in conservation: lessons from the global decline of amphibian populations. Ecol. Lett., 5, 597–608. [https://onlinelibrary.wiley.com/doi/full/10.1046/j.1461-0248.2002.00352.x].

Critical Ecosystem Partnership Fund (CEPF). 2016. Accessed March 31, 2025. [https://www.cepf.net/our-work/biodiversity-hotspots/north-american-coastal-plain].

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

International Union for Conservation of Nature (IUCN). Other Spatial Downloads. The IUCN Red List of Threatened Species. Accessed April 10, 2025. [https://www.iucnredlist.org/resources/other-spatial-downloads].

Noss, Reed & Platt, William & Sorrie, Bruce & Weakley, Alan & Peet, Robert & Costanza, Jennifer. (2013). How a global biodiversity hotspot, the North American Coastal Plain, went unrecognized.

Tracey D. Tuberville, John D. Willson, Michael E. Dorcas, and J. Whitfield Gibbons "Herpetofaunal Species Richness of Southeastern National Parks," *Southeastern Naturalist* 4(3), 537-569, (1 September 2005). [https://doi.org/10.1656/1528-7092(2005)004[0537:HSROSN]2.0.CO;2].

U.S. Geological Survey. Gap Analysis Project. Accessed February 25, 2025. [https://www.usgs.gov/programs/gap-analysis-project].

Imperiled Mammals

This indicator identifies potential habitat to support mammal Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. This indicator uses 58 species models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.

This indicator was last updated in October 2026.

Reason for Selection

Mammals play key roles in maintaining ecosystem integrity, forest health, productivity, pollination, and other ecosystem services within the Southeast (Trani et al. 2007). As a taxonomic group, mammals are highly diverse, spanning a broad range of sizes, physiologies, life histories, and behaviors. They inhabit a wide array of ecosystems and habitats throughout the region, and some mammal species—such as white-tailed deer (Odocoileus virginianus)—are deeply rooted in the region's cultural heritage.

The species used in this indicator come from the Southeast animal Regional Species of Greatest Conservation Need (RSGCN) list. For their Wildlife Action Plans, each state and territorial fish and wildlife agency identifies a list of species most in need of conservation action, known as Species of Greatest Conservation Need (SGCN). Using the SGCN lists developed for the 2015 Wildlife Action Plans, the Southeast Association of Fish and Wildlife Agencies' (SEAFWA) Wildlife Diversity Committee (WDC) created the RSGCN list to identify regional priority species. The Southeast animal RSGCN list includes 960 species representing amphibians, birds, bumblebees, fish, invertebrates (crayfish and mussels), mammals, and reptiles. In addition, the SEAFWA WDC ranked each species by degree of conservation concern (very high, high, or moderate concern).

This indicator identifies potential habitat for 58 out of 60 mammal species on the RSGCN list. While all RSGCN warrant consideration, this indicator emphasizes places that support potential habitat for range-restricted species and narrow endemics that only occur in very small and often singular places, making them more vulnerable to extinction. For example, the Buxton Woods white-footed deermouse (Peromyscus leucopus buxtoni) only occurs on the southern end of Hatteras Island in North Carolina (Shipp-Pennock et al. 2005). In contrast, the Northern long-eared bat (Myotis septentrionalis) has a much broader range and is found in 13 states within the Southeast region.

To highlight the most important places for range-restricted species, we used a measure called "rarity-weighted richness" that measures the importance of each place to the species that occur there (IUCN 2025). With this approach, a place important for just one particularly rare species may score more highly than another place where multiple common species occur. Rarity-weighted richness is complementary to species richness, which measures the total number of species that occur in a given location.

Input Data

- 2024 Annual National Land Cover Database (NLCD): Land cover collection 1.1
- Elevation-based mask used to separate the marine environment from terrestrial areas: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- USGS Global Islands data based on 2024 Landsat imagery, released 03-01-2023
- Southeast Blueprint 2026 extent
- U.S. Geological Survey (USGS) Southeast Gap Analysis Project (GAP) species distribution models for mammal RSGCN, provided by Dr. Steve Williams on December 18, 2025 (available on request by emailing sgwillia@ncsu.edu)

  The Southeast Gap Analysis Project (SE-GAP) is part of the National GAP, which is sponsored by the USGS. "Gap analysis" is the science of examining how well we are protecting common plants and animals (USGS 2025). Using known species ranges and habitat relationships, SE-GAP models the distributions of terrestrial vertebrate species that breed in the Southeast or use habitat there for an important part of their life history. The original SE-GAP models used 2001 landcover, and they have recently been updated to 2023 landcover. This indicator uses the latest SE-GAP models (65 species-habitat models) for 58 mammal species on the RSGCN list out of 60 mammal RSGCN identified (does not include marine mammals).

  A full list of all species included appears in the Full Species List section below.

Mapping Steps

- Using island name ("Name_USGSO"), extract boundaries for Cape Hatteras ("Hatteras"), Sanibel ("Sanibel"), and Captiva ("North Captiva") Islands from the USGS island boundary data for big islands ("USGSEsriWCMC_GlobalIslandsv2_BigIslands").
- Limit the potential habitat for hispid cotton rat in the GAP data to only the subspecies range for insular hispid cotton rat using boundaries for Sanibel and Captiva Islands, since only this subspecies is an RSGCN. Subtract the potential habitat value of 1 from the mammal richness layer for all hispid cotton rat potential habitat outside the subspecies' smaller range.
- Limit the potential habitat for white-footed deer mouse to only the subspecies range for the Buxton white-footed deer mouse using boundaries for Cape Hatteras Island, since only this subspecies is an RSGCN. Subtract the potential habitat value of 1 from the mammal richness layer for all white-footed deer mouse potential habitat outside the subspecies' smaller range.
- Estimate rarity-weighted richness. This means the smaller the area of potential habitat for the species, the higher the weight. Conceptually, to determine the weight for a particular species, we look at the proportion of all the potential habitat for a species that's represented by a single 30 m pixel. However, there are species with very large ranges and some with very small ranges, which gives ArcGIS a hard time. So, we tweak the general idea of just calculating 1 pixel/(number of pixels of potential habitat). To deal with large areas of potential habitat, we

use 10,000/(number of pixels of potential habitat). If we just use 1, some of the resulting decimals are so small that ArcGIS just rounds them down to zero. To deal with species with very small ranges, we set the maximum weighted value of a species to 1. As a final step, we sum the weights for all species to get the total rarity-weighted richness for a given pixel.

- Clip to the spatial extent of Southeast Blueprint 2026.
- Use the elevation-based mask raster to remove marine areas from the rarity-weighted richness layer by assigning them a value of NoData.
- Reclassify the NLCD to create a developed areas mask. Assign all developed classes a value of 0, and all other classes a value of 1. Developed classes include "Developed, Open Space", "Developed, Low Intensity", "Developed, Medium Intensity", and "Developed, High Intensity".
- To remove developed areas, multiply the mammal rarity-weighted richness layer (with marine areas removed) by the developed mask. While there may be a limited amount of potential habitat for some species in the lower intensity urban classes, GAP tends to greatly overpredict potential habitat for species in urban areas.
- Reclassify the resulting raster into the indicator classes seen below. The ranges of legend values are of the lowest value of the class (e.g., value 5 includes exactly 5%; value 4 includes exactly 10% but not exactly 5%; value 3 includes exactly 20% but not exactly 10%).
- As a final step, clip to the spatial extent of Southeast Blueprint 2026.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Highest importance for range-restricted species (top 5%)
4 = Very high importance for range-restricted species (upper 5-10%)
3 = High importance for range-restricted species (upper 10-20%)
2 = Medium importance for range-restricted species (above average)
1 = Low importance for range-restricted species (below average)
0 = Developed landcover or no potential habitat

Known Issues

- Misclassification results in some potential habitat being underestimated in restoration areas that were previously row crop (e.g., large areas of Catahoula National Wildlife Refuge in LA).
- The GAP data does not include models for two mammals on the RSGCN list: Louisiana black bear (*Ursus americanus luteolus*) and Sherman's short-tailed shrew (*Blarina shermani*).
- GAP provides the following disclaimer for their predicted habitat maps: "While the actual distributions of many species are likely to be habitat limited, suitable habitat will not always be occupied because of population dynamics and species interactions. Furthermore, these maps correspond to midscale characterizations of landscapes, but individual animals may deem areas to be unsuitable because of presence or absence of fine-scale features and

characteristics that are not represented in our models (e.g. snags, vernal pools, shrubby undergrowth)."

- GAP data generally overpredicts potential habitat, especially for RSGCN subspecies with limited ranges. To address this issue, we constrained certain species' ranges to better represent potential habitat for the subspecies, like the subspecies of the hispid cotton rat, which only occurs on Sanibel and Captiva Islands off the coast of Florida. There may be some species-habitat models for which we did not reduce ranges appropriately, and we will work to improve this in future updates.
- GAP species-habitat distribution maps predict suitable landcover and environmental conditions for individual species based on their ranges and life history requirements. This data tends to overpredict habitat and cannot always separate development from potential habitat.
- The depiction of potential habitat for Florida bonneted bat over southern Lake Okeechobee is very coarse and both over- and underestimates important habitat in that area. This is the reason for the coarse polygon of high indicator value over southern Lake Okeechobee. This polygon misses some important habitat but also tends to overvalue habitat with it.
- Because this indicator is intended to represent terrestrial and freshwater ecosystems, we attempted to remove any predicted mammal habitat in the marine environment. However, in aquatic systems, it is difficult to determine where freshwater ends and saltwater begins, as that transition is highly dynamic and influenced by many factors including topography, instream flow, tides, storms, and more. To define the extent of this indicator, we used an elevation-based approach that includes all areas greater than -1 m in elevation (i.e., all areas above 1 m below sea level), then backfilled to capture low-lying coastal areas that are disconnected from the ocean. We intentionally erred on the side of including more terrestrial and freshwater areas so as not to exclude small islands that provide important habitat for species like beach mice, as well as coastal freshwater lakes (like Lake Mattamuskeet in NC). This is an imperfect methodology but produced better results than some of the other alternatives we explored.

Full Species List

Table 4. List of mammal RSGCN included in this indicator.

	Scientific Name	**Common Name**
1	*Baiomys taylori*	Northern pygmy mouse
2	*Canis rufus*	Red wolf
3	*Chaetodipus hispidus*	Hispid pocket mouse
4	*Corynorhinus rafinesquii* (includes *macrotis* & *rafinesquii)*	Rafinesque's big-eared bat (Eastern & Mountain pop.)
5	*Corynorhinus townsendii ingens*	Ozark big-eared bat
6	*Corynorhinus townsendii virginianus*	Virginia big-eared bat
7	*Dipodomys elator*	Texas kangaroo rat
8	*Eumops floridanus*	Florida bonneted bat
9	*Geomys breviceps* (includes *breviceps)*	Pocket gopher (Baird's & Oak Ridge)

	Scientific Name	Common Name
10	*Geomys bursarius ozarkensis*	Ozark pocket gopher
11	*Geomys pinetis* (includes *pinetis)*	Southeastern pocket gopher
12	*Geomys streckeri*	Strecker's pocket gopher
13	*Glaucomys sabrinus coloratus*	Carolina Northern flying squirrel
14	*Glaucomys sabrinus fuscus*	West Virginia Northern flying squirrel
15	*Lasiurus cinereus*	Hoary bat
16	*Lasiurus intermedius* (includes *floridanus)*	Northern yellow bat
17	*Lasiurus xanthinus*	Western yellow bat
18	*Microtus chrotorrhinus carolinensis*	Southern rock vole
19	*Microtus pennsylvanicus dukecampbelli*	Florida salt marsh vole
20	*Microtus pinetorum* (ssp. 1)	Pine vole (Florida woodland vole)
21	*Mormoops megalophylla*	Ghost-faced bat
22	*Myotis austroriparius*	Southeastern myotis
23	*Myotis californicus*	California myotis
24	*Myotis grisescens*	Gray bat
25	*Myotis leibii*	Eastern small-footed bat
26	*Myotis lucifugus*	Little brown bat
27	*Myotis septentrionalis*	Northern long-eared bat
28	*Myotis sodalis*	Indiana bat
29	*Myotis velifer*	Cave myotis
30	*Neofiber alleni*	Round-tailed muskrat
31	*Neotoma floridana haematoreia*	Southern Appalachian woodrat
32	*Neotoma floridana smalli*	Key Largo woodrat
33	*Neotoma magister*	Allegheny woodrat
34	*Neovison vison* (only subspecies *evergladensis, halilimnetes* & *lutensis)*	Florida mink subspecies (only Everglades, Gulf salt marsh & Atlantic salt marsh)
35	*Odocoileus virginianus clavium*	Key deer
36	*Onychomys arenicola*	Mearn's grasshopper mouse
37	*Oryzomys palustris* (only subspecies *natator* & *sanibeli)*	Florida subspecies of rice rat (only silver & Sanibel Island marsh)
38	*Perimyotis subflavus*	Tri-colored bat
39	*Peromyscus gossypinus* (includes *allapaticola)*	Cotton deermouse (key largo)
40	*Peromyscus leucopus buxtoni*	Buxton Woods white-footed deermouse
41	*Peromyscus leucopus easti*	Pungo white-footed deermouse
42	*Peromyscus nasutus*	Northern rock mouse

	Scientific Name	Common Name
43	*Peromyscus polionotus* (includes *allophrys, ammobates, leucocephalus, niveiventris, peninsularis, phasma* & *trissyllepsis)*	Old-field deermouse & beach mice (includes Choctawhatchee, Alabama, Santa Rosa, Southeastern, St. Andrew, Anastasia Island & Perdido Key)
44	*Peromyscus truei comanche*	Palo Duro mouse
45	*Podomys floridanus*	Florida mouse
46	*Puma concolor coryi*	Florida panther
47	*Sciurus niger avicennia*	Big Cypress fox squirrel
48	*Sciurus niger bachmani*	Bachman's fox squirrel
49	*Sciurus niger cinereus*	Delmarva fox squirrel
50	*Sciurus niger niger*	Southern fox squirrel
51	*Sigmodon hispidus* (only subspecies *exsputus* & *insulicola)*	Florida subspecies of cotton rat (only Lower Keys & insular)
52	*Sorex dispar blitchi*	Long-tailed or rock shrew
53	*Sorex palustris*	American water shrew
54	*Spilogale putorius* (includes *interrupta)*	Spotted skunk (Eastern & Plains)
55	*Sylvilagus obscurus*	Appalachian cottontail
56	*Sylvilagus palustris hefneri*	Lower Keys marsh rabbit
57	*Sylvilagus robustus*	Davis Mountain cottontail
58	*Vulpes velox*	Swift fox

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

International Union for Conservation of Nature (IUCN). Other Spatial Downloads. The IUCN Red List of Threatened Species. Accessed April 10, 2025. [https://www.iucnredlist.org/resources/other-spatial-downloads].

Sayre, R., 2023, Global Islands: U.S. Geological Survey data release. [https://doi.org/10.5066/P91ZCSGM].

Shipp-Pennock MA., Webster WD., and Freshwater DW. 2005. Systematics of the white-footed mouse (*Peromyscus leucopus*) in the mid-Atlantic region. Journal of Mammalogy 86(4):803-813. [https://academic.oup.com/jmammal/article-abstract/86/4/803/880584].

Trani MK., Ford MW, and Chapman BR., editors. 2007. The Land Manager's Guide to the Mammal's of the South. The Nature Conservancy, Southeastern Region, Durham, North Carolina, USA. 546 pages. [https://www.nrs.fs.usda.gov/pubs/jrnl/2007/nrs_2007_trani-lndmgr-full_009.pdf].

U.S. Geological Survey. Gap Analysis Project. Accessed February 25, 2025. [https://www.usgs.gov/programs/gap-analysis-project].

Intact Habitat Cores

This indicator represents the size of large, unfragmented patches of natural habitat. It identifies minimally disturbed natural areas at least 100 acres in size and greater than 200 meters wide. This indicator originates from Esri's green infrastructure data.

This indicator was last updated in October 2024.

Reason for Selection

Large areas of intact natural habitat are favorable for conservation of numerous species, including reptiles and amphibians, birds, and large mammals. The Esri Green Infrastructure data covers the entire United States and has been used in other broad-scale conservation planning efforts, so using this existing data helps align the Blueprint with other conservation efforts and reduce duplication of effort. We chose to use "Core Size (acres)" as the metric for this indicator. Other evaluation attributes included in this index, such as the default "Core Score", were less suitable because they were calculated using inputs that are duplicative of other indicators.

Input Data

- 2021 National Land Cover Database (NLCD): Land cover
- Southeast Blueprint 2024 extent
- Esri's Intact Habitat Cores 2023, accessed 2-16-2024: Core Size (Acres); to download, select "Open in ArcGIS Desktop" and make a local copy.

According to Esri's data description for the 2023 intact habitat cores update: "This layer represents modeled Intact Habitat Cores, or minimally disturbed natural areas at least 100 acres in size and greater than 200 meters wide. Esri created these data following a methodology outlined by the Green Infrastructure Center Inc. These data were generated using 2019 National Land Cover Data. Cores were derived from all "natural" landcover classes and excluded all "developed" and "agricultural" classes including crop, hay and pasture lands. The resulting cores were tested for size and width requirements (at least 100 acres in size and greater than 200 meters wide) and then converted into unique polygons."

Mapping Steps

- Convert the Esri Intact Habitat Cores 2023 polygons to a 30 m raster using the values in the "Acres" field. We used the feature layer map service as the input in the Polygon to Raster function in the code.

- Reclassify the above raster into 4 classes, seen in the final indicator values below.
- Use NLCD to remove zero values in deep marine areas, which are outside the scope of this terrestrial indicator. Use a conditional statement to assign NoData to any area with a pixel value >0 in the NLCD.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Large core (>10,000 acres)
2 = Medium core (>1,000-10,000 acres)
1 = Small core (>100–1,000 acres)
0 = Not a core

Known Issues

- The core analysis for this indicator is based on the 2019 NLCD, not the more recent 2021 NLCD. Esri has shared the scripts and input data used to create this layer, which may also help update this indicator in the future.
- Even small dirt roads serve as hard boundaries for habitat cores. While this makes sense for some species, this indicator likely underestimates the effective size of the patch for some more mobile animals.
- Waterbodies like reservoirs are also considered part of habitat cores, so this layer likely overestimates the effective size of the habitat core for most species.
- Many intact habitat cores have a speckling of small altered areas inside of them. In some cases, like in areas of west TX with concentrated oil wells, there can be many alterations in a gridded pattern across the entire core. This indicator underestimates the cumulative impacts of interior alterations—especially when the small altered footprints are densely packed in a grid within a habitat core.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Esri Green Infrastructure Center. Data Description: Detailed Description and Methodology for Intact Habitat Cores. PDF. Last updated June 30, 2023. [https://nation.maps.arcgis.com/home/item.html?id=047d9b05e0c842b1b126bc0767acfd5e].

Esri Green Infrastructure Center, Inc. 2023. Intact Habitat Cores (2023). [https://www.arcgis.com/home/item.html?id=b404b86a079a48049cb50272df23267a].

Landscape Condition

This indicator represents natural areas with limited human alteration while also considering the naturalness of the surrounding landscape. Examples of human alteration include urban development and intense agricultural use. This indicator combines data from multiple sources, including the National Land Cover Database; various data on grasslands, mines, quarries, and reservoirs; ideas from the Florida Critical Lands and Waters Identification Project's approach for evaluating land use intensity; and more.

This indicator was last updated in November 2025.

Reason for Selection

A high degree of naturalness across the landscape benefits species diversity as well as ecosystem services such as pollinator habitat, increased water infiltration, and reduced soil erosion. Though much of the Southeast has experienced human alteration at some point, natural landcover across the wider landscape provides many benefits. It allows species to disperse during different life stages, better adapt to a changing climate by accessing refugia, and freely move between different habitats. Natural landscapes can also complement existing protected areas and help increase resilience to extreme weather events such as flooding and hurricanes (Kremen and Merenlender 2018).

This indicator is loosely based on an approach for evaluating land use intensity as part of the "landscape integrity" metric developed by the University of Florida Center for Landscape Conservation Planning for the Florida Critical Lands and Waters Identification Project (CLIP) (Oetting et al. 2016).

Input Data

- National Land Cover Database (NLCD): 2021 Land Cover, 2021 U.S. Forest Service (USFS) Tree Canopy Cover
- Elevation-based mask used to separate the marine environment from terrestrial areas: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Global-scale mining polygons (Version 2); download the geopackage; see the layer in an online viewer; read a journal article about the data development
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0]
- 2020 LANDFIRE Biophysical Settings (BPS) [LF 2.2.0]
- Grassland and savanna extent (Southeast Blueprint indicator) - intermediary code output
- Southeast Blueprint reservoirs mask: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed

Mapping Steps

- Reclassify the 2021 NLCD landcover into 3 alteration classes where 3 is natural, 2 is altered, and 1 is heavily altered. Assign a value of 1 to all pixels with a landcover class of "Developed, High Intensity" or "Developed, Medium Intensity". Assign a value of 2 to all pixels with a landcover class of "Developed, Open Space", "Developed, Low Intensity", "Hay/Pasture", or "Cultivated Crops". Assign a value of 3 to everything else.
- Use LANDFIRE BPS to identify historical grasslands in the western part of the region that wouldn't typically have trees by selecting the following values in the BPS_NAME field: 'Southern Blackland Tallgrass Prairie', 'Central Mixedgrass Prairie', 'Chihuahuan Loamy Plains Desert Grassland', 'Southeastern Great Plains Tallgrass Prairie', 'Western Great Plains Sand Prairie', and 'Western Great Plains Shortgrass Prairie'. Then use the 2021 NLCD USFS Tree Canopy Cover layer to identify current treeless areas by selecting all canopy cover values <1. Combine these layers to identify parts of historical treeless grasslands that now have trees, then convert those pixels from 3 (natural) into 2 (altered).
- Convert the mining polygons to raster. Use the resulting raster to convert pixels within mine footprints classified as 3 (natural) into 2 (altered).
- Extract the "Quarries-Strip Mines-Gravel Pits-Well and Wind Pads" class from the LANDFIRE EVT data. Reproject and align with NLCD. There are groups of location errors in this class in specific parts of South Alabama and West Mississippi. To remove these location errors, remove quarry pixels classified in NLCD as evergreen, mixed, or deciduous forest (41, 42, 43).
- Use this improved depiction of "Quarries-Strip Mines-Gravel Pits-Well and Wind Pads" to convert pixels from 3 (natural) to 2 (altered). We use the mining polygons and LANDFIRE quarries class to attempt to distinguish altered areas of bare earth or rock (such as quarries, industrial areas, mines, and oil and gas well drilling pads) from natural areas of bare earth or rock (such as rock outcroppings and beaches).
- Extract the "Northeastern North American Temperate Forest Plantation" class from the LANDFIRE EVT data. Reproject, align with NLCD, and covert these areas in these pixels that are classified as 3 (natural) into 2 (altered). This class only occurs in the Appalachians and Interior Plateau subregions and addresses a past known issue of the Blueprint overprioritizing pine plantations on the Cumberland Plateau. It does not impact pine plantations in other subregions like the Coastal Plain.
- Reclassify the reservoirs layer to change NoData values to 1 so that all reservoirs receive a value of 0, and all non-reservoirs receive a value of 1. The reservoir layer was previously clipped to the Southeast Blueprint extent, so this change ensures that the circular focal window analysis in one of the following steps can look beyond the Southeast Blueprint boundary to properly calculate scores.
- Use the reservoirs layer to convert reservoir pixels from 3 (natural) to 2 (altered).
- Create a raster with the known and likely grassland classes (values 5, 6, and 7) from the grasslands and savannas indicator. Use this raster to convert grassland areas classified as 2 (altered) based on NLCD landcover to 3 (natural). NLCD often misclassifies natural grasslands and savannas as hay/pasture, which would otherwise lower their landscape condition score.

- Use the elevation-based mask raster to remove marine areas from the reclassified landcover layer by assigning them a value of NoData.
- Many species and ecological processes operate at multiple scales. To account for this, estimate the average amount of alteration using a circular moving window (or neighborhood) analysis at 4 different scales: approximately 0.22 acres (single pixel), approximately 10 acres, approximately 100 acres, and approximately 1,000 acres. Then average the values across all scales. This results in continuous values ranging from 1 to 3.
- Bin the continuous values into the following categories seen in the final indicator values below: 1 (heavily altered): 1 to <1.5; 2 (altered): 1.5 to <2; 3 (partly natural): 2 to <2.5; 4 (mostly natural): 2.5 to <2.9; 5 (natural): 2.9 to <2.99; 6 (very natural): 2.99 to 3. These breaks align with different levels of alteration. For example, an average value of 2.99 reflects a very natural landscape where only 1% of the area is altered and everything else is natural. That amounts to one altered pixel (which scores a 2) for every 99 natural pixels (which score a 3). An average value of 2 represents a partly natural landscape with an equal number of heavily altered and natural areas. For example, a landscape with an equal number of altered pixels (scoring a 2), heavily altered pixels (scoring a 1), and natural pixels (scoring a 3) would have a value of 2.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Very natural landscape
5 = Natural landscape
4 = Mostly natural landscape
3 = Partly natural landscape
2 = Altered landscape
1 = Heavily altered landscape

Known Issues

- This indicator underestimates landscape condition in many areas composed of native grasses and forbs on private lands. NLCD landcover often classifies these areas as “Hay/Pasture”. Most of the areas classified as “Hay/Pasture” are partly altered areas, but some are functioning as natural grasslands either with or without grazing. The indicator does use known and likely grassland data to correct for this issue, but the known grassland data is missing many grassland areas in the Southeast and the likely grassland prediction miss many areas on private lands.
- This indicator may overestimate landscape condition around places like quarries, mines, wind pads, and other industrial sites. The indicator considers these areas to have the same level of alteration as pine plantations, crops, pastures, low-density development, and open space in

developed areas. Treating this range of landcover types as equally altered does not reflect their varying levels of human modification compared to natural areas. Future improvements may consider expanding alteration classes to better capture these nuances.

- This indicator does not account for variation in habitat condition due to invasive species.
- In arid parts of Southwest TX, this indicator underprioritizes some naturally open areas. The combination of LANDFIRE EVT and NLCD landcover does a fairly good job of distinguishing natural from altered open areas but occasionally misclassifies a small set of natural areas that should score higher.
- This indicator overestimates landscape condition in some areas in the Everglades Headwaters National Wildlife Refuge proclamation boundaries. Many, but not all, pastures in that area function similarly to native grasslands.
- This indicator may overestimate landscape condition in places with thin linear alterations (e.g., railroads, thin bridges, pipelines). NLCD can sometimes classify those pixels as natural landcover categories—in part due to the small area they cover within a 30 m pixel.
- Because the reservoir layer used in this indicator was previously clipped to the Blueprint extent, and the landscape condition moving window analysis looks beyond the Blueprint extent, this indicator will overestimate condition along the edges of the Blueprint in places where reservoirs are present within approximately 1 mi outside the Blueprint boundary.
- Because this indicator is intended to represent terrestrial and freshwater ecosystems, we attempted to remove the marine environment so that the ocean does not inflate landscape condition scores in coastal areas. For example, we did not want nearby "unaltered" ocean pixels to raise the score of an otherwise heavily developed barrier island. However, in aquatic systems, it is difficult to determine where freshwater ends and saltwater begins, as that transition is highly dynamic and influenced by many factors including topography, instream flow, tides, storms, and more. To define the extent of this indicator, we used an elevation-based approach that includes all areas greater than -1 m in elevation (i.e., all areas above 1 m below sea level), then backfilled to capture low-lying coastal areas that are disconnected from the ocean. We intentionally erred on the side of including more terrestrial and freshwater areas so as not to exclude small islands and coastal freshwater lakes (like Lake Mattamuskeet in NC). This is an imperfect methodology but produced better results than some of the other alternatives we explored.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

Kremen, Claire, and Adina M. 2018. Merenlender. Landscapes that work for biodiversity and people. Science 362.6412. Eaau6020. [https://www.science.org/doi/10.1126/science.aau6020].

Landscape Fire and Resource Management Planning Tools (LANDFIRE), Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published 2023-05-01. LANDFIRE 2022 Existing Vegetation Type (EVT) CONUS. LF 2022. Raster digital data. Sioux Falls, SD. [https://landfire.gov/evt.php].

Maus, V., Giljum, S., Gutschlhofer, J. et al. A global-scale data set of mining areas. Sci Data 7, 289 (2020). [https://www.nature.com/articles/s41597-020-00624-w].

Oetting J, Hoctor T, and Volk M. 2016. Critical Lands and Waters Identification Project (CLIP): Version 4.0 Technical Report. Accessed Nov 10, 2022. [https://www.fnai.org/PDFs/CLIP_v4_technical_report.pdf].

U.S. Census Bureau. Feature Catalog for the 2018 United States 1:500,000 Cartographic Boundary Files (Shapefile). 2019-05. Current State and Equivalent (national). [https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Mississippi Alluvial Valley Forest Birds - Protection

This indicator prioritizes areas for new land protection within the Mississippi Alluvial Valley (MAV) based on benefits to forest breeding birds that need large interior cores of bottomland hardwood habitat (Swainson's warbler, cerulean warbler, swallow-tailed kite). The model considers core size, the amount of existing protected land within a forest patch, proximity to reforestation priorities, and risk of conversion to agriculture based on flooding frequency. The highest scores go to drier, unprotected forest patches with cores at least 2,000 ha (~5,000 ac) in size that are adjacent to complementary reforestation priority areas also identified by the Lower Mississippi Valley Joint Venture (LMVJV). This indicator originates from the LMVJV MAV forest breeding bird protection priorities.

This indicator was last updated in October 2023.

Reason for Selection

This indicator reflects the importance of conserving large forested tracts in the MAV that provide important habitat for interior forest breeding birds and other species. Agriculture and development have greatly reduced the extent of the bottomland hardwood forests that historically dominated the MAV, while changes in hydrology and forest management have altered the structure and composition of the remaining forests. This indicator also promotes consistency with the forest protection priorities of the Lower Mississippi Valley Joint Venture, which serve as a key resource for land trusts and other

conservation partners in the MAV looking to advance bird conservation objectives through strategic forest protection.

Input Data

- Mississippi Alluvial Valley (MAV) Forest Protection Model from the Lower Mississippi Valley Joint Venture (LMVJV)

  This model identifies forest patches that are a priority for future protection "based on benefits to forest bird conservation afforded by forest patch area, geographic location, and hydrologic condition" (Elliott et al. 2020). It excludes already protected lands. The model first identifies as meriting protection existing forest patches in the MAV with interior cores (<250 m from a hostile edge) that are either ≥1,600 ha in size or adjacent to high priority areas from the complementary reforestation model. It then prioritizes based on the percent of the patch that is unprotected, also factoring in core size, proximity to reforestation priorities, and flood frequency. The model awards the highest scores to unprotected patches that are ≥2,000 ha (~5,000 acres), the least flood-prone (and therefore at higher risk of agricultural conversion), and adjacent to reforestation priorities.

  The model characteristics are based on the habitat needed to support a source population of three species of interior forest breeding birds: Swainson's warbler, swallow-tailed kite, and cerulean warbler. These species received the highest overall priority score in the 1999 Partners in Flight Plan for the MAV based on their level of conservation concern. Each bird is used to represent an ecological suite of species with similar habitat needs. Swainson's warbler also represents prothonotary warbler, hooded warbler, wood thrush, and Acadian flycatcher; cerulean warbler also represents Kentucky warbler, summer tanager, yellow-billed cuckoo, and eastern wood-pewee; and swallow-tailed kite also represents red-shouldered hawk, broad-winged hawk, pileated woodpecker, and Cooper's hawk (Mini and Elliott 2022).

- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Reproject the Mississippi Alluvial Valley (MAV) Forest Protection Model to NAD 1983 Contiguous USA Albers (EPSG 5070).
- Reclassify the LMVJV source data as seen in the legend below.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = Highest priority forest breeding bird habitat patch for future protection (score >90-100)
9 = (score >80-90)
8 = (score >70-80)
7 = (score >60-70)
6 = (score >50-60)
5 = (score >40-50)
4 = (score >30-40)
3 = (score >20-30)
2 = (score >10-20)
1 = Lowest priority (score >0-10)
0 = Not a priority (score = 0)

Known Issues

- The forest bird protection model identifies large forest patches using 2011 Landsat data, which does not reflect more recent changes in landcover. As a result, this indicator may include patches fragmented by recent development that no longer meet the protection criteria. Conversely, this indicator may also overlook patches reforested in the last decade that now meet the protection criteria.
- Similarly, the forest bird protection model uses an older version of the Protected Areas Database of the United States (PAD-US 2.0) and therefore may miss more recently conserved lands. As a result, the indicator likely undervalues forest patches near newly protected areas, or protected areas added for the first time in the more recent versions of PAD-US.
- Because birds are highly mobile and migratory, the forest bird protection model does not prioritize opportunities to improve the connectivity of the conservation estate. This dataset is not intended to identify high priority linkages between protected areas that may support isolated populations of less mobile species like black bear. The forest bird protection model also does not emphasize the value of creating new foundational protected areas in places that don't currently have any. As these opportunities are outside the scope of the source data, this indicator largely does not capture them.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Elliott AB, Mini AE, McKnight SK, Twedt DJ. Conservation–Protection of Forests for Wildlife in the Mississippi Alluvial Valley. Forests. 2020; 11(1):75. https://doi.org/10.3390/f11010075. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5e15ea9decbb473fb91c6d30/1597419918742/MAV+Forest+Protection].

Mini, Anne, and Blaine Elliott. March 17, 2022. Forest conservation priorities for landbirds in the Mississippi Alluvial Valley. Webinar hosted by the Southeast Conservation Adaptation Strategy. [https://www.youtube.com/watch?v=ggYjN3XdfMM&t=819s].

Mississippi Alluvial Valley Forest Birds - Reforestation

This indicator prioritizes areas for reforestation within the Mississippi Alluvial Valley (MAV) based on benefits to three species of forest breeding birds that need large interior cores of bottomland hardwood habitat (Swainson's warbler, cerulean warbler, swallow-tailed kite). The model considers the core size, number of cores, and percent of local forest cover that would result from reforestation, as well as risk of conversion to agriculture based on flooding frequency. The highest scores go to drier areas where reforestation would create new forest patches containing interior cores at least 2,000 ha (~5,000 ac) in size. It originates from the Lower Mississippi Valley Joint Venture MAV forest breeding bird reforestation priorities.

This indicator was last updated in October 2023.

Reason for Selection

This indicator reflects the importance of restoring large forested tracts in the MAV to provide quality habitat for interior forest breeding birds and other species. Agriculture and development have greatly reduced the extent of the bottomland hardwood forests that historically dominated the MAV, while changes in hydrology and forest management have altered the structure and composition of the remaining forests. This indicator also promotes consistency with the reforestation priorities of the Lower Mississippi Valley Joint Venture (LMVJV), which serve as a key resource for land trusts and other conservation partners in the MAV looking to advance bird conservation objectives through strategic forest management and restoration.

Input Data

- MAV Forest Breeding Bird Reforestation Model from the Lower Mississippi Valley Joint Venture (LMVJV)

  This model prioritizes places where reforestation would reduce forest fragmentation and increase the area of interior forest cores (<250 m from a hostile edge). Its primary objective was to increase the number of forest patches containing interior cores >2,000 ha (~5,000 ac) in size, but it also sought to increase the number and size of cores >5,000 ha (~12,000 acres). At the same time, the model prioritized areas where reforestation would achieve at least 60% forest cover within local (320 km^2) landscapes. Finally, it prioritized higher-elevation bottomland hardwood forests that are less flood-prone and therefore at higher risk of agricultural conversion) (LMVJV 2015).

  The model characteristics are based on the habitat needed to support a source population of three species of interior forest breeding birds: Swainson's warbler, swallow-tailed kite, and cerulean warbler. These species received the highest overall priority score in the 1999 Partners

in Flight Plan for the MAV based on their level of conservation concern. Each bird is used to represent an ecological suite of species with similar habitat needs. Swainson's warbler also represents prothonotary warbler, hooded warbler, wood thrush, and Acadian flycatcher; cerulean warbler also represents Kentucky warbler, summer tanager, yellow-billed cuckoo, and eastern wood-pewee; and swallow-tailed kite also represents red-shouldered hawk, broad-winged hawk, pileated woodpecker, and Cooper's hawk (Mini and Elliott 2022).

- Mississippi Alluvial Valley (MAV) Forest Protection Model from the Lower Mississippi Valley Joint Venture (LMVJV)
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Reproject the MAV Forest Breeding Bird Reforestation Model to NAD 1983 Contiguous USA Albers (EPSG 5070).
- Use the MAV Forest Protection Model to calculate the modeling extent of this indicator. The reforestation layer doesn't have zero values to indicate places that are within the modeling extent but are not a priority for reforestation because they are already forested. Assign a value of 0 to any pixel with a value ≥0 in the protection layer that did not already have a value in the reforestation layer. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = Reforestation most likely to contribute to forest breeding bird habitat needs (≥90th percentile)
9 = Reforestation most likely to contribute to forest breeding bird habitat needs (80th to <90th percentile)
8 = Reforestation more likely to contribute to forest breeding bird habitat needs (70th to <80th percentile)
7 = Reforestation less likely to contribute to forest breeding bird habitat needs (60th to <70th percentile)
6 = Reforestation least likely to contribute to forest breeding bird habitat needs (50th to <60th percentile)
5 = Reforestation least likely to contribute to forest breeding bird habitat needs (40th to <50th percentile)
4 = Reforestation least likely to contribute to forest breeding bird habitat needs (30th to <40th percentile)
3 = Reforestation least likely to contribute to forest breeding bird habitat needs (20th to <30th percentile)
2 = Reforestation least likely to contribute to forest breeding bird habitat needs (10th to <20th percentile)
1 = Reforestation least likely to contribute to forest breeding bird habitat needs (<10th percentile)
0 = Not a priority for reforestation

Known Issues

- Most important areas in this indicator represent priorities for reforestation, as opposed to identifying areas currently in good or ideal ecological condition.
- While analysis by the LMVJV has found that it is most important to work in areas scoring higher than 8, we included the full dataset in this indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Lower Mississippi Valley Joint Venture. 2015. MAV Forest Breeding Bird Decision Support Model – Update 2015 documentation. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5bec7942b8a045d24a276e8c/1542224195084/LMVJV_FBBDSM_2015_Summary.pdf].

Lower Mississippi Valley Joint Venture. MAV Reforestation Forest Breeding Bird Decision Support Model webpage. [https://www.lmvjv.org/mav-breedingbird].

Mini, Anne, and Blaine Elliott. March 17, 2022. Forest conservation priorities for landbirds in the Mississippi Alluvial Valley. Webinar hosted by the Southeast Conservation Adaptation Strategy. [https://www.youtube.com/watch?v=ggYjN3XdfMM&t=819s].

Twedt, D.J., D. Pashley, C. Hunter, A. Mueller, C. Brown, and R. Ford. 1999. Partners in Flight Bird Conservation Plan for the Mississippi Alluvial Valley. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5d3095a9e852150001d7aacd/1563465138676/MAV_PIF_PLAN+1999.pdf].

Playas

This indicator represents the condition and location of playas, which are round, shallow depressions found primarily in the western Great Plains that serve as temporary wetlands by collecting water from rainfall and runoff. It defines a healthy playa as one that is not farmed, hydrologically modified, within a wind farm, or impacted by sediment accumulation due to agriculture. It also considers the increased benefits to wildlife provided by clusters of nearby playas, compared to more sparsely distributed playas. This indicator originates from the Playa Lakes Joint Venture probable playas dataset.

This indicator was last updated in October 2023.

Reason for Selection

The unique wet-dry cycle that characterizes this important wetland ecosystem makes playas a biodiversity hotspot within the Great Plains. Playas support a rich community of wetland plants and invertebrates, providing habitat and food for migrating waterfowl, shorebirds, and resident prairie birds, as well as many mammal and amphibian species. Healthy playas “provide a sustainable water source for communities and rain-fed operations, water filtration, flood control, livestock forage, and recreation” (PLJV 2022). They play a particularly critical role in groundwater recharge and filtration that improve the quantity and quality of water in the region’s aquifers. Sediment accumulation due to agriculture in and around playas, as well as modification from construction and hydrologic alteration, have significantly degraded and continue to threaten the habitat and ecosystem services that playas provide (PLJV website).

Input Data

- Base Blueprint 2022 extent
- Playa Lakes Joint Venture probable playas dataset v5, accessed 5-25-2022

  This dataset is a regional compilation of several data sources including the National Wetlands Inventory, Soil Survey Geographic database, and satellite imagery. Specific details on methods are in the release notes. This dataset defines a healthy playa as one that is not farmed, has no hydrological modifications, is not within a wind farm, and is not sedimented. It

defines playa clusters by identifying areas with either high playa density or high playa surface area, according to duck abundance data collected on playas.

- Playa Lakes Joint Venture (JV) boundary extracted from the North American Joint Venture Boundary dataset
- Southeast Blueprint 2023 extent

Mapping Steps

- From the Playa Lakes JV probable playas v5 data, convert polygons to raster and use the fields "healthy" and "cluster" to identify healthy playas and those that are part of a larger cluster.
- Assign zero values to all pixels not identified as a playa above. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Convert the Playa Lakes JV boundary to raster, giving it a value of 1.
- Limit the playas data (values of 0, 1, 2, and 3) to the Playa Lakes JV boundary by taking the playa raster times the JV raster.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Healthy playa and part of a larger cluster
2 = Healthy playa
1 = Other playa
0 = Not identified as a playa

Known Issues

- This indicator underrepresents playas, especially those smaller than 2 acres, in the eastern half of Beaver, Oklahoma. NWI data were missing from that area and the additional data sources used tend to underestimate playas, especially those less than 2 acres in size.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Playa Lakes Joint Venture. Playas of the high plains: Providing water for wildlife and people. [https://pljv.org/docs/What_Are_Playas.pdf].

Playa Lakes Joint Venture probable playas dataset v5. Playa Lakes Joint Venture. Accessed May 25, 2022. [https://pljv.org/playas/playa-tools/].

Playa Lakes Joint Venture website. 2022. Playas Benefit Wildlife and People. [https://pljv.org/playas/].

U.S. Fish and Wildlife Service. FWS Migratory Bird North American Joint Ventures Boundaries. [https://gis-fws.opendata.arcgis.com/datasets/d34bd6ebd33543d68a00fc7d412890fa_0/about].

Potential Access to Parks

This cultural resource indicator prioritizes places to create new parks that would fill gaps in access to open space within socially vulnerable communities in urban areas. It identifies areas where residents currently lack access to parks within a 10-minute walk (accounting for walkable road networks and access barriers like highways and fences), then prioritizes based on park need using demographic and environmental metrics. This indicator originates from the Trust for Public Land's ParkServe park priority areas and the Center for Disease Control's Social Vulnerability Index.

This indicator was last updated in November 2025.

Reason for Selection

Protected natural areas help foster a conservation ethic by providing opportunities for people to connect with nature, and also support ecosystem services like offsetting heat island effects (Greene and Millward 2017, Simpson 1998), water filtration, stormwater retention, and more (Hoover and Hopton 2019). In addition, parks, greenspace, and greenways can help improve physical and psychological health in communities (Gies 2006). However, parks are not equitably distributed within easy walking distance for everyone. This indicator also complements the urban park size indicator by capturing the value of potential new parks.

Input Data

- The Trust for Public Land (TPL) ParkServe database, accessed 12-31-2024: Park priority areas (ParkServe_ParkPriorityAreas, ParkServe_10MWServiceAreas, ParkServe_Parks)

  From the TPL ParkServe documentation:

  The ParkServe database maintains an inventory of parks for every urban area in the U.S., including Puerto Rico. This includes all incorporated and Census-designated places that lie within any of the country's 3,000+ census-designated urban areas. All populated areas in a city that fall outside of a 10-minute walk service area are assigned a level of park priority, based on a comprehensive index of six equally weighted demographic and environmental metrics:

- Population density
- Density of low-income households – which are defined as households with income less than 75 percent of the urban area median household income
- Density of people of color
- Community health – a combined index based on the rate of poor mental health and low physical activity from the 2020 CDC PLACES census tract dataset
- Urban heat islands – surface temperature at least 1.25o greater than city mean surface temperature from The Trust for Public Land, based on Landsat 8 satellite imagery
- Pollution burden - Air toxics respiratory hazard index from 2020 EPA EJScreen

The 10-minute walk

For each park, we create a 10-minute walkable service area using a nationwide walkable road network dataset provided by Esri. The analysis identifies physical barriers such as highways, train tracks, and rivers without bridges and chooses routes without barriers.

- Centers for Disease Control (CDC) Social Vulnerability Index 2022 (United States): RPL_Themes

  Social vulnerability refers to the capacity for a person or group to "anticipate, cope with, resist and recover from the impact" of a natural or anthropogenic disaster such as extreme weather events, oil spills, earthquakes, and fires. Socially vulnerable populations are more likely to be disproportionately affected by emergencies (Wolkin et al. 2015).

  In this indicator, we use the "RPL_THEMES" attribute from the Social Vulnerability Index, described here. "The Geospatial Research, Analysis, and Services Program (GRASP) at Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry developed the Social Vulnerability Index (SVI). The SVI is a dataset intended to help state, local, and tribal disaster management officials identify where the most socially vulnerable populations occur (Agency for Toxic Substances and Disease Registry [ATSDR] 2018)" (Flanagan et al. 2018).

  "The SVI database is regularly updated and includes 15 census variables (ATSDR 2018). Each census variable was ranked from highest to lowest vulnerability across all census tracts in the nation with a nonzero population. A percentile rank was calculated for each census tract for each variable. The variables were then grouped among four themes.... A tract-level percentile rank was also calculated for each of the four themes. Finally, an overall percentile rank for each tract as the sum of all variable rankings was calculated. This process of percentile ranking was then repeated for the individual states" (Flanagan et al. 2018).

- Southeast Blueprint 2025 extent

Mapping Steps

- Remove the ParkServe 10 minute walk service areas and the ParkServe parks from the ParkServe park priority areas. This removes areas that are already being served by an existing park.
- Convert the resulting layer to a raster using the ParkRank field. Note: The ParkRank scores are calculated using metrics classified relative to each city. Each city contains park rank values that range from 1-3. For the purposes of this indicator, we chose to target potential park areas to improve equity. Because the ParkRank scores are relative for each city, a high score in one city is not necessarily comparable to a high score from another city. In an effort to try to bring more equity into this indicator, we also use the CDC Social Vulnerability Index (SVI) to narrow down the results.
- Reclassify the ParkServe raster to make NoData values 0.
- Convert the SVI layer from vector to raster based on the "RPL_Themes" field. In the source geodatabase, the field name is "RPL_Themes" and the alias for that field is "Overall percentile ranking".
- To limit the ParkRank layer to areas with high SVI scores, first identify census tracts with an "RPL_Themes" field value >0.65. Make a new raster that assigns a value of 1 to census tracts that score >0.65, and a value of 0 to everything else. Take the resulting raster times the ParkRank layer.
- Reclassify this raster into the 4 classes seen in the final indicator below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Very high priority for a new park to serve people who lack nearby park access
2 = High priority for a new park to serve people who lack nearby park access
1 = Moderate priority for a new park to serve people who lack nearby park access
0 = Not identified as a priority for a new park to serve people who lack nearby park access (within urban areas)

Known Issues

- This indicator could overestimate park need in areas where existing parks are missing from the ParkServe database. TPL regularly updates ParkServe to incorporate the best available park data. If you notice missing parks or errors in the park boundaries or attributes, you can submit corrections through the ParkReviewer tool or by contacting TPL staff.

- The indicator does not incorporate inequities in access to larger versus smaller parks. In predicting where new parks would benefit nearby people who currently lack access, this indicator treats all existing parks equally.
- Within a given area of high park need, the number of people served by the creation of a new park depends on its size and how centrally located it is. This indicator does not account for this variability. Similarly, while creating a new park just outside an area of high park need would create access for some people on the edge, the indicator does not capture the benefits of new parks immediately adjacent to high-need areas. For a more granular analysis of new park benefits, ParkServe's ParkEvaluator tool allows you to draw a new park, view its resulting 10-minute walk service area, and calculate who would benefit.
- Beyond considering distance to a park and whether it is open to the public, this indicator does not account for other factors that might limit park access, such as park amenities or public safety. The TPL analysis excludes private or exclusive parks that restrict access to only certain individuals (e.g., parks in gated communities, fee-based sites). The TPL data includes a wide variety of parks, trails, and open space as long as there is no barrier to entry for any portion of the population.
- This indicator identifies areas where parks are needed but does not consider whether a site is available to become a park. We included areas of low intensity development in order to capture vacant lots, which can serve as new park opportunities. However, as a result, this indicator also captures some areas that are already used for another purpose (e.g., houses, cemeteries, and businesses) and are unlikely to become parks. In future updates, we would like to use spatial data depicting vacant lots to identify more feasible park opportunities.
- This indicator underestimates places in rural areas where many people within a socially vulnerable census tract would benefit from a new park. ParkServe covers incorporated and census-designated places within census-designated urban areas, which leaves out many rural areas. We acknowledge that there are still highly socially vulnerable communities in rural areas that would benefit from the development of new parks. However, based on the source data, we were not able to capture those places in this version of the indicator.
- Some high-scoring areas may have access to nearby greenways and trails that can serve many of the roles of parks. This indicator does not account for access to nearby greenways and trails.

Other Things to Keep in Mind

- The zero values in this indicator contain three distinct types of areas that we were unable to distinguish between in the legend: 1) Areas that are not in a community analyzed by ParkServe (ParkServe covers incorporated and census-designated places within census-designated urban areas); 2) Areas in a community analyzed by ParkServe that were not identified as a priority; 3) Areas that ParkServe identifies as a priority but do not meet the SVI threshold used to represent areas in most need of improved equitable access.
- This indicator is intended to generally help identify potential parks that can increase equitable access but should not be solely used to inform the creation of new parks. As the social equity component relies on information summarized by census tract, it should only be used in

conjunction with local knowledge and in discussion with local communities (NRPA 2021, Manuel-Navarete et al. 2004).

- This indicator only includes park priority areas that fall within the 65th percentile or above from the Social Vulnerability Index. We did not perform outreach to community leaders or community-led organizations for feedback on this threshold.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Centers for Disease Control and Prevention/ Agency for Toxic Substances and Disease Registry/ Geospatial Research, Analysis, and Services Program. CDC/ATSDR Social Vulnerability Index [2022] Database [US]. [https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html].

Flanagan BE, Hallisey EJ, Adams E, Lavery A. Measuring Community Vulnerability to Natural and Anthropogenic Hazards: The Centers for Disease Control and Prevention's Social Vulnerability Index. J Environ Health. 2018;80(10):34-36. [https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7179070/].

Gies E. (2006). The health benefits of parks: how parks help keep Americans fit & healthy. The Trust for Public Land. Accessed February 16, 2022. [https://cloud.tpl.org/pubs/benefits_HealthBenefitsReport.pdf].

Greene, C. S., & Millward, A. A. (2017). Getting closure: The role of urban forest canopy density in moderating summer surface temperatures in a large city. Urban ecosystems, 20(1), 141-156. [https://link.springer.com/article/10.1007/s11252-016-0586-5].

Hoover, F. A., & Hopton, M. E. (2019). Developing a framework for stormwater management: leveraging ancillary benefits from urban greenspace. Urban ecosystems, 22(6), 1139-1148. [https://link.springer.com/article/10.1007/s11252-019-00890-6].

Manuel-Navarrete D, Kay JJ, Dolderman D. Ecological integrity discourses: linking ecology with cultural transformation. Human Ecology Review 2004, 11:215–229. [https://www.jstor.org/stable/24707715].

National Recreation and Park Association. "NRPA Park Metrics." 2021. Accessed April 5, 2021. [https://www.nrpa.org/publications-research/research-papers/agency-performance-review/].

Simpson, J. R. (1998). Urban forest impacts on regional cooling and heating energy use: Sacramento County case study. Journal of Arboriculture, 24, 201-214. [https://www.fs.usda.gov/research/treesearch/61731].

Wolkin A, Patterson JR, Harris S, et al. Reducing Public Health Risk During Disasters: Identifying Social Vulnerabilities. J Homel Secur Emerg Manag. 2015;12(4):809-822. [https://doi.org/10.1515/jhsem-2014-0104].

Resilient Terrestrial Sites

This indicator depicts an area's capacity to maintain species diversity and ecosystem function in the face of climate change. It measures two factors that influence resilience. The first, landscape diversity, reflects the number of microhabitats and climatic gradients created by topography, elevation, and hydrology. The second, local connectedness, reflects the degree of habitat fragmentation and strength of barriers to species movement. Highly resilient sites contain many different habitat niches that support biodiversity, and allow species to move freely through the landscape to find suitable microclimates as the climate changes. This indicator originates from The Nature Conservancy's Resilient Land data.

This indicator was last updated in November 2025.

Reason for Selection

Resilience scores quantify a combination of landscape diversity and local connectedness, stratified by geophysical setting and ecoregion. These measures represent the number of microclimates available to species and the current state of the landscape. This builds on research from Anderson and Ferree (2010), who showed geophysical diversity and elevation range were associated with biodiversity in the Eastern United States. Resilience emphasizes diverse landscapes where species are likely to be able to move and adjust to changing conditions.

Input Data

- The Nature Conservancy's (TNC) combined terrestrial and coastal resilient sites for the conterminous U.S. in geodatabase format (resilient_sites_national.zip), accessed 12-3-2024
- Southeast Blueprint 2025 extent

Mapping Steps

- Clip the TNC data to the bounding box around the Southeast Blueprint 2025 extent.
- To extract the terrestrial component of the analysis, reclassify the "type_des" field to assign "Terrestrial Resilience" a value of 1 and everything else a value of NoData.
- Reclassify the "CONUS_Legend" field into the final indicator values:
    - 11 Most resilient = 7
    - 12 More resilient = 6
    - 13 Slightly more resilient = 5
    - 17 Average/median resilient (terrestrial or coastal) = 4
    - 18 Slightly less resilient (terrestrial or coastal) = 3
    - 19 Less resilient (terrestrial or coastal) = 2
    - 20 Least resilient (terrestrial or coastal) = 1
    - 91 Developed = 0

- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Most resilient
6 = More resilient
5 = Slightly more resilient
4 = Average/median resilience
3 = Slightly less resilient
2 = Less resilient
1 = Least resilient
0 = Developed

Known Issues

- This indicator does not account for the occurrence and timing of disturbance processes, particularly fire. Without fire, resilient sites in many terrestrial ecosystems will not serve as biodiversity hotspots. This is particularly problematic in pine and prairie ecosystems in the Piedmont and Coastal Plain.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Anderson, M.G., A. Barnett, M. Clark, C. Ferree, A. Olivero Sheldon, J. Prince. 2016. Resilient Sites for Terrestrial Conservation in Eastern North America. The Nature Conservancy, Eastern Conservation Science. [https://easterndivision.s3.amazonaws.com/Resilient_Sites_for_Terrestrial_Conservation.pdf].

Anderson, M.G., Ferree, C.E., 2010. Conserving the stage: climate change and the geophysical underpinnings of species diversity. PLoS One 5, e11554. [https://doi.org/10.1371/journal.pone.0011554].

The Nature Conservancy. Datasets for the Conterminous U.S.: Resilient Sites (Terrestrial and Coastal). Esri File Geodatabase format. Accessed December 3, 2024. [https://www.conservationgateway.org/ConservationPractices/ClimateChange/Pages/RCN-Downloads.aspx].

River Cane Restoration

This indicator represents priority places for river cane restoration and co-stewardship near the lands of federally recognized Tribes. It combines data from multiple sources, including Natural Resources Conservation Service soils data, the Environmental Protection Agency's estimated floodplain, lands of federally recognized Tribes lands from the U.S. Census Bureau, and more.
This indicator was last updated in November 2025.

Reason for Selection

This indicator prioritizes places for river cane restoration within and near the lands of federally recognized Tribes and within protected areas. River cane (*Arundinaria gigantea*) is a bamboo native to the Southeast United States. It historically formed large stands called canebrakes that could stretch for miles within the floodplain and riparian areas. Often mistaken for invasive Asian bamboo, river cane is not a rare species, but river cane habitats and ecosystems are fragmented. Canebrakes have declined to less than 2% of their historical extent (Noss 1995), and large, intact canebrakes that exceed more than a few acres are exceedingly rare. In addition to its significant ecological value, river cane is also considered a cultural keystone species for many Tribes and Indigenous communities.

Cultural keystone species are species whose existence have significantly contributed to the cultural identity of a people (Garibaldi et al. 2004). Historically, Indigenous people stewarded, managed, and harvested river cane to use for many aspects of daily life. Canebrakes have been described as a "supermarket" in terms of the breadth of resources offered (Kniffen et al. 1987). River cane served as the primary material for baskets, tools, fuel and candles, building materials, weaponry, religious practices, and more (Anderson and Oakes 2011). It also served as a food source, and seeds were ground into flour (Kniffen et al. 1987). The significant decline of river cane—particularly canebrakes in sufficiently good condition for harvest—has impacted the health and habitat of many species, and has endangered aspects of Indigenous cultures. For example, in 2012 the Cherokee Nation listed river cane on their Culturally Protected Species list (Chavez 2012).

Canebrakes can serve as important habitat for many native species, including wood thrush *(Hylocichla mustelina)*, Bachman's warbler *(Vermivora bachmanii)*, Swainson's warbler *(Limnothlypis swainsonii)*, timber rattlesnake (*Crotalus horridus*), and numerous pollinators. In addition, river cane rhizomes help secure soils to control erosion, especially to stabilize streambanks. Research also suggests that incorporating river cane into riparian buffer restoration filters surface runoff, effectively reducing nutrient loads into streams, even at riparian buffer widths of just 10 m (Schoonover et. al. 2005, Schoonover and Williard 2007).

Finally, this indicator reflects the potential for relationships between Indigenous Nations and land holders, particularly of public lands, to build co-management practices that prioritize healthy river cane ecosystems. Indigenous leadership and involvement in river cane restoration efforts can enhance effectiveness and impact—benefiting Tribal nations, local communities, and protected area land managers (Wall et. al. 2025, Lyver et. al. 2015).

This indicator also promotes consistency with the Tribal Trust responsibilities held by the federal government to Tribal nations to protect Indigenous resources and lands, and to advance the preservation goals of many Indigenous governments in the Southeast whose cultural survival and land health rely heavily on river cane ecosystems.

Input Data

- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

  The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- Natural Resources Conservation Service (NRCS) Gridded Soil Survey Geographic (gSSURGO) Database: 30 m CONUS soils data, accessed March 2025
- Protected Areas Database of the United States (PAD-US): PAD-US 3.0 national geodatabase - Combined Proclamation Marine Fee Designation Easement; PAD-US 4.0 national geodatabase - Combined Proclamation Marine Fee Designation Easement
- Southeast Floodplain Inundation Frequency dataset developed by Yvonne Allen with the U.S. Fish and Wildlife Service (last updated 4-2-2025)
- U.S. Census Bureau 2021 TIGER/Line National U.S. American Indian/Alaska Native/Native Hawaiian (AIANNH) Areas shapefile; learn more about TIGER/Line data
- 2021 NOAA Coastal Change Analysis Program (C-CAP) regional 30 m CONUS landcover (accessed April 2025)
- Stable coastal wetlands (Southeast Blueprint indicator)
- Southeast Blueprint 2025 extent (continental only)

Mapping Steps

- Create an enhanced floodplain layer to serve as a foundation for potential river cane restoration/management areas. Combine the EPA 100-year floodplain with frequently flooded soil areas from gSSURGO, then use inundation frequency data to remove areas that are too frequently flooded.
    - Using Extract by Mask, clip the gSSURGO soils data to the Southeast Blueprint continental extent.
    - Join the "muaggatt" table to the soils data using the ArcGIS Add Join function, then make a copy of the raster to preserve the join.
    - Reclassify the joined raster using the "flood frequency dominate condition" field. Assign a value of 1 to the "Very frequent", "Frequent", or "Occasional" classes, and assign a value of 0 to the "Very rare", "Rare", or "None" classes.

    - Reclassify the joined raster again using the “flood frequency maximum” field. Assign a value of 1 to the “Very frequent”, “Frequent”, or “Occasional” classes and assign a value of 0 to the “Very rare”, “Rare”, or “None” classes.
    - Combine the two soil flood frequency rasters produced above with the EPA 100-year floodplain raster using the ArcGIS Cell Statistics “MAXIMUM” function. This enhances the EPA floodplain layer by adding additional potentially flooded areas.
    - Reproject the inundation frequency data and convert it to 30 m using the ArcGIS Project Raster function with bilinear resampling type.
    - Extract pixels with an inundation frequency score >5 from the resampled inundation layer using the ArcGIS Spatial Analyst Conditional function. This is a draft threshold to remove areas from the enhanced floodplain layer that may be too wet for river cane. In the output raster, assign a value of 0 to pixels with an inundation frequency score >5 and assign all other pixels a value of 1.
    - Remove frequently inundated pixels from the enhanced floodplain layer using the ArcGIS Spatial Analyst Times function.
- Remove coastal wetlands from the improved floodplain layer to define potential river cane restoration/management areas.
    - Make a copy of the NOAA C-CAP raster. Using the ArcGIS Spatial Analyst Conditional function, assign a value of 0 to the C-CAP estuarine wetland classes (16, 17, 18), barren land class (20), and unconsolidated shore class (19). All other pixels retain the value from the improved floodplain layer created above (either 1 or 0).
    - Using the ArcGIS Times function, use the reclassified C-CAP data to remove the specified landcover classes from the improved floodplain layer.
    - Reclassify the stable coastal wetland indicator to extract non-wetland areas. Assign a value of 1 to pixels scored as NoData or 0, and a value of 0 to pixels scored as 1 or 2.
    - Using the ArcGIS Times function, use the reclassified stable coastal wetlands indicator to further refine the improved floodplain layer by removing coastal wetlands.
    - The resulting layer represents potential river cane restoration/management areas.
- Because of river cane’s unique value to Tribal communities, use Tribal lands to elevate potential restoration/management areas that are on or near Tribal lands.
    - Make a copy of the census AIANNH layer. Extract the lands of federally recognized Tribes by using the ArcGIS Select function to pull out polygons with a AIANNHR value of “F”.
    - Convert the resulting federally recognized Tribal lands layer to a raster. Assign them a value of 1,000 in the output raster.
    - Buffer the federally recognized Tribal lands by 60 miles.
    - Convert the buffered federally recognized Tribal lands to raster. Assign them a value of 100 in the output raster.
- Protected areas offer an opportunity to restore and manage river cane. Use protected areas to elevate potential restoration/management areas on protected lands.
    - Prepare the protected areas data, starting with PAD-US 4.0. To exclude areas that do not meet the intent of this indicator, remove areas with location designations (Loc_Ds) of ‘School Trust Land’, ‘School Lands’, ‘School Land’, ‘State Land Board’, or ‘3201’.

    - Remove areas with the designation type (Des_Tp) of 'Military' or "Proclamation'. Military lands are not primarily managed for conservation. The proclamation category represents the approved boundary of public lands, within which land protection is authorized to occur, but not all lands within the proclamation boundary are necessarily currently in a conserved status.
    - Based on feedback from expert reviewers, add back in Army Corps of Engineers lands that were removed with the rest of the military lands.
    - Remove areas with the owner name of 'BOEM'. These Outer Continental Shelf lease blocks serve as the "legal definition for BOEM offshore boundary coordinates...for leasing and administrative purposes" (BOEM).
    - Remove areas with a category of 'Proclamation' (see explanation above).
    - PAD-US 4.0 is missing state wildlife management area boundaries in Oklahoma. Extract those from PAD-US 3.0 by using a combination of a state name of 'Oklahoma' and local designation of 'State Wildlife Management Area'.
    - Merge the selected polygons from PAD-US 4.0 and PAD-US 3.0, then convert to raster and reclassify to assign a value of 10 to protected areas.
- Rank potential river cane restoration/management areas based on their proximity to Tribal lands and protected status.
    - Combine the following rasters using the ArcGIS Cell Statistics "MAXIMUM" function: potential river cane restoration/management areas (1), Tribal lands (1,000), a 60 mi buffer around Tribal lands (100), and protected lands (10). Lands within the extent of the Southeast Blueprint but outside the identified potential river cane restoration/management areas retain a value of 0.
    - Reclassify the above raster to assign ranks based on proximity to Tribal lands and protected status, as seen in the final legend values below.
- As a final step, clip to the extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Potential river cane restoration/management area on Tribal land
4 = Potential river cane restoration/management area on protected land near Tribal lands
3 = Potential river cane restoration/management area near Tribal lands
2 = Potential river cane restoration/management area on protected land
1 = Potential river cane restoration/management area
0 = Not identified as a river cane restoration/management area

Known Issues

- We removed areas that were identified as estuarine wetland classes or coastal wetlands. By removing these areas, we may have also removed some places where river cane restoration could occur. Other types of cane, like switch cane (*Arundinaria tecta*) can grow in coastal areas.
- We removed areas that were highly inundated or inundated all of the time because they were likely too wet for river cane to occur. However, we are still likely are overpredicting river cane in wetlands or highly saturated areas. Research in western North Carolina suggests that river cane is not a wetlands plant and prefers well drained soils (Tanner et. al. 2011).
- There are historical records of river cane being used by Tribes in New Mexico. We extended this model to include all of Texas and Oklahoma to capture restoration opportunities in places where river cane may have historically occurred but has since been eradicated (Roger Cain, personal communication).
- We are likely overpredicting restoration opportunities for river cane in southern Florida, which is outside river cane's documented historical extent. However, after talking with experts during the external review process, we chose to not limit this indicator based on historical extent. We also did not consult with any federally recognized Tribes in peninsular Florida about this indicator, we and may refine our approach in southern Florida in the future as we receive more input.

Recommendations for Co-Management

This indicator reflects the potential for relationships between Indigenous Nations and land holders, particularly of public lands, to build co-management practices that prioritize healthy river cane ecosystems. Working with Tribes to help steward river cane can support a larger network of conservation efforts for this culturally and ecologically significant species. Below are general steps that can help interested land managers establish a co-stewardship strategy to manage river cane with federally recognized Tribes.

- State fish and wildlife agencies:
    - Consult your agency's guidance on Tribal engagement.
    - Refer to your State Wildlife Action Plan (SWAP): Since 2005, every state and territorial fish and wildlife agency has produced a SWAP, which is updated every 10 years. These plans identify Species of Greatest Conservation Need (SGCN), or species most in need of conservation action. The 2015 West Virginia SWAP lists river cane as a plant SGCN and it is referenced it in at least eight other SWAPs (NC, SC, GA, MS, LA, OK, AR, KY)—highlighted for its value for animal SGCN, its historical importance, and the need for further research on its benefits to wildlife. Some SWAPs also identify river cane restoration as a key conservation strategy to support other SGCN.
- Private organization and land holders: Refer to the literature cited below and to the Rivercane Restoration Alliance for guidance.
    - Identify nearby federally recognized Tribes at Search Federally Recognized Tribes | Indian Affairs (bia.gov).

- Contact the relevant Tribal government offices in your area, such as those within the departments of natural resources or cultural and historic preservation. Consider providing detailed information, including potential access information.
- Establish a collaborative plan for Tribal member access to river cane for harvesting or co-management.
- Note that engagement may take time and involve sensitivities around knowledge practices.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Anderson, M.K. and Oakes, T. 2011. Plant Guide for Giant Cane *Arundinaria gigantea*. USDA Natural Resources Conservation Service, National Plant Data Team. Greensboro, NC 27401

Chavez, Will. August 14, 2012. River cane added to Culturally Protected Species List. Cherokee Phoenix. [https://www.cherokeephoenix.org/culture/river-cane-added-to-culturally-protected-species-list/article_f0953e8b-5c29-5380-b2b8-b5ad39260c18.html].

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Garibaldi, A. and N. Turner. 2004. Cultural keystone species: implications for ecological conservation and restoration. Ecology and Society 9(3): 1. [http://www.ecologyandsociety.org/vol9/iss3/art1/].

Kniffen, F.B., H.F. Gregory, and G.A. Stokes. 1987. Tribes of Louisiana from 1542 to the Present. Louisiana State University Press, Baton Rouge.

Lyver, POB., Akins, A., Phipps, H., Kahui V., Towns, DR., Moller, H. 2016. Key biocultural values to guide restoration action and planning in New Zealand. Restoration Ecology. Volume 24, Issue 3, 314–323.

Noss, R., Scott, M., and LaRoe, E.T. 199). Endangered eco-systems of the United States: A preliminary assessment of loss and degradation. 128.

Schoonover, J.E., Williard, K.W.J., Zaczek, J.J., Mangun J.C., and Carver, A.D. 2005. Nutrient Attenuation in Agricultural Surface Runoff by Riparian Buffer Zones in Southern Illinois, USA. Agroforestry Systems 64, 169–180. [https://doi.org/10.1007/s10457-004-0294-7].

Schoonover, J.E. and Williard, K.W.J. 2007. Ground Water Nitrate Reduction in Giant Cane and Forest Riparian Buffer Zones. JAWRA Journal of the American Water Resources Association. 39. 347-354. [https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1752-1688.2003.tb04389.x].

Tanner, B.R., Kinner, D.A., Griffith, A.D., Young R. S., and Sorrell, L.M. 2011.Presence of Arundinaria gigantea (river cane) on numerous non-wetland sites suggests improper ecological classification of the species. Wetlands Ecology and Management 19, 521–532 (2011). [https://doi.org/10.1007/s11273-011-9233-3].

Wall, J., Moola, F., Lukawiecki, J. 2025. Indigenous-led conservation improves outcomes in protected areas. Nature Reviews Biodiversity. [https://doi.org/10.1038/s44358-025-00054-9].

South Atlantic Forest Birds

This indicator is an index of habitat suitability for twelve upland hardwood and forested wetland bird species (wood thrush, whip-poor-will, American woodcock, red-headed woodpecker, Chuck-will's widow, hooded warbler, Kentucky warbler, Acadian flycatcher, Northern parula, black-throated green warbler, prothonotary warbler, Swainson's warbler) based on patch size and other ecosystem characteristics such as proximity to water and proximity to forest and ecotone edge. It originates from Southeast Gap Analysis Project and Designing Sustainable Landscapes bird habitat models.

This indicator was last updated in October 2023.

Reason for Selection

This indicator represents ecologically important thresholds in forest patch size and other characteristics, such as distance to forest and ecotone edge. Many species are sensitive to these features of the landscape, but forest birds' habitat relationships are the most extensively studied and have available habitat models. The needs of these forest bird species are increasingly restrictive at higher index values, reflecting better quality forested wetland and upland hardwood habitat.

Input Data

- South Atlantic Blueprint 2021 extent
- Southeast Blueprint 2023 extent
- 2019 National Land Cover Database (NLCD): Land cover
- Base Blueprint 2022 extent
- The following Southeast GAP species models:

  Chuck-will's widow (bCWWI)
  Red-headed woodpecker (bRHWO)
  Wood thrush (bWOTH)
  Whip-poor-will (bWPWI)
  American woodcock (bAMWO)
  Hooded warbler (bHOWA)

Kentucky warbler (bKEWA)
Acadian flycatcher (bACFL)
Black-throated green warbler (bBTNW)
Northern parula (bNOPA)
Prothonotary warbler (bPROW)
Swainson's warbler (bSWWA)

- Select models from the South Atlantic Migratory Bird Initiative (SAMBI) Designing Sustainable Landscapes (DSL) avian habitat project:

  Prothonotary warbler (bPROW)
  Swainson's warbler (bSWWA)

Mapping Steps

The index of forest birds is based on scores representing increasingly restrictive limitations of potential habitat for a suite of species. Species-based constraints are:

- If a pixel is identified in the wood thrush (1 ha minimum patch), whip-poor-will (no more than 250 m into forest interior), red-headed woodpecker (interior patch within 500 m of early successional habitat), Chuck-will's widow (interior patch within 500 m of early successional habitat), or American woodcock (within 125 m of ecotone edge) Southeast GAP models, give it a value of 1. If not, give it a value of 0.
- If a pixel is identified in the hooded warbler (15 ha minimum patch) or Kentucky warbler (17 ha minimum patch in wet hardwoods) Southeast GAP models, give it a value of 2. If not, it retains the value from step 1.
- If a pixel is identified in the Acadian flycatcher (40 ha minimum patch) Southeast GAP model, give it a value of 3. If not, it retains the value from step 2.
- If a pixel is identified in the Northern parula (70 ha minimum patch) Southeast GAP model, black-throated green warbler (100 ha minimum patch) Southeast GAP model, or the prothonotary warbler (70 ha minimum patch within 120 m of water) Southeast GAP or DSL model, give it a value of 4. If not, it retains the value from step 3.
- If a pixel is identified in the Swainson's warbler (350 ha minimum patch within 250 m of water) Southeast GAP or DSL model, give it a value of 5. If not, it retains the value from step 4.
- If a pixel is identified in the NLCD as low, medium, or high intensity developed, assign it a value of 0. If not, it retains the value from step 5.
- Clip to the spatial extent of the 2021 South Atlantic Blueprint. The data outside this area was not reviewed by the indicator team who created it, so we decided not to use those data.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Very large patches near water (potential for presence of Swainson's warbler)
4 = Large patches often near water (potential for presence of Northern parula, black-throated green warbler, or prothonotary warbler)
3 = Medium patches (potential for presence of Acadian flycatcher)
2 = Small patches often near water (potential for presence of hooded warbler or Kentucky warbler)
1 = Very small patches or near open areas (potential for presence of wood thrush, whip-poor-will, red-headed woodpecker, Chuck-will's widow, or American woodcock)
0 = Less potential for presence of forest bird index species

Known Issues

- This indicator predicts less habitat for Swainson's warbler and prothonotary warbler in the Apalachicola basin because one of the two models used for these species (DSL) does not cover this area.
- It overpredicts habitat in areas that have been converted to other land uses since the release of 2001 GAP landcover data.
- Thresholds used in this indicator focus primarily on species presence and larger patches are likely needed for source populations of any of these species.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Homer, Collin G., Dewitz, Jon A., Jin, Suming, Xian, George, Costello, C., Danielson, Patrick, Gass, L., Funk, M., Wickham, J., Stehman, S., Auch, Roger F., Riitters, K. H., Conterminous United States land cover change patterns 2001–2016 from the 2016 National Land Cover Database: ISPRS Journal of Photogrammetry and Remote Sensing, v. 162, p. 184–199. [https://doi.org/10.1016/j.isprsjprs.2020.02.019].

Southeast Gap Analysis Project. SE-GAP Vertebrate Predicted Habitat. [http://www.basic.ncsu.edu/segap/DataServer.html].

Williams SG, Rubino MJ. 2012. Designing Sustainable Landscapes Project. Biodiversity and Spatial Information Center, North Carolina State University. [http://www.basic.ncsu.edu/dsl/hab.html]. Accessed April 2014.

Williams SG, Rubino MJ, Silvano AL, Elliott MJ, Cook K, Smith S. 2010. Southeast Gap Analysis Project. Biodiversity and Spatial Information Center, North Carolina State University; Alabama Gap Project, Alabama Cooperative Fish & Wildlife Research Unit, Auburn University; Natural Resources and Spatial Analysis Laboratory, University of Georgia. Accessed April 2014. [http://www.basic.ncsu.edu/segap/].

South Atlantic Low-Urban Historic Landscapes

This cultural resource indicator is an index of sites on the National Register of Historic Places surrounded by limited urban development. It identifies significant historical places that remain connected to their context in the natural world. It uses the National Land Cover Database and historic places data from the National Park Service and various state historic resource agencies.

This indicator was last updated in October 2023.

Reason for Selection

Low-urban historic landscapes indicate significant cultural landscapes whose cultural context has been less impacted by urban development. Cultural landscapes are "properties [that] represent the combined works of nature and of man" (UNESCO 2012). Loss of natural habitat within these cultural landscapes reduces their overall historical and cultural value.

Input Data

- South Atlantic Blueprint 2021 extent
- Southeast Blueprint 2023 extent
- 2016 National Land Cover Database (NLCD): Land cover
- The following National Register of Historic Places data from the National Park Service and various State Historic Preservation Officers:

  The National Register of Historic Places, accessed March 2021: All point layers from the NRIS_CR_Standards_Public geodatabase, including historic buildings, districts, objects, sites and structures. We use this data only in Alabama and Georgia, where no improved state-specific layer is available.

  Some states maintain their own, improved versions of the National Register of Historic Places, while other states rely on the nationwide version maintained by the National Park Service. In Virginia, North Carolina, Florida, and South Carolina, we obtained the following state-specific point and polygon data for places on the National Register:

  - Virginia listed historic districts boundaries
  - Virginia listed site points
  - North Carolina local district boundaries
  - North Carolina NCHPO_NR_SL_DOE_Boundaries
  - North Carolina NCHPO points
  - Florida national register
  - South Carolina NRHP_Districts-Areas_Non-Restricted

- SC_NRHP_Structures_points Non-Restricted

The National Register of Historic Places reflects what Americans value in their historic built environment. It is the collection of our human imprint on the landscape that records through time our changing relationship with the landscape, bridging between modern life and our history by providing, as closely as possible, experiences that evoke our empathy and understanding of previous eras.

Mapping Steps

- Define urban areas using the following classes from the 2019 NLCD - Developed, High Intensity; Developed, Medium Intensity; Developed, Low Intensity; Developed, Open Space. Classify all urban pixels as 1 and all other pixels as 0.
- Calculate the percent urban in a 270 m radius circle for each pixel using the ArcGIS Focal Statistics function. Since the NLCD data resolution is 30 m pixels, use 270 m (9 pixels) to approximate a 250 m radius. Retain all pixels that are <50% urban within a 270 m radius.
- Create a historic places layer as follows:
    - The North Carolina NCHPO points file contains points for both state-level and national-level historic places. To make these data comparable with data from other states, remove the state-specific points using information from the attribute table (remove any point that has a blank value for the YearNR field).
    - Merge together the state-specific point layers (NC, SC, and VA) and the National Park Service-maintained National Register of Historic Places point layers (AL and GA) and buffer by 100 m.
    - Merge all polygon data and buffered point data into one layer and convert to a 30 m raster.
- Use the historic places raster to remove areas that fall outside of the historic places.
- Reclassify the above raster into 3 classes, seen in the final indicator values below.
- Clip to the spatial extent of the South Atlantic Blueprint.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Historic place with nearby low-urban buffer
1 = Historic place with nearby high-urban buffer
0 = Not in the National Register of Historic Places

Known Issues

- There are spatial mapping errors for some of the historic areas.

- Some historic areas with cultural importance are not necessarily captured in the National Register of Historic Places.
- The approach to measuring urban growth doesn't capture degradation to historic places that were historically in larger cities (e.g., courthouses and other downtown buildings). It also doesn't distinguish between historic places that have always been urban and historic places that used to be low-urban.
- This layer underrepresents some historic areas in Georgia and Alabama because we only used the point data from the National Register of Historic Places maintained by the National Park Service. We omit the polygon layers because they contain many GIS errors and often overestimate the extent of historic sites.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Homer, Collin G., Dewitz, Jon A., Jin, Suming, Xian, George, Costello, C., Danielson, Patrick, Gass, L., Funk, M., Wickham, J., Stehman, S., Auch, Roger F., Riitters, K. H., Conterminous United States land cover change patterns 2001–2016 from the 2016 National Land Cover Database: ISPRS Journal of Photogrammetry and Remote Sensing, v. 162, p. 184–199. [https://doi.org/10.1016/j.isprsjprs.2020.02.019].

Stutts M. 2014. National Register of Historic Places. National Register properties are located throughout the United States and their associated territories around the globe. [https://irma.nps.gov/DataStore/Reference/Profile/2210280].

UNESCO (2012) Operational Guidelines for the Implementation of the World Heritage Convention [1]. UNESCO World Heritage Centre. Paris. Page 14. [https://whc.unesco.org/archive/opguide12-en.pdf].

Urban Park Size

This cultural resource indicator measures the size of parks and beaches in the urban environment. It originates from the Protected Areas Database of the United States, OpenStreetMap, U.S. Census Bureau urban areas, and the National Land Cover Database.

This indicator was last updated in October 2024.

Reason for Selection

Protected natural areas in urban environments provide urban residents a nearby place to connect with nature and offer refugia for some species. They help foster a conservation ethic by providing opportunities for people to connect with nature, and also support ecosystem services like offsetting heat island effects (Greene and Millward 2017, Simpson 1998), water filtration, stormwater retention, and more (Hoover and Hopton 2019). In addition, parks, greenspace, and greenways can help improve

physical and psychological health in communities (Gies 2006). Urban park size complements the equitable access to potential parks indicator by capturing the value of existing parks.

Input Data

- Southeast Blueprint 2024 extent
- Southeast Blueprint 2024 subregions
- FWS National Realty Tracts, accessed 12-13-2023
- Protected Areas Database of the United States (PAD-US): PAD-US 3.0 national geodatabase - Combined Proclamation Marine Fee Designation Easement, accessed 12-6-2023
- 2020 Census Urban Areas from the U.S. Census Bureau's urban-rural classification; download the data, read more about how urban areas were redefined following the 2020 census
- OpenStreetMap data "multipolygons" layer, accessed 12-5-2023

  A polygon from this dataset is considered a beach if the value in the "natural" tag attribute is "beach". Data for coastal states (VA, NC, SC, GA, FL, AL, MS, LA, TX) were downloaded in .pbf format and translated to an ESRI shapefile using R code. OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- 2021 National Land Cover Database (NLCD): Percent developed imperviousness
- 2023 NOAA coastal relief model: volumes 2 (Southeast Atlantic), 3 (Florida and East Gulf of Mexico), 4 (Central Gulf of Mexico), and 5 (Western Gulf of Mexico), accessed 3-27-2024

Mapping Steps

- Create a seamless vector layer to constrain the extent of the urban park size indicator to inland and nearshore marine areas <10 m in depth. The deep offshore areas of marine parks do not meet the intent of this indicator to capture nearby opportunities for urban residents to connect with nature. Shallow areas are more accessible for recreational activities like snorkeling, which typically has a maximum recommended depth of 12-15 meters. This step mirrors the approach taken in the Caribbean version of this indicator.
    - Merge all coastal relief model rasters (.nc format) together using QGIS "create virtual raster".
    - Save merged raster to .tif and import into ArcPro.
    - Reclassify the NOAA coastal relief model data to assign areas with an elevation of land to -10 m a value of 1. Assign all other areas (deep marine) a value of 0.
    - Convert the raster produced above to vector using the "RasterToPolygon" tool.
    - Clip to 2024 subregions using "Pairwise Clip" tool.
    - Break apart multipart polygons using "Multipart to single parts" tool.
    - Hand-edit to remove deep marine polygon.
    - Dissolve the resulting data layer.
    - This produces a seamless polygon defining land and shallow marine areas.

- Clip the census urban area layer to the bounding box of NoData surrounding the extent of Southeast Blueprint 2024.
- Clip PAD-US 3.0 to the bounding box of NoData surrounding the extent of Southeast Blueprint 2024.
- Remove the following areas from PAD-US 3.0, which are outside the scope of this indicator to represent parks:
    - All School Trust Lands in Oklahoma and Mississippi (Loc Des = “School Lands” or “School Trust Lands”). These extensive lands are leased out and are not open to the public.
    - All tribal and military lands (“Des_Tp” = ‘TRIBL’ or “Des_Tp” = ‘MIL’). Generally, these lands are not intended for public recreational use.
    - All BOEM marine lease blocks (“Own_Name” = ‘BOEM’). These Outer Continental Shelf lease blocks do not represent actively protected marine parks, but serve as the “legal definition for BOEM offshore boundary coordinates…for leasing and administrative purposes” (BOEM).
    - All lands designated as “proclamation” (“Des_Tp” = ‘PROC’). These typically represent the approved boundary of public lands, within which land protection is authorized to occur, but not all lands within the proclamation boundary are necessarily currently in a conserved status.
- Retain only selected attribute fields from PAD-US to get rid of irrelevant attributes.
- Merged the filtered PAD-US layer produced above with the OSM beaches and FWS National Realty Tracts to produce a combined protected areas dataset.
- The resulting merged data layer contains overlapping polygons. To remove overlapping polygons, use the Dissolve function.
- Clip the resulting data layer to the inland and nearshore extent.
- Process all multipart polygons (e.g., separate parcels within a National Wildlife Refuge) to single parts (referred to in Arc software as an “explode”).
- Select all polygons that intersect the census urban extent within 0.5 miles. We chose 0.5 miles to represent a reasonable walking distance based on input and feedback from park access experts. Assuming a moderate intensity walking pace of 3 miles per hour, as defined by the U.S. Department of Health and Human Service’s physical activity guidelines, the 0.5 mi distance also corresponds to the 10-minute walk threshold used in the equitable access to potential parks indicator.
- Dissolve all the park polygons that were selected in the previous step.
- Process all multipart polygons to single parts (“explode”) again.
- Add a unique ID to the selected parks. This value will be used in a later step to join the parks to their buffers.
- Create a 0.5 mi (805 m) buffer ring around each park using the multiring plugin in QGIS. Ensure that “dissolve buffers” is disabled so that a single 0.5 mi buffer is created for each park.
- Assess the amount of overlap between the buffered park and the census urban area using “overlap analysis”. This step is necessary to identify parks that do not intersect the urban area, but which lie within an urban matrix (e.g., Umstead Park in Raleigh, NC and Davidson-

Arabia Mountain Nature Preserve in Atlanta, GA). This step creates a table that is joined back to the park polygons using the UniqueID.

- Remove parks that had ≤10% overlap with the urban areas when buffered. This excludes mostly non-urban parks that do not meet the intent of this indicator to capture parks that provide nearby access for urban residents. Note: The 10% threshold is a judgement call based on testing which known urban parks and urban National Wildlife Refuges are captured at different overlap cutoffs and is intended to be as inclusive as possible.
- Calculate the GIS acres of each remaining park unit using the Add Geometry Attributes function.
- Buffer the selected parks by 15 m. Buffering prevents very small and narrow parks from being left out of the indicator when the polygons are converted to raster.
- Reclassify the parks based on their area into the 7 classes seen in the final indicator values below. These thresholds were informed by park classification guidelines from the National Recreation and Park Association, which classify neighborhood parks as 5-10 acres, community parks as 30-50 acres, and large urban parks as optimally 75+ acres (Mertes and Hall 1995).
- Assess the impervious surface composition of each park using the NLCD 2021 impervious layer and the Zonal Statistics "MEAN" function. Retain only the mean percent impervious value for each park.
- Extract only parks with a mean impervious pixel value <80%. This step excludes parks that do not meet the intent of the indicator to capture opportunities to connect with nature and offer refugia for species (e.g., the Superdome in New Orleans, LA, the Astrodome in Houston, TX, and City Plaza in Raleigh, NC).
- Extract again to the inland and nearshore extent.
- Export the final vector file to a shapefile and import to ArcGIS Pro.
- Convert the resulting polygons to raster using the ArcPy Feature to Raster function and the area class field.
- Assign a value of 0 to all other pixels in the Southeast Blueprint 2024 extent not already identified as an urban park in the mapping steps above. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Use the land and shallow marine layer and "extract by mask" tool to save the final version of this indicator.
- Add color and legend to raster attribute table.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = 75+ acre urban park
5 = 50 to <75 acre urban park
4 = 30 to <50 acre urban park
3 = 10 to <30 acre urban park
2 = 5 to <10 acre urban park
1 = <5 acre urban park
0 = Not identified as an urban park

Known Issues

- This indicator does not include park amenities that influence how well the park serves people and should not be the only tool used for parks and recreation planning. Park standards should be determined at a local level to account for various community issues, values, needs, and available resources.
- This indicator includes some protected areas that are not open to the public and not typically thought of as "parks", like mitigation lands, private easements, and private golf courses. While we experimented with excluding them using the public access attribute in PAD, due to numerous inaccuracies, this inadvertently removed protected lands that are known to be publicly accessible. As a result, we erred on the side of including the non-publicly accessible lands.
- The NLCD percent impervious layer contains classification inaccuracies. As a result, this indicator may exclude parks that are mostly natural because they are misclassified as mostly impervious. Conversely, this indicator may include parks that are mostly impervious because they are misclassified as mostly natural. We tested many Southeast parks for misclassification issues, and the 80% average impervious threshold seemed to best capture edge cases like Old Fourth Ward Park in Atlanta, GA (where NLCD overestimates impervious) and City Plaza in Raleigh, NC (where NLCD underestimates impervious). However, some parks are likely still represented incorrectly in this indicator.
- This indicator includes beaches from OpenStreetMap, which is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the boundary of a park) or incorrect tags (e.g., labelling an area as a park that is not actually a park). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new parks to improve the accuracy and coverage of this indicator in the future.

Other Things to Keep in Mind

- This indicator calculates the area of each park using the park polygons from the source data. However, simply converting those park polygons to raster results in some small parks and narrow beaches being left out of the indicator. To capture those areas, we buffered parks and beaches by 15 m and applied the original area calculation to the larger buffered polygon, so as

not to inflate the area by including the buffer. As a result, when the buffered polygons are rasterized, the final indicator has some areas of adjacent pixels that receive different scores. While these pixels may appear to be part of one contiguous park or suite of parks, they are scored differently because the park polygons themselves are not actually contiguous.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

Gies E. (2006). The health benefits of parks: how parks help keep Americans fit & healthy. The Trust for Public Land. Accessed February 16, 2022. [https://www.tpl.org/wp-content/uploads/2014/03/benefits_HealthBenefitsReport.pdf].

Greene, C. S., & Millward, A. A. (2017). Getting closure: The role of urban forest canopy density in moderating summer surface temperatures in a large city. Urban ecosystems, 20(1), 141-156. [https://link.springer.com/article/10.1007/s11252-016-0586-5].

Hoover, F. A., & Hopton, M. E. (2019). Developing a framework for stormwater management: leveraging ancillary benefits from urban greenspace. Urban ecosystems, 22(6), 1139-1148. [https://link.springer.com/article/10.1007/s11252-019-00890-6].

Mertes, James D. and James R. Hall. 1995. Park, recreation, open space, and greenway guidelines. A project of the National Recreation and Park Association and American Academy for Park and Recreation Administration. Alexandria, VA: NPRA.

NOAA National Centers for Environmental Information. (2023). Coastal Relief Models (CRMs) [Data set]. NOAA National Centers for Environmental Information. [doi: 10.25921/5ZN5-KN44].

Simpson, J. R. (1998). Urban forest impacts on regional cooling and heating energy use: Sacramento County case study. Journal of Arboriculture, 24, 201-214. [https://www.fs.usda.gov/research/treesearch/61731].

U.S. Census Bureau, Geography Division, Spatial Data Collection and Products Branch. June 1, 2017. 2010 Census Urban Area National. [https://catalog.data.gov/dataset/tiger-line-shapefile-2017-2010-nation-u-s-2010-census-urban-area-national].

U.S. Department of Health and Human Services. Physical Activity Guidelines for Americans, 2nd edition. Washington, DC: U.S. Department of Health and Human Services; 2018. [https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf].

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2022, Protected Areas Database of the United States (PAD-US) 3.0: U.S. Geological Survey data release. [https://doi.org/10.5066/P9Q9LQ4B].

Yang, Limin, Jin, Suming, Danielson, Patrick, Homer, Collin G., Gass, L., Bender, S.M., Case, Adam, Costello, C., Dewitz, Jon A., Fry, Joyce A., Funk, M., Granneman, Brian J., Liknes, G.C.,Rigge, Matthew B., Xian, George. 2018. A new generation of the United States National Land Cover Database—Requirements, research priorities, design, and implementation strategies:ISPRS Journal of Photogrammetry and Remote Sensing, v. 146, p. 108–123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

West Coastal Plain & Ouachitas Forested Wetland Birds

This indicator is an index of habitat suitability for five forested wetland bird species (Acadian flycatcher, Kentucky warbler, yellow-throated warbler, prothonotary warbler, red-shouldered hawk) within bottomland hardwood forests and riparian areas in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region. It uses metrics like patch size, dispersal distance, and distance to water to assess the potential for habitat to support sustainable populations of these birds. This indicator originates from the Lower Mississippi Valley Joint Venture's forested wetland decision support model for the WGCPO region.

This indicator was last updated in October 2023.

Reason for Selection

Some factors related to patch size and landscape composition that are important to forested wetland species are not well-represented by other indicators. Forested wetlands and associated bottomland hardwoods represent a unique and imperiled habitat in the region that supports birds and many other wildlife species. Maintaining the structure and integrity of these forested wetlands has conservation implications that extend beyond the West Gulf Coastal Plain/Ouachitas region due to the ecosystem's importance for birds migrating to and from Mexico, the West Indies, and Central and South America. This indicator also promotes consistency with the forest management and protection priorities of the Lower Mississippi Valley Joint Venture.

Input Data

- WGCPO Forested Wetland Decision Support Model; detailed description of methods is available under "Decision support model" in the West Gulf Coastal Plain/Ouachitas Forested Wetland Plan, Version 1.0

  This decision support model first characterized the landscape that was available to each forested wetland bird umbrella species for breeding. It identified landscapes with enough

potentially suitable habitat to support a minimum viable population based on the amount of woody wetland landcover, minimum viable population size and area requirements, and natal dispersal distance. Then, it prioritized that potential breeding landscape based on each species' unique habitat needs: distance to water, percent forest cover, flood tolerance and flood preference, and presence of bald cypress-tupelo floodplain forest. Lastly, it combined each species-specific habitat suitability model into the final composite index used in this indicator.

- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Reproject the WGCPO Forested Wetland Decision Support Model source data to NAD 1983 Contiguous USA Albers (EPSG 5070).
- Reclassify the source data as seen in the legend below. The WGCPO Forested Wetland Decision Support Model layer from the LMVJV has values ranging from 0 to 0.933365. Classify the data into five groups based on percentile priority within WGCPO regional boundary for the six umbrella species (Acadian flycatcher, Kentucky warbler, yellow-throated warbler, prothonotary warbler, red-shouldered hawk) as shown in the final indicator values below. Assign all pixels with an actual value of 0 in the WGCPO model a value of 0 in the indicator. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = High habitat suitability for forested wetland bird umbrella species (Acadian flycatcher, Kentucky warbler, yellow-throated warbler, prothonotary warbler, red-shouldered hawk) (score >80)
4 = Medium-high habitat suitability (score >60-80)
3 = Medium habitat suitability (score >40-60)
2 = Medium-low habitat suitability (score >20-40)
1 = Low habitat suitability for forested wetland bird umbrella species (score >0-20)
0 = Not suitable for forested wetland bird umbrella species (score = 0)

Known Issues

- The indicator does not account for important components of stand structure that impact habitat quality for these species. Key factors include: a) large tree diameter (>23 cm dbh), b) density of large trees (>40 trees >50 dbh/ha), c) low tree density (250-300/ha), d) mid-story cover (open), e) understory cover (open), f) understory cover (dense), g) moderate to well-developed canopy (60–70%), and h) small cavities (<10 inch diameter) or snag density of 5 snags/ha.
- The indicator does not account for variation in water quality.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

West Gulf Coastal Plain/Ouachitas Forested Wetland Landbird Plan. 2017. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5bfc99b221c67c2cdd60ab54/1543281078237/WGCPO_BLH_Riparian_Plan_Oct2017.pdf].

West Coastal Plain & Ouachitas Open Pine Birds

This indicator identifies areas with pine trees that, if managed for open condition, could support a population of three umbrella bird species (brown-headed nuthatch, Bachman's sparrow, red-cockaded woodpecker). It evaluates potential habitat in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region based on each species' habitat needs and population dynamics, prioritizing opportunities to restore and manage habitat to benefit open pine birds. Final scores reflect both the selectiveness of the species and whether an area meets the habitat requirements through one large patch, or clusters of smaller patches in sufficiently close proximity for breeding pairs to disperse. This indicator updates the Lower Mississippi Valley Joint Venture's open pine decision support model for the WGCPO region.

This indicator was last updated in October 2023.

Reason for Selection

The three bird species used in this indicator (brown-headed nuthatch, Bachman's sparrow, red-cockaded woodpecker) represent the structure and function of open pine ecosystems in the West Gul Coastal Plain. Open pine and other grassland birds are experiencing significant declines and are currently off-track for meeting the SECAS 10% goal, so it is important that the Blueprint capture opportunities to restore open pine habitat. This indicator also promotes consistency with the priorities of the Lower Mississippi Valley Joint Venture.

Input Data

- Update to the WGCPO Open Pine Decision Support Model

The Lower Mississippi Valley Joint Venture created the original data and methods in 2011 as the WGCPO Open Pine Decision Support Model. The original model used 2001 National Land Cover Database (NLCD) landcover as an input. This layer re-runs the model using the same methods, but an updated 2019 version of the NLCD. This model represents 3 species: brown-headed nuthatch (BHNU), Bachman's sparrow (BACS), and red-cockaded woodpecker (RCWO). It highlights patches of pine trees that are large enough to support a population of the umbrella species but are not necessarily in the open condition required to provide quality habitat. Its goal is to prioritize existing pine forests for open pine management actions.

Additional information about model assumptions

- The brown-headed nuthatch analysis assumed a density of 3 ha/pair, a minimum viable population of 28 pairs, a suitable habitat requirement of 84 ha, and a dispersal potential of 0.92 km (½ dispersal potential of 0.46 km).
- The Bachman's sparrow analysis assumed a density of 3 ha/pair, a minimum viable population of 50 pairs, a suitable habitat requirement of 150 ha, and a dispersal potential of 3 km (½ dispersal potential of 1.5 km).
- The red-cockaded woodpecker analysis assumed a density of 50 ha/pair, a minimum viable population of 20 pairs, a suitable habitat requirement of 1,000 ha, and a dispersal potential of 8 km (½ dispersal potential of 4 km).

Table 5. Model assumptions for each species in West Coastal Plains and Ouachitas open pine birds.

	D (ha/pair)	MVP (pairs)	Suitable habitat required (ha)	Dispersal potential (km)	½ Dispersal potential (km)
Red-cockaded woodpecker	50	20	1,000	8	4
Bachman's sparrow	3	50	150	3	1.5
Brown-headed nuthatch	3	28	84	0.92	0.46

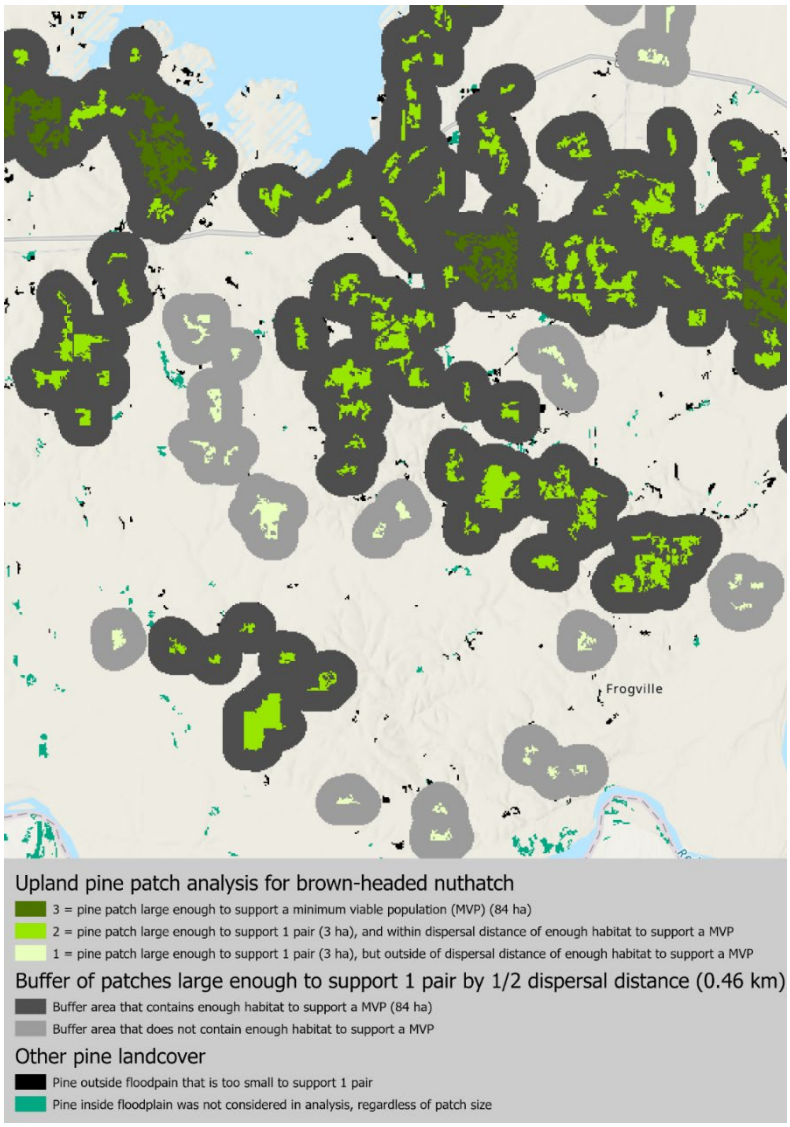


Figure 2. Example of analysis process on one of the umbrella species (brown-headed nuthatch).

- 2019 National Land Cover Database (NLCD): Land cover
- West Gulf Coastal Plain/Ouachitas floodplain map produced by the Lower Mississippi Valley Joint Venture (LMVJV) (contact the LMVJV for access)
- West Gulf Coastal Plain/Ouachitas Bird Conservation Region (WGCPO BCR) (2010 version, contact the LMVJV for access)
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Buffer the WGCPO BCR by 10 km to create the study area.
- Clip the 2019 NLCD to the study area.

- Pull out the evergreen and mixed forest classes from NLCD (values 42 and 43) to use as an estimate of the location of current pine forests.
- Remove all pine forests pixels that occur in the floodplain, using the West Gulf Coastal Plain/Ouachitas floodplain layer.
- Perform a patch analysis on the remaining pine pixels to determine the size of contiguous patches.
- Convert patch size from number of pixels to hectares.
- At this point, the model performs a series of the same analysis steps but modifies the thresholds to be specific for each species.
    - Remove all patches of forest that are not large enough to support at least one pair of birds. The BHNU and BACS analysis removed patches smaller than 3 ha, while the RCWO analysis removed patches smaller than 50 ha.
    - Buffer (enlarge) each patch by ½ the dispersal distance for that species. If the buffer of one patch intersects the buffer of another patch, then we assume the two patches are in close enough proximity that birds have the ability to disperse from one to another. The BHNU analysis buffered by 0.46 km, the BACS analysis buffered by 1.5 km, and the RCWO analysis buffered by 4 km.
    - Perform a new patch analysis on the buffers of the original patches. This identifies clusters of patches or areas that contain patches that are all close enough together that birds could disperse from one to another (to another, to another, etc.).
    - Analyze each cluster of patches to determine if they contain enough pine forest to support a minimum viable population (MVP) for the species. This analysis is done using the patches, so we exclude any patches of forest that aren't big enough to individually support one pair. The amount of forest needed to support an MVP was calculated by multiplying the hectares needed to support 1 pair by the number of pairs that represent an MVP of each species. The BHNU analysis used a minimum habitat amount of 84 ha (3 ha/pair times the MVP of 29 pairs = 84 ha), the BACS analysis used a minimum habitat amount of 150 ha (3 ha/pair times the MVP of 50 pairs = 150 ha), and the RCWO analysis used a minimum habitat amount of 1,000 ha (50 ha/pair times the MVP of 20 pairs = 1,000 ha).
    - If a cluster of patches contains enough pine forest to support at least one MVP, then all the original patches contained in that cluster of patches are identified as potential targets for open pine management.
- Reclassify the BHNU layer produced above so that:
    - 0 = Not an upland pine patch or an upland pine patch too small to support 1 pair
    - 1 = Patch large enough to support 1 pair, but outside of dispersal distance of enough habitat to support MVP
    - 2 = Patch is large enough to support 1 pair, and is within dispersal distance of enough habitat to support MVP
    - 3 = Patch alone is large enough to support MVP
- Reclassify BACS layer above so that:
    - 0 = Not an upland pine patch or an upland pine patch too small to support 1 pair

    - 10 = Patch large enough to support 1 pair, but outside of dispersal distance of enough habitat to support MVP
    - 20 = Patch is large enough to support 1 pair, and is within dispersal distance of enough habitat to support MVP
    - 30 = Patch alone is large enough to support MVP
- Reclassify RCWO layer above so that:
    - 0 = Not an upland pine patch or an upland pine patch too small to support 1 pair
    - 100 = Patch large enough to support 1 pair, but outside of dispersal distance of enough habitat to support MVP
    - 200 = Patch is large enough to support 1 pair, and is within dispersal distance of enough habitat to support MVP
    - 300 = Patch alone is large enough to support MVP
- Add three species layers above together using the Cell Statistics "SUM" function. The result is a raster with habitat values for BHNU in the hundreds place, BACS in the tens, and RCWO in the ones.
- Reclassify to create the legend values seen below:
    - 6 = 333
    - 5 = 331, 332
    - 4 = 312, 321, 322
    - 3 = 222
    - 2 = 122, 220, 221
    - 1 = 120, 121, 210
    - 0 = 0, 110
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Pine patch large enough to support a population of all 3 umbrella bird species (brown-headed nuthatch, Bachman's sparrow, red-cockaded woodpecker) if managed in open condition
5 = Pine patch large enough to support a population of 2 umbrella bird species if managed in open condition
4 = Pine patch large enough to support a population of 1 umbrella bird species if managed in open condition
3 = Pine patch part of a cluster of nearby patches able to support a population of all 3 umbrella bird species if managed in open condition
2 = Pine patch part of a cluster of nearby patches able to support a population of 2 umbrella bird species if managed in open condition
1 = Pine patch part of a cluster of nearby patches able to support a population of 1 umbrella bird species if managed in open condition
0 = Pine patch too small and isolated to support a population of any umbrella bird species or not an upland pine patch

Known Issues

- This indicator does not capture areas where planting new pine stands in existing agricultural areas would benefit open pine birds.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Lower Mississippi Valley Joint Venture. October 2011. West Gulf Coastal Plains/Ouachitas Open Pine Landbird Plan. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5bedd2650ebbe83079c0c7b1/1542312550791/WGCPO_Landbird_Open_Pine_Plan_Oct_2011.pdf].

Southeast Conservation Adaptation Strategy. 2021. Recent Trends in Southeastern Ecosystems (2021): Measuring Progress toward the SECAS Goal. [https://secassoutheast.org/pdf/SECAS-goal-report-2021.pdf].

U.S. Geological Survey (USGS). Published June 2021. National Land Cover Database (NLCD) 2019 Land Cover Conterminous United States. Sioux Falls, SD. [https://doi.org/10.5066/P9KZCM54].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

West Gulf Coast Mottled Duck Nesting

This indicator depicts marshes and grasslands along the coast of Louisiana and Texas that are important for mottled duck nesting, based on key biological parameters such as patch size, landcover type, and distance to brood rearing habitat. It originates from a mottled duck decision support tool developed by the Gulf Coast Joint Venture.

This indicator was last updated in October 2023.

Reason for Selection

The mottled duck is a non-migratory waterfowl species endemic to the Gulf coast that relies on a unique mix of coastal marshes and nearby grassland habitat for foraging, nesting, and raising young. The West Gulf Coast mottled duck population has declined significantly since the mid-1990s due to habitat loss from urban growth, agricultural development, saltwater intrusion, and invasive species (Moon et al. 2021, Krainyk et al. 2019, Moon et al. 2021). Mottled ducks are considered an indicator of coastal marsh health and function (Moon et al. 2015), and other indicators in the Blueprint were not capturing the important ecological role of grassland habitats near wetlands in the Western Gulf Coast.

Input Data

- Spatially Explicit Decision Support Tool for Guiding Habitat Conservation for Western Gulf Coast Mottled Ducks: Currently suitable nesting habitat

  Because mottled ducks use coastal marsh and agricultural landscapes somewhat differently, this model identified suitable nesting habitat in two different zones: agricultural and coastal. In agricultural areas, nesting cover is usually found in “fallow fields, low-intensity grazing systems, hay fields, native prairies, and other grasslands” with less than 30% woody cover (Krainyk et al. 2019). Here, the model defined suitable nesting habitat as patches of suitable agricultural or grassland landcover that are at least 40 acres in size, with a 0.025 m/m^2 edge-to-interior ratio, and no more than a mile from currently suitable brood-rearing habitat.

  In coastal marshes, “mottled ducks prefer nesting habitat characterized by tall, dense stands of grass, usually located in elevated sites within 150 meters of water” or “on narrow ridges or levees that are surrounded by marsh” (Krainyk et al. 2019). Here, the model defined suitable nesting habitat as patches of suitable agricultural, grassland, or wetland landcover that are at least 40 acres in size and no more than a mile from currently suitable brood-rearing habitat (very similar to the parameters used in the agricultural zone, but with the addition of wetland landcover and removal of the edge-to-interior ratio requirement).

- MODU_range_byInitiaveArea (contact Blair Tirpak for access at blair_tirpak@fws.gov)
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Divide the currently suitable nesting habitat layer into 10 equal area classes using the Slice tool in ArcPro.
- Create an extent raster with a value of 0 from the MODU_range_byInitiativeArea shapefile. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Mosaic the 10-class raster with the extent raster.
- Reclassify the resulting value field into the final indicator values seen below.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = 90th-100th percentile of suitable mottled duck nesting habitat
9 = 80th-90th percentile of suitable mottled duck nesting habitat
8 = 70th-80th percentile of suitable mottled duck nesting habitat
7 = 60th-70th percentile of suitable mottled duck nesting habitat
6 = 50th-60th percentile of suitable mottled duck nesting habitat
5 = 40th-50th percentile of suitable mottled duck nesting habitat
4 = 30th-40th percentile of suitable mottled duck nesting habitat
3 = 20th-30th percentile of suitable mottled duck nesting habitat
2 = 10th-20th percentile of suitable mottled duck nesting habitat
1 = 0-10th percentile of suitable mottled duck nesting habitat
0 = Not identified as suitable mottled duck nesting habitat (within TX and LA)

Known Issues

- This indicator represents a perspective on grasslands through the lens of a single species that requires grasslands to be in proximity to low salinity wetlands. As a result, it potentially excludes some larger blocks of intact grasslands.
- Mottled ducks also nest within some emergent coastal marshes; therefore, this indicator might include significant contiguous blocks of emergent wetlands in some areas, especially where the salinities don't exceed intermediate ranges.

- This indicator defines suitability for mottled duck based solely on landscape context and does not consider grassland condition (e.g., it does not differentiate overgrazed areas, grasslands dominated by exotics, etc.).

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Krainyk, Anastasia and Bart M. Ballard. 2014. Decision Support Tool: Prioritization of Mottled Duck (*Anas fulvigula*) Habitat for Conservation and Management in the Western Gulf Coast. Final report. [https://www.sciencebase.gov/catalog/item/547cb566e4b0bdc51793037c].

Krainyk, Anastasia, B. Ballard, M.G. Brasher, B.C. Wilson, M. W. Parr, C. K. Edwards. Decision support tool: Mottled duck habitat management and conservation in the Western Gulf Coast. Journal of Environmental Management, v. 203, p43-52. [https://doi.org/10.1016/j.jenvman.2018.09.054].

Moon, J.A. and S.E. Lehnen, K.L. Metzger, M.A. Squires, M.G. Brasher, B.C. Wilson, W.C. Conway, D.A. Haukos, B.E. Davis, F.C. Rohwer, E.M. Wehland, B.M. Ballard. Projected impact of sea-level rise and urbanization on mottled duck (*Anas fulvigula*) habitat along the Gulf Coast of Louisiana and Texas through 2100. Ecological Indicators, Volume 132, 2021, 108276, ISSN 1470-160X, [https://doi.org/10.1016/j.ecolind.2021.108276].

Moon, J.A., Haukos, D.A. and Conway, W.C. (2017), Seasonal survival of adult female mottled ducks. Jour. Wild. Mgmt., 81: 461-469. [https://doi.org/10.1002/jwmg.21221].

Moon, Jena, D. Haukos, W. Conway. 2015. Mottled Duck *(Anas fulvigula)* Movements in the Texas Chenier Plain Region. Journal of the Southeastern Association of Fish and Wildlife Agencies 2:255-261. [https://seafwa.org/sites/default/files/journal-articles/37%252520Moon%252520et%252520al%252520255-261.pdf].

Ringelman, K. M., Bonczek, E. S., Marty, J. R., Booth, A. R., & Dopkin, A. L. (2021). Survival of western Gulf Coast Mottled Ducks (*Anas fulvigula*) in the path of a Category 4 hurricane. Ecology and Evolution, 11, 15477– 15483. [https://doi.org/10.1002/ece3.8276].

Freshwater

Atlantic Migratory Fish Habitat

This indicator measures the condition of migratory fish habitat along the Atlantic coast within each catchment, using metrics of water quality, aquatic connectivity, habitat fragmentation, flow alteration, and more. Areas of excellent fish habitat are already in good condition and face few threats. Restoration opportunity areas are doing well in some respects, but restoration projects could significantly improve them. Degraded areas of opportunity face many challenges, and restoration projects are unlikely to increase available fish habitat unless particularly large in scope and scale. This indicator originates from the Atlantic Coast Fish Habitat Partnership's fish habitat conservation area mapping and prioritization project. It applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).

This indicator was last updated in October 2023.

Reason for Selection

This indicator represents aquatic connectivity between fresh and salt water in Atlantic drainages. It incorporates both physical barriers to connectivity and indirect barriers related to habitat quality. It also promotes consistency with the priorities of the Atlantic Coast Fish Habitat Partnership.

Input Data

- Atlantic Coast Fish Habitat Partnership (ACFHP) Fish Habitat Conservation Area Mapping and Prioritization Project: South Atlantic and Mid-Atlantic Diadromous Analysis
- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

  The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Convert the South Atlantic Diadromous Analysis from vector to 30 m raster using the FINALSCORE field.
- Convert the Mid-Atlantic Diadromous Analysis from vector to 30 m raster using the TotalPoints field.
- Combine the above rasters using the ArcPy Spatial Analyst Cell Statistics "MAXIMUM" function.

- Reclassify the above raster into 8 classes, seen in the final indicator values below.
- Clip to the EPA estimated floodplain.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

8 = Final score of 80 (areas of excellent fish habitat)
7 = Final score of 70 (areas of excellent fish habitat)
6 = Final score of 60 (restoration opportunity areas)
5 = Final score of 50 (restoration opportunity areas)
4 = Final score of 40 (restoration opportunity areas)
3 = Final score of 30 (restoration opportunity areas)
2 = Final score of 20 (restoration opportunity areas)
1 = Final score of 10 (degraded areas of opportunity)
0 = Final score of 0 (degraded areas of opportunity)

Known Issues

- This indicator under and overrepresents migratory fish habitat in some areas. The South Atlantic and Mid-Atlantic Diadromous Analysis did not include fish presence and fishing data because of inconsistent sampling methods across the study area and because this data was unavailable in many shallow water habitats.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Martin, Erik, Kat Hoenke, and Lisa Havel. Atlantic Coast Fish Habitat Partnership. Fish Habitat Conservation Area Mapping and Prioritization Project: A Prioritization of Atlantic Coastal, Estuarine, and Diadromous Fish Habitats for Conservation. August 2020. [https://www.atlanticfishhabitat.org/wp-content/uploads/2020/08/ACFHP-Mapping-and-Prioritization-Final-Report.pdf].

Floodplain Inundation

This indicator uses the frequency of inundation to measure the hydrologic function of the floodplain. It is based on Sentinel-2 satellite imagery and methods developed by Yvonne Allen of the U.S. Fish and Wildlife Service.

This indicator was last updated in November 2025.

Reason for Selection

"Although floodplains support high levels of biodiversity and some of the most productive ecosystems on Earth, they are also among the most converted and threatened ecosystems and therefore have recently become the focus of conservation and restoration programs across the United States and globally" (Opperman et al. 2010). The various aquatic indicators in the Blueprint depict different aspects of healthy, functional, and connected aquatic ecosystems, including landcover, barriers to fish passage, and more.

This indicator measures the frequency of inundation in order to assess hydrologic function within the floodplain. The timing, extent, and duration of floodplain inundation—in other words, when, where, and how often flooding occurs, and how long it lasts—has a significant impact on the quality of fish and wildlife habitat, the diversity of aquatic species, and the supply of ecosystem services (Allen 2016).

In a well-functioning floodplain system, water can move from lakes and mainstem rivers out onto the adjacent floodplain. Variation in high and low flow drives natural variation in the parts of the floodplain that are wet or dry at any given time. These fluctuations influence the availability and quality of food, spawning, and nursery habitat for aquatic species; the structure, composition, and productivity of wetlands and swamp forests; the availability of nutrients such as nitrogen; and the downstream impacts of flooding on natural and human communities (Allen 2016). Regular, but not persistent, inundation is essential for many floodplain-dependent species, including numerous fish (e.g., alligator gar), migratory waterfowl, and more.

When portions of the floodplain remain constantly wet or constantly dry, this often signals a loss of hydrologic function from human alteration. Outside of lakes and mainstem rivers, which naturally experience persistent inundation, constant flooding can result from dams, dikes, levees, and stream incision due to erosion. Restricting water to certain parts of the floodplain leaves other parts constantly dry, disrupting seasonal variability in the ebb and flow of water.

Input Data

- Southeast Floodplain Inundation Frequency dataset developed by Yvonne Allen with the U.S. Fish and Wildlife Service (last updated 4-2-2025)

  This Southeast-wide map of floodplain inundation frequency was created following the methods described in Allen 2016, then modified to use Google Earth Engine (Gorelick et. al

2017) and Sentinel-2 data from 2017 to present instead of Landsat data from 1980s to present. Sentinel-2 offers higher resolution and higher imaging frequency. This dataset measures the relative frequency of floodplain inundation on a scale from 0-100. The inundation frequency score is an index reflecting the number of satellite scenes in which a given location appeared to be flooded, out of the total number of available images for that site. Places that are constantly flooded, such as main river channels, receive a score of 90-100, while places that never flood receive a score of 0. Intermediate scores depict varying levels of periodic to frequent flood inundation.

- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data (Estimated_floodplain_CONUS.zip)
- Southeast Blueprint 2025 extent

Mapping Steps

- Reclassify the inundation frequency data into the following bins:
    - 0 = inundation index of 0 (no observed inundation)
    - 1 = inundation index of 91-100 (persistent inundation)
    - 2 = inundation index of 2-5 (occasional inundation)
    - 3 = inundation index of 6-20 (regular inundation)
    - 4 = inundation index of 21-90 (frequent inundation)
- Reclassify the EPA floodplain layer to assign areas within the 100-year floodplain a value of 1, and areas outside the floodplain a value of zero.
- Combine the reclassified inundation frequency score and the reclassified EPA floodplain layer. Add a value of 1 to the categorized inundation frequency value, then multiply by the reclassified floodplain raster. This assigns a value of zero to all areas outside the floodplain, while all areas within the floodplain receive their previous score + 1.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Frequent inundation within the floodplain (flooded in 21-90% of days with available data)
4 = Regular inundation within the floodplain (flooded in 6-20% of days with available data)
3 = Occasional inundation within the floodplain (flooded in 2-5% of days with available data)
2 = Persistent inundation within the floodplain (flooded in 91-100% of days with available data)
1 = No detected inundation within the floodplain (flooded in 0% of days with available data)
0 = Not identified as a floodplain

Known Issues

- When the Sentinel-2 satellite images were mosaiced together using Google Earth Engine, an error occurred where not all available images were used for all sites. This primarily affects two areas around 100 km^2 in size near Georgia and Florida (e.g., near Augusta, GA). This error has been reported to Google Earth Engine and will hopefully be rectified for future updates to this indicator. Patching the missing scenes using Landsat imagery would be another option to address this in subsequent revisions.
- This indicator takes an ecological view of the floodplain, which does not encompass permanent waterbodies, but rather focuses on the adjacent areas that are regularly flooded and dried. In a well-functioning hydrologic system, it is natural for some waterbodies, like mainstem river channels, to remain constantly flooded. However, persistent inundation can also signal hydrologically altered areas like impoundments. Given that other Blueprint aquatic indicators prioritize the open water areas of rivers and lakes, we erred on the side of not overprioritizing impoundments and focused this indicator on the parts of the floodplain that experience regular, but not constant, flooding. As a result, this indicator undervalues the importance of the open water of unaltered rivers and natural lakes.
- The Sentinel-2 imagery only goes back to 2017, which may not be long enough to see the full range of flood conditions that may occur. Between Sentinel-2 A and B, the same location is imaged every 5 days (each of the two satellites covers the same site every 10 days). However, this is an improvement over Landsat, which flies the same location every 16 days. Combining Landsat and Sentinel-2 imagery in the future could be a future option to explore help extend the covered hydroperiod further back to the 1980's, but Landsat does not provide the same resolution as the Sentinel-2 imagery.
- In the area just south of Guadalupe Mountains National Park in West Texas, this indicator depicts the floodplain as a series of straight lines that poorly match the actual floodplain. This is due to an error in the EPA 100-year floodplain map used in this indicator.
- While this indicator generally includes the open water area of reservoirs, some open water portions of reservoirs (e.g., Kerr Lake in NC/VA) are missing from the EPA estimated 100-year floodplain dataset.
- Due to the extent of the source data, this indicator includes some estuaries and parts of the nearshore marine environment that are not typically considered as part of a functional floodplain system. However, these areas receive a relatively low score of 2, as they are constantly inundated, so should not overly influence the Blueprint priorities.
- This indicator does not account for some types of human alteration to hydrology, including rice fields, some modifications to natural flow regimes due to dam releases, and others.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Allen, Y. “Landscape Scale Assessment of Floodplain Inundation Frequency Using Landsat Imagery,” *River Research and Applications* 32, no. 7 (September 1, 2016): 1609–20. [https://doi.org/10.1002/rra.2987].

Gorelick, Noel & Matt Hancher, Mike Dixon, Simon Ilyushchenko, David Thau, Rebecca Moore, Google Earth Engine: Planetary-scale geospatial analysis for everyone, Remote Sensing of Environment, Volume 202, 2017, Pages 18-27, ISSN 0034-4257. [https://doi.org/10.1016/j.rse.2017.06.031].

Opperman, J.J., Luster, R., McKenney, B.A., Roberts, M. and Meadows, A.W. (2010), Ecologically Functional Floodplains: Connectivity, Flow Regime, and Scale. JAWRA Journal of the American Water Resources Association, 46: 211-226. [https://doi.org/10.1111/j.1752-1688.2010.00426.x].

Gulf Migratory Fish Connectivity

This indicator captures how far upstream migratory fish in the Gulf of America have been observed. How far upstream migratory fish can travel reflects not just the presence of dams and other barriers, but also the presence of measures like fish ladders that allow specific species to access habitat upstream of dams. This indicator originates from The Nature Conservancy’s Southeast Aquatic Connectivity Assessment Project and U.S. Fish and Wildlife Service critical habitat data. It applies to the Environmental Protection Agency’s estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).

This indicator was last updated in October 2023.

Reason for Selection

Migratory fish presence reflects uninterrupted connections between freshwater, estuarine, and marine ecosystems. Aquatic connectivity benefits diadromous fish and is considered a high priority for the integrity of aquatic ecosystems. Larger diadromous fish, like sturgeon, are often more sensitive to disruptions in aquatic connectivity. Smaller fish can make better use of fish ladders and other fish passage measures than larger fish.

Input Data

- Southeast Aquatic Connectivity Assessment Project (SEACAP); see the final report for more information

  SEACAP developed linear spatial data on the presence of priority diadromous species. These layers are modified versions of the NHDPlus Version 2. These data were altered to contain presence of Alabama shad using data from the Atlantic States Marine Fisheries Commission (produced for the ASMFC by the Biodiversity and Spatial Information Center at North Carolina State University, Alexa McKerrow), and expert knowledge of the SEACAP Workgroup.

SEACAP also developed a functional river network layer (final SEACAP report, page 9). A functional river network is defined by those stream reaches that are accessible to a hypothetical fish within that network. The functional river network is defined by lines (streams). SEACAP also calculated "functional catchments," which are polygons that represent the catchment area that is associated with each of those functional networks.

Note: A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics. To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

- National Oceanic and Atmospheric Administration (NOAA) Gulf Sturgeon Critical Habitat
- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- U.S. Geological Survey (USGS) Watershed Boundary Dataset (WBD), accessed 8-11-2020: HUC6s, HUC12s; download the data
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Combine all the linework for Gulf sturgeon using the ArcPy Data Management Merge function. This includes line data from SEACAP and the NOAA critical habitat. Add and calculate a field showing that these are sturgeon lines.
- Combine all the linework for the Alabama shad, American shad, or striped bass from SEACAP using the ArcPy Data Management Merge function. Add and calculate a field showing these lines represent the above species.
- Assign the values from the two sets of linework above to HUC12s using two separate ArcPy Analysis Spatial Join functions.

- Add and calculate a new field. If it intersects a sturgeon line, give it a value of 2. Otherwise, if it intersects the other species linework, give it a value of 1.
- Covert the HUC12s from polygons to a 30 m raster using the field above.
- Convert the polygon layers from the Gulf sturgeon critical habitat to 30 m rasters and give those pixels a value of 2.
- Combine the two rasters above using the ArcPy Spatial Analyst Cell Statistics "MAXIMUM" function.
- Clip the resulting layer to the EPA estimated floodplain.
- Use the HUC6 layer to remove from the resulting raster areas outside the Gulf drainage where those 4 species ranges occur. The Atlantic drainages are represented in the Blueprint by the Atlantic Migratory Fish Habitat Indicator.
- Use the HUC6 layer to add zero values to the above raster representing the Gulf range of the 4 species listed above. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Presence of Gulf sturgeon
1 = Presence of Alabama shad, American shad, or striped bass
0 = Not identified as Gulf migratory fish habitat (east of the Mississippi River)

Known Issues

- This indicator does not account for smaller dams/culverts that serve as barriers to fish passage.
- Where the SEACAP linear spatial data interests a dam, the indicator can extend to reservoirs that are not accessible to fish due to fish passage barriers (e.g., Ross R. Barnett Reservoir in MS).
- The EPA Estimated Floodplain layer sometimes misses the small, linear connections made by artificial canals, especially when they go through areas that wouldn't naturally be part of the floodplain. As a result, some areas (like lakes) that are connected via canals may appear to be disconnected, but still receive high scores.
- While this indicator generally includes the open water area of reservoirs, some open water portions of reservoirs are missing from the estimated floodplain dataset.
- Estuaries where Gulf sturgeon are not present are often underprioritized because data for the other species do not extend into the estuaries.

- This indicator does not account for instream habitat quality, which can also be a barrier to fish passage.
- This indicator likely underestimates the value of some areas for American eel. That species is not included in the indicator due to a lack of integrated regionwide data depicting how far upstream American eels have been observed.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Martin, E. H, Hoenke, K., Granstaff, E., Barnett, A., Kauffman, J., Robinson, S. and Apse, C.D. 2014. SEACAP: Southeast Aquatic Connectivity Assessment Project: Assessing the ecological impact of dams on Southeastern rivers. The Nature Conservancy, Eastern Division Conservation Science, Southeast Aquatic Resources Partnership. [https://secassoutheast.org/pdf/SEACAP_Report.pdf].

Imperiled Aquatic Species

This indicator measures the number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed within each 12-digit HUC subwatershed. It includes approximately 580 species of fish, mussels, crayfish, amphibians, and reptiles. This indicator originates from state Natural Heritage Program data collected by the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).

This indicator was last updated in October 2024.

Reason for Selection

According to the Southeastern Freshwater Biodiversity Conservation Strategy, "the Southeastern United States is a global hotspot of freshwater biodiversity, supporting almost two-thirds of the country's fish species, over 90% of the US total species of mussels and nearly half of the global total for crayfish species. More than a quarter of this region's species are found nowhere else in the world. Unfortunately, this region is also a hotspot for imperilment. The number of imperiled freshwater fish species in the Southeast has risen 125% in the past 20 years" (RBC and TNACI 2023). This indicator identifies areas with abundant rare and endemic aquatic species that would benefit from conservation action. It captures patterns of rare and endemic species diversity not well-represented by other freshwater indicators.

The Southeast Aquatic Resources Partnership (SARP) tracks the number of aquatic animal species in different conservation categories, including Species of Greatest Conservation Need (SGCN), Regional

SGCN (RSGCN), and threatened or endangered. To depict the aquatic priorities of state fish and wildlife agencies, this indicator previously used SGCN, which are identified in each State Wildlife Action Plan (SWAP) as most in need of conservation action. In 2024, this indicator switched to using RSGCN. RSGCN are regional priority species derived from all Southeast SWAPs using a set of consistent criteria, such as level of conservation concern, regional stewardship responsibility, and biological or ecological significance (Terwilliger Consulting 2019). Using RSGCN provides more clarity and focus for regional conservation and enhanced opportunities for multi-state collaboration. It also accounts for differences in the ways aquatic species are identified as SGCN in each SWAP, which caused the indicator to overprioritize some states because they had a more comprehensive SGCN list, but not necessarily more aquatic biodiversity.

Input Data

- Southeast Blueprint 2024 extent
- Southeast Aquatic Resources Partnership (SARP) RSGCN HUC12 summaries

  SARP provided these summaries on May 8, 2024 as a spreadsheet containing the ID number for each HUC12 watershed and an attribute for the number of aquatic animal RSGCN observed in that watershed. This dataset is based on state Natural Heritage Program occurrence records for fishes, mussels, snails, crayfish, and amphibians. It was last updated in April 2024.

  More information on this dataset is available in Appendix I. A full list of all species included appears in the Full Species List section below (note: the detailed species list was generated in May 2025, compared to April 2024 for the data used in the indicator; it is possible the species list in the documentation may include a handful of species not included in the indicator if new Heritage observations were incorporated during that time).

- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

  The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- National Hydrography Dataset Plus (NHDPlus) Version 2.1 medium resolution catchments (note: V2.1 is just the current sub-version of the dataset generally called NHDPlusV2); view the user guide

Catchments

A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.

To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

- U.S. Geological Survey (USGS) Watershed Boundary Dataset (WBD), accessed 12-2-2021: HUC12s; download the data

Mapping Steps

- Join the RSGCN count table to the HUC12 spatial data.
- Convert to a 30 m raster, where the values represent the number of RSGCN.
- Reclassify the species count values to the 1-9 indicator values below.
- Mask the resulting raster to the EPA estimated floodplain. Assign a value of 0 to all areas outside the EPA floodplain. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Mask the resulting floodplain-masked raster to the NHDPlus medium resolution catchments layer to remove values in the nearshore environment.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

9 = 8+ aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed
8 = 7 aquatic animal RSGCN observed
7 = 6 aquatic animal RSGCN observed
6 = 5 aquatic animal RSGCN observed
5 = 4 aquatic animal RSGCN observed
4 = 3 aquatic animal RSGCN observed
3 = 2 aquatic animal RSGCN observed
2 = 1 aquatic animal RSGCN observed
1 = 0 aquatic animal RSGCN observed
0 = Not identified as a floodplain

Known Issues

- As this indicator is based on occurrence records, poorly surveyed areas may be scored too low. Therefore, this data does not imply absence of species.
- This indicator overprioritizes a number of watersheds in Mississippi because 21 aquatic species in SARP's database were incorrectly attributed as RSGCN, despite not actually being on the Southeast RSGCN list. These species were observed in 396 HUC12 subwatersheds within or overlapping Mississippi. For most of the watersheds, the score is only inflated by 1 species, but where these species co-occur, some watersheds received a score increase of as many as 10 species. This will be corrected in the next update to this indicator.
- While this indicator generally includes the open water area of reservoirs, some open water portions of reservoirs (e.g., Kerr Lake in NC/VA) are missing from the estimated floodplain dataset.
- Small headwaters and creeks are not included in this indicator because the EPA estimated floodplain dataset does not include them.
- This indicator may underprioritize areas important for aquatic plants. A list of Southeastern plant Regional Species of Greatest Conservation Need has recently been developed, but at this point SARP is only collecting information on aquatic animal species. We will explore ways to incorporate aquatic plant species in the future.
- In the area just south of Guadalupe Mountains National Park in West Texas, this indicator depicts the floodplain as a series of straight lines that poorly match the actual floodplain. This is due to an error in the EPA floodplain map used in this indicator.
- The catchment boundaries are inconsistent in how far they extend toward the ocean. As a result, this indicator does not consistently apply to estuaries, coastal areas, and barrier islands.

Full Species List*

Table 6. The amphibian RSGCN with observations included in this indicator.

	Scientific Name	Common name
1	*Ambystoma barbouri*	Streamside salamander
2	*Ambystoma bishopi*	Reticulated flatwoods salamander
3	*Ambystoma cingulatum*	Frosted flatwoods salamander
4	*Ambystoma mabeei*	Mabee's salamander
5	*Amphiuma pholeter*	One-toed amphiuma
6	*Amphiuma tridactylum*	Three-toed amphiuma
7	*Anaxyrus quercicus*	Oak toad
8	*Aneides aeneus*	Green salamander
9	*Desmognathus abditus*	Cumberland dusky salamander
10	*Desmognathus aeneus*	Seepage salamander
11	*Desmognathus auriculatus*	Southern dusky salamander
12	*Desmognathus folkertsi*	Dwarf black-bellied Salamander
13	*Desmognathus marmoratus*	Shovel-nosed Salamander
14	*Desmognathus organi*	Northern pygmy salamander
15	*Desmognathus santeetlah*	Santeetlah dusky salamander
16	*Desmognathus welteri*	Black mountain salamander
17	*Desmognathus wrighti*	Southern pygmy salamander
18	*Eurycea aquatica*	Brown-backed Salamander
19	*Eurycea hillisi*	Hillis's dwarf salamander
20	*Eurycea junaluska*	Junaluska salamander
21	*Eurycea wallacei*	Georgia blind salamander
22	*Eurycea wilderae*	Blue Ridge two-lined salamander
23	*Lithobates capito*	Gopher frog
24	*Lithobates heckscheri*	River frog
25	*Necturus alabamensis*	Black warrior waterdog
26	*Necturus beyeri*	Gulf Coast waterdog
27	*Necturus lewisi*	Neuse River waterdog
28	*Necturus punctatus*	Dwarf waterdog
29	*Notophthalmus perstriatus*	Striped newt
30	*Plethodon amplus*	Blue Ridge gray-cheeked salamander
31	*Plethodon aureolus*	Tellico salamander
32	*Plethodon chattahoochee*	Chattahoochee slimy salamander
33	*Plethodon cheoah*	Cheoah Bald salamander
34	*Plethodon hubrichti*	Peaks of Otter salamander
35	*Plethodon meridianus*	South Mountain gray-cheeked salamander
36	*Plethodon petraeus*	Pigeon Mountain salamander
37	*Plethodon punctatus*	Cow Knob salamander

	Scientific Name	Common name
38	*Plethodon shenandoah*	Shenandoah salamander
39	*Plethodon shermani*	Red-legged salamander
40	*Plethodon ventralis*	Southern zigzag salamander
41	*Plethodon virginia*	Shenandoah Mountain salamander
42	*Plethodon wehrlei*	Wehrle's salamander
43	*Plethodon welleri*	Weller's salamander
44	*Pseudacris ornata*	Ornate chorus frog
45	*Scaphiopus hurterii*	Hurter's spadefoot toad
46	*Siren lacertina*	Greater siren
47	*Stereochilus marginatus*	Many-lined salamander
48	*Urspelerpes brucei*	Patch-nosed salamander

Table 7. The crayfish RSGCN with observations included in this indicator.

	Scientific Name	Common Name
1	*Barbicambarus cornutus*	Bottlebrush crayfish
2	*Barbicambarus simmonsi*	Tennessee bottlebrush crayfish
3	*Cambarellus diminutus*	Least dwarf crayfish
4	*Cambarellus lesliei*	Angular dwarf crawfish
5	*Cambarus bouchardi*	Big South Fork crayfish
6	*Cambarus brimleyorum*	Valley River crayfish
7	*Cambarus callainus*	Big sandy crayfish
8	*Cambarus catagius*	Greensboro burrowing crayfish
9	*Cambarus chaugaensis*	Chauga crayfish
10	*Cambarus clivosus*	Short Mountain crayfish
11	*Cambarus conasaugaensis*	Mountain crayfish
12	*Cambarus coosawattae*	Coosawattee crayfish
13	*Cambarus cracens*	Slenderclaw crayfish
14	*Cambarus cymatilis*	Conasauga blue burrower
15	*Cambarus doughertyensis*	Dougherty burrowing crayfish
16	*Cambarus eeseeohensis*	Grandfather Mountain crayfish
17	*Cambarus elkensis*	Elk River crayfish
18	*Cambarus englishi*	Tallapoosa crayfish
19	*Cambarus extraneus*	Chickamauga crayfish
20	*Cambarus fasciatus*	Etowah crayfish
21	*Cambarus georgiae*	Little Tennessee crayfish
22	*Cambarus halli*	Slackwater crayfish
23	*Cambarus hamulatus*	Prickly cave crayfish
24	*Cambarus harti*	Piedmont blue burrower
25	*Cambarus hiwasseensis*	Hiwassee crayfish
26	*Cambarus howardi*	Chattahoochee crayfish

	Scientific Name	Common Name
27	*Cambarus jezerinaci*	Spiny scale crayfish
28	*Cambarus jonesi*	Alabama cave crayfish
29	*Cambarus laconensis*	Lacon exit cave crayfish
30	*Cambarus lenati*	Broad River stream crayfish
31	*Cambarus manningi*	Greensaddle crayfish
32	*Cambarus nerterius*	West Virginia cave crayfish
33	*Cambarus obeyensis*	Obey crayfish
34	*Cambarus parrishi*	Hiwassee headwater crayfish
35	*Cambarus pauleyi*	Meadow River mudbug
36	*Cambarus pecki*	Phantom cave crayfish
37	*Cambarus pristinus*	Pristine crayfish
38	*Cambarus pyronotus*	Fireback crayfish
39	*Cambarus reburrus*	French Broad crayfish
40	*Cambarus scotti*	Chattooga River crayfish
41	*Cambarus smilax*	Greenbrier crayfish
42	*Cambarus sp. 1*	Emory River crayfish
43	*Cambarus speciosus*	Beautiful crayfish
44	*Cambarus speleocoopi*	Sweet Home Alabama crayfish
45	*Cambarus spicatus*	Broad River spiny crayfish
46	*Cambarus stockeri*	Cocoa crayfish
47	*Cambarus strigosus*	Lean crayfish
48	*Cambarus subterraneus*	Delaware County cave crayfish
49	*Cambarus tartarus*	Oklahoma cave crayfish
50	*Cambarus truncatus*	Oconee burrowing crayfish
51	*Cambarus tuckasegee*	Tuckasegee stream crayfish
52	*Cambarus unestami*	Blackbarred crayfish
53	*Cambarus veitchorum*	White Spring cave crayfish
54	*Cambarus veteranus*	Guyandotte river crayfish
55	*Cambarus williami*	Brawleys Fork crayfish
56	*Creaserinus hortoni*	Hatchie burrowing crayfish
57	*Creaserinus oryktes*	Flatwoods digger
58	*Distocambarus carlsoni*	Mimic crayfish
59	*Distocambarus devexus*	Broad River burrowing crayfish
60	*Distocambarus hunteri*	Saluda burrowing crayfish
61	*Distocambarus youngineri*	Newberry burrowing crayfish
62	*Fallicambarus burrisi*	Burrowing bog crayfish
63	*Fallicambarus dissitus*	Pine hills digger
64	*Fallicambarus macneesei*	Old prairie digger
65	*Faxonella beyeri*	Sabine fencing crawfish
66	*Faxonella blairi*	Blair’s fencing crawfish

	Scientific Name	Common Name
67	*Faxonella creaseri*	Ouachita fencing crawfish
68	*Faxonius bisectus*	Crittenden crayfish
69	*Faxonius burri*	Blood River crayfish
70	*Faxonius cooperi*	Flint River crayfish
71	*Faxonius jeffersoni*	Louisville crayfish
72	*Faxonius macrus*	Neosho midget crayfish
73	*Faxonius maletae*	Kisatchie painted crawfish
74	*Faxonius margorectus*	Livingston crayfish
75	*Faxonius menae*	Mena crayfish
76	*Faxonius nana*	Midget crayfish
77	*Faxonius saxatilis*	Kiamichi crayfish
78	*Faxonius shoupi*	Nashville crayfish
79	*Faxonius taylori*	Crescent crayfish
80	*Faxonius wrighti*	Hardin crayfish
81	*Orconectes barri*	Cumberland Plateau cave crayfish
82	*Orconectes packardi*	Appalachian cave crayfish
83	*Orconectes pellucidus*	Mammoth Cave crayfish
84	*Procambarus barbiger*	Jackson prairie crayfish
85	*Procambarus braswelli*	Waccamaw crayfish
86	*Procambarus brazoriensis*	Brazoria crayfish
87	*Procambarus capillatus*	Capillaceous crayfish
88	*Procambarus elegans*	Elegant creek crawfish
89	*Procambarus escambiensis*	Escambia crayfish
90	*Procambarus geminus*	Twin crawfish
91	*Procambarus lagniappe*	Lagniappe crayfish
92	*Procambarus lecontei*	Mobile crayfish
93	*Procambarus machardyi*	Caddo chimney crawfish
94	*Procambarus medialis*	Pamlico crayfish
95	*Procambarus nechesae*	Neches crayfish
96	*Procambarus nigrocinctus*	Blackbelted crayfish
97	*Procambarus pearsei*	Carolina Sandhills crayfish
98	*Procambarus pentastylus*	Calcasieu creek crawfish
99	*Procambarus pictus*	Black creek crayfish
100	*Procambarus planirostris*	Flatnose crayfish
101	*Procambarus steigmani*	Parkhill prairie crayfish
102	*Procambarus viaeviridis*	Vernal crayfish

Table 8. The fish RSGCN with observations included in this indicator.

	Scientific Name	**Common Name**
1	*Acipenser brevirostrum*	Shortnose sturgeon
2	*Acipenser fulvescens*	Lake sturgeon
3	*Acipenser oxyrinchus desotoi*	Gulf sturgeon
4	*Acipenser oxyrinchus oxyrinchus*	Atlantic sturgeon
5	*Alosa alabamae*	Alabama shad
6	*Alosa mediocris*	Hickory shad
7	*Alosa sapidissima*	American shad
8	*Ambloplites cavifrons*	Roanoke bass
9	*Amblyopsis rosae*	Ozark cavefish
10	*Amblyopsis spelaea*	Northern cavefish
11	*Ameiurus serracanthus*	Spotted bullhead
12	*Ammocrypta clara*	Western sand darter
13	*Ammocrypta vivax*	Scaly sand darter
14	*Anguilla rostrata*	American eel
15	*Atractosteus spatula*	Alligator gar
16	*Awaous banana*	River goby
17	*Campostoma anomalum michauxi*	Central stoneroller
18	*Campostoma ornatum*	Mexican stoneroller
19	*Carcharhinus obscurus*	Dusky shark
20	*Carcharhinus plumbeus*	Sandbar shark
21	*Carpiodes sp. cf. cyprinus*	Carolina quillback
22	*Chologaster cornuta*	Swampfish
23	*Chrosomus cumberlandensis*	Blackside dace
24	*Chrosomus tennesseensis*	Tennessee dace
25	*Clinostomus funduloides ssp. 1*	Smoky dace
26	*Cottus baileyi*	Black sculpin
27	*Cottus paulus*	Pygmy sculpin
28	*Crystallaria asprella*	Crystal darter
29	*Crystallaria cincotta*	Diamond darter
30	*Ctenogobius claytonii*	Mexican goby
31	*Cycleptus elongatus*	Blue sucker
32	*Cycleptus meridionalis*	Southeastern blue sucker
33	*Cyprinella analostana*	Satinfin shiner
34	*Cyprinella caerulea*	Blue shiner
35	*Cyprinella callitaenia*	Bluestripe shiner
36	*Cyprinella camura*	Bluntface shiner
37	*Cyprinella labrosa*	Thicklip chub
38	*Cyprinella lepida*	Plateau shiner
39	*Cyprinella proserpina*	Proserpine shiner

	Scientific Name	Common Name
40	*Cyprinella pyrrhomelas*	Fieryblack shiner
41	*Cyprinella sp.*	Nueces river shiner
42	*Cyprinella xaenura*	Altamaha shiner
43	*Cyprinodon bovinus*	Leon Springs pupfish
44	*Cyprinodon elegans*	Comanche Springs pupfish
45	*Cyprinodon eximius*	Conchos pupfish
46	*Dionda argentosa*	Manantial roundnose minnow
47	*Dionda diaboli*	Devils River minnow
48	*Dionda episcopa*	Roundnose minnow
49	*Dionda serena*	Nueces roundnose minnow
50	*Elassoma alabamae*	Spring pygmy sunfish
51	*Elassoma boehlkei*	Carolina pygmy sunfish
52	*Elassoma gilberti*	Gulf coast pygmy sunfish
53	*Elassoma okatie*	Bluebarred pygmy sunfish
54	*Enneacanthus chaetodon*	Blackbanded sunfish
55	*Enneacanthus obesus*	Banded sunfish
56	*Epinephelus drummondhayi*	Speckled hind
57	*Epinephelus itajara*	Atlantic goliath grouper
58	*Epinephelus nigritus*	Warsaw grouper
59	*Erimonax monachus*	Spotfin chub
60	*Erimystax cahni*	Slender chub
61	*Erimystax dissimilis*	Streamline chub
62	*Etheostoma acuticeps*	Sharphead darter
63	*Etheostoma akatulo*	Bluemask (jewel) darter
64	*Etheostoma aquali*	Coppercheek darter
65	*Etheostoma baileyi*	Emerald darter
66	*Etheostoma barbouri*	Teardrop darter
67	*Etheostoma boschungi*	Slackwater darter
68	*Etheostoma brevirostrum*	Holiday darter
69	*Etheostoma brevispinum*	Carolina fantail darter
70	*Etheostoma cervus*	Chickasaw darter
71	*Etheostoma chermocki*	Vermilion darter
72	*Etheostoma chienense*	Relict darter
73	*Etheostoma chlorobranchium*	Greenfin darter
74	*Etheostoma chuckwachatte*	Lipstick darter
75	*Etheostoma cinereum*	Ashy darter
76	*Etheostoma clinton*	Beaded darter
77	*Etheostoma collis*	Carolina darter
78	*Etheostoma corona*	Crown darter
79	*Etheostoma denoncourti*	Golden darter

	Scientific Name	Common Name
80	*Etheostoma ditrema*	Coldwater darter
81	*Etheostoma duryi*	Black darter
82	*Etheostoma etowahae*	Etowah darter
83	*Etheostoma euzonum euzonum*	Arkansas saddled darter
84	*Etheostoma fonticola*	Fountain darter
85	*Etheostoma forbesi*	Barrens darter
86	*Etheostoma fragi*	Strawberry River darter
87	*Etheostoma fricksium*	Savannah darter
88	*Etheostoma grahami*	Rio Grande darter
89	*Etheostoma gutselli*	Tuckasegee darter
90	*Etheostoma hopkinsi*	Christmas darter
91	*Etheostoma inscriptum*	Turquoise darter
92	*Etheostoma jessiae*	Blueside darter
93	*Etheostoma lemniscatum*	Tuxedo darter
94	*Etheostoma mariae*	Pinewoods darter
95	*Etheostoma marmorpinnum*	Marbled darter
96	*Etheostoma maydeni*	Redlips darter
97	*Etheostoma microlepidum*	Smallscale darter
98	*Etheostoma microperca*	Least darter
99	*Etheostoma moorei*	Yellowcheek darter
100	*Etheostoma nebra*	Buck darter
101	*Etheostoma neopterum*	Lollypop darter
102	*Etheostoma nianguae*	Niangua darter
103	*Etheostoma nuchale*	Watercress darter
104	*Etheostoma osburni*	Candy darter
105	*Etheostoma pallididorsum*	Paleback darter
106	*Etheostoma parvipinne*	Goldstripe darter
107	*Etheostoma percnurum*	Duskytail darter
108	*Etheostoma perlongum*	Waccamaw darter
109	*Etheostoma phytophilum*	Rush darter
110	*Etheostoma pseudovulatum*	Egg-mimic darter
111	*Etheostoma pyrrhogaster*	Firebelly darter
112	*Etheostoma raneyi*	Yazoo darter
113	*Etheostoma rubrum*	Bayou darter
114	*Etheostoma rupestre*	Rock darter
115	*Etheostoma sagitta*	Cumberland arrow darter
116	*Etheostoma scotti*	Cherokee darter
117	*Etheostoma sitikuense*	Citico darter
118	*Etheostoma spilotum*	Kentucky arrow darter
119	*Etheostoma striatulum*	Striated darter

	Scientific Name	Common Name
120	*Etheostoma susanae*	Cumberland darter
121	*Etheostoma swannanoa*	Swannanoa darter
122	*Etheostoma tecumsehi*	Shawnee darter
123	*Etheostoma thalassinum*	Seagreen darter
124	*Etheostoma thompsoni*	Gumbo darter
125	*Etheostoma tippecanoe*	Tippecanoe darter
126	*Etheostoma trisella*	Trispot darter
127	*Etheostoma tuscumbia*	Tuscumbia darter
128	*Etheostoma vulneratum*	Wounded darter
129	*Etheostoma wapiti*	Boulder darter
130	*Etheostoma whipplei*	Redfin darter
131	*Etheostoma zonifer*	Backwater darter
132	*Fundulus bifax*	Stippled studfish
133	*Fundulus dispar*	Starhead topminnow
134	*Fundulus euryzonus*	Broadstripe topminnow
135	*Fundulus jenkinsi*	Saltmarsh topminnow
136	*Fundulus julisia*	Barrens topminnow
137	*Fundulus rathbuni*	Speckled killifish
138	*Fundulus sp. cf. diaphanus*	Lake Phelps killifish
139	*Fundulus waccamensis*	Waccamaw killifish
140	*Gambusia heterochir*	Clear Creek gambusia
141	*Gambusia krumholzi*	Spotfin gambusia
142	*Hemitremia flammea*	Flame chub
143	*Hiodon tergisus*	Mooneye
144	*Hippocampus erectus*	Lined seahorse
145	*Hybognathus hayi*	Cypress minnow
146	*Hybognathus placitus*	Plains minnow
147	*Hybopsis amnis*	Pallid shiner
148	*Hybopsis hypsinotus*	Highback chub
149	*Hybopsis lineapunctata*	Lined chub
150	*Hybopsis rubrifrons*	Rosyface chub
151	*Hybopsis sp. 9*	Etowah chub
152	*Hyla andersonii*	Pine barrens treefrog
153	*Ichthyomyzon bdellium*	Ohio lamprey
154	*Ichthyomyzon fossor*	Northern brook lamprey
155	*Ichthyomyzon greeleyi*	Mountain brook lamprey
156	*Ictalurus sp.*	Chihuahua catfish
157	*Lethenteron appendix*	American brook lamprey
158	*Lythrurus lirus*	Mountain shiner
159	*Lythrurus snelsoni*	Ouachita shiner

	Scientific Name	Common Name
160	*Macrhybopsis australis*	Prairie chub
161	*Macrhybopsis etnieri*	Coosa chub
162	*Macrhybopsis gelida*	Sturgeon chub
163	*Macrhybopsis meeki*	Sicklefin chub
164	*Macrhybopsis tetranema*	Peppered chub
165	*Megalops atlanticus*	Tarpon
166	*Menidia audens*	Mississippi silverside
167	*Menidia extensa*	Waccamaw silverside
168	*Microphis brachyurus*	Opossum pipefish
169	*Micropterus cataractae*	Shoal bass
170	*Micropterus chattahoochae*	Chattahoochee bass
171	*Micropterus coosae*	Redeye bass
172	*Micropterus notius*	Suwannee bass
173	*Moxostoma ariommum*	Bigeye jumprock
174	*Moxostoma pappillosum*	V-lip Redhorse
175	*Moxostoma pisolabrum*	Pealip redhorse
176	*Moxostoma robustum*	Robust redhorse
177	*Moxostoma sp. 2*	Sicklefin redhorse
178	*Moxostoma sp. 4*	Brassy jumprock
179	*Narcine bancroftii*	Caribbean electric ray
180	*Notropis albizonatus*	Palezone shiner
181	*Notropis alborus*	Whitemouth shiner
182	*Notropis ariommus*	Popeye shiner
183	*Notropis asperifrons*	Burrhead shiner
184	*Notropis bairdi*	Red River shiner
185	*Notropis bifrenatus*	Bridle shiner
186	*Notropis buccula*	Smalleye shiner
187	*Notropis buchanani*	Ghost shiner
188	*Notropis cahabae*	Cahaba shiner
189	*Notropis candidus*	Silverside shiner
190	*Notropis chalybaeus*	Ironcolor shiner
191	*Notropis chihuahua*	Chihuahua shiner
192	*Notropis edwardraneyi*	Fluvial shiner
193	*Notropis girardi*	Arkansas River shiner
194	*Notropis hypsilepis*	Highscale shiner
195	*Notropis mekistocholas*	Cape fear shiner
196	*Notropis melanostomus*	Blackmouth shiner
197	*Notropis ortenburgeri*	Kiamichi shiner
198	*Notropis oxyrhynchus*	Sharpnose shiner
199	*Notropis ozarcanus*	Ozark shiner

	Scientific Name	Common Name
200	*Notropis perpallidus*	Peppered shiner
201	*Notropis potteri*	Chub shiner
202	*Notropis procne*	Swallowtail shiner
203	*Notropis rupestris*	Bedrock shiner
204	*Notropis sabinae*	Sabine shiner
205	*Notropis semperasper*	Roughhead shiner
206	*Notropis suttkusi*	Rocky shiner
207	*Noturus baileyi*	Smoky madtom
208	*Noturus crypticus*	Chucky madtom
209	*Noturus eleutherus*	Mountain madtom
210	*Noturus fasciatus*	Saddled madtom
211	*Noturus flavipinnis*	Yellowfin madtom
212	*Noturus furiosus*	Carolina madtom
213	*Noturus gilberti*	Orangefin madtom
214	*Noturus gladiator*	Piebald madtom
215	*Noturus hildebrandi*	Least madtom
216	*Noturus lachneri*	Ouachita madtom
217	*Noturus munitus*	Frecklebelly madtom
218	*Noturus placidus*	Neosho madtom
219	*Noturus stanauli*	Pygmy madtom
220	*Noturus stigmosus*	Northern madtom
221	*Noturus taylori*	Caddo madtom
222	*Paralichthys lethostigma*	Southern flounder
223	*Percina antesella*	Amber darter
224	*Percina aurantiaca*	Tangerine darter
225	*Percina aurolineata*	Goldline darter
226	*Percina aurora*	Pearl darter
227	*Percina brevicauda*	Coal darter
228	*Percina burtoni*	Blotchside logperch
229	*Percina copelandi*	Channel darter
230	*Percina crassa*	Piedmont darter
231	*Percina crypta*	Halloween darter
232	*Percina cymatotaenia*	Bluestripe darter
233	*Percina gymnocephala*	Appalachia darter
234	*Percina jenkinsi*	Conasauga logperch
235	*Percina kusha*	Bridled darter
236	*Percina lenticula*	Freckled darter
237	*Percina macrocephala*	Longhead darter
238	*Percina nasuta*	Longnose darter
239	*Percina pantherina*	Leopard darter

	Scientific Name	Common Name
240	*Percina rex*	Roanoke logperch
241	*Percina sipsi*	Bankhead darter
242	*Percina squamata*	Olive darter
243	*Percina tanasi*	Snail darter
244	*Percina uranidea*	Stargazing darter
245	*Percina williamsi*	Sickle darter
246	*Phenacobius crassilabrum*	Fatlips minnow
247	*Phenacobius teretulus*	Kanawha minnow
248	*Phenacobius uranops*	Stargazing minnow
249	*Pimephales tenellus parviceps*	Eastern slim minnow
250	*Platygobio gracilis*	Flathead chub
251	*Pristis pectinata*	Smalltooth sawfish
252	*Pristis pristis*	Largetooth sawfish
253	*Pteronotropis euryzonus*	Broadstripe shiner
254	*Pteronotropis hubbsi*	Bluehead shiner
255	*Pteronotropis welaka*	Bluenose shiner
256	*Rachycentron canadum*	Cobia
257	*Salvelinus fontinalis*	Brook trout
258	*Sander sp. cf. vitreus*	Walleye
259	*Satan eurystomus*	Widemouth blindcat
260	*Scaphirhynchus albus*	Pallid sturgeon
261	*Scaphirhynchus platorynchus*	Shovelnose sturgeon
262	*Scaphirhynchus suttkusi*	Alabama sturgeon
263	*Semotilus lumbee*	Sandhills chub
264	*Speoplatyrhinus poulsoni*	Alabama cavefish
265	*Sphyrna lewini*	Scalloped hammerhead
266	*Squalus acanthias*	Spiny dogfish
267	*Umbra limi*	Central mudminnow

Table 9. The mussel RSGCN with observations included in the indicator.

	Scientific Name	Common Name
1	*Actinonaias pectorosa*	Pheasantshell
2	*Alasmidonta atropurpurea*	Cumberland elktoe
3	*Alasmidonta raveneliana*	Appalachian elktoe
4	*Alasmidonta triangulata*	Southern elktoe
5	*Alasmidonta varicosa*	Brook floater
6	*Alasmidonta viridis*	Slippershell mussel
7	*Amblema neislerii*	Fat threeridge
8	*Anodontoides denigrata*	Cumberland papershell
9	*Arcidens wheeleri*	Ouachita rock pocketbook

	Scientific Name	Common Name
10	*Cyclonaias infucata*	Sculptured pigtoe
11	*Cyclonaias kieneriana*	Coosa orb
12	*Cyclonaias kleiniana*	Florida mapleleaf
13	*Cyclonaias petrina*	Texas pimpleback
14	*Cyprogenia aberti*	Ozark fanshell
15	*Cyprogenia stegaria*	Fanshell
16	*Dromus dromas*	Dromedary pearlymussel
17	*Elliptio arca*	Alabama spike
18	*Elliptio arctata*	Delicate spike
19	*Elliptio chipolaensis*	Chipola slabshell
20	*Elliptio folliculata*	Pod lance
21	*Elliptio fraterna*	Brother spike
22	*Elliptio lanceolata*	Yellow lance
23	*Elliptio monroensis*	St. John's elephantear
24	*Elliptio nigella*	Winged spike
25	*Elliptio purpurella*	Inflated spike
26	*Elliptio spinosa*	Altamaha spinymussel
27	*Elliptoideus sloatianus*	Purple bankclimber
28	*Epioblasma ahlstedti*	Duck river dartersnapper
29	*Epioblasma aureola*	Golden riffleshell
30	*Epioblasma brevidens*	Cumberlandian combshell
31	*Epioblasma capsaeformis*	Oyster mussel
32	*Epioblasma curtisii*	Curtis pearlymussel
33	*Epioblasma obliquata*	Catspaw
34	*Epioblasma penita*	Southern combshell
35	*Epioblasma triquetra*	Snuffbox
36	*Epioblasma walkeri*	Tan riffleshell
37	*Fusconaia burkei*	Tapered pigtoe
38	*Fusconaia chunii*	Trinity pigtoe
39	*Fusconaia cor*	Shiny pigtoe
40	*Fusconaia cuneolus*	Fine-rayed Pigtoe
41	*Fusconaia escambia*	Narrow pigtoe
42	*Fusconaia masoni*	Atlantic pigtoe
43	*Fusconaia mitchelli*	False spike
44	*Fusconaia subrotunda*	Longsolid
45	*Hamiota altilis*	Finelined pocketbook
46	*Hamiota australis*	Southern sandshell
47	*Hamiota perovalis*	Orangenacre mucket
48	*Hamiota subangulata*	Shinyrayed pocketbook
49	*Hemistena lata*	Cracking pearlymussel

	Scientific Name	Common Name
50	*Lampsilis abrupta*	Pink mucket
51	*Lampsilis bracteata*	Texas fatmucket
52	*Lampsilis powellii*	Arkansas fatmucket
53	*Lampsilis rafinesqueana*	Neosho mucket
54	*Lampsilis satura*	Sandbank pocketbook
55	*Lampsilis splendida*	Rayed pink fatmucket
56	*Lampsilis streckeri*	Speckled pocketbook
57	*Lampsilis virescens*	Alabama lampmussel
58	*Lasmigona decorata*	Carolina heelsplitter
59	*Lasmigona etowaensis*	Etowah heelsplitter
60	*Lasmigona subviridis*	Green floater
61	*Lemiox rimosus*	Birdwing pearlymussel
62	*Leptodea leptodon*	Scaleshell
63	*Margaritifera hembeli*	Louisiana pearlshell
64	*Margaritifera marrianae*	Alabama pearlshell
65	*Margaritifera monodonta*	Spectaclecase
66	*Medionidus acutissimus*	Alabama moccasinshell
67	*Medionidus conradicus*	Cumberland moccasinshell
68	*Medionidus parvulus*	Coosa moccasinshell
69	*Medionidus penicillatus*	Gulf moccasinshell
70	*Medionidus simpsonianus*	Ochlockonee moccasinshell
71	*Medionidus walkeri*	Suwannee moccasinshell
72	*Obovaria arkansasensis*	Southern hickorynut
73	*Obovaria choctawensis*	Choctaw bean
74	*Obovaria retusa*	Ring pink
75	*Obovaria subrotunda*	Round hickorynut
76	*Obovaria unicolor*	Alabama hickorynut
77	*Parvaspina collina*	James spinymussel
78	*Parvaspina steinstansana*	Tar river spinymussel
79	*Pegias fabula*	Little-winged pearlymussel
80	*Plethobasus cicatricosus*	White wartyback
81	*Plethobasus cooperianus*	Orangefoot pimpleback
82	*Plethobasus cyphyus*	Sheepnose
83	*Pleurobema athearni*	Canoe creek clubshell
84	*Pleurobema clava*	Clubshell
85	*Pleurobema decisum*	Southern clubshell
86	*Pleurobema georgianum*	Southern pigtoe
87	*Pleurobema hanleyianum*	Georgia pigtoe
88	*Pleurobema oviforme*	Tennessee clubshell
89	*Pleurobema perovatum*	Ovate clubshell

	Scientific Name	Common Name
90	*Pleurobema plenum*	Rough pigtoe
91	*Pleurobema pyriforme*	Oval pigtoe
92	*Pleurobema riddellii*	Louisiana pigtoe
93	*Pleurobema rubrum*	Pyramid pigtoe
94	*Pleurobema strodeanum*	Fuzzy pigtoe
95	*Pleurobema taitianum*	Heavy pigtoe
96	*Pleuronaia barnesiana*	Tennessee pigtoe
97	*Pleuronaia dolabelloides*	Slabside pearlymussel
98	*Pleuronaia gibber*	Cumberland pigtoe
99	*Potamilus amphichaenus*	Texas heelsplitter
100	*Potamilus capax*	Fat pocketbook
101	*Potamilus inflatus*	Inflated heelsplitter
102	*Potamilus metnecktayi*	Salina mucket
103	*Pseudodontoideus connasaugaensis*	Alabama creekmussel
104	*Pseudodontoideus subvexus*	Southern creekmussel
105	*Ptychobranchus foremanianus*	Rayed kidneyshell
106	*Ptychobranchus greenii*	Triangular kidneyshell
107	*Ptychobranchus jonesi*	Southern kidneyshell
108	*Ptychobranchus subtentus*	Fluted kidneyshell
109	*Quadrula fragosa*	Winged mapleleaf
110	*Strophitus radiatus*	Rayed creekshell
111	*Theliderma cylindrica*	Rabbitsfoot
112	*Theliderma intermedia*	Cumberland monkeyface
113	*Theliderma sparsa*	Appalachian monkeyface
114	*Thoburnia atripinnis*	Blackfin sucker
115	*Thoburnia hamiltoni*	Rustyside sucker
116	*Toxolasma cylindrellus*	Pale lilliput
117	*Toxolasma pullus*	Savannah lilliput
118	*Trogloglanis pattersoni*	Toothless blindcat
119	*Truncilla cognata*	Mexican fawnsfoot
120	*Truncilla macrodon*	Texas fawnsfoot
121	*Utterbackia peggyae*	Florida floater
122	*Utterbackiana couperiana*	Barrel floater
123	*Utterbackiana hartfieldorum*	Cypress floater
124	*Utterbackiana heardi*	Apalachicola floater
125	*Venustaconcha trabalis*	Tennessee bean
126	*Villosa nebulosa*	Alabama rainbow
127	*Villosa taeniata*	Painted creekshell
128	*Villosa umbrans*	Coosa creekshell
129	*Villosa vanuxemensis*	Mountain creekshell

	Scientific Name	Common Name
130	*Villosa vaughaniana*	Carolina creekshell
131	*Villosa villosa*	Downy rainbow

Table 10. The reptile RSGCN with observations included in the indicator.

	Scientific Name	Common Name
1	*Apalone mutica*	Smooth softshell turtle
2	*Caretta caretta*	Loggerhead sea turtle
3	*Chelonia mydas*	Green sea turtle
4	*Clemmys guttata*	Spotted turtle
5	*Crotalus adamanteus*	Eastern diamond-backed rattlesnake
6	*Dermochelys coriacea*	Leatherback sea turtle
7	*Glyptemys muhlenbergii*	Bog turtle
8	*Graptemys barbouri*	Barbour's map turtle
9	*Graptemys ernsti*	Escambia map turtle
10	*Graptemys nigrinoda*	Black-knobbed map turtle
11	*Graptemys oculifera*	Ringed map turtle
12	*Graptemys ouachitensis*	Ouachita map turtle
13	*Graptemys pearlensis*	Pearl river map turtle
14	*Graptemys pulchra*	Alabama map turtle
15	*Graptemys sabinensis*	Sabine map turtle
16	*Heterodon simus*	Southern hognose snake
17	*Kinosternon baurii*	Striped mud turtle
18	*Lepidochelys kempii*	Kemp's ridley
19	*Macrochelys suwanniensis*	Suwanee alligator snapping turtle
20	*Macrochelys temminckii*	Alligator snapping turtle
21	*Micrurus fulvius*	Harlequin coralsnake
22	*Nerodia floridana*	Florida green water snake
23	*Nerodia sipedon williamengelsi*	Carolina watersnake
24	*Ophisaurus compressus*	Island glass lizard
25	*Ophisaurus mimicus*	Mimic glass lizard
26	*Ophisaurus ventralis*	Eastern glass lizard
27	*Pituophis melanoleucus melanoleucus*	Northern pinesnake
28	*Pituophis melanoleucus mugitus*	Florida pine snake
29	*Pseudemys alabamensis*	Alabama red-bellied turtle
30	*Rhadinaea flavilata*	Pine woods snake
31	*Rhineura floridana*	Florida worm lizard
32	*Sternotherus carinatus*	Razor-backed musk turtle
33	*Sternotherus minor peltifer*	Stripe-necked musk turtle
34	*Trachemys scripta troostii*	Cumberland slider

**SARP generated this species list in May 2025 and it may have been updated with new species observations more recently than the indicator itself, which was generated in April 2024. As a result, some of these species may not be reflected in the current version of the indicator, but will be captured in the next update. SARP and SECAS staff will work to keep the species list and indicator in sync moving forward. We felt the value of communicating the represented species made it worth including the list of species, even if the dates on the data are not an exact match.*

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Southeast Aquatic Resources Partnership. Species Summaries by HUC12. Accessed May 2024.

Terwilliger Consulting, Inc. Regional Species of Greatest Conservation Need in the Southeastern United States. July 2019. [https://secassoutheast.org/pdf/SEAFWA_RSGCN_Final_Report_20190715.pdf].

The River Basin Center and The Tennessee Aquarium Conservation Institute. The Southeastern Freshwater Biodiversity Conservation Strategy. Accessed July 31, 2023. [https://southeastfreshwater.org/].

U.S. Environmental Protection Agency (USEPA) and the U.S. Geological Survey (USGS). 2012. National Hydrography Dataset Plus 2. [https://www.epa.gov/waterdata/get-nhdplus-national-hydrography-dataset-plus-data].

Lakes & Reservoirs

This indicator assesses the condition of lakes and reservoirs based on the amount of natural landcover within the upstream watershed. Higher scores go to lakes and reservoirs with less surrounding urban and agricultural development (including intensive forest management for timber production), which negatively impact drinking water quality and quantity and fish habitat. This indicator uses waterbody and watershed data from the LAGOS-LOCUS dataset, as well as landcover from the National Land Cover Database and LANDFIRE.

This indicator was last updated in November 2025.

Reason for Selection

Freshwater lake biomes, whether naturally occurring or created by artificial structures like dams, are prominent ecosystems on the landscape. They provide important fish habitat as well as drinking water for people (Keith et al. 2022, USGS 2025). The natural landcover surrounding lakes and reservoirs is strongly linked to good drinking water and safe places to recreate (Caldwell et al. 2014). There are many ways to assess the water quality of lakes (e.g., Secchi methods, nutrients loads, etc.); however, those data are not often spatially consistent or available for the full extent of the Southeast Conservation Blueprint. Land use practices upstream of lakes and reservoirs are correlated with (because they strongly influence) the nutrient, sediment, and contaminant loads entering a lake, and land use data are available at a broad scale.

This indicator promotes consistency with the Midwest Glacial Lakes Fish Habitat Partnership's watershed disturbance thresholds (Midwest Glacial Lakes Partnership 2019), and with the Midwest Landscape Initiative's Midwest Conservation Blueprint. It also fills a key gap in the Southeast Blueprint by comparing the overall condition of different lakes and reservoirs. Prior to version 2025, the Blueprint did not attempt to prioritize across lakes and reservoirs and excluded them from the analysis due to a lack of indicator data.

Input Data

- 2023 LANDFIRE Existing Vegetation Type (EVT) CONUS; access the data
- 2021 National Land Cover Database (NLCD): Land cover
- LAGOS-US LOCUS v1.0 (download geopackage)
- Global-scale mining polygons (Version 2); download the geopackage; see the layer in an online viewer; read a journal article about the data development
- Southeast Blueprint 2025 extent

Mapping Steps

- Reclassify the 2021 NLCD to begin separating the disturbed and developed classes from the natural classes. This will be further refined with additional datasets below. Assign the following disturbed classes a value of 0: Developed Open Space, Developed Low Intensity, Developed Medium Intensity, Developed High Intensity, Pasture/Hay, Cultivated Crops. All other classes are considered natural; assign them a value of 1. Also convert all NLCD pixels with a value of 0 to NoData because they are outside the continental United States geography.
- Use a conditional statement to extract silviculture (i.e. intensive forest management for timber production) classes from the 2023 LANDFIRE EVT raster. Assign the following two silviculture classes a value of 0 (disturbed): "Northeastern North American Temperate Forest Plantation class" (9312) and "Southeastern North American Temperate Forest Plantation" (9322). Assign all other classes a value of 1.
- Using the ArcPy Spatial Analyst Times function, combine the two rasters created above. In the resulting raster, silviculture areas from LANDFIRE receive a value of 0 (disturbed). Otherwise, a pixel retains the disturbed (0) or natural (1) values determined by the NLCD landcover.

- Select U.S. areas from the global mining boundaries dataset, then add and calculate a field that will be used to convert this polygon dataset to a raster. Convert to a raster, assigning mine footprints a value of 0. Reclassify the raster output to change the NoData values to 1 so that mines receive a value of 0 and all other areas receive a value of 1.
- Using the ArcPy Spatial Analyst Times function, combine the mining raster created above with the silviculture/landcover disturbance dataset to add mines to the disturbed class.
- Use a conditional statement to extract quarries, strip mines, and gravel pit classes from the 2023 LANDFIRE EVT raster. This creates a raster where the "Quarries-Strip Mines-Gravel Pits-Well and Wind Pads" have a value of 0 and everything else is 1. We use the mining polygons and LANDFIRE quarries class to attempt to distinguish altered areas of bare earth or rock (such as quarries, industrial areas, mines, and oil and gas well drilling pads) from natural areas of bare earth or rock (such as rock outcroppings and beaches).
- There are groups of location errors in this quarry LANDFIRE class in specific parts of South Alabama and West Mississippi. To remove these location errors, assign a value of 1 (natural) to quarry pixels classified in NLCD as evergreen, mixed, or deciduous forest (41, 42, 43). Essentially, use NLCD to override the LANDFIRE classification in places where NLCD considers a quarry pixel to actually be forested. Otherwise, retain the disturbed value from the LANDFIRE quarry class.
- Using the ArcPy Spatial Analyst Times function, combine the corrected "Quarries-Strip Mines-Gravel Pits-Well and Wind Pads" raster created above with the silviculture/landcover/mining disturbance dataset to add quarries, strip mines, gravel wells, and wind pads to the disturbed class.
- Using the lakes layer from the LAGOS dataset, remove lakes and reservoirs from the natural/disturbed landcover raster created above. Especially for lakes or reservoirs with very small lakesheds, we don't want the open water of a lake or reservoir itself (which would otherwise be considered natural) to influence the percent disturbance calculation. Convert the LAGOS lakes layer from polygon to raster and assign all lakes a value of 1. Reclassify the output to convert NoData areas to 0.
- Use a conditional statement to assign a value of 9 in the combined natural/disturbed raster to any pixel identified as a lake or reservoir in the LAGOS dataset. Otherwise, retain the disturbed (0) or natural (1) value assigned above. This distinguishes lakes and reservoirs (9) from the areas outside the contiguous United States (NoData). Export this as a .tif for further analysis in QGIS.
- In QGIS, use the fix geometries tool to repair the geometry of the "ws" layer from the LAGOS download (gis_locus_v1.0.gpkg). The ws layer contains the lakeshed polygons.
- In QGIS, use the Zonal Histogram tool to calculate the area of disturbance (0) and natural (1) in each lakeshed.
- Convert the zonal histogram output geopackage layer to a geodatabase layer, to allow it to work better in ArcGIS.
- Use the CalculateField command to create a field to hold the percent disturbance value for each lakeshed. Calculate the percent of each lakeshed that is disturbed by dividing the disturbed area of each lakeshed by the total area of each lakeshed and multiplying by 100.

Note: this causes a divide by zero error in lakesheds contained entirely in Canada, which are not covered by the CONUS landcover used to create the disturbance layer. Remove those lakesheds from the analysis and rerun the calculation.

- Add a new field and populate it with the legend values for this indicator. This classification method for lake water quality is based on the Midwest Glacial Lakes Partnership thresholds for watershed landcover composition, as seen in their Conservation Planner online tool. Lakesheds with a percent disturbance >60% receive a value of 1, those with a percent disturbance >25% and ≤60% receive a value of 2, and those with a percent disturbance ≤25% receive a value of 3.
- Convert the LAGOS geopackage lake layer to a geodatabase.
- Using the Join Field command, join to the geodatabase lakes layer to the table containing the percent disturbance for the lakeshed.
- Convert the lakes layer to a raster using the PolygonToRaster command, retaining the field holding the numeric values for this indicator.
- Reclassify the above raster to change NoData values to 0. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Convert to NoData all areas not covered by the NLCD because their disturbed condition or natural condition was not assessed.
- Apply the legend values as seen below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Lake or reservoir with low disturbance (0-25%) in upstream watershed
2 = Lake or reservoir with medium disturbance (>25-60%) in upstream watershed
1 = Lake or reservoir with high disturbance (>60%) in upstream watershed
0 = Not identified as a lake or reservoir

Things to Keep in Mind

- We investigated a number of other datasets that provide similar information, but did not use them in this version of the indicator:
    - The approach developed by Meyer et al. (2024) categorizes lakes into eutrophic, mixotrophic, and dystrophic by matching up Landsat reflectance with limnological observations. It then applies those results to an entire lake using the HydroLAKES dataset. However, HydroLAKES excludes lakes less than 10 hectares in size, and many lakes in North Florida are absent in this dataset. The Meyer et al. (2024) approach also relies on point-in-time observations of surface reflectance, which for some lakes can vary enormously by season, environmental conditions, and year. We determined

upstream watershed condition to provide a more stable, robust assessment of the likelihood of nutrient and sediment loading.
    - The SPARROW watershed model does not cover the entire Southeast.

Known Issues

- This indicator underprioritizes the nearshore areas of lakes and reservoirs (i.e. the littoral zone), which have unique importance for wildlife and plants. We attempted to do a shoreline analysis to highlight these areas but ran out of time due to complexities in the process. We will revisit this in future Blueprint updates to improve this indicator.
- Some lakesheds along the U.S. border with Canada and Mexico extend beyond the boundaries of NLCD (i.e., outside the continental United States). Because we do not have landcover information for those portions of the lakeshed, the percent disturbance calculation used in this indicator is only based on the portion of the lakeshed covered by the NLCD. This might cause some lakeshed disturbance scores to be lower or higher than conditions on the ground depending on the condition of those excluded areas.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Caldwell, Peter; Muldoon, Corinne; Ford-Miniat, Chelcy; Cohen, Erika; Krieger, Suzanne; Sun, Ge; McNulty, Steven; Bolstad, Paul V. 2014. Quantifying the role of National Forest System lands in providing surface drinking water supply for the Southern United States. Gen. Tech. Rep. SRS-197. Asheville, NC: U.S. Department of Agriculture Forest Service, Southern Research Station. 135 p. [https://research.fs.usda.gov/treesearch/47706].

Keith, D. A., J. R. Ferrer-Paris, E. Nicholson, M. Bishop, B. A, Polidoro, E. Ramirez-Llodra, M. G. Tozer, J. L. Nel, R. Mac Nally, E. J. Gregr, K. E. Watermeyer, F. Essl, D. Faber-Langendoen, J. Franklin, C. E. R. Lehmann, A. Etter, D. J. Roux, J. S. Stark, J. A. Rowland, N. A. Brummitt, U. C. Fernandez-Arcaya, I. M. Suthers, S. K. Wiser, I. Donohue, L. J. Jackson, R. T. Pennington, N. Pettorelli, A. Andrade, A. Lindgaard, T. Tahvanainen, A. Terauds, M. A. Chadwick, N. J. Murray, J. Moat, P. Pliscoff, I. Zager, and R. T. Kingsford (2022) A function-based typology for Earth's ecosystems Nature 610, 513–518. DOI:10.1038/s41586-022-05318-4. [https://www.nature.com/articles/s41586-022-05318-4].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published December 18, 2024. Existing Vegetation Type (EVT) CONUS. LANDFIRE 2023 Update, LF 2023, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Meyer, M. F., Topp, S. N., King, T. V., Ladwig, R., Pilla, R. M., Dugan, H. A., Eggleston, J. R., Hampton, S. E., Leech, D. M., Oleksy, I. A., Ross, J. C., Ross, M. R. V., Woolway, R. I., Yang, X., Brousil, M. R., Fickas, K. C., Padowski, J. C., Pollard, A. I., Ren, J., & Zwart, J. A. (2024). National-scale remotely sensed lake trophic state from 1984 through 2020. *Scientific data*, *11*(1), 77. [https://doi.org/10.1038/s41597-024-02921-0].

Midwest Glacial Lakes Partnership. 2019. Midwest Glacial Lakes Partnership Conservation Planner. User Guide. Accessed May 20, 2025. [https://midwestglaciallakes.org/wp-content/uploads/2019/05/MGLP-Conservation-Planner-User-Guide-5_7_191.pdf].

Smith, N.J., K.E. Webster, L.K. Rodriguez, K.S. Cheruvelil, and P.A. Soranno. 2021. LAGOS-US LOCUS v1.0: Data module of location, identifiers, and physical characteristics of lakes and their watersheds in the conterminous U.S. ver 1. Environmental Data Initiative. Accessed December 13, 2021. [https://doi.org/10.6073/pasta/e5c2fb8d77467d3f03de4667ac2173ca].

U.S. Geological Survey (USGS). 2025. Water Science School: Lakes and Reservoirs. Accessed 2025-05-08. [https://www.usgs.gov/special-topics/water-science-school/science/lakes-and-reservoirs].

Natural Landcover in Floodplains

This indicator measures the amount of natural landcover in the estimated floodplain of rivers and streams within each catchment. It assesses the stream channel and its surrounding riparian buffer, measuring the percent of unaltered habitat like forests, wetlands, or open water (rather than agriculture or development). This indicator originates from the National Land Cover Database and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).

This indicator was last updated in October 2024.

Reason for Selection

Habitat near rivers and streams is strongly linked to water quality and instream flow (Naiman 1997). Intact vegetated buffers within the floodplain of rivers and streams provide aquatic habitat, improve water quality, reduce erosion and flooding, recharge groundwater, and more (WeConservePA 2014). Natural floodplain landcover is often described as "the first line of defense" for aquatic systems.

Input Data

- 2021 National Land Cover Database (NLCD): Land cover
- Southeast Blueprint 2024 extent
- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

  The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are

based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- National Hydrography Dataset Plus (NHDPlus) Version 2.1 medium resolution catchments (note: V2.1 is just the current sub-version of the dataset generally called NHDPlusV2); view the user guide

  #### Catchments

  A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.

  To learn more about catchments and how they're defined, check out these resources:

  - An article from USGS explaining the differences between various NHD products
  - The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

Mapping Steps

- Clip the 2021 NLCD to the EPA estimated floodplain layer.
- Reclassify the clipped NLCD to identify natural landcover using the following classes: open water, barren land, deciduous forest, evergreen forest, mixed forest, scrub/shrub, grassland/herbaceous, woody wetlands, and emergent wetlands.
- Calculate the percent of riparian natural landcover inside each NHDPlus catchment using ArcPy Spatial Analyst Zonal Statistics "MEAN" function.
- Take the resulting raster times 100 to convert from a decimal to whole number percent.
- Reclassify the above raster into the 1-5 classes seen in the final indicator values below.
- Clip the resulting raster back to the EPA estimated floodplain layer. It is necessary to do this again since the Zonal Statistics function outputs pixel values for the entire catchment. During this step, assign a value of 0 to areas outside the EPA floodplain. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = >90% natural landcover within the estimated floodplain, by catchment
4 = >80-90%
3 = >70-80%
2 = >60-70%
1 = ≤60% natural landcover within the estimated floodplain, by catchment
0 = Not identified as a floodplain

Known Issues

- Small headwaters and creeks are not included in this indicator because the EPA estimated floodplain dataset does not include them.
- This indicator does not account for the accumulated impacts of upstream riparian buffers. Buffers at the headwaters are treated the same as those downstream.
- This indicator does not consider the river or stream size in relation to the estimated floodplain. Aquatic habitat needs may differ based on the river size class. For example, smaller headwater streams may need more natural landcover than larger rivers to maintain aquatic health. It also does not account for variation in buffer quality within the floodplain at a scale below the catchment. This means that within the estimated floodplain, loss of natural habitat adjacent to the river is treated the same as loss farther away.
- While this indicator generally includes the open water area of reservoirs, some open water portions of reservoirs (e.g., Kerr Lake in NC/VA) are missing from the estimated floodplain dataset.
- The catchment boundaries are inconsistent in how far they extend toward the ocean. As a result, this indicator does not consistently apply to estuaries, coastal areas, and barrier islands.
- In the area just south of Guadalupe Mountains National Park in West Texas, this indicator depicts the floodplain as a series of straight lines that poorly match the actual floodplain. This is due to an error in the EPA floodplain map used in this indicator.
- The catchment boundaries cross the United States/Mexico border, but the NLCD impervious data does not; as a result, the values along the United States/Mexico border are only based on the portion of the catchment where there are NLCD impervious values.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Naiman, Robert J., and Henri Decamps. "The Ecology of Interfaces: Riparian Zones." Annual Review of Ecology and Systematics 28 (1997): 621–58. [https://www.jstor.org/stable/2952507].

U.S. Environmental Protection Agency (USEPA) and the U.S. Geological Survey (USGS). 2012. National Hydrography Dataset Plus 2. [https://www.epa.gov/waterdata/get-nhdplus-national-hydrography-dataset-plus-data].

WeConservePA. 2014. ConservationTools.org: The Science Behind the Need for Riparian Buffer Protection. [https://conservationtools.org/guides/131-the-science-behind-the-need-for-riparian-buffer-protection].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Network Complexity

This indicator depicts the number of different stream size classes in a river network not separated by dams or waterfalls. It originates from the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood, also known as the 1% annual chance flood.

This indicator was last updated in October 2024.

Reason for Selection

River networks with a variety of connected stream size classes are more likely to have a wide range of available habitat to support a greater number of species. This will help retain aquatic biodiversity in a changing climate by allowing species to access climate refugia and move between habitats.

Input Data

- Base Blueprint 2022 extent
- Southeast Blueprint 2024 extent
- Southeast Aquatic Resources Partnership's Network Complexity metric

  The Southeast Aquatic Resources Partnership (SARP) developed metrics for their Southeast Aquatic Barrier Prioritization Tool. On June 7, 2023, Brendan Ward with Astute Spruce (software developer working on behalf of SARP) shared high resolution NHDPlus flowlines

with attributes depicting the network complexity attribute for each functional network (see definition of "functional network" below). The network complexity attribute calculates the total number of different stream size classes within each functional network. SARP assigned stream and river reaches to size classes based on total drainage area:

- 1a: Headwaters (<3.861 sq mi)
- 1b: Creeks (≥3.861 and <8.61 sq mi)
- 2: Small Rivers (≥38.61 and <200 sq mi)
- 3a: Medium Tributary Rivers (≥200 and <1,000 sq mi)
- 3b: Medium Mainstem Rivers (≥1,000 and <3,861 sq mi)
- 4: Large Rivers (≥3,861 and <9,653 sq mi)
- 5: Great Rivers (≥9,653 sq mi)

Functional network

SARP compiles the Southeast Aquatic Barrier Inventory from national, regional, state, and local partner databases across the Southeast region. These include the National Inventory of Dams (2018), National Anthropogenic Barrier Dataset (2012), databases from state dam safety programs and other state agencies, information from local partners, and dam locations estimated by SARP. Waterfalls are compiled from national datasets and local partners.

Dams and waterfalls are snapped to hydrologic networks extracted from the National Hydrography Dataset (NHD) - High Resolution Beta version. All dams and waterfalls are treated as "hard" barriers for network connectivity analysis. Aquatic networks are cut at the location of each barrier. All network "loops" (non-primary flowlines) are omitted from the analysis.

An upstream functional network is constructed by traversing upstream from each barrier through all tributaries to the upstream-most origination point or upstream barrier, whichever comes first. Additional functional networks are defined from downstream-most non-barrier termination points, such as marine areas or other downstream termination points.
The total length of all network segments within a functional network is summed to calculate the total network length of each functional network. Each flowline segment within the NHD is assigned to a size class based on total drainage area. This was used to calculate the number of unique size classes per functional network.

- Estimated Floodplain Map of the Conterminous U.S. from the Environmental Protection Agency's (EPA) EnviroAtlas; see this factsheet for more information; download the data

  The EPA Estimated Floodplain Map of the Conterminous U.S. displays "...areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood). These data are based on the Federal Emergency Management Agency (FEMA) 100-year flood inundation maps with the goal of creating a seamless floodplain map at 30-m resolution for the conterminous

United States. This map identifies a given pixel's membership in the 100-year floodplain and completes areas that FEMA has not yet mapped" (EPA 2018).

- National Hydrography Dataset Plus High Resolution (NHDPlus HR) National Release catchments, accessed 11-30-2022; download the data; view the user guide

- NHDPlus Version 2.1 medium resolution catchments (note: V2.1 is just the current sub-version of the dataset generally called NHDPlusV2); view the user guide

Catchments

A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.

To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

Mapping Steps

- Merge the functional network lines from the 11 subregions delivered by SARP into one feature class.
- Convert the combined SARP network complexity values from the high resolution NHDPlus flowlines to a 30 m raster.
- Clip to the Base Blueprint 2022 extent.
- Apply the network complexity values to the NHDPlus HR catchments using the ArcPy Zonal Statistics "MAJORITY" function. This results in a raster where each catchment is assigned the majority network complexity value that intersects the catchment. Most catchments have only one intersecting line, but for catchments with interior dams, the analysis uses the majority network complexity value.
- To define the analysis extent of the indicator, make a copy of the NHDPlus HR catchments and convert it to raster, assigning it a value of 1.
- Clip the network complexity raster to the EPA floodplain layer. During this step, assign a value of 0 to areas outside the EPA floodplain. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Some areas of the floodplain are not scored in the resulting layer because they are missing SARP network complexity values. This is due to the fact that some small reaches, such as braids and loops in the stream network, are not assigned a network complexity value. SARP has to remove loops and braided streams in order to calculate network complexity because the analysis can only accommodate a one-way flow of water. Identify these holes in the

floodplain and fill them in by looking at the network complexity value of the surrounding pixels and assigning the maximum value to the missing catchments in the floodplain. Note: This simplifies a complex series of analysis steps. For more specifics, please consult the code.
- Clip the network complexity raster to the NHDPlus V2.1 medium resolution catchments. This removes estuarine areas that are outside the intended scope of this indicator, particularly on the NC coast.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = 7 connected stream classes
6 = 6 connected stream classes
5 = 5 connected stream classes
4 = 4 connected stream classes
3 = 3 connected stream classes
2 = 2 connected stream classes
1 = 1 connected stream class
0 = Not identified as a floodplain

Known Issues

- This indicator does not include other smaller scale attributes of complexity (e.g., sinuosity, mixtures of riffles/pools/runs) that influence the habitat quality of the connections.
- The EPA Estimated Floodplain layer sometimes misses the small, linear connections made by artificial canals, especially when they go through areas that wouldn't naturally be part of the floodplain. As a result, some areas (like lakes) that are connected via canals may appear to be disconnected, but still receive high scores.
- Small headwaters and creeks are not included in this indicator because the EPA estimated floodplain dataset does not include them.
- While this indicator generally includes the open water area of reservoirs, some open water portions of reservoirs (e.g., Kerr Lake in NC/VA) are missing from the estimated floodplain dataset.
- This indicator likely overestimates the number of connected stream classes in some areas due to missing barriers in the inventory, such as smaller dams or road-stream crossings. It could also underestimate the number of connected stream classes, given the extensive ongoing restoration work to improve aquatic connectivity across the SECAS geography. If you identify a missing barrier or a removed barrier, please let SARP know by emailing Kat Hoenke at kat@southeastaquatics.net. You can learn more about the current inventory of dams and road-stream crossings by visiting https://connectivity.sarpdata.com/.

- SARP did a lot of work to snap the dam locations to the line network, but there are likely still dams (including some large ones) that didn't get snapped correctly due to the large distance between the centerpoint of the dam and the nearest flowline. If you see any of these cases when reviewing the data, please let SARP know (the giveaway is networks that look longer than they should on a map).
- In the area just south of Guadalupe Mountains National Park in West Texas, this indicator depicts the floodplain as a series of straight lines that poorly match the actual floodplain. This is due to an error in the EPA floodplain map used in this indicator.
- Due to issues with the national NHDPlus HR catchments layer, there are a handful of missing catchments (e.g., northwest TX, coastal LA, and eastern NC). These places receive a value of NoData in the indicator and are therefore underprioritized. We are investigating ways to resolve this in future updates.
- This indicator may slightly overvalue network complexity in WV compared to other Southeast states because the coverage of dams and barriers data is not as comprehensive. While the dams and barriers data coverage improved sufficiently for us to use the network complexity indicator across the entire state of WV in 2023 for the first time (whereas it was only used in the southern part of the state in 2022), there is still room for improvement and we anticipate significant progress in the next Blueprint update.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

EPA EnviroAtlas. 2018. Estimated Floodplain Map of the Conterminous U.S. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/EstimatedFloodplains.pdf].

Martin, E. H, Hoenke, K., Granstaff, E., Barnett, A., Kauffman, J., Robinson, S. and Apse, C.D. 2014. SEACAP: Southeast Aquatic Connectivity Assessment Project: Assessing the ecological impact of dams on Southeastern rivers. The Nature Conservancy, Eastern Division Conservation Science, Southeast Aquatic Resources Partnership. [https://secassoutheast.org/pdf/SEACAP_Report.pdf].

Southeast Aquatic Barrier Prioritization Tool. Southeast Aquatic Resources Partnership (SARP). [https://connectivity.sarpdata.com/].

Permeable Surface

This indicator measures the average percent of non-impervious cover within each catchment. It originates from the National Land Cover Database.

This indicator was last updated in October 2024.

Reason for Selection

Impervious cover is strongly linked to water quality, estuary condition, eutrophication, and freshwater inflow. Impervious surface affects not only aquatic habitats and biodiversity, but also human communities. High levels of impervious surface cause more frequent flooding by increasing the volume of stormwater runoff, reduce the amount of available drinking water by preventing groundwater recharge, and pollute waterways where people swim and fish (Chesapeake 2023, USGS 2018, EPA 2018).

The 90% permeable surface threshold (i.e., 10% impervious) is a well-documented signal of major negative changes to aquatic ecosystems (Schueler et al. 2009). The 95% permeable surface threshold (i.e., 5% impervious) has been documented to impact Piedmont fish [tricolor shiner *(Cyprinella trichroistia)*, bronze darter *(Percina palmaris)*, Etowah darter *(Etheostoma etowahae)*] (Wenger et al. 2008) and estuarine species [blue crab *(Callinectes sapidus)*, white perch *(Morone americana)*, striped bass *(M. Saxatilis)* and spot *(Leiostomus xanthurus)*] (Uphoff Jr. et al. 2011).

Input Data

- 2021 National Land Cover Database (NLCD): Percent developed imperviousness
- Southeast Blueprint 2024 extent
- National Hydrography Dataset Plus (NHDPlus) Version 2.1 medium resolution catchments (note: V2.1 is just the current sub-version of the dataset generally called NHDPlusV2); view the user guide

Catchments

A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.
To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

Mapping Steps

- The NLCD impervious surface raster uses a value of 127 as NoData. Use the Set Null function to change these values to NoData so they won't impact the analysis.
- Calculate percent impervious for each NHDPlus catchment using the NLCD 2021 impervious surface layer and the ArcPy Spatial Analyst Zonal Statistics "MEAN" function. This calculates the average of the impervious surface values in each catchment and assigns that value to all the cells inside that catchment.

- Convert percent impervious to percent permeable using the formula [percent permeable = 100 - percent impervious] to maintain consistent scoring across Southeast indicators (high values indicate better ecological condition).
- Reclassify the above raster into 4 classes, seen in the final indicator values below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = >95% of catchment permeable (likely high water quality and supporting most sensitive aquatic species)
3 = >90-95% of catchment permeable (likely declining water quality and supporting most aquatic species)
2 = >70-90% of catchment permeable (likely degraded water quality and not supporting many aquatic species)
1 = ≤70% of catchment permeable (likely degraded instream flow, water quality, and aquatic species communities)

Known Issues

- This indicator may not account for differences in permeability between different types of soils and land uses.
- The catchment boundaries are inconsistent in how far they extend toward the ocean. As a result, this indicator does not consistently apply to estuaries, coastal areas, and barrier islands in different parts of the Southeast.
- The catchment boundaries cross the United States/Mexico border, but the NLCD impervious data does not; as a result, the values along the United States/Mexico border are only based on the portion of the catchment where there are NLCD impervious values.
- The NLCD percent impervious layer contains classification inaccuracies that may cause this indicator to overestimate or underestimate the amount of permeable surface in some catchments.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Chesapeake Bay Program. 2023. Stormwater Runoff. Accessed September 7, 2023. [https://www.chesapeakebay.net/issues/threats-to-the-bay/stormwater-runoff].

Dewitz, J., 2023, National Land Cover Database (NLCD) 2021 Products: U.S. Geological Survey data release. [https://doi.org/10.5066/P9JZ7AO3].

Environmental Protection Agency. EnviroAtlas. Data Fact Sheet. January 2018. Percent of Stream and Shoreline with 15% or More Impervious Cover within 30 Meters. Accessed September 7, 2023. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/ESN/Percstreamw15percentimperviousin30meters.pdf].

Schueler, T., Fraley-McNeal, L., and Cappiella, K. 2009. "Is Impervious Cover Still Important? Review of Recent Research." J. Hydrol. Eng. 14, SPECIAL ISSUE: Impervious Surfaces in Hydrologic Modeling and Monitoring, 309–315. [https://ascelibrary.org/doi/10.1061/%28ASCE%291084-0699%282009%2914%3A4%28309%29].

Uphoff Jr. JH, McGinty M, Lukacovic R, Mowrer J, Pyle B. 2011. Impervious surface, summer dissolved oxygen, and fish distribution in Chesapeake Bay subestuaries: linking watershed development, habitat conditions, and fisheries management. North American Journal of Fisheries Management 31:554-566. [https://dnrweb.dnr.state.md.us/archive/fisheries/2011_Impervious_Surface.pdf].

U.S. Environmental Protection Agency (USEPA) and the U.S. Geological Survey (USGS). 2012. National Hydrography Dataset Plus 2. [https://www.epa.gov/waterdata/get-nhdplus-national-hydrography-dataset-plus-data].

U.S. Geological Survey (USGS). Water Science School. Impervious Surfaces and Flooding. June 5, 2018. Accessed September 7, 2023. [https://www.usgs.gov/special-topics/water-science-school/science/impervious-surfaces-and-flooding].

Wenger, S. J., J. T. Peterson, M. C. Freeman, B. J. Freeman, D. D. Homans. 2008. Stream fish occurrence in response to impervious cover, historic land use and hydrogeomorphic factors Canadian Journal of Fisheries and Aquatic Sciences 65, 1250-1264. [https://cdnsciencepub.com/doi/10.1139/F08-046].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Coastal & Marine

Atlantic Coral & Hardbottom

This indicator predicts the presence of coral and hardbottom in the Atlantic Ocean based on direct observations, distribution models, and known locations of artificial reefs and shipwrecks. The models use hardbottom observations and a suite of environmental predictors including measures of depth, seafloor topography and substrate, oceanography, and geography. This indicator combines data from multiple sources, including The Nature Conservancy's South Atlantic Bight Marine Assessment, several National Oceanic and Atmospheric Administration datasets, Florida state data, and more.

This indicator was last updated in October 2024.

Reason for Selection

Hardbottom provides an anchor for important seafloor habitats such as deep-sea corals, plants, and sponges. Hardbottom and these associated communities provide important habitat structure for many invertebrate and fish species (NOAA 2018). Hardbottom areas serve as fish nursery, spawning, and foraging grounds, supporting commercially valuable fisheries like snapper and grouper (NCDEQ 2016). According to Dunn and Halpin (2009), "hardbottom habitats support high levels of biodiversity and are frequently used as a surrogate for it in marine spatial planning." Human-created hardbottom (e.g., artificial reefs) is also known to provide additional habitat that is quickly colonized to provide a suite of ecosystem services commonly associated with naturally occurring hardbottom (Wu et al. 2019).

Input Data

- Southeast Blueprint 2024 extent
- Southeast Blueprint 2024 subregions

Coral & hardbottom

- Mapping and Geomorphic Characterization of the Vast Cold-Water Coral Mounds of the Blake Plateau; data provided prior to official release on 6-14-2023 by Dr. Derek Sowers with Ocean Exploration Trust (derek@oceanexplorationtrust.org); read the published journal article about this research; read more about the mapping expedition; read an abstract describing this work from the 2024 Ocean Sciences Meeting; read a white paper about the survey
- The Nature Conservancy's (TNC) South Atlantic Bight Marine Assessment (SABMA); chapter 3 of the final report provides more detail on the seafloor habitats analysis
- NOAA deep-sea coral and sponge locations, accessed 5-10-2024 on the NOAA Deep-Sea Coral & Sponge Map Portal
- Florida coral and hardbottom habitats, accessed 12-19-2023
- National Oceanic and Atmospheric Administration (NOAA) Characterizing Spatial Distributions of Deep-sea Corals and Hardbottom Habitats in the U.S. Southeast Atlantic; read the final report; data shared prior to official release on 2-4-2022 by Matt Poti with the NOAA National Centers for Coastal Ocean Science (NCCOS) (matthew.poti@noaa.gov)

- Predictive Modeling and Mapping of Hardbottom Seafloor Habitats off the Southeast U.S: unpublished NOAA data and draft final report entitled *Assessment of Benthic Habitats for Fisheries Management* provided on 1-28-2021 by Matt Poti with NOAA NCCOS (matthew.poti@noaa.gov)

Shipwrecks & artificial reefs

- NOAA wrecks and obstructions layer (shapefile), accessed 5-1-2023 on the Marine Cadastre
- NOAA artificial reefs, accessed 5-9-2024 on the Marine Cadastre, provided by the NOAA Office for Coastal Management
- Florida Fish and Wildlife Conservation Commission (FWC): Artificial Reefs in Florida (.xlsx), accessed 5-9-2024

Defining inland extent & split with Gulf

- Marine Ecoregions Level III from the Commission for Environmental Cooperation North American Environmental Atlas, accessed 12-8-2021
- 2023 NOAA coastal relief model: volumes 2 (Southeast Atlantic), 3 (Florida and East Gulf of Mexico), 4 (Central Gulf of Mexico), and 5 (Western Gulf of Mexico), accessed 3-27-2024

Mapping Steps

- Create a seamless vector layer to constrain the extent of the Atlantic coral and hardbottom indicator to marine and estuarine areas <1 m in elevation. This defines how far inland it extends.
    - Merge together all coastal relief model rasters (.nc format) using the create virtual raster tool in QGIS.
    - Save the merged raster to .tif format and import it into ArcPro.
    - Reclassify the NOAA coastal relief model data to assign a value of 1 to areas from deep marine to 1 m elevation. Assign all other areas (land) a value of 0.
    - Convert the raster produced above to vector using the raster to polygon tool.
    - Clip to the 2024 Blueprint subregions using the pairwise clip tool.
    - Hand-edit to remove terrestrial polygons (one large terrestrial polygon and the Delmarva peninsula).
    - Dissolve the resulting data layer to produce a seamless polygon defining marine and estuarine areas <1 m in elevation.
    - Hand-edit to select all but the main marine polygon and delete.
- Define the extent of the Atlantic version of this indicator to separate it from the Gulf. This split reflects the extent of the different datasets available to represent coral and hardbottom habitat in the Atlantic and Gulf, rather than a meaningful ecological transition.
    - Use the select tool to select the Florida Keys class from the Level III marine ecoregions ("NAME_L3 = 'Florida Keys'").
    - Buffer the "Florida Keys" Level III marine ecoregion by 2 km to extend it far enough inland to intersect the inland edge of the <1 m elevation layer.

    - Reclassify the two NOAA Atlantic hardbottom suitability datasets to give all non-NoData pixels a value of 0. Combine the reclassified hardbottom suitability datasets to define the total extent of these data. Convert the raster extent to vector and dissolve to create a polygon representing the extent of both NOAA hardbottom datasets.
    - Union the buffered ecoregion with the combined NOAA extent polygon created above. Add a field and use it to dissolve the unioned polygons into one polygon. This leaves some holes inside the polygon, so use the eliminate polygon part tool to fill in those holes, then convert the polygon to a line.
    - Hand-edit to extract the resulting line between the Atlantic and Gulf.
    - Hand-edit to use this line to split the <1 m elevation layer created earlier in the mapping steps to create the separation between the Atlantic and Gulf extent.
- From the NOAA deep-sea coral and sponge point locations, select all points. Buffer the point locations by 150 m, convert to raster, and assign them a value of 8. The buffer distance used here, and later for shipwrecks and artificial reefs, follows guidance from the Army Corps of Engineers for setbacks around artificial reefs and fish havens (Riley et al. 2021).
- From the FL FWC coral and hardbottom data, convert to raster and assign the "coral reef" class a value of 8 and the "hardbottom" and "hardbottom with seagrass" classes a value of 7.
- From the TNC SABMA data, pull out observed hardbottom polygons that contain a value of "01. mapped hardbottom area" in the TEXT_DESC field. Convert to a raster and assign a value of 7.
- Combine the NOAA regional artificial reef dataset with the Florida FWC artificial reefs. Buffer the points by 150 m and convert to raster, assigning all buffered points a value of 6.
- From the wrecks database, extract locations having "high" and "medium" confidence (positionQuality = "high" and positionQuality = "medium"). Buffer by 150 m and convert to raster, assigning them a value of 5.
- From the Blake Plateau dataset, pull out peaks, ridges, and slopes from the landform data and assign them all a value of 4.
- Reclassify both NOAA hardbottom suitability datasets into 5 quantiles. Assign the top quantile a value of 3, the second-highest quantile a value of 2, and the middle quantile a value of 1. Assign the lower two quantiles a value of NoData. They are not used in the indicator due to the relatively low likelihood of hardbottom presence (<40% probability).
- Combine the two NOAA hardbottom suitability datasets and use the newer data from the "Characterizing Spatial Distributions of Deep-sea Corals and Hardbottom Habitat in the U.S. Southeast Atlantic" project where pixels overlap. Snap and project the result based on the Southeast Blueprint 2024 extent.
- Combine all the layers produced above using the Cell Statistics "MAXIMUM" function.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

8 = Confirmed hardbottom-associated species (corals, sponges)
7 = Confirmed natural hardbottom
6 = Artificial reefs
5 = Shipwrecks
4 = Predicted cold-water coral mounds (Blake Plateau)
3 = Highest probability of hardbottom (>80th percentile)
2 = High probability of hardbottom (>60th-80th percentile)
1 = Medium probability of hardbottom (>40th-60th percentile)
0 = Not identified as hardbottom

Known Issues

- This indicator likely underpredicts hardbottom suitability in shallow waters. While this indicator includes new hardbottom suitability models based on recent hardbottom observations for deep waters (depths of 50 m or below), the underlying NOAA data available for shallow waters were developed in 2014.
- While this layer has a 30 m resolution, both NOAA hardbottom datasets were coarser than that. We downsampled 100 m pixels and 92 m pixels to 30 m.
- This indicator underestimates shallower hardbottom habitat (<200 m depth) north of the NC/VA state line because the study area of the shallower hardbottom suitability dataset was restricted only to the South Atlantic marine environment and did not cover the northern portion of the SECAS marine area. The indicator also underestimates deeper hardbottom habitat north of approximately 37.5°N latitude because the study area of the deeper hardbottom suitability dataset does not perfectly align with the SECAS marine area and leaves an area of NoData.
- This indicator likely overpredicts hardbottom in the "3 = highest probability of hardbottom" class because the sampling data used to generate these hardbottom probability models is not sufficient to finely resolve the relative likelihood of hardbottom within this bin. While there is a high probability of hardbottom within these areas, there is likely more variation than the model depicts (i.e., the probabilities are somewhat "smeared").

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Commission for Environmental Cooperation. 2009. Marine Ecoregions of North America, 2008. Vector digital data. Montréal, Québec, Canada. [https://www.cec.org/north-american-environmental-atlas/marine-ecoregions/].

Conley, M.F., M.G. Anderson, N. Steinberg, and A. Barnett, eds. 2017. The South Atlantic Bight Marine Assessment: Species, Habitats and Ecosystems. The Nature Conservancy, Eastern Conservation Science. [https://easterndivision.s3.amazonaws.com/Marine/SABMA/SABMA_Report_11April2018.pdf].

Deep Sea Coral Research and Technology Program (DSCRTP) (2016). Observations of Deep-Sea Coral and Sponge Occurrences from the NOAA National Deep-Sea Coral and Sponge Database, 1842-Present, version 20230620-0 (NCEI Accession 0145037). NOAA National Centers for Environmental Information. Dataset. Accessed December 20, 2023. [https://www.ncei.noaa.gov/archive/accession/0145037].

Dunn DC, Halpin PN (2009) Rugosity-based regional modeling of hard-bottom habitat. Marine Ecology Progress Series 377:1-11. [https://doi.org/10.3354/meps07839].

Florida Fish and Wildlife Conservation Commission. Artificial Reefs: Download Reef Locations. Microsoft Excel file. Accessed December 19, 2023. [https://myfwc.com/fishing/saltwater/artificial-reefs/locate/].

Florida Fish and Wildlife Conservation Commission-Fish and Wildlife Research Institute. Coral and Hard Bottom Habitats Florida. Vector digital data. Published March 12, 2015. Accessed December 19, 2023. [https://geodata.myfwc.com/datasets/myfwc::coral-and-hard-bottom-habitats-in-florida/about].

Gulf of Mexico Fishery Management Council. October 2010. Final Report: 5-Year Review of the Final Generic Amendment Number 3 Addressing Essential Fish Habitat Requirements, Habitat Areas of Particular Concern, and Adverse Effects of Fishing in the Fishery Management Plans of the Gulf of Mexico. [https://gulf-council-media.s3.amazonaws.com/uploads/2025/03/EFH-5-Year-Review-Final-10-10.pdf].

National Oceanographic and Atmospheric Administration. Deep Sea Coral Research and Technology Program 2018 Report to Congress. December 2018. [https://coast.noaa.gov/coral_storage/library/NOAA/DSCRTP/Other/Reports_To_Congress/2018/DSCRTP2018_Report_to_Congress.pdf].

National Oceanographic and Atmospheric Administration. Unpublished Draft: Assessment of Benthic Habitats for Fisheries Management. May 2015. [https://secassoutheast.org/pdf/DRAFT_Benthic-Habitats-for-Fisheries-Management-Final-Report-April-2015-with-MP-additions.pdf].

NCDEQ (North Carolina Department of Environmental Quality) 2016. North Carolina Coastal Habitat Protection Plan Source Document. Morehead City, NC. Division of Marine Fisheries. 475 p. [https://deq.nc.gov/media/26813/open].

Neilson, B. J., & Ferry, P. S. (1978) A Water Quality Study of the Estuarine James River. Special Reports in Applied Marine Science and Ocean Engineering (SRAMSOE) No.131. Virginia Institute of Marine Science, College of William and Mary. [https://doi.org/10.21220/V52157].

NOAA National Centers for Environmental Information. (2023). Coastal Relief Models (CRMs) [Data set]. NOAA National Centers for Environmental Information. [doi: 10.25921/5ZN5-KN44].

Office for Coastal Management, 2023: Wrecks and Obstructions. NOAA National Centers for Environmental Information. [https://www.fisheries.noaa.gov/inport/item/70439].

Poti M, Goyert HF, Salgado EJ, Bassett R, Coyne M, Winship AJ, Etnoyer PJ, Hourigan TF, Coleman HM, Christensen J. 2022. Data synthesis and predictive modeling of deep-sea coral and hardbottom habitats offshore of the southeastern US: guiding efficient discovery and protection of sensitive benthic areas. New Orleans (LA): US Department of the Interior, Bureau of Ocean Energy Management. 224 p. Contract No.: M16PG00010. Report No.: OCS Study BOEM 2022- 038. [https://espis.boem.gov/final%20reports/BOEM_2022-038.pdf].

Riley, K.L., Wickliffe, L.C., Jossart, J.A., MacKay, J.K., Randall, A.L., Bath, G.E., Balling, M.B., Jensen, B.M., and Morris, J.A. Jr. 2021. An Aquaculture Opportunity Area Atlas for the U.S. Gulf of Mexico. NOAA Technical Memorandum NOS NCCOS 299. Beaufort, NC. 545 pp. [https://doi.org/10.25923/8cb3-3r66].

Sowers, Derek, Larry Mayer, Giuseppe Masetti, Erik Cordes, Ryan Gasbarro, Elizabeth Lobecker, Kasey Cantwell, Shannon Hoy, Michael White, Sam Candio, Mashkoor Malik, and Matt Dornback. Mapping and Geomorphic Characterization of the Vast Cold-Water Coral Mounds of the Blake Plateau. Accessed June 14, 2023. [https://secassoutheast.org/pdf/ISDSC8_Sowers_et_al_Poster_Final.pdf].

Sowers, D.C.; Mayer, L.A.; Masetti, G.; Cordes, E.; Gasbarro, R.; Lobecker, E.; Cantwell, K.; Candio, S.; Hoy, S.; Malik, M.; et al. Mapping and Geomorphic Characterization of the Vast Cold-Water Coral Mounds of the Blake Plateau. Geomatics 2024, 4, 17-47. [https://doi.org/10.3390/geomatics4010002].

Wu, Zhongin, James R. Tweedley, Neil R. Loneragan, Xiumei Zhang. 2019. Artificial reefs can mimic natural habitats for fish and macroinvertrbrates in temperate coastal waters of the Yellow Sea. Ecological Engineering, vol 139. [https://doi.org/10.1016/j.ecoleng.2019.08.009].

Atlantic Deep-Sea Coral Richness

This indicator measures the number of deep-sea coral genera predicted to occur in the Atlantic Ocean at depths of approximately 50 m or below. It is based on coral observations and a suite of environmental predictors including measures of depth, seafloor topography and substrate, oceanography, and geography. This indicator combines probability models for 24 deep-sea coral genera to predict overall richness. Deep-sea corals provide valuable habitat structure that supports a wide range of invertebrate and fish species, and higher coral diversity typically creates more complex habitats occupied by more species. This indicator originates from a National Oceanic and Atmospheric Administration project characterizing the spatial distributions of deep-sea corals and hardbottom habitats in the U.S. Southeast Atlantic.

This indicator was last updated in October 2023.

Reason for Selection

Many deep-sea corals form tree-like shapes and complex reefs that provide valuable three-dimensional habitat structure for many fish and invertebrate species. The presence of more coral genera typically creates more complex habitats that support more species. In tropical coral reef communities, higher levels of coral diversity and topographic complexity have been shown to promote higher diversity of fish species (Komyakova et al. 2013). Deep-sea corals support commercially important fisheries such as grouper, snapper, sea bass, rockfish, shrimp, and crab. Because most deep-sea corals grow very slowly, they are highly vulnerable to damage from trawling and energy development, as well as ocean acidification due to climate change (NOAA 2018).

Input Data

- Southeast Blueprint 2023 subregions: Marine (combined Atlantic & Gulf)
- Southeast Blueprint 2023 extent
- National Oceanic and Atmospheric Administration (NOAA) Characterizing Spatial Distributions of Deep-sea Corals and Hardbottom Habitats in the U.S. Southeast Atlantic; read the final report; data shared prior to official release on 2-4-2022 by Matt Poti with the NOAA National Centers for Coastal Ocean Science (matthew.poti@noaa.gov)

  This dataset provides probability models for 24 deep-sea coral genera *(Eunicella, Enallopsammia, Cladocora, Chrysogorgia, Callogorgia, Bathypathes, Antipathes, Anthothela, Acanthogorgia, Acanella, Thesea, Tanacetipathes, Stylasteridae, Stichopathes, Solenosmilia, Plumarella, Paramuricea, Paragorgia, Oculina, Nicella, Muricea, Madrepora, Lophelia, Leiopathes),* as well as a combined genus richness layer that counts the average number of unique genera predicted to occur in each 100 m pixel. Matt Poti provided a summarized vector version of the continuous genus richness raster layer that mirrors the way the Bureau of Safety and Environmental Enforcement displays the Gulf version of these data. This classification is based on natural breaks in the data distribution and uses 10 bins. We further collapsed the bins from 10 to 5 for simplicity.

Mapping Steps

- Convert the categorical text descriptions in the provided genus richness shapefile into the indicator values.
- Reclassify to collapse the 10 classes into 5 by combining the “0-0.5” and “0.5-1” classes and assigning a value of 1, combining the “1-1.5” and “1.5-2” classes and assigning a value of 2, combining the “2-2.5” and “2.5-3” classes and assigning a value of 3, combining the “3-3.5” and “3.5-4” classes and assigning a value of 4, and combining the “4-5” and “>5” classes and assigning a value of 5.
- Convert the shapefile to a raster.
- Clip to the Southeast Blueprint 2023 marine subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Highest predicted average genus richness (>4)
4 = High predicted average genus richness (>3-4)
3 = Medium predicted average genus richness (>2-3)
2 = Low predicted average genus richness (>1-2)
1 = Lowest predicted average genus richness (0-1)

Known Issues

- While this layer has a 30 m resolution, the NOAA deep-sea coral models were coarser than that. We downsampled 100 m pixels to 30 m.
- This indicator underprioritizes areas with low survey effort for the variables used to predict deep-sea corals. This is especially true for seafloor geology, curvature, and aspect. Sharp linear features and shapes in this indicator, when not aligned with a shelf break, are a symptom of places with low survey effort for these variables.
- The source data does not consistently cover all areas deeper than 50 m within the Atlantic marine subregion. According to the final report for the NOAA project, “The study area included waters between 50–3,500 m depth within BOEM’s Straits of Florida, South Atlantic, and Mid-Atlantic Planning Areas and extended from Florida to Delaware. Locations of underwater visual surveys used to compile the presence-absence database for this study did not span this entire depth range across the study area. Therefore, the depth range of the study area extent varied with latitude.”

Other Things to Keep in Mind

- Atlantic and Gulf deep-sea coral richness are intended to serve as complementary indicators and are based on very similar NOAA source data. Because of the different deep-sea coral communities present in the Atlantic and the Gulf, the data provider recommended different thresholds for what level of genus richness qualifies as highest, high, medium, etc. to ensure the Blueprint captures the most important deep-sea coral areas within each region.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Komyakova V, Munday PL, Jones GP. Relative importance of coral cover, habitat complexity and diversity in determining the structure of reef fish communities. PLoS One. 2013 Dec 13;8(12):e83178. [https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3862682/#B11].

National Oceanographic and Atmospheric Administration. Deep Sea Coral Research and Technology Program 2018 Report to Congress. December 2018. [https://coast.noaa.gov/coral_storage/library/NOAA/DSCRTP/Other/Reports_To_Congress/2018/DSCRTP2018_Report_to_Congress.pdf].

Poti M, Goyert HF, Salgado EJ, Bassett R, Coyne M, Winship AJ, Etnoyer PJ, Hourigan TF, Coleman HM, Christensen J. 2022. Data synthesis and predictive modeling of deep-sea coral and hardbottom habitats offshore of the southeastern US: guiding efficient discovery and protection of sensitive benthic areas. New Orleans (LA): US Department of the Interior, Bureau of Ocean Energy Management. 224 p. Contract No.: M16PG00010. Report No.: OCS Study BOEM 2022- 038. [https://espis.boem.gov/final%20reports/BOEM_2022-038.pdf].

Atlantic Estuarine Fish Habitat

This indicator measures the condition of estuarine fish habitat along the Atlantic coast using metrics of water quality, marsh edges, seagrass and oyster reefs, fragmentation, human development, and more. Areas of excellent fish habitat are already in good condition and face few threats. Restoration opportunity areas are doing well in some respects, but restoration projects could significantly improve them. Degraded areas of opportunity face many challenges, and restoration projects are unlikely to increase available fish habitat unless particularly large in scope and scale. This indicator originates from the Atlantic Coast Fish Habitat Partnership's fish habitat conservation area mapping and prioritization project.

This indicator was last updated in October 2023.

Reason for Selection

This indicator captures many factors influencing Atlantic estuarine fish habitat quality and promotes consistency with the priorities of the Atlantic Coast Fish Habitat Partnership. Estuarine habitat provides critical food resources and serves as a refuge from predators for many recreational and commercial fish, crabs and small animals (Naturally Resilient Communities 2017). Submerged aquatic vegetation in coastal and estuarine ecosystems is a food source for waterfowl and, along with marsh edges and seagrass, serves as important nursery habitat for many species (Dennison et al. 1993). Oyster reefs and beds also help to improve surrounding water quality, buffer coasts from waves, and reduce erosion (Naturally Resilient Communities 2017).

Input Data

- Atlantic Coast Fish Habitat Partnership (ACFHP) Fish Habitat Conservation Area Mapping and Prioritization Project: South Atlantic, Mid-Atlantic, and South Florida Estuarine Analysis
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Convert the South Atlantic Estuarine Analysis from vector to a 30 m raster using the FINALSCORE field.
- Convert the Mid-Atlantic Estuarine Analysis from vector to a 30 m raster using the TotalPoints field.
- Convert the South Florida Estuarine Analysis from vector to a 30 m raster using the FINALSCORE field.
- Combine the above rasters using the ArcPy Spatial Analyst Cell Statistics “MAXIMUM” function.
- Reclassify the above raster into 9 classes, seen in the final indicator values below.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

8 = Final score of 80 (areas of excellent fish habitat)
7 = Final score of 70 (areas of excellent fish habitat)
6 = Final score of 60 (restoration opportunity areas)
5 = Final score of 50 (restoration opportunity areas)
4 = Final score of 40 (restoration opportunity areas)
3 = Final score of 30 (restoration opportunity areas)
2 = Final score of 20 (restoration opportunity areas)
1 = Final score of 10 (degraded areas of opportunity)
0 = Final score of 0 (degraded areas of opportunity)

Known Issues

- This indicator underestimates scores in parts of Georgia and North Florida that weren't well-surveyed for oyster reefs at the time of the ACFHP assessment.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Dennison, William C., Robert J. Orth, Kenneth A. Moore, J. Court Stevenson, Virginia Carter, Stan Kollar, Peter W. Bergstrom, Richard A. Batiuk, Assessing Water Quality with Submersed Aquatic Vegetation: Habitat requirements as barometers of Chesapeake Bay health, BioScience, Volume 43, Issue 2, February 1993, Pages 86–94. [https://doi.org/10.2307/1311969].

Martin, Erik, Kat Hoenke, and Lisa Havel. Atlantic Coast Fish Habitat Partnership. Fish Habitat Conservation Area Mapping and Prioritization Project: A Prioritization of Atlantic Coastal, Estuarine, and Diadromous Fish Habitats for Conservation. August 2020. [https://www.atlanticfishhabitat.org/wp-content/uploads/2020/08/ACFHP-Mapping-and-Prioritization-Final-Report.pdf].

Naturally Resilient Communities. 2017. Solution: Oyster Reefs. [https://nrcsolutions.org/wp-content/uploads/2017/03/NRC_Solutions_Oyster_Reefs.pdf].

Atlantic Marine Birds

This indicator identifies important areas in the Atlantic Ocean for birds that feed exclusively or mainly at sea. It uses seasonal predictions of relative abundance for 19 species of marine birds (Audubon's shearwater, band-rumped storm petrel, black-capped petrel, black scoter, Bonaparte's gull, bridled tern, brown pelican, common loon, common tern, Cory's shearwater, great shearwater, Manx shearwater, Northern gannet, parasitic jaeger, red-throated loon, royal tern, sooty shearwater, sooty tern, white-winged scoter) based on sightings from boat-based surveys and marine environmental data like fronts, primary productivity, and ocean currents. This indicator originates from Duke University's Marine-life Data and Analysis Team marine bird models.

This indicator was last updated in October 2023.

Reason for Selection

Marine birds help identify key areas of ocean productivity and overall ocean health. According to Parsons et al. 2008, seabirds are often considered useful ecological indicators for the marine environment because they "essentially represent the top of the food chain" and therefore are likely to signal changes in lower trophic levels and the surrounding environment. Seabird populations also respond to anthropogenic pressures such as "overexploitation of their food resources and pollution from industrial discharge." Long-term monitoring has "generated high-quality data on population counts and demographic parameters" and seabirds "are considered to be of international importance and have high resonance with the public and policy-makers" (Parsons et al. 2008). This indicator complements the Atlantic marine mammals indicator by providing finer spatial resolution and stronger connections to forage fish productivity.

Input Data

- Southeast Blueprint 2023 extent
- 2019 National Land Cover Database (NLCD): Land cover
- Seasonal avian abundance models (Version 2.0) from Duke University's Marine-life Data and Analysis Team (Avian abundance ZIP), accessed 6-8-2021

  These models use aggregated survey information and oceanographic variables to predict the relative abundance of marine birds across the entire U.S. Atlantic. Potential species to include in this indicator began with Tier 1 and Tier 2 priority species in Bird Conservation Region 27 (the Southeastern Coastal Plain) in the Southeast United States Regional Waterbird Conservation Plan. This plan did not include waterfowl species, so additional waterfowl were added from priority species identified by the Northwest Atlantic Birds at Sea Conservation Cooperative (now the Atlantic Marine Bird Cooperative). We narrowed down this larger combined list in two ways. First, we removed species from the list that didn't have spatial models available. Then, we removed species that had models with poor predictive performance and/or large spatial errors.

Species with seasonal models used in this indicator are: Audubon's shearwater (fall, spring, summer), band-rumped storm-petrel (summer), black-capped petrel (spring, summer, winter), black scoter (fall, spring, winter), Bonaparte's gull (spring, winter), bridled tern (fall, summer), brown pelican (fall), common loon (spring, summer, winter), common tern (fall, spring, summer), Cory's shearwater (fall, spring, summer), great shearwater (fall, spring, summer), Manx shearwater (fall, spring, summer), Northern gannet (spring, summer, winter), parasitic jaeger (fall, spring, summer), red-throated loon (spring, winter), royal tern (fall, spring, summer), sooty shearwater (fall, spring, summer), sooty tern (spring, summer) and white-winged scoter (fall, spring, winter).

Mapping Steps

- To identify important areas for each species, use the core area algorithm (CAZMAX) in Zonation 5. Include each seasonal abundance layer as a separate input and weight them equally. To account for boundary effects, run all the models across the full extent of the input data (entire U.S. Atlantic).
- Reproject the resulting data to Albers Equal Area.
- Resample the data to 30 m cell size using the nearest neighbor method (the source data is 2 km cells).
- Convert from a floating point raster with a range of 0-1 to an integer raster ranging from 0-100.
- Reclassify to values of 1-10 based on increments of 10 to produce the indicator values seen below.
- Add a zero class for land by reclassifying the NLCD to identify areas of land included in the marine mammal models, which are outside the scope of this indicator. Assign a value of 0 to all pixels that are not a value of 0 or 11 in the NLCD.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = >90th percentile of importance for marine bird index species (across the full East Coast study area)
9 = >80th-90th percentile of importance
8 = >70th-80th percentile of importance
7 = >60th-70th percentile of importance
6 = >50th-60th percentile of importance
5 = >40th-50th percentile of importance
4 = >30th-40th percentile of importance
3 = >20th-30th percentile of importance
2 = >10th-20th percentile of importance
1 = ≤10th percentile of importance
0 = Land

Known Issues

- Models are likely underpredicting the importance of areas in the eastern part of the Atlantic marine ecosystem. Survey effort was very low in that area and many input models did not extend their predictions into the eastern area.
- This indicator does not capture fine resolution patterns nearshore. Model predictions are fairly coarse and do not capture finer variations in relative abundance nearshore and near estuaries.
- This indicator does not cover the estuarine ecosystem. Many of these marine bird species could also be excellent indicators in the open water portion of the estuarine ecosystem; however, spatial models covering the full area of the estuarine ecosystem are not available at this time.
- While this layer has a 30 m resolution, the source data was coarser than that. We downsampled 2 km pixels to 30 m.
- There are a small number of pixels on the eastern edge of the Atlantic marine subregion, east of central and southern Florida, where the Duke models have no data. As a result, those areas are not scored in this indicator.

Other Things to Keep in Mind

- We ran the Zonation analysis across the full East Coast study area because some of the important areas for some species during some months were mostly in an area north of the Blueprint area and just barely extended into the northernmost tip of the Southeast Blueprint marine area. If we ran Zonation only within the Southeast Blueprint Atlantic marine subregion, the indicator would have overprioritized those areas, since they would appear to be very rare features, when in actuality they are common features that just more frequently occur further north. While doing the Zonation run across the full East Coast study area addressed this problem, it means that the percentile calculations are for the full East Coast

study area, which included areas outside the extent of the final indicator. As a result, the various classes within the indicator do not cover equal areas within the indicator's extent, as you might expect with a percentile-based indicator—they cover equal areas within the full East Coast study area, and then are clipped down to produce the indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Curtice, C., Cleary J., Shumchenia E., Halpin P.N. 24 June 2019. Marine-life Data and Analysis Team (MDAT) technical report on the methods and development of marine-life data to support regional ocean planning and management. Prepared on behalf of the Marine-life Data and Analysis Team (MDAT). [https://seamap.env.duke.edu/models/mdat/MDAT-Technical-Report.pdf].

Hunter, W.C., Golder, W., Melvin, S., Wheeler, J., 2006. Southeast United States Regional Waterbird Conservation Plan. U.S. Fish and Wildlife Service, Atlanta, GA. [https://static1.squarespace.com/static/5bb3865d2727be6f94acf2fc/t/5c79a2efe5e5f0214c34c48c/1551475442564/SE_US_Waterb_Plan_2006.pdf].

Moilanen, A., Lehtinen, P., Kohonen, I., Virtanen, E., Jalkanen, J. and Kujala, H. 2022. Novel methods for spatial prioritization with applications in conservation, land use planning and ecological impact avoidance. Methods in Ecology and Evolution. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13819].

Parsons, Matt, Ian Mitchell, Adam Butler, Norman Ratcliffe, Morten Frederiksen, Simon Foster, James B. Reid, Seabirds as indicators of the marine environment, ICES Journal of Marine Science, Volume 65, Issue 8, November 2008, Pages 1520–1526. [https://doi.org/10.1093/icesjms/fsn155].

Atlantic Marine Mammals

This indicator identifies important areas in the Atlantic Ocean for dolphins, whales, and seals. It incorporates density predictions for 20 marine mammals species or species groups (Atlantic spotted dolphin, Atlantic white-sided dolphin, Clymene dolphin, common bottlenose dolphin, Cuvier's beaked whale, dwarf and pygmy sperm whales, fin whale, harbor porpoise, humpback whale, mesoplodont beaked whales, North Atlantic right whale, pantropical spotted dolphin, pilot whales, Risso's dolphin, rough-toothed dolphin, seals, short-beaked common dolphin, sperm whale, striped dolphin, unidentified beaked whales) based on sightings from boat-based and aerial surveys and data on oceanographic conditions. It uses marine mammal models developed by the Duke Marine Lab.

This indicator was last updated in October 2023.

Reason for Selection

Marine mammals help identify key areas of ocean productivity and overall ocean health, are regularly monitored, and resonate with a variety of audiences. Marine mammals are often used as ocean health indicators due to their long lifespans, feeding at high trophic levels, and large blubber stores that can serve as repositories for anthropogenic chemicals and toxins (Bossart 2011). This indicator complements the Atlantic marine birds indicator by identifying important habitats in areas too deep for marine birds to reach.

Input Data

- Southeast Blueprint 2023 extent
- 2019 National Land Cover Database (NLCD): Land cover
- Habitat-based Marine Mammal Density Models for the U.S. Atlantic (last updated 6-20-2022), accessed 12-12-2023

  These models use aggregated survey information, distance-sampling, and oceanographic variables to predict cetacean density throughout the region. We used data for all available species with the exception of blue whale, common minke whale, false killer whale, Fraser's dolphin, killer whale, melon-headed whale, Northern bottlenose whale, pygmy killer whale, sei whale, spinner dolphin, and white-beaked dolphin. Those species had significant spatial problems in the Southeast.

  The indicator used monthly predictions for 11 species: Atlantic spotted dolphin, Atlantic white-sided dolphin, common bottlenose dolphin, fin whale, harbor porpoise, humpback whale (2009-2019), North Atlantic right whale (2003-2019), Risso's dolphin, seals, short-beaked common dolphin, and sperm whale. It used yearly predictions for 9 rarer species/species groups: Clymene dolphin, Cuvier's beaked whale, dwarf and pygmy sperm whales, mesoplodont beaked whales, pantropical spotted dolphin, pilot whales, rough-toothed dolphin, striped dolphin, and unidentified beaked whales.

Mapping Steps

- To identify important areas for each species, use the core area algorithm (CAZMAX) in Zonation 5. Include each monthly density layer for each marine mammal species as a separate input and weight them equally. To account for boundary effects, run all the models across the full extent of the input data (entire U.S. Atlantic).
- Reproject the resulting source data to Albers Equal Area.
- Resample to a 30 m cell size using the nearest neighbor method (the source data is 5 km cells).
- Convert from a floating point raster with a range of 0-1 to an integer raster ranging from 0-100.
- Reclassify to values from 1-10 based on increments of 10.
- Add in a zero class for land by reclassifying NLCD to identify areas of land included in the marine mammal models, which are outside the scope of this indicator. Assign a value of 0 to all pixels that are not a value of 0 or 11 in the NLCD.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = >90th percentile of importance for marine mammal index species (across the full East Coast study area)
9 = >80th-90th percentile of importance
8 = >70th-80th percentile of importance
7 = >60th-70th percentile of importance
6 = >50th-60th percentile of importance
5 = >40th-50th percentile of importance
4 = >30th-40th percentile of importance
3 = >20th-30th percentile of importance
2 = >10th-20th percentile of importance
1 = ≤10th percentile of importance
0 = Land

Known Issues

- While this layer has a 30 m resolution, the source data was coarser than that. We downsampled 5 km pixels to 30 m.
- There are a small number of pixels on the eastern edge of the Atlantic marine subregion just south of southern Florida where the Duke models have no data. As a result, those areas are not scored in this indicator.

Other Things to Keep in Mind

- We ran the Zonation analysis across the full East Coast study area because some of the important areas for some species during some months were mostly in an area north of the Blueprint area and just barely extended into the northernmost tip of the Southeast Blueprint marine area. If we ran Zonation only within the Southeast Blueprint Atlantic marine subregion, the indicator would have overprioritized those areas, since they would appear to be very rare features, when in actuality they are common features that just more frequently occur further north. While doing the Zonation run across the full East Coast study area addressed this problem, it means that the percentile calculations are for the full East Coast study area, which included areas outside the extent of the final indicator. As a result, the various classes within the indicator do not cover equal areas within the indicator's extent, as you might expect with a percentile-based indicator—they cover equal areas within the full East Coast study area, and then are clipped down to produce the indicator.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

Bossart, G. D. "Marine Mammals as Sentinel Species for Oceans and Human Health." Veterinary Pathology Online 48, no. 3 (May 1, 2011): 676–90. [https://doi.org/10.1177/0300985810388525].

Moilanen, A., Lehtinen, P., Kohonen, I., Virtanen, E., Jalkanen, J. and Kujala, H. 2022. Novel methods for spatial prioritization with applications in conservation, land use planning and ecological impact avoidance. Methods in Ecology and Evolution. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13819].

Roberts, J., Best, B., Mannocci, L. et al. Habitat-based cetacean density models for the U.S. Atlantic and Gulf of Mexico. Sci Rep 6, 22615 (2016). [https://doi.org/10.1038/srep22615].

Roberts JJ, Yack TM, Halpin PN (2023) Marine mammal density models for the U.S. Navy Atlantic Fleet Training and Testing (AFTT) study area for the Phase IV Navy Marine Species Density Database (NMSDD). Document version 1.3. Report prepared for Naval Facilities Engineering Systems Command, Atlantic by the Duke University Marine Geospatial Ecology Lab, Durham, NC. [https://seamap.env.duke.edu/seamap-models-files/Duke/Reports/AFTT_Marine_Mammal_Density_Models_2022_v1.3.pdf].

Coastal Shoreline Condition

This indicator assesses shoreline condition based on the presence of hardened structures like jetties, groins, and riprap, as well as other human development. Natural shorelines in harder-to-develop coastal areas receive the highest shoreline condition scores, while hardened shorelines receive the lowest scores. This indicator originates from the National Oceanic and Atmospheric Administration's Environmental Sensitivity Index dataset.

This indicator was last updated in October 2023.

Reason for Selection

Human development along shorelines, along with jetties, groins, seawalls, revetments, and other structures, provide a measure of overall habitat alteration and shoreline condition. Human infrastructure along shorelines generally stabilizes barrier islands, impeding natural beach migration and barrier island rollover processes. Groins, seawalls, jetties, and revetments have resulted in narrowing of beaches, or greater beach loss, compared to unstructured beaches (Dugan et al. 2008, Hall and Pilkey 1991, Mohanty et al. 2012, Pilkey and Wright III 1988). Jetties also alter sand transport and may result in downdrift erosion (Bruun 1995). Hardened structures landward of tidal wetlands can cause "coastal squeeze" by accelerating erosion during storms and preventing inland migration in response to sea level rise (Gittman 2015).

In addition, hardened shorelines, particularly seawalls, generally support lower levels of biodiversity (Gittman et al. 2015, 2016). Studies funded by NOAA's National Centers for Coastal Ocean Science found that shoreline hardening has a negative impact on amounts of submerged aquatic vegetation (SAV) and on fish density and egg-laying (NOAA NCOS 2015, 2013).

Input Data

- National Oceanic and Atmospheric Administration (NOAA) Environmental Sensitivity Index (ESI) National Shoreline dataset: These data classify shorelines into types of natural habitat and hardened structures to identify resources that are at risk if an oil spill occurs nearby.
- Spatial designations of the Coastal Barrier Resources System (CBRS), as a result of the Coastal Barrier Resources Act (1982) and subsequent amendments (last updated 3-15-2019), accessed 1-31-2022

  The CBRS was used to set boundaries of relatively undeveloped coastal areas that are more difficult to develop. The Act incentivizes the conservation of mostly intact coastal barrier landforms—such as barrier islands, beaches and dunes, and fringing mangroves, and others—to protect valuable natural resources and minimize the loss of human life and property due to hurricanes. It restricts federal expenditures that encourage development in these areas, such as flood insurance.

- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent

Mapping Steps

- Convert the CBRS polygons to a 30 m raster. Partially armored and natural shoreline pixels will receive a bump in their score if they fall within a CBRS polygon.
- From the ESI data, extract saline estuarine shorelines, where LINE field is not 'H' and ENVIR field is 'E'.
- The GENERALIZED_ESI_TYPE field contains 1-3 shoreline types, listed landward to seaward. If the field contains only '1: Armored', consider it fully armored and assign it the lowest value (0) in the indicator.
- If GENERALIZED_ESI_TYPE contains multiple shoreline types and any of them are armored, consider it partially armored. Assign these pixels a value of 1 (if they are outside the CBRS) or 2 (if they are inside the CBRS) in the indicator.
- Consider all other GENERALIZED_ESI_TYPE fields as natural shoreline. Assign them values of 3 (if they are outside the CBRS) or 4 (if they are inside the CBRS) in the indicator.
- Convert the entire ESI linework to a 30 m raster.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = Natural and harder to develop
3 = Natural
2 = Partially armored and harder to develop
1 = Partially armored
0 = Armored

Known Issues

- This indicator overestimates shoreline condition in areas near hardened structures. It only predicts low values in places that are currently hardened and not in nearby areas that can also be impacted by that hardening.
- This indicator overestimates shoreline condition in areas with active beach renourishment. Beach renourishment negatively affects some beach and dune species but is not captured by this indicator.
- This indicator does not take into account protected natural areas (e.g., National Wildlife Refuges such as St. Marks in Florida) that are unlikely to be developed but are not within the CBRS.
- In areas where source data is outdated, this indicator sometimes overestimates shoreline condition because it does not capture more recent shoreline alteration (such as the addition of new residential docks and boat slips along the Texas coast). This dataset is a compilation of individual state inputs that vary in age, ranging back to 1996. Some states, like Florida and Texas, maintain their own ESI data separate from the national NOAA ESI, but differences in data structure and formatting made it difficult to combine those with the national ESI data this year. We hope to incorporate more recent state data in future Blueprint updates.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Bruun, P., 1995. The development of downdrift erosion. J. Coast. Res., 1242-1257. [https://www.jstor.org/stable/4298427].

Dugan, J.E., Hubbard, D.M., Rodil, I.F., Revell, D.L., Schroeter, S., 2008. Ecological effects of coastal armoring on sandy beaches. Marine Ecology 29, 160-170. [https://onlinelibrary.wiley.com/doi/full/10.1111/j.1439-0485.2008.00231.x].

Gittman, Rachel K., Steven B. Scyphers, Carter S. Smith, Isabelle P. Neylan, Jonathan H. Grabowski, Ecological Consequences of Shoreline Hardening: A Meta-Analysis, BioScience, Volume 66, Issue 9, 01 September 2016, Pages 763-773. [https://doi.org/10.1093/biosci/biw091].

Gittman, R.K., Fodrie, F.J., Popowich, A.M., Keller, D.A., Bruno, J.F., Currin, C.A., Peterson, C.H. and Piehler, M.F. (2015), Engineering away our natural defenses: an analysis of shoreline hardening in the US. Frontiers in Ecology and the Environment, 13: 301-307. [https://doi.org/10.1890/150065].

Hall, M.J., Pilkey, O.H., 1991. Effects of hard stabilization on dry beach width for New Jersey. J. Coast. Res., 771-785. [https://journals.flvc.org/jcr/article/view/78532/75937].

Mohanty, P.K., Patra, S.K., Bramha, S., Seth, B., Pradhan, U., Behera, B., Mishra, P., Panda, U.S., 2012. Impact of groins on beach morphology: a case study near Gopalpur Port, east coast of India. J. Coast. Res. 28, 132-142. [https://bioone.org/journals/journal-of-coastal-research/volume-28/issue-1/JCOASTRES-D-10-00045.1/Impact-of-Groins-on-Beach-Morphology--A-Case-Study/10.2112/JCOASTRES-D-10-00045.1.full].

National Oceanic and Atmospheric Administration National Centers for Coastal Ocean Science. August 10, 2015. Hardened Shorelines Make it Hard for Submerged Aquatic Vegetation. [https://coastalscience.noaa.gov/news/hardened-shorelines-make-hard-submerged-aquatic-vegetation/].

National Oceanic and Atmospheric Administration National Centers for Coastal Ocean Science. January 23, 2013. Project Finds Fish Prefer Natural Shorelines. [https://coastalscience.noaa.gov/news/riprap-project-finds-fish-prefer-natural-shorelines-human-response-to-sea-level-rise-at-issue/].

National Oceanic and Atmospheric Administration Office of Response and Restoration, Seattle, Washington. May 2017. National Environmental Sensitivity Index Shoreline: GULF/ATLANTIC ESI: ESI Shoreline Types – Lines. [https://response.restoration.noaa.gov/esi_download].

Pilkey, O.H., Wright III, H.L., 1988. Seawalls versus beaches. J. Coast. Res., 41-64. [https://www.jstor.org/stable/25735351].

Rice, T.M., 2012a. Inventory of Habitat Modifications to Tidal Inlets in the Continental U.S. Coastal Migration and Wintering Range of the Piping Plover *(Charadrius melodus)*. Appendix 1b, in Comprehensive Conservation Strategy for the Piping Plover *(Charadrius melodus)* in its Coastal Migration and Wintering Range in the Continental United States. U.S. Fish and Wildlife Service, East Lansing, Michigan.

U.S. Fish and Wildlife Service. March 2019. John H. Chafee Coastal Barrier Resources System Approximate Polygons Vector Digital Dataset. Falls Church, VA. Accessed January 31, 2022. [https://www.fws.gov/program/coastal-barrier-resources-act/maps-and-data].

Estuarine Coastal Condition

This indicator combines measures of water quality, sediment quality, contaminants in fish tissue, and benthic community condition to create an overall index of coastal estuarine condition. It originates from the Environmental Protection Agency's National Coastal Condition Assessment data.

This indicator was last updated in October 2023.

Reason for Selection

This index measures the condition of the nation's estuaries following standard national methodologies and is synthesized by the U.S. Environmental Protection Agency (EPA) roughly every five years. Estuaries serve as important nursery habitat for wildlife, including many species of fish and shellfish eaten as seafood. They also improve water quality by filtering out sediments and pollutants, provide recreational opportunities, and support coastal economies (NOAA 2019).

The estuaries and surrounding coastal habitats of the Southeast coast "serve as important stopovers for migratory birds along the Atlantic Flyway as well as nurseries for fish and other animals. The highly productive ecosystems of Southeast estuarine waters contribute to commercial and recreational fishing. The Southeast Coast provides a wealth of economic and ecosystem services that sustain local economies and quality of life. These services include storm-surge and sea-level protection, maritime transportation and trade, commercial and recreational fisheries, and tourism."

Input Data

- Environmental Protection Agency (EPA) National Coastal Condition Assessment (NCCA): 2010 data and 2015 data; download NCCA data

  The NCCA surveys the estuaries and nearshore waters of the Great Lakes using various indicators of ecological condition and human health risk. We use four ecological indicators (USEPA 2015, 2021):

  1. The water quality/eutrophication index - includes measures of phosphorus, nitrogen, water clarity, chlorophyll a, and dissolved oxygen
  2. The sediment quality index - includes sediment contaminant levels and sediment toxicity to live organisms
  3. The benthic/biological condition index – measures the condition of the community of macroinvertebrates (worms, mollusks and crustaceans) living in the sediment, based on diversity, abundance and sensitivity to pollution
  4. The fish tissue contaminants/ecological fish tissue quality index - measures the concentrations of metals and organic contaminants in fish and estimates potential harm to the wildlife that eat them

  This indicator is scored in the same way as EPA's scoring: 1 = poor, 3 = fair, and 5 = good. The 2015 assessment reported no strong regional trends in coastal condition between 2010 and

2015, so we combined both assessments to increase the density of the point data used to create the interpolation.

- NOAA coastal relief model: Shallow water bathymetry derived by extracting the area with depths from 0-10 m; we converted this spatial extent to a shapefile and used it to constrain the interpolation of the mean estuarine condition index
- Southeast Blueprint 2023 extent

Mapping Steps

- The 2010 and 2015 point data contain separate fields for the water quality index, the sediment quality index, the fish tissue index, and the benthic index. Assign numeric values based on EPA's scoring system. If a point scores poor in these fields, assign it a value of 1. If a point scores fair in these fields, assign it a value of 3. If a point scores good in these fields, assign it a value of 5.
- Add and calculate a field to contain the average of the four indices for all sites sampled in either 2010 or 2015. Note: Sites are drawn randomly so are not usually sampled in both timesteps.
- Import the tabular data into ArcPro and join to the table of EPA sampling locations.
- Interpolate the mean EPA coastal condition index value using Spline with Barriers. To avoid interpolating estuarine scores over land areas, use as the barriers input the shallow water extent from the NOAA coastal relief model. If a shallow water bathymetry polygon does not contain an EPA sampling location, that polygon is classified as 0 in a later step because we don't know anything about its condition. Note: This operation had to be accomplished in two steps (Gulf and Atlantic regions) because of file size limitations.
- Merge the Gulf and Atlantic outputs and clip the resulting raster to the 0-10 m shallow water layer from the NOAA coastal relief model.
- Reclassify the above raster into the 5 classes seen in the final indicator values below:
    - 0.3087-2 = 1 (Note: 0.3087 is the minimum value over the entire raster)
    - 2-2.4 = 2
    - 2.4-3.7 = 3
    - 3.7-4 = 4
    - 4-5 = 5
- Intersect the shallow water bathymetry polygon with the EPA sampling locations. If a shallow water extent polygon does not contain an EPA sampling location, use the Calculate Field function to classify that polygon as 0 because we don't know anything about its condition. Assign a value of 1 to all other polygons that do contain EPA sampling locations.
- Convert the resulting vector layer to raster.
- Multiply this new raster by the mean coastal condition index raster to produce a result that shows the mean coastal condition index where that value is known, and 0 for shallow estuaries lacking an EPA condition score. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Good
4 = Fair to good
3 = Fair
2 = Poor to fair
1 = Poor
0 = Shallow estuary not assessed for condition

Known Issues

- The CCA was designed to support regional interpretations of coastal condition, not site-level interpolation as used in this indicator. As a result, estimates for areas far from NCCA sampling sites are highly uncertain.
- This indicator uses only two snapshots in time in a highly dynamic system.
- There is typically a significant time lag between the year the data is collected, and the year the data is published (allowing time for analysis, quality assurance, compiling the report, etc.). As a result, even the most recent available NCCA data (collected in 2015) is somewhat outdated.
- There is an error in the interpolation off the coast of Pensacola, FL that creates an unusual spatial pattern in the indicator. However, this pattern is not reflected in the final Blueprint as the entire area scores highest priority.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

National Oceanic and Atmospheric Administration (NOAA). Last updated February 1, 2019. Life in an estuary. [https://www.noaa.gov/education/resource-collections/marine-life/life-in-estuary].

U.S. Environmental Protection Agency (EPA). 2012. National Coastal Condition Report IV. Office of Research and Development/Office of Water. Washington D.C. EPA-842-R-10-003. [https://www.epa.gov/national-aquatic-resource-surveys/national-coastal-condition-reports].

U.S. Environmental Protection Agency (EPA). August 2021. National Coastal Condition Assessment: A Collaborative Survey of the Nation's Estuaries and Great Lakes Nearshore Waters (National Coastal Condition Assessment 2015). EPA 841-R-21-001. [https://www.epa.gov/system/files/documents/2021-09/nccareport_final_2021-09-01.pdf].

U.S. Environmental Protection Agency (EPA). Last updated June 3, 2022. Southeast Coast Estuaries: National Coastal Condition Assessment 2015 (Setting). [https://www.epa.gov/national-aquatic-resource-surveys/southeast-coast-estuaries-national-coastal-condition-assessment].

U.S. Environmental Protection Agency (EPA). Office of Water and Office of Research and Development. (2015). National Coastal Condition Assessment 2010 (EPA 841-R-15-006). Washington, DC. December 2015.

Gulf Coral & Hardbottom

This indicator predicts the presence of coral and hardbottom in the Gulf of America based on direct observations, acoustic surveys, and known locations of artificial reefs and shipwrecks. It combines data from multiple sources, including Bureau of Ocean Energy Management seismic water bottom anomalies, usSEABED sediment data, several National Oceanic and Atmospheric Administration datasets, various state layers, and more.

This indicator was last updated in October 2024.

Reason for Selection

Hardbottom provides an anchor for important seafloor habitats such as deep-sea corals, plants, and sponges. Hardbottom is also sometimes associated with chemosynthetic communities that form around cold seeps or hydrothermal vents. In these unique ecosystems, micro-organisms that convert chemicals into energy form the base of complex food webs (Love et al. 2013). Hardbottom and associated species provide important habitat structure for many fish and invertebrates (NOAA 2018). Hardbottom areas serve as fish nursery, spawning, and foraging grounds, supporting commercially valuable fisheries like snapper and grouper (NCDEQ 2016).

According to Dunn and Halpin (2009), “hardbottom habitats support high levels of biodiversity and are frequently used as a surrogate for it in marine spatial planning.” Artificial reefs are also known to provide additional habitat that is quickly colonized to provide a suite of ecosystem services commonly associated with naturally occurring hardbottom (Wu et al. 2019). We did not include active oil and gas structures as human-created hardbottom. Although they provide habitat, because of their temporary nature, risk of contamination, and contributions to climate change, they do not have the same level of conservation value as other artificial structures.

Input Data

- Southeast Blueprint 2024 extent
- Southeast Blueprint 2024 subregions

Coral & hardbottom

- usSEABED Gulf sediments, accessed 12-14-2023; download the data; view and read more about the data on the National Oceanic and Atmospheric Administration (NOAA) Gulf Data Atlas (select Physical --> Marine geology --> 1. Dominant bottom types and habitats)

- Bureau of Ocean Energy Management (BOEM) Gulf seismic water bottom anomalies, accessed 12-20-2023
- The Nature Conservancy's (TNC) South Atlantic Bight Marine Assessment (SABMA); chapter 3 of the final report provides more detail on the seafloor habitats analysis
- NOAA deep-sea coral and sponge locations, accessed 12-20-2023 on the NOAA Deep-Sea Coral & Sponge Map Portal
- Florida coral and hardbottom habitats, accessed 12-19-2023

Shipwrecks & artificial reefs

- NOAA wrecks and obstructions layer, accessed 12-12-2023 on the Marine Cadastre
- Louisiana Department of Wildlife and Fisheries (LDWF) Artificial Reefs: Inshore Artificial Reefs, Nearshore Artificial Reefs, Offshore and Deepwater Artificial Reefs (Google Earth/KML files), accessed 12-19-2023
- Texas Parks and Wildlife Department (TPWD) Artificial Reefs, accessed 12-19-2023; download the data from The Artificial Reefs Interactive Mapping Application (direct download from interactive mapping application)
- Mississippi Department of Marine Resources (MDMR) Artificial Reef Bureau: Inshore Reefs, Offshore Reefs, Rigs to Reef (lat/long coordinates), accessed 12-19-2023
- Alabama Department of Conservation and Natural Resources (ADCNR) Artificial Reefs: Master Alabama Public Reefs v2023 (.xls), accessed 12-19-2023
- Florida Fish and Wildlife Conservation Commission (FWC): Artificial Reefs in Florida (.xlsx), accessed 12-19-2023

Defining inland extent & split with Atlantic

- Marine Ecoregions Level III from the Commission for Environmental Cooperation North American Environmental Atlas, accessed 12-8-2021
- 2023 NOAA coastal relief model: volumes 2 (Southeast Atlantic), 3 (Florida and East Gulf of Mexico), 4 (Central Gulf of Mexico), and 5 (Western Gulf of Mexico), accessed 3-27-2024
- National Oceanic and Atmospheric Administration (NOAA) Characterizing Spatial Distributions of Deep-sea Corals and Hardbottom Habitats in the U.S. Southeast Atlantic; read the final report; data shared prior to official release on 2-4-2022 by Matt Poti with the NOAA National Centers for Coastal Ocean Science (NCCOS) (matthew.poti@noaa.gov)
- Predictive Modeling and Mapping of Hardbottom Seafloor Habitats off the Southeast U.S: unpublished NOAA data and draft final report entitled *Assessment of Benthic Habitats for Fisheries Management* provided on 1-28-2021 by Matt Poti with NOAA NCCOS (matthew.poti@noaa.gov)

Mapping Steps

Note: Most of the mapping steps were accomplished using the graphical modeler in QGIS 3.34. Individual models were created to combine data sources and assign ranked values. These models were combined in a single model to assemble all the data sources and create a summary raster.

- Create a seamless vector layer to constrain the extent of the Atlantic coral and hardbottom indicator to marine and estuarine areas <1 m in elevation. This defines how far inland it extends.
    - Merge together all coastal relief model rasters (.nc format) using the create virtual raster tool in QGIS.
    - Save the merged raster to .tif format and import it into ArcPro.
    - Reclassify the NOAA coastal relief model data to assign a value of 1 to areas from deep marine to 1 m elevation. Assign all other areas (land) a value of 0.
    - Convert the raster produced above to vector using the raster to polygon tool.
    - Clip to the 2024 Blueprint subregions using the pairwise clip tool.
    - Hand-edit to remove terrestrial polygons (one large terrestrial polygon and the Delmarva peninsula).
    - Dissolve the resulting data layer to produce a seamless polygon defining marine and estuarine areas <1 m in elevation.
    - Hand-edit to select all but the main marine polygon and delete.
- Define the extent of the Gulf version of this indicator to separate it from the Atlantic. This split reflects the extent of the different datasets available to represent coral and hardbottom habitat in the Atlantic and Gulf, rather than a meaningful ecological transition.
    - Use the select tool to select the Florida Keys class from the Level III marine ecoregions ("NAME_L3 = 'Florida Keys'").
    - Buffer the "Florida Keys" Level III marine ecoregion by 2 km to extend it far enough inland to intersect the inland edge of the <1 m elevation layer.
    - Reclassify the two NOAA Atlantic hardbottom suitability datasets to give all non-NoData pixels a value of 0. Combine the reclassified hardbottom suitability datasets to define the total extent of these data. Convert the raster extent to vector and dissolve to create a polygon representing the extent of both NOAA hardbottom datasets.
    - Union the buffered ecoregion with the combined NOAA extent polygon created above. Add a field and use it to dissolve the unioned polygons into one polygon. This leaves some holes inside the polygon, so use the eliminate polygon part tool to fill in those holes, then convert the polygon to a line.
    - Hand-edit to extract the resulting line between the Gulf and Atlantic.
    - Hand-edit to use this line to split the <1 m elevation layer created earlier in the mapping steps to create the separation between the Gulf and Atlantic extent.
- From the BOEM seismic water bottom anomaly data, extract the following shapefiles: anomaly_confirmed_relic_patchreefs.shp, anomaly_Cretaceous.shp, anomaly_relic_patchreefs.shp, seep_anomaly_confirmed_buried_carbonate.shp, seep_anomaly_confirmed_carbonate.shp, seep_anomaly_confirmed_organisms.shp, seep_anomaly_positives.shp, seep_anomaly_positives_confirmed_gas.shp, seep_anomaly_positives_confirmed_oil.shp, seep_anomaly_positives_possible_oil.shp, seep_anomaly_confirmed_corals.shp, seep_anomaly_confirmed_hydrate.shp.
- To create a class of confirmed BOEM features, merge anomaly_confirmed_relic_patchreefs.shp, seep_anomaly_confirmed_organisms.shp,

seep_anomaly_confirmed_corals.shp, and seep_anomaly_confirmed_hydrate.shp and assign a value of 6.

- To create a class of predicted BOEM features, merge the remaining extracted shapefiles and assign a value of 3.
- From usSEABED sediments data, use the field "gom_domnc" to extract polygons: rock (dominant and subdominant) receives a value of 2 and gravel (dominant and subdominant) receives a value of 1.
- From the wrecks database, extract locations having "high" and "medium" confidence (positionQuality = "high" and positionQuality = "medium"). Buffer these locations by 150 m and assign a value of 4. The buffer distance used here, and later for coral locations, follows guidance from the Army Corps of Engineers for setbacks around artificial reefs and fish havens (Riley et al. 2021).
- Merge artificial reef point locations from FL, AL, MS and TX. Buffer these locations by 150 m. Merge this file with the three LA artificial reef polygons and assign a value of 5.
- From the NOAA deep-sea coral and sponge point locations, select all points. Buffer the point locations by 150 m and assign a value of 7.
- From the FWC coral and hardbottom dataset polygon locations, fix geometries, reproject to EPSG=5070, then assign coral reefs a value of 7, hardbottom a value of 6, hardbottom with seagrass a value of 6, and probable hardbottom a value of 3. Hand-edit to remove an erroneous hardbottom polygon off of Matagorda Island, TX, resulting from a mistake by Sheridan and Caldwell (2002) when they digitized a DOI sediment map. This error is documented on page 6 of the Gulf of Mexico Fishery Management Council's 5-Year Review of the Final Generic Amendment Number 3.
- From the TNC SABMA data, fix geometries and reproject to EPSG=5070, then select all polygons with TEXT_DESC = '01. mapped hard bottom area' and assign a value of 6.
- Union all of the above vector datasets together—except the vector for class 6 that combines the SABMA and FL data—and assign final indicator values. Class 6 had to be handled separately due to some unexpected GIS processing issues. For overlapping polygons, this value will represent the maximum value at a given location.
- Clip the unioned polygon dataset to the buffered marine subregions.
- Convert both the unioned polygon dataset and the separate vector layer for class 6 using GDAL "rasterize".
- Fill NoData cells in both rasters with zeroes and, using Extract by Mask, mask the resulting raster with the Gulf indicator extent. Adding zero values helps users better understand the extent of this indicator and to make this indicator layer perform better in online tools.
- Use the raster calculator to evaluate the maximum value among the two rasters described above (one for class 6 and one for all other classes).
- Export the final dataset as an unsigned 8-bit integer raster to ArcPro. Create the raster attribute table using the legend values below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2024.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Confirmed hardbottom-associated species (e.g., corals, sponges, patch reef, chemosynthetic communities)
6 = Confirmed natural hardbottom
5 = Artificial reefs
4 = Shipwrecks
3 = Probable natural hardbottom (fine resolution)
2 = Rock (coarse resolution)
1 = Gravel (coarse resolution)
0 = Not identified as hardbottom

Known Issues

- While active oil and gas platforms also provide artificial hardbottom habitat that supports diverse marine life (Claisse et al. 2014), we did not include active drilling sites in this version of the indicator. This decision reflects an attempt to balance the positive structural habitat value of oil and gas platforms with the negative impacts of climate change caused by fossil fuel combustion (e.g., coral bleaching and ocean acidification) (NOAA 2023) and the negative impacts of a potential oil spill (Daley 2019). We intend to further investigate these tradeoffs in future revisions.
- The shipwrecks class does not account for variation in the condition of submerged structures, such as “black reef” phenomena caused by pollution and invasive species growth around contaminated shipwrecks (Degnarain 2020).
- This indicator may underprioritize hardbottom and coral habitat in the eastern Gulf of America due to limited data availability. According to the Florida State University Marine Lab, “Most deep-sea research in the Gulf of Mexico has targeted areas important to the oil and gas industry...This research has resulted in significant contributions to the discovery and in some cases the protection of hard bottom and deep-sea reefs. In contrast, the eastern Gulf of Mexico remains relatively unexplored. In fact, it is perhaps the least explored region of the Gulf. Therefore, there is a limited amount of data on their distributions...” This issue particularly impacts the confirmed hardbottom-associated species and predicted fine-resolution hardbottom classes because the BOEM seismic anomalies dataset is limited to the BOEM central and western planning areas in the Gulf of America. As a result, this dataset does not cover state and federal waters around Florida, or other state waters across the Gulf.
- This indicator underrepresents coral and hardbottom habitat within three Gulf deepwater coral Habitat Areas of Particular Concern (Steamboat Lumps, The Edges, Madison Swanson) where finer scale data on coral and hardbottom is lacking. We hope to improve coverage within these HAPCs in future updates.

- While this layer has a 30 m resolution, the usSEABED dataset is coarser than that. We downsampled 2 km grid cells to 30 m.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Alabama Department of Conservation and Natural Resources. Artificial Reefs 2023. Microsoft Excel file. Accessed December 19, 2023. [https://www.outdooralabama.com/saltwater-fishing/artificial-reefs].

Claisse JT, Pondella DJ, Love M, Zahn LA, Williams CM, Williams JP, Bull AS (2014) Oil platforms off California are among the most productive marine fish habitats globally. Proc Natl Acad Sci USA 111: 15462−15467 [https://www.pnas.org/doi/10.1073/pnas.1411477111].

Commission for Environmental Cooperation. 2009. Marine Ecoregions of North America, 2008. Vector digital data. Montréal, Québec, Canada. [https://www.cec.org/north-american-environmental-atlas/marine-ecoregions/].

Conley, M.F., M.G. Anderson, N. Steinberg, and A. Barnett, eds. 2017. The South Atlantic Bight Marine Assessment: Species, Habitats and Ecosystems. The Nature Conservancy, Eastern Conservation Science.
[https://easterndivision.s3.amazonaws.com/Marine/SABMA/SABMA_Report_11April2018.pdf].

Daley, Jason. Deepwater Horizon Site is Now a Sticky Wasteland Populated by Sickly Crabs. September 19, 2019. Smithsonian Magazine. [https://www.smithsonianmag.com/smart-news/deepwater-horizon-site-wasteland-populated-sickly-crabs-180973181/].

Deep Sea Coral Research and Technology Program (DSCRTP) (2016). Observations of Deep-Sea Coral and Sponge Occurrences from the NOAA National Deep-Sea Coral and Sponge Database, 1842-Present, version 20230620-0 (NCEI Accession 0145037). NOAA National Centers for Environmental Information. Dataset. [https://www.ncei.noaa.gov/archive/accession/0145037]. Accessed December 20, 2023.

Degnarain, Nashan. Black Reef Risk: How Sunken Metal Shipwrecks Attract Coral-Destroying Invasive Species. August 20, 2020. Forbes. [https://www.forbes.com/sites/nishandegnarain/2020/08/20/black-reef-risk-how-sunken-metal-shipwrecks-attract-coral-destroying-invasive-species/?sh=55ae49c4129d].

Dunn DC, Halpin PN (2009) Rugosity-based regional modeling of hard-bottom habitat. Marine Ecology Progress Series 377:1-11. [https://doi.org/10.3354/meps07839].

Florida Fish and Wildlife Conservation Commission-Fish and Wildlife Research Institute. Coral and Hard Bottom Habitats Florida. Vector digital data. Published March 12, 2015. Accessed December 19, 2023. [https://geodata.myfwc.com/datasets/myfwc::coral-and-hard-bottom-habitats-in-florida/about].

Florida Fish and Wildlife Conservation Commission. Artificial Reefs: Download Reef Locations. Microsoft Excel file. Accessed December 19, 2023. [https://myfwc.com/fishing/saltwater/artificial-reefs/locate/].

Florida State University Marine Lab. Coleman and Koenig Research: Hard Bottom Reefs of the West Florida Shelf. "Oil Platforms in the Gulf of Mexico." Accessed March 20, 2024. [https://marinelab.fsu.edu/labs/coleman/research/shelf-edge/].

Jenkins C. Dominant Bottom Types and Habitats In Gulf of Mexico Data Atlas [Internet]. Stennis Space Center (MS): National Centers for Environmental Information; 2011. [5 screens]. [https://gulfatlas.noaa.gov].

Louisiana Department of Wildlife and Fisheries. Artificial Reefs: Inshore Artificial Reefs, Nearshore Artificial Reefs, Offshore and Deepwater Artificial Reefs. Google Earth/KML files. Accessed December 19, 2023. [https://www.wlf.louisiana.gov/page/artificial-reefs].

Love, M., Baldera, A., Yeung, C. & Robbins C. (2013). The Gulf of Mexico Ecosystem: A Coastal and Marine Atlas. New Orleans, LA: Ocean Conservancy, Gulf Restoration Center. [https://oceanconservancy.org/wp-content/uploads/2017/05/gulf-atlas.pdf].

Mississippi Department of Marine Resources. Artificial Reef Bureau. Inshore Reefs, Offshore Reefs, Rigs to Reef. Lat/long coordinates. Accessed December 19, 2023. [https://dmr.ms.gov/artificial-reef/].

National Oceanographic and Atmospheric Administration. Deep Sea Coral Research and Technology Program 2018 Report to Congress. December 2018. [https://coast.noaa.gov/coral_storage/library/NOAA/DSCRTP/Other/Reports_To_Congress/2018/DSCRTP2018_Report_to_Congress.pdf].

National Oceanographic and Atmospheric Administration. How does climate change affect coral reefs? National Ocean Service website. Last updated January 20, 2023. Accessed August 9, 2023. [https://oceanservice.noaa.gov/facts/coralreef-climate.html].

National Oceanographic and Atmospheric Administration. Unpublished Draft: Assessment of Benthic Habitats for Fisheries Management. May 2015. [https://secassoutheast.org/pdf/DRAFT_Benthic-Habitats-for-Fisheries-Management-Final-Report-April-2015-with-MP-additions.pdf].

NCDEQ (North Carolina Department of Environmental Quality) 2016. North Carolina Coastal Habitat Protection Plan Source Document. Morehead City, NC. Division of Marine Fisheries. 475 p. [https://deq.nc.gov/media/26813/open].

NOAA National Centers for Environmental Information. (2023). Coastal Relief Models (CRMs) [Data set]. NOAA National Centers for Environmental Information. [doi: 10.25921/5ZN5-KN44].

Office for Coastal Management, 2023: Wrecks and Obstructions. NOAA National Centers for Environmental Information. [https://www.fisheries.noaa.gov/inport/item/70439].

Poti M, Goyert HF, Salgado EJ, Bassett R, Coyne M, Winship AJ, Etnoyer PJ, Hourigan TF, Coleman HM, Christensen J. 2022. Data synthesis and predictive modeling of deep-sea coral and hardbottom habitats offshore of the southeastern US: guiding efficient discovery and protection of sensitive benthic areas. New Orleans (LA): US Department of the Interior, Bureau of Ocean Energy Management. 224 p. Contract No.: M16PG00010. Report No.: OCS Study BOEM 2022- 038. [https://espis.boem.gov/final%20reports/BOEM_2022-038.pdf].

Riley, K.L., Wickliffe, L.C., Jossart, J.A., MacKay, J.K., Randall, A.L., Bath, G.E., Balling, M.B., Jensen, B.M., and Morris, J.A. Jr. 2021. An Aquaculture Opportunity Area Atlas for the U.S. Gulf of Mexico. NOAA Technical Memorandum NOS NCCOS 299. Beaufort, NC. 545 pp. [https://doi.org/10.25923/8cb3-3r66].

Texas Parks and Wildlife Department. The Artificial Reefs Interactive Mapping Application. Accessed December 19, 2023. [https://tpwd.maps.arcgis.com/apps/webappviewer/index.html?id=644a45d5b75b4881b1fbb6c42bdd09d4].

Wu, Zhongin, James R. Tweedley, Neil R. Loneragan, Xiumei Zhang. 2019. Artificial reefs can mimic natural habitats for fish and macroinvertebrates in temperate coastal waters of the Yellow Sea. Ecological Engineering, vol 139. [https://doi.org/10.1016/j.ecoleng.2019.08.009].

Gulf Deep-Sea Coral Richness

This indicator measures the number of different deep-sea coral genera predicted to occur in the Gulf of America at depths of 50 m or below. It is based on coral observations and a suite of environmental predictors including measures of depth, seafloor topography and substrate, oceanography, and geography. This indicator combines probability models for 28 deep-sea coral genera to predict overall richness. It originates from a National Oceanic and Atmospheric Administration project characterizing the spatial distributions of deep-sea corals and chemosynthetic communities in the U.S. Gulf.

This indicator was last updated in October 2023.

Reason for Selection

Many deep-sea corals form tree-like shapes and complex reefs that provide valuable three-dimensional habitat structure for many fish and invertebrate species. They support commercially important fisheries such as grouper, snapper, sea bass, rockfish, shrimp, and crab. The presence of more diverse types of coral typically creates more complex habitats that support more species. Because most deep-sea corals grow very slowly, they are highly vulnerable to damage from trawling and energy development, as well as ocean acidification due to climate change (NOAA 2018).

Input Data

- Southeast Blueprint 2023 subregions: Marine (combined Atlantic & Gulf)
- Southeast Blueprint 2023 extent
- National Oceanic and Atmospheric Administration (NOAA) Characterizing Spatial Distributions of Deep-sea Corals and Chemosynthetic Communities in the U.S. Gulf; read the final report; data shared prior to official release on 2-1-2023 by Matt Poti with the NOAA National Centers for Coastal Ocean Science (matthew.poti@noaa.gov).

  This dataset provides probability models for 28 deep-sea coral genera *(Acanella, Antipathes, Bathypathes, Bebryce, Caliacis, Callogorgia, Cheliodonisis, Enallopsammia, Hypnogorgia, Keratoisis, Leiopathes, Lepidisis, Lophelia, Madracis, Madrepora, Nicella, Oculina, Paramuricea, Placogorgia, Plumapathes, Scleracis, Solenosmilia, Stauropathes, Stichopathes, Swiftia, Tanacetipathes, Thesea, Villogorgia)*, as well as a combined genus richness layer that counts the average number of unique genera predicted to occur in each 100 m pixel. Matt Poti provided a summarized vector version of the continuous genus richness raster layer that mirrors the way the Bureau of Safety and Environmental Enforcement displays these data. This classification is based on natural breaks in the data distribution and uses 10 bins. We further collapsed the bins from 10 to 5 for simplicity.

Mapping Steps

- Reclassify to collapse the 10 classes in the provided genus richness shapefiles into 5 by combining the “0-0.5” and “0.5-1” classes and assigning a value of 1, combining the “1-2” and “2-3” classes and assigning a value of 2, combining the “3-4” and “4-5” classes and assigning a value of 3, combining the “5-6” and “6-7” classes and assigning a value of 4, and combining the “7-8” and “>8” classes and assigning a value of 5.
- Convert the provided genus richness shapefile from vector to raster.
- Clip to the Southeast Blueprint 2023 marine subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Highest predicted average genus richness (>7)
4 = High predicted average genus richness (>5-7)
3 = Medium predicted average genus richness (>3-5)
2 = Low predicted average genus richness (>1-3)
1 = Lowest predicted average genus richness (0-1)

Known Issues

- While this layer has a 30 m resolution, the NOAA deep-sea coral models were coarser than that. We downsampled 100 m pixels to 30 m.
- Areas with low survey effort for the variables used to predict deep-sea corals are underprioritized. This is especially true for seafloor geology, curvature, and aspect. Sharp linear features and shapes in this indicator, when not aligned with a shelf break, are a symptom of places with low survey effort for these variables.

Other Things to Keep in Mind

- Atlantic and Gulf deep-sea coral richness are intended to serve as complementary indicators based on similar NOAA source data. Because of the different deep-sea coral communities present in the Atlantic and the Gulf, the data provider recommended different thresholds for what level of genus richness qualifies as highest, high, medium, etc. to ensure the Blueprint captures the most important deep-sea coral areas within each region.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Goyert HF, Bassett R, Christensen J, Coleman, H, Coyne M, Etnoyer PJ, Frometa J, Hourigan, TF, Poti M, Salgado EJ, Williams, B, Winship AJ. 2020. Characterizing spatial distributions of deep-sea corals and chemosynthetic communities in the US Gulf of Mexico through data synthesis and predictive modeling. New Orleans (LA): US Department of the Interior, Bureau of Ocean Energy Management. OCS Study BOEM 2021-027. 317 p. [https://espis.boem.gov/final%20reports/BOEM_2021-027.pdf].

Komyakova V, Jones GP, Munday PL. Strong effects of coral species on the diversity and structure of reef fish communities: A multi-scale analysis. PLoS One. 2018 Aug 13;13(8):e0202206. [https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0202206].

National Oceanographic and Atmospheric Administration. Deep Sea Coral Research and Technology Program 2018 Report to Congress. December 2018. [https://coast.noaa.gov/coral_storage/library/NOAA/DSCRTP/Other/Reports_To_Congress/2018/DSCRTP2018_Report_to_Congress.pdf].

Gulf Marine Birds

This indicator identifies important areas in the Gulf of America for birds that feed exclusively or mainly at sea. It uses habitat suitability models for 24 species of marine birds (black-capped petrel, black tern, Bonaparte's gull, brown booby, brown noddy, brown pelican, bridled tern, band-rumped storm-petrel, Cory's shearwater, common loon, common tern, great shearwater, herring gull, laughing gull, masked booby, magnificent frigatebird, northern gannet, parasitic jaeger, royal tern, sandwich tern, sargasso shearwater, sooty tern, Wilson's storm-petrel, and pomarine jaeger) based on sightings from vessel-based and aerial surveys. This indicator originates from the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) seabird models.

This indicator was last updated in October 2026.

Reason for Selection

Marine birds help identify key areas of ocean productivity and are an indicator of overall ocean health. According to Parsons et al. (2008), seabirds are often considered useful ecological indicators for the marine environment because they "essentially represent the top of the food chain," making them more likely to signal changes in lower trophic levels and the surrounding environment. Seabird populations also respond to anthropogenic pressures such as fisheries bycatch, "overexploitation of their food resources, pollution from marine debris, and industrial discharge (Parsons et al. 2008, NABCI 2025)." Because seabirds move across vast areas, they often span or cross multiple jurisdictions in a relatively short time period, making them species of shared interest across federal, state, and international entities (Jodice 2019). Despite the abundance of seabird species present in the Gulf, data gaps in species composition and distribution still exist (Gleason et al. 2025), which this indicator seeks to address.

This indicator complements the Gulf marine mammals indicator by using other data from the same GoMMAPPS initiative to represent stronger connections to forage fish productivity.

Input Data

- Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) seabird models for 24 species: download the data (accessed 2-18-2026); read the BOEM GoMMAPPS final report describing survey methods; read the technical report describing modeling methods

  The GoMMAPPS seabird project was a collaborative effort between numerous federal agencies and research institutions. Its goal was to collect broad-scale information on the distribution and abundance of 24 seabird species in the Gulf to inform both spatially explicit density estimates by season. The program had two main components: 1) aerial and vessel-based

surveys collecting data characterizing the distribution, abundance, and diversity of birds in the northern Gulf from 2017-2023, and 2) Model-based analyses of these and other data to interpret the influences of natural and human-associated variables on seabird populations (including MaxEnt models for the 24 bird species, which were used in this indicator).

- Atlantic marine birds (Southeast Blueprint indicator)
- Gulf marine mammals (Southeast Blueprint indicator)
- Southeast Blueprint 2026 extent

Mapping Steps

- Sum all individual species rasters to generate a count of bird species predicted in each 4.67 km pixel, then convert to integer. Converting to integer reduces file size and overall processing time.
- Resample to a 30 m resolution and snap to the Southeast Blueprint extent.
- Clip to the Blueprint extent to remove areas outside of U.S. waters.
- Species predictions extend into the Atlantic Ocean and cover portions of the Florida Keys that overlap with the Blueprint's Atlantic marine birds indicator. Remove areas that overlap with the Atlantic marine birds indicator.
- After removing overlap areas with the Atlantic marine bird indicator, there are still a large number of nearshore pixels in the Atlantic and Gulf that are not included in the Gulf marine mammals indicator. That Gulf marine mammals indicator is also based on GoMMAPPS data and explicitly removed nearshore areas where the models don't perform well. For consistency, remove from the Gulf marine birds indicator any pixels that weren't also in the Gulf marine mammals indicator. This removes nearshore pixels where the models don't perform well and remaining pixels in the Atlantic where the models were not intended to be used.
- To stay consistent with how these data are used in the Bureau of Safety and Environmental Enforcement (BSEE) Environmental Sensitivity Index and other applications of the data, reclassify the results into three quantiles to represent low, medium, and high categories based on species richness. Those are the final indicator categories as seen below.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = High marine bird richness (9-16 of 24 modeled species)
2 = Medium marine bird richness (6-8 of 24 modeled species)
1 = Low marine bird richness (0-5 of 24 modeled species)

Known Issues

- While this layer has a 30 m resolution, the source data was coarser than that. We down-sampled 4.67 km pixels to 30 m.
- This indicator does not capture fine resolution patterns nearshore. Model predictions are fairly coarse and do not capture finer variations in relative abundance nearshore and near estuaries.
- Since the models only used three years of data, it does not include annual variation in species composition and abundance patterns that didn't occur during those years.
- The models of individual species were not scaled to each other. Each species was treated the same and not weighted by protected status, disturbance sensitivity, or number of observations.
- Although surveys occurred during all four seasons, survey effort was not uniformly distributed across each season. Aerial surveys tended to occur in April-October and only one vessel-based survey effort took place during a winter month.
- None of the variables used in the models were available from the ship, so environmental data used in analyses were aggregated from existing products integrating satellite and in-situ data used in an oceanographic model.

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There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

BSEE-USCG Offshore Information for Area Contingency Planning, Offshore Environmental Sensitivity Index (ESI) Atlas for the GOM, Technical Document #6. [https://www.bsee.gov/sites/bsee.gov/files/2023-08/TD6_OffshoreESIAtlas_14June2023.pdf].

Gleason. J.S, A.L. Sussman, K.L. Davis, J.C. Haney, K.M. Hixon, P.G.R. Jodice, J.E. Lyons, P.E. Michael, Y. G. Satge, E.D. Silverman, E.F. Zipkin, and R.R. Wilson. 2025. Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS): seabird surveys in the northern Gulf of America, 2017-2020. New Orleans (LA): US Department of the Interior, Bureau of Ocean Energy Management. 301p. Obligation No.: M17PG00011. Report No.: BOEM 2025-026. [https://espis.boem.gov/Final%20Reports/BOEM_2025-026.pdf].

Gleason, J.S., R.R. Wilson, P.G.R. Jodice, Y.G. Satgé, P E. Michael, K.M. Hixson, A.L. Sussman, and J.C. Haney. 2022. Seabird visual surveys using line-transect methods collected from NOAA vessels in the northern Gulf of Mexico for the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) project from 2017-07-21 to 2019-09-25 (NCEI Accession 0247206). U.S. Department of the Interior, Bureau of Ocean Energy Management. NOAA National Centers for Environmental Information. Unpublished Dataset. Accessed 04-13-2026. [https://doi.org/10.25921/afrq-h385].

Jodice, P. G. R., E.M. Adams, J. Lamb, Y. Satge, and J.S. Gleason. 2019. GoMAMN Strategic Bird Monitoring Guidelines: Seabirds. Pages 129-170 in R.R Wilson, A.M.V. Fournier, J.S. Gleason, J.E. Lyons, and M.S. Woodrey (Editors), Strategic Bird Monitoring Guidelines for the Northern Gulf of Mexico. Mississippi Agricultural and Forestry Experiment Station Research Bulletin 1228, Mississippi State University. 324 pp. [https://gomamn.org/wp-content/uploads/2020/02/chapter6-1.pdf].

North American Bird Conservation Initiative. 2025. The State of the Birds, United States of America, 2025. [https://stateofthebirds.org].

Parsons, Matt, Ian Mitchell, Adam Butler, Norman Ratcliffe, Morten Frederiksen, Simon Foster, James B. Reid, Seabirds as indicators of the marine environment, ICES Journal of Marine Science, Volume 65, Issue 8, November 2008, Pages 1520–1526. [https://doi.org/10.1093/icesjms/fsn155].

Wilson, R.R, J.S. Gleason, J.E. Lyons, E.D. Silverman, A.L. Sussman, E.F. Zipkin, and K.L. Davis. 2022. Seabird visual surveys using line-transect methods collected from USFWS aircraft in the Gulf of Mexico for the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) project from 2018-01-31 to 2020-02-12 (NCEI Accession 0247205). [https://doi.org/10.25921/vyg0-tv44].

Gulf Marine Mammals

This indicator identifies important areas in the Gulf of America for dolphins and whales. It incorporates monthly density predictions for 13 marine mammal species or species groups (Atlantic spotted dolphin, beaked whales, blackfish [which includes killer whale, melon-headed whale, false killer whale, pygmy killer whale], bottlenose dolphin, Bryde's whale, clymene dolphin, pantropical spotted dolphin, pilot whales, pygmy/dwarf sperm whales, Rice's whale, Risso's dolphin, sperm whale, spinner dolphin) based on sightings from boat-based and aerial surveys and data on oceanographic conditions. It uses marine mammal models developed by the National Oceanic and Atmospheric Administration as part of the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS).

This indicator was last updated in October 2023.

Reason for Selection

Marine mammals help identify key areas of ocean productivity and overall ocean health in the Gulf of America, are regularly monitored, and resonate with a variety of audiences. Marine mammals are often used as ocean health indicators due to their long lifespans, feeding at a high trophic levels, and large blubber stores that can serve as repositories for anthropogenic chemicals and toxins (Bossart 2011).

Input Data

- Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS): GoMMAPPS marine mammal spatial density model outputs (version 2.2), accessed 12-14-2022

Based on ship-based and aerial line-transect surveys conducted in the U.S. waters of the Gulf of America between 2003 and 2019, the National Oceanic and Atmospheric Administration (NOAA) Southeast Fisheries Science Center developed spatial density models (SDMs) for cetacean and sea turtle species for the entire Gulf of America. SDMs were developed using a generalized additive modeling framework to determine the relationship between species abundance and environmental variables (monthly averaged oceanographic conditions during 2015-2019).

- 2019 National Land Cover Database (NLCD): Land cover
- Southeast Blueprint 2023 subregions: Marine (combined Atlantic & Gulf)
- Southeast Blueprint 2023 extent

Mapping Steps

- Replace all values of -9999 with 0.
- Convert to monthly rasters for each species/species group using the following fields: “Jan_n”, “Feb_n”, “Mar_n”, “Apr_n”, “May_n”, “Jun_n”, “Jul_n”, “Aug_n”, “Sep_n”, “Oct_n”, “Nov_n”, and “Dec_n”. The pygmy/dwarf sperm whale model only includes data for April through November. The rest of the species/species group models include data for all months. Use the Southeast Blueprint 2023 marine subregion for pixel size and snap.
- Use the beaked whale data and the NLCD to create a mask to define the extent of the Zonation analysis. The beaked whale data represents the full sample area for the other species in GoMMAPPS. The area covered by the marine mammal models overlaps with land in a few areas. This mask removes from the analysis all landcover classes that are not open water (not a value of 11 in the NLCD) within the extent of the NLCD. The resulting Zonation mask covers open water areas where there are both modeled data for marine mammals and NLCD data to remove land.
- To identify important areas for each species, use the core area algorithm (CAZMAX) in Zonation 5. First, include all species but do separate runs by season: Spring (Mar, Apr, May), Summer (Jun, Jul, Aug), Fall (Sep, Oct, Nov), and Winter (Dec, Jan, Feb). We did this for two reasons: 1) Some conservation decisions in the marine environmental are seasonal and 2) Computational limitations prevented a full run for all species and months. Then do a final Zonation run with the seasonal results as inputs. This creates a single layer that accounts for different species and monthly variation while also producing seasonal intermediary products to help with specific marine decisions.
- Reproject the Zonation results to Albers Equal Area.
- Convert from a floating point raster with a range of 0-1 to an integer raster ranging from 0-100.
- Reclassify to values of 1-10 based on increments of 10 to produce the indicator values seen below.
- Use the NLCD and the modeling extent of the source data to identify areas of land not used in the analysis and assign those pixels a value of 0, since they are outside the scope of this marine indicator.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

10 = >90th percentile of importance for marine mammal index species (across larger analysis area)
9 = >80th-90th percentile of importance
8 = >70th-80th percentile of importance
7 = >60th-70th percentile of importance
6 = >50th-60th percentile of importance
5 = >40th-50th percentile of importance
4 = >30th-40th percentile of importance
3 = >20th-30th percentile of importance
2 = >10th-20th percentile of importance
1 = ≤10th percentile of importance
0 = Land

Known Issues

- While this layer has a 30 m resolution, the source data was coarser than that. We downsampled hexagons with an area of 40 km^2 to 30 m pixels.
- The indicator may underrepresent striped dolphin habitat. While model results are available for this species, there were processing issues that prevented the use of models for this species in the indicator.
- Some important areas for Rice's whale may be underprioritized. The Blueprint currently covers habitat predicted by the GoMMAPPS model for Rice's whale, but NOAA core distribution polygons cover a much larger area.

Other Things to Keep in Mind

- We ran the Zonation analysis across open water areas where there were both marine mammal models and NLCD data present to discriminate between land and water. We did this for multiple reasons. We didn't run Zonation across the full area covered by the GoMMAPPS data because the full files were very large and required long processing times. We also anticipated that Zonation would not have been able to computationally handle the full area. We extended the Zonation run beyond U.S. waters to try to account for areas of high mammal density just south of the Blueprint's Gulf marine boundary. As a result, the various classes within the indicator do not cover equal areas within the indicator's extent, as you might expect with a percentile-based indicator—they cover equal areas within the full analysis area, and then are clipped down to produce the indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Bossart, G. D. "Marine Mammals as Sentinel Species for Oceans and Human Health." Veterinary Pathology Online 48, no. 3 (May 1, 2011): 676–90. [https://doi.org/10.1177/0300985810388525].

Litz J, Aichinger Dias L, Rappucci G, Martinez A, Soldevilla M, Garrison L, Mullin K, Barry K, Foster M. 2022. Cetacean and sea turtle spatial density model outputs from visual observations using line-transect survey methods aboard NOAA vessel and aircraft platforms in the Gulf of Mexico from 2003-06-12 to 2019-07-31 (NCEI Accession 0256800). NOAA National Centers for Environmental Information. Dataset. [https://doi.org/10.25921/efv4-9z56].

Moilanen, A., Lehtinen, P., Kohonen, I., Virtanen, E., Jalkanen, J. and Kujala, H. 2022. Novel methods for spatial prioritization with applications in conservation, land use planning and ecological impact avoidance. Methods in Ecology and Evolution. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13819].

Gulf Sea Turtles

This indicator identifies important areas in the Gulf of America for sea turtles. It incorporates monthly density predictions for four species (green, Kemp's ridley, leatherback, and loggerhead sea turtles) based on sightings from boat-based and aerial surveys and data on oceanographic conditions. It uses sea turtle models developed by the National Oceanic and Atmospheric Administration as part of the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS).

This indicator was last updated in October 2024.

Reason for Selection

As a keystone species, even at diminished population levels, sea turtles play an important role in ocean ecosystems by maintaining healthy seagrass beds and coral reefs, providing key habitat for other marine life, helping to balance marine food webs, and facilitating nutrient cycling from water to land (Wilson 2010). Sea turtles use large areas of the ocean for feeding and reproduction, making them a good indicator of ocean productivity and overall ocean health. For example, Kemp's ridley sea turtles nesting in southern Texas consistently forage in areas near the Yucatán Peninsula, the Gulf coast of Florida, the Mississippi River Delta, and the Texas-Louisiana shelf (Gredzens and Shaver 2020).

Input Data

- Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) - GoMMAPPS sea turtle spatial density model outputs (version 2.2)

Based on ship-based and aerial line-transect surveys conducted in the U.S. waters of the Gulf of America between 2003 and 2019, the NOAA Southeast Fisheries Science Center developed spatial density models (SDMs) for cetacean and sea turtle species for the entire Gulf of America. SDMs were developed using a generalized additive modeling framework to determine the relationship between species abundance and environmental variables (monthly averaged oceanographic conditions during 2015-2019).

- Southeast Blueprint 2023 subregions – marine (combined Atlantic & Gulf)
- Southeast Blueprint 2023 extent
- 2019 National Land Cover Database (NLCD): Land cover

Mapping Steps

- Replace all values of -9999 with 0.
- Convert to monthly rasters for each species using the following fields: “Jan_n”, “Feb_n”, “Mar_n”, “Apr_n”, “May_n”, “Jun_n”, “Jul_n”, “Aug_n”, “Sep_n”, “Oct_n”, “Nov_n”, and “Dec_n”. Use the marine subregion for pixel size, snap, and extent.
- Use the loggerhead sea turtle data and the NLCD to create a mask to define the extent of the Zonation analysis. The loggerhead data represents the full sample area for the other species in GoMMAPPS. The area covered by the sea turtle models overlaps with land in a few areas. This mask removes from the analysis all landcover classes that are not open water (not a value of 11 in the NLCD) within the extent of NLCD. The resulting Zonation mask covers open water areas where there is both modeled data for sea turtles and NLCD data to remove land.
- To identify important areas for each species, use the core area algorithm (CAZMAX) in Zonation 5. Include each monthly density layer as a separate input and weight them equally.
- Reproject the Zonation results data to Albers Equal Area.
- Convert from a floating point raster with a range of 0-1 to an integer raster ranging from 0-100.
- Reclassify to produce the indicator values seen below so that 0-65 is 1, 66-70 is 2, 71-80 is 3, 81-90 is 4, and 91-100 is 5. The variation in values from Zonation below 65 was less helpful than the other higher classes so we classified all values from 65 and below as 1.
- Use the NLCD and the modeling extent of the source data to identify areas of land not used in the analysis and assign those pixels a value of 0, since they are outside the scope of this marine indicator.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = >90th percentile of importance for sea turtle index species (across larger analysis area)
4 = >80th-90th percentile of importance
3 = >70th-80th percentile of importance
2 = >65th-70th percentile of importance
1 = ≤65th percentile of importance
0 = Land

Known Issues

- While this layer has a 30 m resolution, the source data was coarser than that. We downsampled hexagons with an area of 40 km^2 to 30 m pixels.

Other Things to Keep in Mind

- We ran the Zonation analysis across open water areas where there were both sea turtle models and NLCD data present to discriminate between land and water. We did this for multiple reasons. We didn't run Zonation across the full area covered by the GoMMAPPS data because the full files were very large and required long processing times. We also anticipated that Zonation would not have been able to computationally handle the full area. We extended the Zonation run beyond U.S. waters to try to account for areas of high mammal density just south of the Blueprint's Gulf subregion. As a result, the various classes within the indicator do not cover equal areas within the indicator's extent, as you might expect with a percentile-based indicator—they cover equal areas within the full analysis area, and then are clipped down to produce the indicator.

Disclaimer: Comparing with Older Indicator Versions

- There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Gredzens C, Shaver DJ. 2020. Satellite tracking can inform population-level dispersal to foraging grounds of post-nesting Kemp's ridley sea turtles. Frontiers in Marine Science, section Marine Megafauna, Special Theme Issue Research Topic: Advances in Understanding Sea Turtle Use of the Gulf of Mexico. [https://doi.org/10.3389/fmars.2020.00559].

Litz J, Aichinger Dias L, Rappucci G, Martinez A, Soldevilla M, Garrison L, Mullin K, Barry K, Foster M. 2022. Cetacean and sea turtle spatial density model outputs from visual observations using line-transect survey methods aboard NOAA vessel and aircraft platforms in the Gulf of Mexico from 2003-06-12 to 2019-07-31 (NCEI Accession 0256800). NOAA National Centers for Environmental Information. Dataset. [https://doi.org/10.25921/efv4-9z56].

Moilanen A, Lehtinen P, Kohonen I, Virtanen E, Jalkanen J, Kujala H. 2022. Novel methods for spatial prioritization with applications in conservation, land use planning and ecological impact avoidance. Methods in Ecology and Evolution. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13819].

Wilson, E.G., Miller, K.L., Allison, D. and Magliocca, M. 2010. Why Healthy Oceans Need Sea Turtles: The Importance of Sea Turtles to Marine Ecosystems. [https://oceana.org/wp-content/uploads/sites/18/Why_Healthy_Oceans_Need_Sea_Turtles_0.pdf].

Island Habitat

This indicator represents important habitat for coastal island-dependent species across the Southeast. The highest scores go to island critical habitat for six threatened and endangered animal and plant species: piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, and Bartram's hairstreak butterfly. This indicator uses U.S. Fish and Wildlife Service critical habitat data and island boundaries from the U.S. Geological Survey and Esri.

This indicator was last updated in October 2023.

Reason for Selection

Islands provide important habitat for many species, including birds, sea turtles, mammals, insects, and plants. Their relative isolation from disturbance and mainland predators can make them important breeding habitat for coastal birds and sea turtles (as represented by piping plover and loggerhead sea turtle). Their unique ecology and isolation can also make them important habitat for some mammals, plants and insects that are only found on islands (as represented by Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, and Bartram's hairstreak butterfly). As a barrier that can protect the mainland from major storms, they also help protect ecosystems and human communities from extreme weather events.

The critical habitat included in this indicator refers to areas with specific physical or biological features that are essential to conserving a federally threatened or endangered species and may require special management or protection.

Input Data

- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent
- Island boundaries from the Global Island Explorer provided by the U.S. Geological Survey (USGS) and Esri, accessed 5-13-2022 (note: USGS Global Islands data have been updated since this release; access the latest version)
- Critical habitat provided by the U.S. Fish and Wildlife Survey, accessed 6-23-2022

Mapping Steps

- Clip the Global Island Explorer data to the Base Blueprint 2022 extent and merge small and large islands.
- From the critical habitat data (CRITHAB_POLY.shp), select the following species based on 'comname': piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, Bartram's hairstreak butterfly.
- Clip the critical habitat for the selected species to the merged islands layer.
- Convert the islands and selected species data to raster and clip to the spatial extent of Base Blueprint 2022.
- Add zero values to help users better understand the extent of this indicator and to make it perform better in online tools. Buffer the island shapefile by 40 km to make a continuous buffer along the coast, with no gaps. Use this to create a raster of zeros for that buffer.
- Combine rasters so parts of islands with selected species critical habitat get a value of 2, other island pixels get a value of 1, and all other areas in the buffer described above get a value of 0.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Island critical habitat for any of six threatened and endangered species (piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, or Bartram's hairstreak butterfly)
1 = Other island area
0 = Not a coastal island

Known Issues

- This indicator underestimates piping plover critical habitat in parts of the Chandeleur Islands off the coast of Louisiana. Locations of barrier islands can be highly dynamic, and the island boundaries and critical habitat data did not agree on the locations of some parts of the islands.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Manca, F. Muller-Karger, B. Nyberg, R. Parsons, J. Saarinen, J. Steiner, and A. Reed. 2018. A new 30 meter resolution global shoreline vector and associated global islands database for the development of standardized global ecological coastal units. Journal of Operational Oceanography–A Special Blue Planet Edition. [https://doi.org/10.1080/1755876X.2018.1529714].

Sayre, R., S. Noble, S. Hamann, R. Smith, D. Wright, S. Breyer, K. Butler, K. Van Graafeiland, C. Frye, D. Karagulle, D. Hopkins, D. Stephens, K. Kelly, Z, basher, D. Burton, J. Cress, K. Atkins, D. van Sistine, B. Friesen, B. Allee, T. Allen, P. Aniello, I Asaad, M. Costello, K. Goodin, P. Harris, M. Kavanaugh, H. Lillis, E. Manca, F. Muller-Karger, B. Nyberg, R. Parsons, J. Saarinen, J. Steiner, and A. Reed. 2018. A new 30 meter resolution global shoreline vector and associated global islands database for the development of standardized global ecological coastal units. Journal of Operational Oceanography–A Special Blue Planet Edition. [https://doi.org/10.1080/1755876X.2018.1529714].

U.S. Geological Survey and Esri. Global Island Explorer. Accessed May 13, 2022. [https://www.sciencebase.gov/catalog/item/63bdf25dd34e92aad3cda273].

U.S. Fish and Wildlife Service. Critical Habitat. Accessed June 23, 2022. [https://ecos.fws.gov/ecp/report/table/critical-habitat.html].

Marine Highly Migratory Fish

This indicator identifies important foraging and spawning areas for highly migratory fish in the Atlantic Ocean and Gulf of America. It uses physical capture and satellite tag observations, remote sensing of environmental variables, and physical oceanographic data to analyze the habitat preferences of three species (skipjack tuna, bluefin tuna, and blue shark) at various life stages. The indicator originates from European Commission Joint Research Centre global fish models.

This indicator was last updated in October 2023.

Reason for Selection

Highly migratory fish species travel long distances and often cross domestic and international boundaries (NOAA Fisheries). They are often the focus of sport and commercial fishing, and the management of their populations can have far-reaching impacts. As top predators in the marine food chain, highly migratory fish are good indicators for areas of high ocean productivity. In particular, sharks and tunas play an important role in marine food webs by helping regulate populations of lower-level predators that, if left unchecked, could compromise the integrity of marine ecosystems across the Gulf and Atlantic (Baum 2009). In effect, the places where these species feed and spawn are also important for other types of marine life.

Input Data

- Southeast Blueprint 2023 extent
- Southeast Blueprint 2023 subregions: Marine (combined Atlantic & Gulf)

- European Commission global fish models, accessed 6-14-2023: Bluefin tuna (adult feeding, adult spawning, juvenile feeding), skipjack tuna (adult feeding), blue shark (adult female foraging, adult male foraging, large juvenile female foraging, large juvenile male foraging, small juvenile foraging)

  Note: While these data are global in scope, they use large amounts of data from U.S. Atlantic and Gulf waters and include a spawning model specific to the Gulf of America.

Mapping Steps

- Clip all global input data to the Southeast Blueprint 2023 marine subregion.
- Reproject the blue shark data to match the tuna data.
- Use the Southeast Blueprint 2023 marine subregion to make a Zonation mask for tuna and blue shark. The species data are global, and we wanted to limit the analysis just to the Southeast Blueprint marine subregion.
- To identify important areas for tunas and blue shark, use the core area algorithm (CAZMAX) in Zonation 5. For bluefin and skipjack tuna, include each monthly density layer for each tuna species and life stage for 2015, 2016, and 2017 and weight them equally. For blue shark, include each monthly density layer for each life stage for 2015, 2016, and 2017 and weight them equally. We did two separate runs, one for tunas and one for blue shark, due to differences in the input datasets. We chose 2015-2017 so we had matching years across the species and focused on more recent conditions to reduce the overall impact of climate change on the estimates.
- Reproject the Zonation results to Albers Equal Area and resample to 30 m pixels.
- Convert the tuna and blue shark layers from floating point rasters with a range of 0-1 to integer rasters ranging from 0-100.
- Reclassify each raster to produce the indicator values seen below so that 0-30 is 1, 31-40 is 2, 41-50 is 3, 51-60 is 4, 61-70 is 5, 71-80 is 6, 81-90 is 7, and 91-100 is 8. The variation in values from Zonation below 30 was less helpful than the other higher classes so we classified all values from 30 and below as 1. This was primarily due to large low probability areas for tuna in part of the Atlantic that all had the same low value.
- Combine the reclassified tuna and blue shark rasters into a single indicator by using the Cell Statistics “MAXIMUM” function.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

8 = >90th percentile of importance for bluefin and skipjack tuna or blue shark
7 = >80th-90th percentile of importance
6 = >70th-80th percentile of importance
5 = >60th-70th percentile of importance
4 = >50th-60th percentile of importance
3 = >40th-50th percentile of importance
2 = >30th-40th percentile of importance
1 = ≤30th percentile of importance

Known Issues

- While this layer has a 30 m resolution, the source data was coarser than that. We downsampled 1/24° pixels (~4 km) to 30 m.
- This indicator doesn't fully represent the specialized habitat requirements of other highly migratory fish not used in the indicator. The indicator focuses on highly productive currents and eddies. While these are important for many highly migratory species, especially for feeding, this approach likely misses important areas for some species.

Other Things to Keep in Mind

- We weren't able to run all the tuna and blue shark data together in Zonation, so we ran them separately and took the maximum value of the two resulting outputs. As a result, the various classes within the indicator do not cover equal areas within the indicator's extent, as you might expect with a percentile-based indicator—they cover equal areas within the underlying tunas run and shark run and are then combined.
- While these are global models, we did the Zonation run just within the Blueprint marine subregion. We could explore running them over a larger ecological region in the future. For this year, given we were already running into processing challenges and didn't have an alternative subglobal subregion in mind for these species, we stuck with the Blueprint marine subregion.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Baum, J.K., and B. Worm, "Cascading top-down effects of changing oceanic predator abundances," Journal of Animal Ecology, 78: 699-714 (2009), p. 11. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/j.1365-2656.2009.01531.x].

Druon, Jean-Noel (2019): GMIS - Favourable feeding habitat of adult Atlantic bluefin tuna (ABFT) Monthly 1998-2017 (frequency of occurrence, %). European Commission, Joint Research Centre (JRC) [Dataset] PID: [https://data.europa.eu/89h/60a0a5b0-b63c-473f-b8bc-207ea037eb3b].

Druon, Jean-Noel (2019): GMIS - Favourable feeding habitat of juvenile Atlantic bluefin tuna (ABFT) Monthly 2003-2017 (frequency of occurrence, %). European Commission, Joint Research Centre (JRC) [Dataset] PID: [https://data.europa.eu/89h/8f566919-c88f-4c2d-84dc-7862800d5385].

Druon, Jean-Noel (2019): GMIS - Favourable feeding habitat of skipjack tuna (SKJT) Monthly 1998-2017 (frequency of occurrence, %). European Commission, Joint Research Centre (JRC) [Dataset] PID: [https://data.europa.eu/89h/98b2254b-aca4-4f0b-9777-4814d1e635ad].

Druon, Jean-Noel (2019): GMIS - Favourable spawning habitat of adult Atlantic bluefin tuna (ABFT) Monthly 1998-2017 (frequency of occurrence, %). European Commission, Joint Research Centre (JRC) [Dataset] PID: [https://data.europa.eu/89h/a60cc972-2a78-461b-b8bd-9aae008e9c03].

European Commission, Joint Research Centre (JRC) (2022): Foraging habitat of blue shark (Prionace glauca) - Global Ocean MONTHLY - 2003-2018 (% of daily favorable occurrence, 1/24° by 1/24°). European Commission, Joint Research Centre (JRC) [Dataset] PID: [https://data.europa.eu/89h/6062ddae-2478-409a-89bf-8121902dc70e].

NOAA Fisheries: National Oceanic and Atmospheric Administration – Highly Migratory Species. Accessed June 16, 2023. PID: [https://www.fisheries.noaa.gov/highly-migratory-species].

Resilient Coastal Sites

This indicator depicts the capacity of coastal habitats to migrate to adjacent lowlands in order to sustain biodiversity and natural services under increasing inundation from sea level rise. It is based on the physical and condition characteristics of current tidal complexes, their predicted migration space, and surrounding buffer areas. These characteristics include marsh complex size, shared edge with migration space, sediment balance, water quality, natural landcover, landform diversity, and more. This indicator originates from The Nature Conservancy's Resilient Coastal Sites project.

This indicator was last updated in November 2025.

Reason for Selection

The resilient coastal sites indicator seeks to capture features of salt marshes that are important for salt marsh species and ecosystem function both now and in the future. Many of these characteristics, like water quality and landform diversity, also serve as indicators of a site's potential future habitat quality once the salt marsh is fully inundated and transitions to a new estuarine or marine habitat.

Input Data

- Southeast Blueprint 2025 extent
- The Nature Conservancy's (TNC) combined terrestrial and coastal resilient sites for the conterminous U.S. in geodatabase format (resilient_sites_national.zip), accessed 12-3-2024

Mapping Steps

- To extract the coastal component of the analysis, reclassify the "type_des" field to assign "Marsh Migration Space", "Tidal Complex", "Additional Marsh Migration Space", and "In sea level rise zone" a value of 1, and everything else a value of NoData.
- Clip the TNC data to the bounding box around the Southeast Blueprint 2025 extent.
- Reclassify the numeric resilience field into the standard deviation-based classes that TNC uses to display the resilience data in their viewer. Note: the "sea level rise area" class denotes locations that are not part of a current tidal complex or future marsh migration space but are at risk of inundation under 6 feet of SLR. Because they are not current or future marsh, they are outside the scope of the coastal resilience analysis. TNC also does not score them in their terrestrial resilience analysis because they are likely to be inundated in the future.
    - -3502 = 0 (sea level rise area)
    - -4000 to ≤ -3503 or -3501 to ≤ -2000 = 1 (least resilient)
    - -2001 to ≤ -1000 = 2 (less resilient)
    - -1001 to ≤ -500 = 3 (slightly less resilient)
    - -501 to ≤ 499 = 4 (average/median resilience)
    - 500 to ≤ 999 = 5 (slightly more resilient)
    - 1000 to ≤ 1999 = 6 (more resilient)
    - 2000 to ≤ 3260 = 7 (most resilient)
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Most resilient
6= More resilient
5 = Slightly more resilient
4 = Average/median resilience
3 = Slightly less resilient
2 = Less resilient
1 = Least resilient
0 = Sea level rise area

Known Issues

- This indicator does not account for the occurrence and timing of disturbance processes, like prescribed fire, which impact resilience.
- Accretion rates were not incorporated into the marsh migration data.
- TNC assessed sea level rise under a slightly different range of scenarios in the Northeast and Mid-Atlantic region, compared to the Gulf of Mexico and South Atlantic regions. In the Gulf and South Atlantic, they used four scenarios: 1.5 ft, 3 ft, 4 ft, and 6.5 ft. In the Northeast and Mid-Atlantic, they used 1 ft increments ranging from 1-6 feet. This means that the data used in the portion of coastal Virginia covered by the Northeast and Mid-Atlantic analysis assesses marsh migration space under a slightly less extreme maximum sea level rise scenario (6 ft vs. 6.5 ft). As a result, this indicator likely underestimates the extent of resilient areas in this portion of coastal Virginia.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Anderson, M.G. and Barnett, A. 2017. Resilient Coastal Sites for Conservation in the Northeast and Mid-Atlantic US. The Nature Conservancy, Eastern Conservation Science. [https://easterndivision.s3.amazonaws.com/coastal/Resilient_Coastal_Sites_for_Conservation_NE_Mid_Atlantic.pdf].

Anderson, M.G. and Barnett, A. 2019. Resilient Coastal Sites for Conservation in the Gulf of Mexico US. The Nature Conservancy, Eastern Conservation Science. [https://secassoutheast.org/pdf/GulfOfMexico_Resilient_Coastal_Sites_31Oct2019.pdf].

Anderson, M.G. and Barnett, A. 2019. Resilient Coastal Sites for Conservation in the South Atlantic US. The Nature Conservancy, Eastern Conservation Science. [https://secassoutheast.org/pdf/SouthAtlantic_Resilient_Coastal_Sites_31Oct2019.pdf].

Anderson, M.G., Ferree, C.E., 2010. Conserving the stage: climate change and the geophysical underpinnings of species diversity. PLoS One 5, e11554. [https://doi.org/10.1371/journal.pone.0011554].

The Nature Conservancy. Datasets for the Conterminous U.S.: Resilient Sites (Terrestrial and Coastal). Esri File Geodatabase format. Accessed December 3, 2024. [https://www.conservationgateway.org/ConservationPractices/ClimateChange/Pages/RCN-Downloads.aspx].

Seagrass

This indicator represents the presence of seagrass in the Atlantic Ocean and Gulf of America. It originates from the National Oceanic and Atmospheric Administration's Marine Cadastre.

This indicator was last updated in October 2023.

Reason for Selection

Seagrasses provide food and habitat for a range of marine and estuarine wildlife, including fish, sea turtles, shrimp, crabs, oysters, and more. They also produce oxygen, filter water, sequester carbon, control erosion, and buffer storms. Seagrasses serve as an important indicator of the overall health of coastal ecosystems because they are sensitive to water quality and require sufficiently clear water for sunlight to penetrate (NPS 2021, NOAA 2021, NWF 2021). In addition, seagrass meadows are being increasingly recognized for their value as a carbon sink in the face of climate change (McKenzie et al. 2020).

Input Data

- Southeast Blueprint 2023 extent
- Base Blueprint 2022 extent
- NOAA seagrass layer, accessed 5-30-2023 on the Marine Cadastre

Mapping Steps

- Convert the polygons from the Marine Cadastre seagrass layer to raster, snapping to the 2022 Base Blueprint extent.
- The data were originally classified into various biotic subclasses of "Aquatic Vascular Vegetation" and "Benthic Macroalgae." Because these classes were not applied consistently across different states, collapse these categories into a single category of seagrass presence.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

1 = Seagrass present

Known Issues

- Because the data used in this indicator were opportunistically collected based on available state data, this indicator underestimates seagrass presence in some areas and misses some known seagrass locations (e.g., the north shore of Lake Pontchartrain in LA and areas farther offshore of central FL in the Gulf of America).

- Because the source data does not provide any information about the extent of surveyed areas, this indicator cannot distinguish between places where seagrass was surveyed and not found (true absences) vs. places that were not surveyed. Ideally, we would prefer to depict areas where seagrass was surveyed and not found as zeroes and unsurveyed areas as NoData.
- Due to the variety of data sources combined in this indicator, there is significant variation across the Southeast in how rigorously seagrass data is validated. Water turbidity can also influence the quality of validation. NOAA characterizes the level of validation in NC as good, the east coast of FL (particularly the St. John's River Water Management District) as excellent, the west coast of FL (particularly the Southwest Florida Water Management District) as good, the Big Bend of FL as more challenging due to turbidity, the area immediately surrounding the FL Keys as good, other parts of Florida Bay as challenging due to turbidity, Mobile Bay in AL as good, LA as less consistent, Matagorda in TX as somewhat good, and Laguna Madre in TX as excellent. This indicator is more reliable in areas with strong validation.
- Similarly, variation in the minimum mapping unit used by each data provider causes differences in the size of seagrass patches captured in different parts of the Southeast. For example, some surveys map patches as small as 20 m^2, while others do not map patches smaller than 1 acre. As a result, this indicator may underpredict small seagrass patches in areas where survey efforts used a coarse minimum mapping unit.
- In estuarine areas near river deltas (e.g., portions of Mobile Bay in AL), this indicator captures freshwater species of submerged aquatic vegetation in addition to true seagrass.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

McKenzie, Len J. and Lina M. Nordlund, Benjamin L. Jones, Leanne C. Cullen-Unsworth, Chris Roelfsema, and Richard K. F. Unsworth. The global distribution of seagrass meadows. 2020. Environmental Research Letters 15 074041. [https://doi.org/10.1088/1748-9326/ab7d06].

National Park Service. Seagrass Meadows. Accessed December 22, 2021. [https://www.nps.gov/subjects/oceans/seagrass-meadows.htm].

The National Wildlife Federation. Seagrasses. Accessed December 22, 2021. [https://www.nwf.org/Educational-Resources/Wildlife-Guide/Plants-and-Fungi/Seagrasses].

NOAA Office for Coastal Management, 2021: Seagrasses. NOAA National Centers for Environmental Information. [https://www.fisheries.noaa.gov/inport/item/56960].

South Atlantic Beach Birds

This indicator is an index of habitat suitability for four shorebird species (American oystercatcher, Wilson's plover, least tern, piping plover) in the South Atlantic, based on observed abundance. It assesses beaches and nearby onshore habitats. This indicator combines bird data from the U.S. Geological Survey and state waterbird biologists in FL, GA, SC, and NC.

This indicator was last updated in October 2023.

Reason for Selection

The relative use of beach habitat by shorebird species for nesting, foraging, and breeding is an indicator of beach health and quality. Shorebird populations are highly responsive to threats like sea level rise, changes in freshwater inflow, shoreline alteration and loss, human disturbances, and contaminants. In particular, the American oystercatcher "has been proposed as a 'sentinel' bio-indicator of ecosystem integrity because of the depth of life history information available, its specialized dependence on oysters and associated marine invertebrates, and known reproductive responses to a variety of natural and anthropogenic pressures" (Ogden et al. 2014). As a result of these pressures, North American shorebird populations are experiencing "consistent, steep population loss" (Ronenberg et al. 2019). The species included in this index are already monitored by state and Federal agencies and collectively represent a variety of coastal ecosystem features (e.g., nesting habitat availability and quality, fish and marine invertebrate populations).

Input Data

- South Atlantic Blueprint 2021 extent
- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent
- 2019 National Land Cover Database (NLCD): Land cover
- 2023 U.S. Census TIGER/Line state boundaries, accessed 5-15-2023; download the data
- The following beach bird datasets:

  #### Wilson's plover & American oystercatcher

  Betsy Von Holle (University of Central Florida) led a project that included state waterbird biologists in the South Atlantic: Tim Keyes, Felicia Sanders, Sara Schweitzer, and Janell Brush. They mapped habitat suitability based on nest, or breeding pair, density per beach segment, as was used for Von Holle et al.'s sea turtle research. The approach is documented in Von Holle et al. 2018.

  The following nesting years were used for the analysis:

  - American oystercatcher: FL (2005-2011), GA (2010-2011), SC (2008), NC (2007)
  - Wilson's plover: FL (2005-2011), GA (2010-2011), SC (2009-2011), NC (2007)

Piping plover & least tern

A previous post-doctoral researcher, Bradley Pickens, processed the following datasets by ranking least tern nest abundance and piping plover individuals by quantile (0-6). They were exported as 90 m rasters.

Least tern

The point locations and number of least tern nests, or breeding pairs, were provided by waterbird biologists from each state's natural resource department (Tim Keyes, Felicia Sanders, Sara Schweitzer, and Janell Brush). All least tern locations were buffered by 1 km. Although least tern do not actively forage on the beach itself, the buffer characterizes habitat selected by least tern (i.e., beach width, predator abundance, etc.) and accounts for interannual variability in nesting locations. Among years, data showed least tern often shifted the location of nests in the general vicinity of previous years.

Piping plover (winter distribution)

The 2011 winter population census of piping plover was provided by the U.S. Geological Survey, as the international census is repeated every five years. Locations were buffered with a 2 km radius. Although home range estimates exist for piping plover (Cohen et al. 2008, Drake et al. 2001), these measures depict primarily linear habitats. We used a 2 km buffer, as this is similar to the mean linear distance of 4.2 km that piping plover moved during winter in Texas (Drake et al. 2001). The resulting buffer was also substantiated by maps in Cohen et al. (2008).

Mapping Steps

- Von Holle et al. used a categorical habitat suitability ranking based on six quantiles of nest density, or breeding pair density, for each species on the Atlantic Coast. Convert to numeric scores by assigning a value of 0 to polygons rated as "none", a value of 1 to those rated as "low", a value of 2 to those rated as "moderate", a value of 3 to those rated as "moderately high", a value of 4 to those rated as "high", a value of 5 to those rated as "very high", and a value of 6 to those rated "extremely high". Convert the polygons to a separate raster for each species (Wilson's plover and American oystercatcher), assigning those numeric values.
- Resample and reproject the least tern and piping plover source data from 90 m pixels to 30 m pixels so it can be used in Zonation in a later step.
- Create a mask to define the extent of the Zonation run that will be used to create this indicator. Use the Cell Statistics "MAXIMUM" function to combine all 4 bird rasters (the two Von Holle rasters created in the previous step, and the 90 m piping plover and least tern rasters). Include in the mask all pixels with a value >0 on any of the bird rasters.
- To create the beach bird index, use the software program Zonation v4 with the core-area algorithm and without the edge removal option (removal rule = 1, edge removal = 0, warp = 1). Input the mask and the rasters for all four species. Zonation produces a continuous ranking of all pixels within the mask based on their importance for the 4 beach-nesting bird species.
- Reclassify the output Zonation rank raster into 5 classes, seen in the final indicator values below. This layer represents the relative use of habitat for beach nesting birds in the South

Atlantic. Generally, areas with higher values in this layer are considered to have greater relative abundance of beach nesting birds than areas with lower values.

- To create an analysis extent for the indicator, buffer the polygons provided by Von Holle by 25 km. Assign a value of 0 to all pixels within the analysis extent that do not already receive a score elsewhere in the indicator. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the South Atlantic Blueprint 2021 extent.
- Further limit the extent of the zero class by removing pixels from Virginia, because the source data did not cover Virginia. Also, remove deep marine areas that are not covered by a value of 0 or greater in the 2019 NLCD, where this terrestrial indicator does not apply.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = >80th percentile of importance for bird index species (American oystercatcher, Wilson's plover, least tern, and piping plover)
4 = >60th-80th percentile of importance
3 = >40th-60th percentile of importance
2 = >20th-40th percentile of importance
1 = ≤20th percentile of importance
0 = Open water or not identified as a priority for bird index species

Known Issues

- Beach bird survey data are summarized by beach segment and do not account for variations in density within those segments.
- Volunteers often collect beach bird data in discrete time frames and survey effort may differ by location. Some areas may not have been surveyed or nests may have been missed. Therefore, this data does not imply absence of species.
- Red knot is not included due to lack of data.
- This indicator may underestimate beach bird use of dune areas inland of beach segments because of inconsistencies in the extent of the various bird models.
- The spatial resolution of the source data for least tern and piping plover has been degraded. While the indicator has a 30 m resolution, it was not created directly from the spatially precise input data for those two birds. Here, we resampled the 90 m resolution data used in the previous version of the indicator back down to 30 m resolution so that we could use it in Zonation.

- The indicator has an extraneous “tail” of zero values at its southern edge. This is an artifact of how the buffer of zero values was created. Since zero values do not influence Zonation, this does not affect the final Blueprint priorities. It was discovered too late to fix in this update cycle. We hope to make a larger improvement to this indicator in the future to better depict important areas for beach birds across more of the region’s coastline.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

Cohen, J.B., Karpanty, S.M., Catlin, D.H., Fraser, J.D., Fischer, R.A., 2008. Winter Ecology of Piping Plovers at Oregon Inlet, North Carolina. Waterbirds 31, 472-479. [https://doi.org/10.1675/1524-4695-31.3.472].

Drake, K.R., Thompson, J.E., Drake, K.L., Zonick, C., 2001. Movements, habitat use, and survival of nonbreeding Piping Plovers. The Condor 103, 259-267. [https://doi.org/10.1093/condor/103.2.259].

Moilanen, A., L. Meller, J. Leppänen, F.M. Pouzols, H. Kujala, A. Arponen. 2014. Zonation Spatial Conservation Planning Framework and Software V4.0, User Manual. [https://github.com/cbig/zonation-core/releases/download/4.0.0/zonation_manual_v4_0.pdf].

Ogden, John C., John D. Baldwin, Oron L. Bass, Joan A. Browder, Mark I. Cook, Peter C. Frederick, Peter E. Frezza, Rafael A. Galvez, Ann B. Hodgson, Kenneth D. Meyer, Lori D. Oberhofer, Ann F. Paul, Pamela J. Fletcher, Steven M. Davis, Jerome J. Lorenz. Waterbirds as indicators of ecosystem health in the coastal marine habitats of southern Florida: 1. Selection and justification for a suite of indicator species, Ecological Indicators, Volume 44, 2014, Pages 148-163, ISSN 1470-160X, [https://doi.org/10.1016/j.ecolind.2014.03.007].

Rosenberg, Kenneth & Dokter, Adriaan & Blancher, Peter & Sauer, John & Smith, Adam & Smith, Paul & Stanton, Jessica & Panjabi, Arvind & Helft, Laura & Parr, Michael & Marra, Peter. (2019). Decline of the North American avifauna. Science. 366. [https://www.science.org/doi/10.1126/science.aaw1313].

U.S. Department of Commerce, U.S. Census Bureau, Geography Division, Spatial Data Collection and Products Branch. TIGER/Line Shapefile, 2023, U.S. Current State and Equivalent National. 2023. [https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html].

Von Holle, B., Irish, J. L., Spivy, A., Weishampel, J. F., Meylan, A., Godfrey, M. H., Dodd, M., Schweitzer, S.H., Keyes, T., Sanders, F., Chaplin, M. K. 2018. Effects of future sea level rise on coastal habitat. The Journal of Wildlife Management 9999. [https://wildlife.onlinelibrary.wiley.com/doi/epdf/10.1002/jwmg.21633].

South Atlantic Maritime Forest

This indicator depicts the maritime forest currently present in the South Atlantic. It originates from LANDFIRE landcover.

This indicator was last updated in October 2023.

Reason for Selection

Overall acreage of existing maritime forest provides an indicator of whether maritime forests being inundated by sea level rise are being replaced or restored somewhere else. Since maritime forest has been substantially reduced from its historical extent, protecting the remaining acreage is important (Bellis 1995). This ecosystem supports a unique suite of plants that tolerate wind, salt, and flooding, as well as many species of birds, mammals, and reptiles. It also helps buffer the coastline from storms (NOAA 2021). Maritime forest is also well monitored and resonates with a diversity of audiences.

Input Data

- Base Blueprint 2022 subregions
- Southeast Blueprint 2023 extent
- 2016 LANDFIRE Existing Vegetation Type (EVT), accessed 3-31-2021; access the data

Mapping Steps

- Extract the LANDFIRE existing vegetation type classes with the following names, which are considered maritime forest: Central Atlantic Coastal Plain Maritime Forest, East Gulf Coastal Plain Maritime Forest, Southern Atlantic Coastal Plain Maritime Forest, Southern Atlantic Coastal Plain Dune and Maritime Grassland.
- Add zero values to help users better understand the extent of this indicator and to make this indicator layer perform better in online tools. Use the expand tool to buffer the maritime forest pixels by 40 km. We chose this distance because it was large enough to make a continuous buffer along the coast, with no gaps. Shrink the expanded areas back down by 30 km. We added this step to better reflect that maritime forest exists near the marine shoreline.
- Clip to the 'Atlantic Coastal Plain' and East Gulf Coastal Plain' subregions. LANDFIRE depicts maritime forest classes that also cover parts of Louisiana, Alabama, and Florida that are not represented in this indicator, but the data has not yet been vetted in those areas. Since the original indicator was developed for the South Atlantic region, expert review for other regions is needed before they are used in the Blueprint.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

1 = Maritime forest
0 = Not identified as maritime forest

Known Issues

- The LANDFIRE existing vegetation map underpredicts the extent of maritime forest in several known locations, including Kitty Hawk Woods and Nags Head Woods in the Outer Banks of North Carolina.
- We include the Southern Atlantic Coastal Plain Dune and Maritime Grassland existing vegetation classes from LANDFIRE because some known maritime forests in the South Atlantic are misclassified as this type. This may result in overpredicting maritime forest in some dune and grassland areas.
- This indicator underpredicts maritime forest along the Gulf Coast of Florida.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Bellis, Vincent. Ecology of Maritime Forests of the Southern Atlantic Coast: A Community Profile. May 1995. Biological Report 30. U.S. Department of the Interior. National Biological Service. [https://www.fws.gov/media/ecology-maritime-forests-southern-atlantic-coast-community-profile].

National Oceanic and Atmospheric Administration. National Ocean Service. January 4, 2021. What is a maritime forest? [https://oceanservice.noaa.gov/facts/maritime-forest.html].

U.S. Geological Survey, Earth Resources Observation and Science (EROS) Center. 2020. LANDFIRE Remap 2016 Existing Vegetation Type (EVT) CONUS: LF Remap. Accessed March 31, 2021. [https://www.landfire.gov].

Stable Coastal Wetlands

This indicator uses remote sensing to calculate the unvegetated-vegetated ratio of tidal wetlands, which compares how much of a wetland is not covered by plants (e.g., sediment, rocks, open water) to how much is covered by plants. Marshes that maintain a higher proportion of vegetation tend to be more stable and resilient to threats like sea level rise, erosion, and coastal development. This indicator originates from a U.S. Geological Survey project on an unvegetated to vegetated ratio for coastal wetlands.

This indicator was last updated in October 2023.

Reason for Selection

Threats like sea level rise, erosion, and development are driving widespread salt marsh loss. Tidal marshes protect coastlines from storms, filter pollution to improve water quality, and provide important habitat for birds, fish, and shellfish. Studies show that the unvegetated-vegetated ratio (UVVR) of tidal marshes serves as a good surrogate for marsh degradation processes like sediment loss and conversion to open water (Ganju 2017, Wasson et al. 2019). In general, "...marshes with high vegetative cover and lower open-water area tend to trap and retain sediment, while marshes that are losing plant cover will further lose sediment and convert to open water" (Wasson et al. 2019).

Examining snapshots of the UVVR and how it changes over time can differentiate between relatively stable and more vulnerable marshes. This can help land managers determine which marshes are the best candidates for restoration to help them persist in the face of changing conditions (USGS 2017). Recent research also shows that UVVR correlates with elevation measurements that detect vertical changes due to accretion, subsidence, or compaction—meaning that stable marshes with high vegetative cover tend to better keep pace with sea level rise. This suggests that marshes with low vegetative cover and an unstable UVVR tend to be "both horizontally and vertically vulnerable". It also shows that "horizontal integrity is a prerequisite for vertical stability: a marsh can only maintain elevation if the plain is intact with minimal unvegetated area" (Ganju et al. 2023).

Input Data

- Base Blueprint 2022 extent
- Southeast Blueprint 2023 extent
- National Wetlands Inventory
- An unvegetated to vegetated ratio (UVVR) for coastal wetlands

  This dataset computes the UVVR for coastal wetlands using Landsat data from 2014-2018. The average and standard deviation of the UVVR across that five-year period can be used to characterize both a marsh's overall stability and its variability from year to year. Figure 3 below, from Ganju et al. 2022, identifies the threshold used in this indicator to define stable, low uncertainty tidal marshes (<0.15 average UVVR and <0.15 standard deviations from the average UVVR).

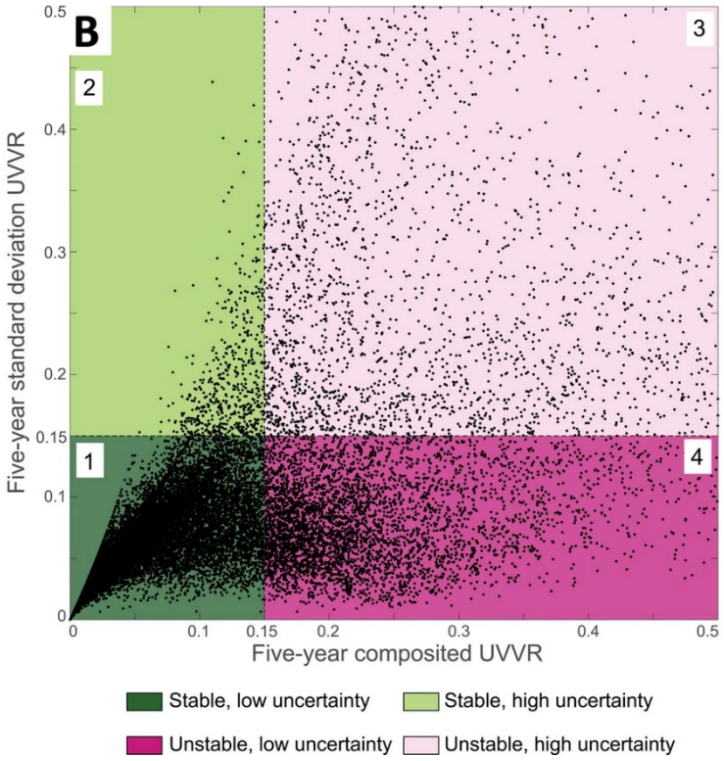


Figure 3. Scatterplot of five-year composited and standard deviation of UVVR, used to identify areas that have crossed the nominal 0.15 stability threshold. 1) Indicates stable, low uncertainty pixels that are consistently below the stability threshold of 0.15 (i.e., good) and don't fluctuate much; 2) indicates stable, high uncertainty pixels that are below the stability threshold of 0.15, but there are big annual outliers (e.g., overwash event, dieback); 3) indicates unstable, high uncertainty pixels that are above the stability threshold of 0.15 (i.e., bad) and there are large annual fluctuations (water levels, controlled burn); and 4) indicates unstable, low uncertainty pixels that are consistently above the stability threshold of 0.15, and they don't fluctuate much (Ganju 2022).

Mapping Steps

- Calculate the mean and standard deviation of the UVVR 2014-2018 imagery.
- Extract areas with mean and standard deviation values <0.15, which represent stable vegetated areas. Reclassify the resulting raster to assign stable coastal wetlands a value of 2.
- Extract fresh and saline tidal wetlands from NWI (note: because of the complex way the NWI is classified, the query to extract these classes is too long to include here, so consult the code for more specifics). Buffer by 30 m and convert to raster. Clip the stable vegetated areas produced in the previous step to the NWI tidal wetlands.
- Reclassify the standard deviation rasters produced in the first step to assign a value of 0 to the study area. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Cip the study area raster to the NWI raster to create a coastal wetland extent raster and reclassify to assign it a value of 1.

- Use the Cell Statistics "MAXIMUM" function to combine the rasters produced above. This produces a new raster with the indicator values seen below, where each pixel is assigned the highest value it received in any of the previous steps.
- Clip to the spatial extent of Base Blueprint 2022.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Stable coastal wetlands
1 = Other coastal wetlands
0 = Not identified as coastal wetlands

Known Issues

- The age of the NWI data is inconsistent across the Gulf and Atlantic coast.
- Newly created wetlands, such as the Wax Lake Delta in Louisiana, are missing from this indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Couvillion, B.R., Ganju, N.K., and Defne, Z., 2021, An Unvegetated to Vegetated Ratio (UVVR) for coastal wetlands of the Conterminous United States (2014-2018): U.S. Geological Survey data release. [https://doi.org/10.5066/P97DQXZP].

Ganju, N., Defne, Z., Kirwan, M. et al. Spatially integrative metrics reveal hidden vulnerability of microtidal salt marshes. Nat Commun 8, 14156 (2017). [https://doi.org/10.1038/ncomms14156].

Ganju, Neil. May 19, 2022. Developing standardized geospatial metrics for salt marsh management and restoration. Webinar hosted by the Southeast Conservation Adaptation Strategy. [https://www.youtube.com/watch?v=9ULO2ZZNVxc].

Ganju, N.K., Couvillion, B.R., Defne, Z. et al. Development and Application of Landsat-Based Wetland Vegetation Cover and Unvegetated-Vegetated Marsh Ratio (UVVR) for the Conterminous United States. Estuaries and Coasts (2022). [https://doi.org/10.1007/s12237-022-01081-x].

Ganju, N. K., Defne, Z., & Fagherazzi, S.. (2020). Are elevation and open-water conversion of salt marshes connected?. Geophysical Research Letters, 47, e2019GL086703. [https://doi.org/10.1029/2019GL086703].

Ganju, N.K., Defne, Z., Schwab, C. et al. Horizontal Integrity a Prerequisite for Vertical Stability: Comparison of Elevation Change and the Unvegetated-Vegetated Marsh Ratio Across Southeastern USA Coastal Wetlands. Estuaries and Coasts (2023). [https://doi.org/10.1007/s12237-023-01221-x].

U. S. Fish and Wildlife Service. October 2020. National Wetlands Inventory. U.S. Department of the Interior, Fish and Wildlife Service, Washington, D.C. [https://www.fws.gov/program/national-wetlands-inventory/data-download].

U.S. Geological Survey. January 24, 2017. “New technique quickly predicts salt marsh vulnerability.” [https://www.usgs.gov/programs/cmhrp/news/new-technique-quickly-predicts-salt-marsh-vulnerability].

Wasson, Kerstin, Neil K Ganju, Zafer Defne, Charlie Endris, Tracy Elsey-Quirk, Karen M Thorne, Chase M Freeman, Glenn Guntenspergen, Daniel J Nowacki and Kenneth B Raposa. (2019). Understanding tidal marsh trajectories: evaluation of multiple indicators of marsh persistence. Environmental Research Letters 14 124073. [https://iopscience.iop.org/article/10.1088/1748-9326/ab5a94/].

Caribbean

Terrestrial

Caribbean Greenways & Trails

This cultural resource indicator measures both the natural condition and connected length of greenways and trails in the U.S. Caribbean to characterize the quality of the recreational experience. Natural condition is based on the amount of impervious surface surrounding the path. Connected length captures how far a person can go without leaving a dedicated path, based on common distances for walking, running, and biking. This indicator originates from OpenStreetMap data and the National Oceanic and Atmospheric Administration's Coastal Change Analysis Program landcover.

This indicator was last updated in October 2023.

Reason for Selection

This indicator captures the recreational value and opportunities to connect with nature provided by greenways and trails. Greenways and trails provide many well-established social and economic benefits ranging from improving human health, reducing traffic congestion and air and noise pollution, increasing property values, and generating new jobs and business revenue (ITRE 2018). The locations of greenways and trails are regularly updated through the open-source database OpenStreetMap.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2012 NOAA Coastal Change Analysis Program (C-CAP) land cover files for the U.S. Virgin Islands (St. Thomas, St. John, and St. Croix are provided as separate rasters), accessed 11-10-2022; learn more about C-CAP high resolution land cover and change products
- 2010 NOAA C-CAP land cover files for Puerto Rico, accessed 11-10-2022; learn more about C-CAP high resolution land cover and change products
- OpenStreetMap data "lines" layer, accessed 2-26-2023

  A line from this dataset is considered a potential greenway/trail if the "highway" tag attribute is eitherbridleway, cycleway, footway, or path. In OpenStreetMap, a highway refers to "any road, route, way, or thoroughfare on land which connects one location to another and has been paved or otherwise improved to allow travel by some conveyance, including motorized vehicles, cyclists, pedestrians, horse riders, and others (but not trains)". OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

Mapping Steps

The greenways and trails indicator score reflects both the natural condition and connected length of the greenway/trail.

Natural condition

Natural condition is based on the amount of impervious surface surrounding the greenway/trail. Since perceptions of a greenway's "naturalness" are influenced both by the immediate surroundings adjacent to the path, and the greater viewshed, natural condition is calculated by averaging two measurements: local impervious and nearby impervious.

Local impervious is defined as the percent impervious surface of the 30 m pixel that intersects the trail. Nearby impervious is defined as the average impervious surface within a 300 m radius circle surrounding the path (note: along a 300 m stretch of trail, we only count the impervious surface within a 45 m buffer on either side of the trail, since pixels nearer the trail have a bigger impact on the greenway/trail experience). The natural classes are defined as follows:

3 = Mostly natural: Average of local and nearby impervious is ≤1%
2 = Partly natural: Average of local and nearby impervious is >1 and <10%
1 = Developed: Average of local and nearby impervious is ≥10%

- To create a percent impervious layer, start by converting the C-CAP landcover rasters for Puerto Rico (2 m resolution) and the U.S. Virgin Islands (separate downloads for St. Thomas, St. John, and St. Croix with 2.4 m resolution) from .img format to .tif using the Copy Raster function.
- For each individual C-CAP layer, use the ArcPy Conditional function to make a binary raster assigning the impervious class a value of 100 (representing fully impervious) and all other classes a value of 0 (representing fully permeable). This mimics the data format of the 2019 National Land Cover Database (NLCD) used in the continental Southeast permeable surface indicator, which provides a continuous impervious surface value ranging from 0 to 100. Use the Focal Statistics function to calculate the percent of cells in a 30 m square that are identified as impervious in the C-CAP data, then reproject and resample the result to a 30 m resolution.
- Use the Cell Statistics "MAXIMUM" function to combine the resulting four 30 m C-CAP impervious rasters. This creates an approximation of the percent developed impervious score from the 2019 NLCD.

Connected length

The connected length of the path is calculated using the entire extent of the potential greenways/trails dataset. A trail is considered connected to another trail if it is within 2 m of the other trail. Length thresholds are defined by typical lengths of three common recreational greenway activities: walking, running, and biking. The 40 km threshold for biking is based on the standard triathlon biking segment of 40 km (~25 mi). Because a 5K is the most common road race distance, the

running threshold is set at 5 km (~3.1 mi) (Running USA 2017). The 1.9 km (1.2 mi) walking threshold is based on the average walking trip on a summer day (U.S. DOT 2002).

- Using the statistics software R, download the OpenStreetMap data for Puerto Rico and the US Virgin Islands.
- Select all lines from the OpenStreetMap data that have a highway tag of either footway, cycleway, bridleway, or path. These are all considered potential trails.
- Removed all lines marked as private.
- Identify lines from the potential trails that are tagged as sidewalks. Assign them a value of 1 in the indicator.

Final scores

If the potential greenway/trail was tagged as a sidewalk in the "other tags" field, it is given a value of 1 to separate sidewalks from what most people think of as a trail or greenway. If a pixel does not intersect a potential greenway/trail but is covered by the C-CAP landcover data, it is coded with a value of 0. Clip to the Caribbean Blueprint 2023 subregion. As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Table 11. Greenways and trails indicator scoring.

		Trail length				**Sidewalk**
		L ≥ 40 km	40 km > L ≥ 5 km	5 km > L ≥ 1.9 km	L < 1.9 km	
Average of local and nearby impervious	Imp ≤ 1%	7	6	5	4	1
	1% < Imp < 10%	6	5	4	3	
	Imp ≥ 10%	5	4	3	2	
Sidewalk		1				

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km
5 = Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km
4 = Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km
3 = Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km
2 = Developed and connected for <1.9 km
1 = Sidewalk
0 = Not identified as a trail, sidewalk, or other path

Known Issues

- This indicator sometimes misclassifies sidewalks as greenways and trails because they are not tagged as sidewalks in the OpenStreetMap data.
- This indicator occasionally misclassifies driveways as "sidewalks and other paths" in places where they are not correctly tagged as private in OpenStreetMap. These typically appear as isolated pixels receiving a score of 1 on the indicator.
- OpenStreetMap does not provide a complete inventory of greenways and trails in the U.S. Caribbean. Paths that are missing from the source data will be underprioritized in this indicator. For example, some trails are missing within National Wildlife Refuges.
- This indicator includes trails and sidewalks from OpenStreetMap, which is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the path of a greenway) or incorrect tags (e.g., mislabeling a path as a footway that is actually a road for vehicles). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new greenways and trails to improve the accuracy and coverage of this indicator in the future.
- This indicator sometimes underestimates greenway length when connections route under bridges or along abandoned dirt roads. Some of these issues have been fixed through active testing and improvement, but some likely remain.
- Some greenways and trails continue along roadways that allow motorized vehicles, which are excluded from this indicator. As a result, certain trails may appear incomplete because the indicator only captures the sections dedicated for cyclists, pedestrians, and horseback riders.
- When calculating nearby impervious for one greenway, if there's another greenway within 300 m, impervious surface from the different but overlapping greenway buffer area is also used to compute natural condition. This is an unintended issue with the analysis methods. Investigation into potential fixes is ongoing.
- The indicator doesn't currently include areas where future greenways are planned.
- This indicator doesn't include Mona Island, even though there are important and popular trails, due to the lack of landcover data.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

American Planning Association. 2018. Recommendations for Future Enhancements to the Blueprint. [https://secassoutheast.org/pdf/Recommendations-for-Future-Enhancements-to-the-Blueprint-FINAL.pdf].

Institute for Transportation Research and Education (ITRE) & Alta Planning and Design. February 2018. Evaluating the Economic Impact of Shared Use Paths in North Carolina: 2015-2017 Final Report. [https://itre.ncsu.edu/wp-content/uploads/2018/03/NCDOT-2015-44_SUP-Project_Final-Report_optimized.pdf].

National Oceanic and Atmospheric Administration, Office for Coastal Management. "C-CAP Land Cover Files for Puerto Rico and US Virgin Islands". Coastal Change Analysis Program (C-CAP) High-Resolution Land Cover. Charleston, SC: NOAA Office for Coastal Management. Accessed November 2022. [https://www.coast.noaa.gov/htdata/raster1/landcover/bulkdownload/hires/].

OpenStreetMap. Highways. Data extracted through Geofabrik downloads. Accessed February 26, 2023. [https://wiki.openstreetmap.org/wiki/Highways].

Running USA. 23 March 2017. U.S. Road Race Trends. Road race finisher total experiences slight year-over-year decline in 2016. [https://web.archive.org/web/20170404232619/https://www.runningusa.org/2017-us-road-race-trends].

U.S. Department of Transportation. National Highway Traffic Safety Administration and the Bureau of Transportation Statistics. 2002. National Survey of Pedestrian & Bicyclist Attitudes and Behaviors: Highlights Report. [https://www.bts.gov/sites/bts.dot.gov/files/docs/browse-statistical-products-and-data/bts-publications/archive/203331/entire-1.pdf].

Yang, Limin, Jin, Suming, Danielson, Patrick, Homer, Collin G., Gass, L., Bender, S.M., Case, Adam, Costello, C., Dewitz, Jon A., Fry, Joyce A., Funk, M., Granneman, Brian J., Liknes, G.C., Rigge, Matthew B., Xian, George. 2018. A new generation of the United States National Land Cover Database—Requirements, research priorities, design, and implementation strategies: ISPRS Journal of Photogrammetry and Remote Sensing, v. 146, p. 108–123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].

Caribbean Habitat Patch Size (Large Islands)

This indicator represents the size of natural habitat patches on large islands in the U.S. Caribbean that are unfragmented by roads, urban development, or agriculture. It uses LANDFIRE landcover and road data, mimicking Esri's intact habitat cores approach from their green infrastructure data.

This indicator was last updated in October 2023.

Reason for Selection

Large areas of intact natural habitat are favorable for conservation of numerous species, including reptiles and amphibians, birds, and large mammals. Historically, many Caribbean islands experienced dramatic habitat alteration across much of their land mass from intensive colonial agriculture (such as sugar cane cultivation) and later by urbanization and other anthropogenic modifications (Fitzpatrick et al. 2007, CEPF 2023). As a result, contiguous natural habitat patches are particularly important for supporting native species, restoring native vegetation, and maintaining ecosystem services. This indicator roughly follows the same approach for evaluating patch size used by the intact habitat cores indicator in the continental part of the Southeast Blueprint, but lowers the highest patch size threshold, recognizing that the maximum patch size on small Caribbean islands is lower than in the continental Southeast and on large islands (Esri 2017).

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Caribbean island size: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) and Operational Roads (Roads) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data

Mapping Steps

- Reclassify the LANDFIRE EVT data into natural and unnatural classes. All classes in "EVT_NAME" that start with "Quarries", "Developed", or "Agriculture" are considered unnatural. "Caribbean bush fruit and berries", which captures sun coffee plantations, is also considered unnatural, while "tropical agroforestry plantation", which captures shade coffee, is considered natural. Water is classified as "natural".
- Reclassify the Caribbean island size layer to assign all islands a value of 1 and assign the ocean a value of NoData.
- Multiply the reclassified LANDFIRE natural/unnatural layer and island extent data. This makes the ocean NoData, but retains freshwater, smaller salt ponds, and enclosed brackish water areas, as well as terrestrial areas on the islands.
- Reclassify primary, secondary, and tertiary roads in the LANDFIRE roads layer as barriers.
- Combine the natural/unnatural raster and roads raster to identify natural areas without roads.

- Use region group to find the size of patch that each pixel belongs to.
- Reclassify based on the final indicator values seen below.
- Use the island extent layer to limit indicator to only large islands, excluding small and medium islands, which are captured in a different indicator.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Very large patch (>10,000 acres)
4 = Large patch (>1,000-10,000 acres)
3 = Medium patch (>100-1,000 acres)
2 = Small patch (>10-100 acres)
1 = Very small patch (≤10 acres)
0 = Developed or agriculture

Known Issues

- Some small dirt roads serve as hard boundaries for habitat cores. While this makes sense for some species, this indicator likely underestimates the effective size of the patch for some more mobile animals.
- This indicator doesn't account for variation in habitat condition within the patch.
- Some docks are overprioritized due to landcover misclassification. LANDFIRE sometimes misclassifies docks as forests, so they can get included in habitat patches for this indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Critical Ecosystem Partnership Fund (CEPF). 2023. Caribbean Islands – Threats. [https://www.cepf.net/our-work/biodiversity-hotspots/caribbean-islands/threats].

Esri Green Infrastructure Center, Inc. 2017. Esri Green Infrastructure Data Description. [https://www.arcgis.com/sharing/rest/content/items/5561da06127d48a982d6b91929f29666/data].

Fitzpatrick, Scott & Keegan, William. (2007). Human Impacts and Adaptations in the Caribbean Islands: an Historical Ecology Approach. Earth and Environmental Science Transactions of the Royal Society of Edinburgh. 98. 29 - 45. 10.1017/S1755691007000096. [https://www.cambridge.org/core/journals/earth-and-environmental-science-transactions-of-royal-society-of-edinburgh/article/abs/human-impacts-and-adaptations-in-the-caribbean-islands-an-historical-ecology-approach/68BFF9386CA9BDAD3FB44E5C937AC92F].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Operational Roads Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Caribbean Habitat Patch Size (Small Islands)

This indicator represents the size of natural habitat patches on small islands in the U.S. Caribbean that are unfragmented by roads, urban development, or agriculture. It uses LANDFIRE landcover and road data, mimicking Esri's intact habitat cores approach from their green infrastructure data.

This indicator was last updated in October 2023.

Reason for Selection

Large areas of intact natural habitat are favorable for conservation of numerous species, including reptiles and amphibians, birds, and large mammals. Historically, many Caribbean islands, especially smaller ones, experienced dramatic habitat alteration across much of their land mass from intensive colonial agriculture (such as sugar cane cultivation) and later by urbanization and other anthropogenic modifications (Fitzpatrick et al. 2007, CEPF 2023). As a result, contiguous natural habitat patches are particularly important for supporting native species, restoring native vegetation, and maintaining ecosystem services. This indicator roughly follows the same approach for evaluating patch size used by the intact habitat cores indicator in the continental part of the Southeast Blueprint (Esri 2017), but lowers the highest patch size threshold, recognizing that the maximum patch size on small Caribbean islands is lower than in the continental Southeast and on large islands.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Caribbean island size: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Southeast Blueprint 2023 extent

- 2020 LANDFIRE Existing Vegetation Type (EVT) and Operational Roads (Roads) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data

Mapping Steps

- Reclassify the LANDFIRE EVT data into natural and unnatural classes. All classes in "EVT_NAME" that start with "Quarries", "Developed", or "Agriculture" are considered unnatural. "Caribbean bush fruit and berries", which captures sun coffee plantations, is also considered unnatural, while "tropical agroforestry plantation", which captures shade coffee, is considered natural. Water is classified as "natural".
- Reclassify the Caribbean island size layer to assign all islands a value of 1 and assign the ocean a value of NoData.
- Multiply the reclassified LANDFIRE natural/unnatural layer and island extent data. This makes the ocean NoData, but retains freshwater, smaller salt ponds, and enclosed brackish water areas, as well as terrestrial areas on the islands.
- Reclassify primary, secondary, and tertiary roads in the LANDFIRE roads layer as barriers.
- Combine the natural/unnatural raster and roads raster to identify natural areas without roads.
- Use region group to find the size of patch that each pixel belongs to.
- Reclassify based on the final indicator values seen below.
- Make a small island mask using the island extent layer for "small" and "medium" islands. Use this mask to include only habitat patches within that mask, excluding large islands, which are captured in a different indicator.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = Large patch (>1,000 acres)
3 = Medium patch (>100-1,000 acres)
2 = Small patch (>10-100 acres)
1 = Very small patch (≤10 acres)
0 = Developed or agriculture

Known Issues

- Some small dirt roads serve as hard boundaries for habitat cores. While this makes sense for some species, this indicator likely underestimates the effective size of the patch for some more mobile animals.
- This indicator doesn't account for variation in habitat condition within the patch.

- Some docks are overprioritized due to landcover misclassification. LANDFIRE sometimes misclassifies docks as forests, so they can get included in habitat patches for this indicator.
- In Eastern St. Croix, where it's particularly dry, this indicator can underestimate or overestimate patch size. LANDFIRE has trouble differentiating pasture from natural thornscrub in this area. Pasture isn't considered natural in this indicator, so any areas of thornscrub misclassified as pasture would underestimate patch size, while areas of pasture classified as thornscrub would overestimate patch size.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Critical Ecosystem Partnership Fund (CEPF). 2023. Caribbean Islands – Threats. [https://www.cepf.net/our-work/biodiversity-hotspots/caribbean-islands/threats].

Esri Green Infrastructure Center, Inc. 2017. Esri Green Infrastructure Data Description. [https://www.arcgis.com/sharing/rest/content/items/5561da06127d48a982d6b91929f29666/data].

Fitzpatrick, Scott & Keegan, William. (2007). Human Impacts and Adaptations in the Caribbean Islands: an Historical Ecology Approach. Earth and Environmental Science Transactions of the Royal Society of Edinburgh. 98. 29 - 45. 10.1017/S1755691007000096. [https://www.cambridge.org/core/journals/earth-and-environmental-science-transactions-of-royal-society-of-edinburgh/article/abs/human-impacts-and-adaptations-in-the-caribbean-islands-an-historical-ecology-approach/68BFF9386CA9BDAD3FB44E5C937AC92F].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Operational Roads Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Caribbean Island Habitat

This indicator represents the importance of island habitat in the U.S. Caribbean for federally listed and other imperiled species based on the presence of imperiled and invasive animals. This indicator uses species data from Island Conservation's Threatened Island Biodiversity Database, U.S. Fish and Wildlife Service critical habitat, and the Puerto Rico and U.S. Virgin Islands Gap Analysis Project.

This indicator was last updated in October 2023.

Reason for Selection

Islands provide important habitat for many species of birds, reptiles, amphibians, mammals, insects, and plants. Due to their isolation, islands tend to be ecologically unique, making them hotspots for species diversity. In addition, the relative isolation of islands from disturbance and mainland predators can make them important breeding habitat for coastal birds and sea turtles. However, these factors also increase islands' vulnerability to invasive species (IUCN 2018).

According to Bernie 2015, "islands warrant a unique level of attention for biodiversity conservation because they make up only a small percentage of land area but are known for their many endemic species." This rich diversity is also disproportionately threatened, with "61% of all extinct species and 37% of all critically endangered species confined to islands" (Bernie 2015).

The critical habitat included in this indicator refers to areas with specific physical or biological features that are essential to conserving a federally threatened or endangered species and may require special management or protection.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Caribbean island size: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Southeast Blueprint 2023 extent
- Critical habitat provided by the U.S. Fish and Wildlife Survey, accessed 6-23-2022; download the data
- Threatened Island Biodiversity (TIB) Database, accessed 11-29-2022

  This database includes terrestrial vertebrate species that breed on islands and appear on the IUCN Red List as either Critically Endangered or Endangered. It also includes seabird species listed as Vulnerable and terrestrial vertebrates listed as Extinct in the Wild.

- Puerto Rico Gap Analysis Project (GAP) predicted vertebrate species distributions: data provided by Dr. Bill Gould with the Caribbean Climate Hub on 4-4-2022 (contact william.a.gould@usda.gov for more information); read the final report
- U.S. Virgin Islands GAP predicted vertebrate species distributions: data and report appendices provided by Dr. Bill Gould with the Caribbean Climate Hub on 2-6-2023 (contact william.a.gould@usda.gov for more information); read the final report

Mapping Steps

- From the TIB Database, in the "Invasive Species on Island" sheet, select islands where "Region_Archipelago" is "Greater Antilles (Puerto Rican Islands)", "US Virgin Islands", "US Virgin Islands (St. Croix Islands)", or "US Virgin Islands (St. Thomas Islands)".
- Count the number of invasives species records per island, ignoring any row where "Common_Name" is "NONE".
- From the TIB Database, in the "Threatened Species on Island" sheet, select islands where "Region_Archipelago" is "Greater Antilles (Puerto Rican Islands)", "US Virgin Islands", "US Virgin Islands (St. Croix Islands)", or "US Virgin Islands (St. Thomas Islands)".
- Count the number of threated species records per island where "Present_Breeding_Status" is "Confirmed" or "Potential Breeding".
- Combine the invasive and threatened species data using "Island_GID_Code" and fill in invasive or threatened species counts with 0 if islands didn't have a record in either of the sheets.
- Include only islands < 50 km^2 (smaller than St. Thomas) using the "Corrected_Area_KM2" field. The input data on imperiled species and invasive animals did not include species distributions within individual islands. That was less of an issue on smaller islands but made the source data less informative for larger islands.
- Extract "Corrected_Latitude" and "Corrected_Longitude" of islands from the "Invasive Species on Islands" sheet, then create shapefile of the summarized data.
- Spatially join the summarized data from TIB Database to the Caribbean island size layer.
- Many species in the island biodiversity database have habitat on large islands, but the database doesn't provide sufficiently detailed species distribution information within those islands to be used as an indicator. To address this, use predicted habitat models from Puerto Rico and U.S. Virgin Islands GAP for species listed in the TIB Database that live on large islands (i.e., >50 $km^{2)}$. These species were: Puerto Rican parrot, Culebra Island giant anole, Puerto Rican nightjar, Mottled coqui, Cricket coqui, Hedrick's coqui, Golden coqui, Locust coqui, Forest coqui, Richmond's coqui, Wrinkled coqui, Virgin Island tree boa, and Desecheo dwarf gecko.
- To prepare the GAP data, begin by fixing issues in two species layers. For Puerto Rican nightjar (*Caprimulgus noctitherus*), convert NoData to 0 to fill in a square of NoData in the center of Puerto Rico. For forest coqui (*Eleutherodactylus portoricensis*), add 1 to the values to make them consistent with the values used in the other layers for presence and absence.
- Sum the values of the species models and subtract by the total number of species to get species counts for each pixel. GAP uses 2 to indicate presence and 1 to indicate absence.
- Combine the Puerto Rico and U.S. Virgin Islands GAP species counts into a single raster. Then snap and reproject based on the Caribbean Blueprint subregion.
- Clip the critical habitat to the Caribbean islands extent layer and convert to raster. Include all critical habitat in the area except for yellow-shoulder blackbird. The large area of poor-quality habitat covered by critical habitat for this species was causing significant overprioritization in this indicator and local reviewers recommended removing it. This still results in important

areas for the species being included based on models for other species but greatly reduces overall overprioritization in many places.
- Combine data from the TIB database, critical habitat, and Puerto Rico and U.S. Virgin Islands GAP as shown in the final indicator values below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

7 = Island area with critical habitat for a threatened or endangered species
6 = Island area with no invasive animals and 2+ imperiled species
5 = Island area with no invasive animals and 1 imperiled species
4 = Island area with no invasive animals
3 = Island area with invasive animals and 2+ imperiled species
2 = Island area with invasive animals and 1 imperiled species
1 = Island area with invasive animals
0 = Not an island

Known Issues

- This indicator underestimates habitat values for some small islands due to a lack of data in the Threatened Island Biodiversity Database. For some areas on large islands, like Vieques National Wildlife Refuge, it underestimates habitat values due to a lack of detailed distribution models for some species.
- This indicator likely underpredicts important habitat for federally listed species. Not all federally listed species have critical habitat designated, and not all of the critical habitat data designated by the U.S. Fish and Wildlife Service is available from the data source used.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Bernie R. Tershy, Kuo-Wei Shen, Kelly M. Newton, Nick D. Holmes, Donald A. Croll, The Importance of Islands for the Protection of Biological and Linguistic Diversity, BioScience, Volume 65, Issue 6, June 2015, Pages 592–597. [https://doi.org/10.1093/biosci/biv031].

Gould, William A.; Alarcón, Caryl; Fevold, Brick; Jiménez, Michael E.; Martinuzzi, Sebastián; Potts, Gary; Quiñones, Maya; Solórzano, Mariano; Ventosa, Eduardo. 2008. The Puerto Rico Gap Analysis Project. Volume 1: Land cover, vertebrate species distributions, and land stewardship. Gen. Tech. Rep. IITF-GTR-39. Río Piedras, PR: U.S. Department of Agriculture, Forest Service, International Institute of Tropical Forestry. 165 p. [https://data.fs.usda.gov/research/pubs/iitf/iitf_gtr39.pdf].

Gould WA, Solórzano MC, Potts GS, Quiñones M, Castro-Prieto J, Yntema LD. 2013. U.S. Virgin Islands Gap Analysis Project – Final Report. USGS, Moscow ID and the USDA FS International Institute of Tropical Forestry, Río Piedras, PR. 163 pages and 5 appendices. [http://www.thinkamap.com/share/IndividualGISdata/PDFs/USVI_FINAL_REPORT.pdf].

IUCN (2018). Guidelines for invasive species planning and management on islands. Cambridge, UK and Gland, Switzerland: IUCN. viii + 40pp. [https://portals.iucn.org/library/sites/library/files/documents/2018-030-En.pdf].

Sayre, R., S. Noble, S. Hamann, R. Smith, D. Wright, S. Breyer, K. Butler, K. Van Graafeiland, C. Frye, D. Karagulle, D. Hopkins, D. Stephens, K. Kelly, Z, basher, D. Burton, J. Cress, K. Atkins, D. van Sistine, B. Friesen, B. Allee, T. Allen, P. Aniello, I Asaad, M. Costello, K. Goodin, P. Harris, M. Kavanaugh, H. Lillis, E. Manca, F. Muller-Karger, B. Nyberg, R. Parsons, J. Saarinen, J. Steiner, and A. Reed. 2018. A new 30 meter resolution global shoreline vector and associated global islands database for the development of standardized global ecological coastal units. Journal of Operational Oceanography–A Special Blue Planet Edition. [https://doi.org/10.1080/1755876X.2018.1529714].

Threatened Island Biodiversity Database Partners. 2018. The Threatened Island Biodiversity Database: developed by Island Conservation, University of California Santa Cruz Coastal Conservation Action Lab, BirdLife International and IUCN Invasive Species Specialist Group. Version 2018. Downloaded on November 2022. [http://tib.islandconservation.org/].

U.S. Fish and Wildlife Service. Critical Habitat. Accessed June 23, 2022. [https://ecos.fws.gov/ecp/report/table/critical-habitat.html].

Caribbean Karst Habitat

This indicator for the U.S. Caribbean represents natural karst areas with limited human alteration from activities such as urban development and intensive agriculture. Karst is a geologically unique landscape where the movement of water through easily dissolved bedrock, particularly limestone, produces distinctive features like caves, sinkholes, and underground rivers. This indicator combines LANDFIRE landcover with karst datasets from the Puerto Rico Department of Natural and Environmental Resources, National Park Service, and U.S. Geological Survey.

This indicator was last updated in October 2023.

Reason for Selection

"Karst is a type of landscape where the dissolving of the bedrock has created sinkholes, sinking streams, caves, springs, and other characteristic features. Karst is associated with soluble rock types such as limestone, marble, and gypsum. In general, a typical karst landscape forms when much of the water falling on the surface interacts with and enters the subsurface through cracks, fractures, and holes that have been dissolved into the bedrock. After traveling underground, sometimes for long distances, this water is then discharged from springs, many of which are cave entrances" (NPS 2022). The Caribbean region has one of the most significant karst landscapes in the world and is recognized by the International Union for Conservation of Nature (IUCN) World Commission of Protected Areas as in need of protection (Day 2010). In Puerto Rico, the unique karst landscape provides habitat for many endemic and rare species, such as the Puerto Rican boa, the Puerto Rican skink, Puerto Rican sharp-shinned hawk, and many federally listed plants (Lugo et al. 2001). In addition, the Puerto Rico karst belt contains the island's largest mature forests, coastal wetlands, estuaries, and underground cave systems. The northern limestone region supplies 22% of Puerto Rico's public drinking water (Lugo et al. 2001).

On St. Croix, the karst landscape supports the largest aquifer in the U.S. Virgin Islands, which covers approximately 25 mi^2 and produces about 67% of the total groundwater withdrawn in the U.S. Virgin Islands (Zack 1986; Dr. Olassee Davis - Ecologist and Extension Specialist with the University of the Virgin Islands, personal communication, July 6, 2023). Caribbean karst is also an important cultural landscape. For example, caves served as refuges for Maroons during colonization (Day 2010). Cave systems within this landscape also provide opportunities for low-impact nature-based tourism (Hall 2014).

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- U.S. Geological Survey Karst in the United States: A digital map compilation and database, last modified 8-1-2014, accessed 5-18-2023
- National Park Service Digital Geologic Map of St. Croix and Buck Island Reef National Monument, U.S. Virgin Islands (published 4-28-2008), accessed 6-22-2023
- Puerto Rico Karst Restricted Zone shapefile, shared on 6-15-2023 by Rossana Vidal Rodriguez with the Puerto Rico Department of Natural and Environmental Resources (PR DNER) (rvidal@drna.pr.gov)

  This is an improved version of the Karst Conservation Zone layer (also known as "zona_amortiguamiento_carso" layer) found in the 2018 Puerto Rico Natural Areas database. The Karst Conservation Zone layer is a 50 m buffer around the Karst Restricted Zone. The delimitation of the Karst Restricted Zone originated from the PR DNER's 2008 Study of Karst .

Mapping Steps

- Create a karst shapefile for St. Thomas and St. John by selecting all features in the USGS Karst file VIcarbonates.shp except where UNIT_NAME = 'Kingshill Marl'. That geologic unit appears in St. Croix and we exclude it because we have better data for St. Croix from the National Park Service.
- Create a karst shapefile for St. Croix from the National Park Service St. Croix geology map by selecting GLG_SYM = 'Mikh' from buisglg.shp. MIkh is the code for Kingshill Marl.
- Merge the karst shapefiles created for St. John, St. Thomas, and St. Croix with the Puerto Rico Karst Restricted Zone and Karst Conservation Zone buffer shapefiles, then convert to raster. This creates a combined karst layer for the U.S. Virgin Islands and Puerto Rico.
- Reclassify the LANDFIRE EVT data into 3 alteration classes where 3 is natural, 2 is altered, and 1 is heavily altered. Assign a value of 1 to all pixels with a value in the EVT_NAME field of "Developed-High Intensity" or "Developed-Medium Intensity". Assign a value of 2 to all pixels with a value in the EVT_LF field of "Developed", "Agriculture", or "Barren", or a value in the EVT_NAME field of "Agriculture-Pasture and Hay" or "Caribbean Bush fruit and berries". Assign a value of 3 to everything else. Classify pixels with a value in the EVT_NAME field of "Open Water" as NoData. Note: The Caribbean bush fruit and berries LANDFIRE class is intended to represent sun-grown coffee, while tropical agroforestry plantation is intended to capture shade-grown coffee (LANDFIRE help desk, personal communication, 2-22-23). As a result, we considered Caribbean bush fruit and berries to be in the middle alteration class and considered tropical agroforestry plantation to be natural for the purposes of this indicator.
- Combine the karst raster with the reclassified LANDFIRE EVT data to produce the final indicator values shown below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Karst with natural landcover
2 = Karst with pasture, crops, or open space in developed area
1 = Karst with developed landcover
0 = Not identified as karst

Known Issues

- Some karst areas are underprioritized in Puerto Rico because they are not included in the Karst Restricted Zone. While the Karst Restricted Zone focuses on the most sensitive karst areas, it does not include all karst on the island.

- Some areas are both under- and overprioritized in St. Thomas. The USGS karst data are not as fine-resolution as other data in this indicator and are likely under- and overpredicting karst in areas along the boundary of the polygons used.
- This indicator likely misses some karst areas in the U.S. Caribbean that are not captured in the input datasets.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Day, Mick. 2010. Human Interaction with Caribbean Karst Landscapes: Past, Present and Future. Acta Carsologica. 39. 137-146. 10.3986/ac.v39i1.119. Accessed June 27, 2023. [https://ojs.zrc-sazu.si/carsologica/article/view/119/109].

Department of Nature and Environment Resources (DNER). 2009. Declaración de evaluación de impacto ambiental estratégica. Estudio del carso. (trans. Strategic Environmental Impact Statement. Study of Karst.) [https://secassoutheast.org/pdf/DIA-E_estudio_carso.pdf].

Hall, A., and Day, M., Ecotourism in the state forest karst of Puerto Rico. April 2014. Journal of Cave and Karst Studies, v. 76, no. 1, p. 30–41. DOI: 10.4311/2011SS0263 [https://legacy.caves.org/pub/journal/PDF/v76/cave-76-01-30r.pdf].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Lugo, Ariel E.; Castro, Leopoldo Miranda; Vale, Abel; López, Tania del Mar; Prieto, Enrique Hernández; Martinó, Andrés García; Rolón, Alberto R. Puente; Tossas, Adrianne G.; McFarlane, Donald A.; Miller, Tom; Rodríguez, Armando; Lundberg, Joyce; Thomlinson, John; Colón, José ; Schellekens, Johannes H.; Ramos, Olga; Helmer, Eileen. 2001. Puerto Rican Karst-A Vital Resource. United States Department of Agriculture Forest Service Gen. Tech. Report WO-65. [https://doi.org/10.2737/WO-GTR-65].

National Park Service. Karst Landscapes. Last updated April 27, 2022. Accessed August 16, 2023. [https://www.nps.gov/subjects/caves/karst-landscapes.htm].

Protected Areas Conservation Action Team. 2018. Puerto Rico Protected Areas Database. December 2018 version. GIS data. San Juan, PR. [https://research.fs.usda.gov/iitf/products/dataandtools/puerto-rico-protected-natural-areas-2018].

Weary, D.J., and Doctor, D.H., Karst in the United States of America: a digital map compilation and database: U.S. Geological Survey Open-file Report 2014-1156. [https://pubs.usgs.gov/of/2014/1156/].

Zack, Allen, Rodriques-Alonso, Teresita, and Roman-Mas, Angel. 1986. U.S. Virgin Islands Ground-water Quality report. U.S. Geological Survey Open-File Report 87-0756. [https://pubs.usgs.gov/of/1987/0756/report.pdf].

Caribbean Landscape Condition

This indicator for the U.S. Caribbean represents natural areas with limited human alteration while also considering the natural landcover of the surrounding landscape. Examples of human alteration include urban development and intense agricultural use. This indicator uses LANDFIRE landcover and ideas from the Florida Critical Lands and Waters Identification Project's approach for evaluating landscape integrity.

This indicator was last updated in October 2023.

Reason for Selection

A high degree of naturalness across the landscape benefits species diversity as well as ecosystem services such as pollinator habitat, increased water infiltration, and reduced soil erosion. Though many Caribbean habitats have experienced human alternation at some point, natural landcover across the wider landscape provides many benefits. It allows species to disperse during different life stages, better adapt to a changing climate by accessing refugia, and freely move between different habitats. Natural landscapes can also complement existing protected areas and help increase resilience to extreme weather events such as flooding and hurricanes (Kremen and Merenlender 2018).

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Caribbean island size: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data

Mapping Steps

- Reclassify the LANDFIRE EVT data into 3 alteration classes where 3 is natural, 2 is altered, and 1 is heavily altered. Assign a value of 1 to all pixels with a value in the EVT_NAME field of "Developed-High Intensity" or "Developed-Medium Intensity". Assign a value of 2 to all pixels with a value in the EVT_LF field of "Developed", "Agriculture", or "Barren", or a value in the EVT_NAME field of "Agriculture-Pasture and Hay" or "Caribbean Bush fruit and berries". Assign a value of 3 to everything else. Classify pixels with a value in the EVT_NAME field of "Open Water" as 3. Note: The Caribbean bush fruit and berries LANDFIRE class is intended to represent sun-grown coffee, while tropical agroforestry plantation is intended to capture

shade-grown coffee (LANDFIRE help desk, personal communication, 2-22-23). As a result, we considered Caribbean bush fruit and berries to be in the middle alteration class and considered tropical agroforestry plantation to be natural for the purposes of this indicator.

- Reclassify the Caribbean island size layer to assign all islands a value of 1 and assign the ocean a value of NoData.
- Multiply the reclassified LANDFIRE natural/altered layer and island extent data. This makes the ocean NoData, but retains freshwater, smaller salt ponds, and enclosed brackish water areas, as well as terrestrial areas on the islands.
- Many species and ecological processes operate at multiple scales. To account for this, estimate the average amount of alteration using a moving window (or neighborhood) analysis at 4 different scales: approximately 0.22 acres (single pixel), approximately 10 acres; approximately 100 acres; and approximately 1,000 acres. Then average the values across all scales. This results in continuous values from 1 to 3.
- Bin the continuous values into the following categories seen in the final indicator values below: 1 (heavily altered): 1 to <1.5; 2 (altered): 1.5 to <2; 3 (partly natural): 2 to <2.5; 4 (mostly natural): 2.5 to <2.9; 5 (natural): 2.9 to 3.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Natural landscape
4 = Mostly natural landscape
3 = Partly natural landscape
2 = Altered landscape
1 = Heavily altered landscape

Known Issues

- This indicator overestimates landscape condition in some marinas and nearby areas because LANDFIRE often classifies marinas as ruderal forests (i.e., a forest type made up of native and non-native early successional species that are the first to colonize a disturbed area).
- This indicator does not account for variation in habitat condition due to invasive species.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Kremen, Claire, and Adina M. 2018. Merenlender. Landscapes that work for biodiversity and people. Science 362.6412. Eaau6020. [https://www.science.org/doi/10.1126/science.aau6020].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Oetting J, Hoctor T, and Volk M. 2016.Critical Lands and Waters Identification Project (CLIP): Version 4.0 Technical Report. Accessed Nov 10, 2022. [https://www.fnai.org/PDFs/CLIP_v4_technical_report.pdf].

Caribbean Low-Urban Historic Landscapes

This cultural resource indicator is an index of sites on the National Register of Historic Places and other historic sites surrounded by limited urban development in the U.S. Caribbean. It identifies significant historical places that remain connected to their context in the natural world. This indicator uses LANDFIRE landcover and historic places data from the Puerto Rico State Historic Preservation Office, OpenStreetMap, and the University of the Virgin Islands.

This indicator was last updated in October 2023.

Reason for Selection

Low-urban historic landscapes indicate significant cultural landscapes whose cultural context has been less impacted by urban development. Cultural landscapes are “properties [that] represent the combined works of nature and of man” (UNESCO 2012). Loss of natural habitat within these cultural landscapes reduces their overall historical and cultural value.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- The following The National Register of Historic Places data for Puerto Rico provided by Eduardo Cancio, Information Systems Specialist with the Puerto Rico State Historic Preservation Office (SHPO) on 2-21-2023 (contact ecancio@prshpo.pr.gov for more information):
    - NRHP_PR_individual_properties.shp
    - NRHP_PR_lineal_districts.shp
    - NRHP_PR_polygonal_districts.shp

The National Register of Historic Places reflects what Americans value in their historic built environment. It is the collection of our human imprint on the landscape that records through time our changing relationship with the landscape, bridging between modern life and our history by providing, as closely as possible, experiences that evoke our empathy and understanding of previous eras.

- OpenStreetMap data "multipolygons" layer, accessed 3-14-2023

  A polygon from this dataset is considered a historic site if the "historic" tag is not null. In OpenStreetMap, a historic feature refers to "features that still exist or of which traces are observable, and that are of historic interest, or where the feature class is generally of historical interest". We only used historic polygons if the name tag is also not null. OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- Select USVI historic districts: Polygon boundaries for the Christiansted National Historic District on St. Thomas and Charlotte Amalie Historic and Architectural Historic District on St. Croix, provided by Nikita Beck with the University of the Virgin Islands on 3-6-2023 (contact nikita.beck@uvi.edu for more information)

Mapping Steps

- Identify urban areas using the following classes from 2020 LANDFIRE EVT: Developed-High Intensity, Developed-Low Intensity, Developed-Medium Intensity, Developed-Open Space, Developed-Roads. Classify all urban pixels as 1 and all other pixels as 0.
- Calculate the percent urban in a 270 m radius circle for each pixel using the ArcGIS Focal Statistics function. Since the LANDFIRE data resolution is 30 m, 270 m (9 pixels) approximates a 250 m radius. Retain all pixels that are <50% urban within a 270 m radius.
- Create a historic places layer by combining the following vector datasets as follows:
  - Buffer National Register point data from the Puerto Rico SHPO by 100 m.
  - Combine National Register polygons from the Puerto Rico SHPO, select USVI historic districts, and OpenStreetMap polygons. Only use OpenStreetMap polygons if both the historic and name columns are null. Buffer the polygons by 30 m.
  - Buffer line data from the Puerto Rico SHPO by 30 m.
  - Merge all buffered point, polygon, and line data into one layer and convert to a 30 m raster representing historic places.
- Use the historic places raster to remove areas that fall outside of the historic places.
- Reclassify the above raster into 3 classes, seen in the final indicator values below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Historic place with nearby low-urban buffer
1 = Historic place with nearby high-urban buffer
0 = Not identified as a historic place

Known Issues

- There are likely spatial mapping errors for some of the historic areas.
- Some historic areas with cultural importance are not captured in the National Register of Historic Places.
- The approach to measuring urban development doesn't capture degradation to historic places that were historically in larger cities (e.g., courthouses and other downtown buildings). It also doesn't distinguish between historic places that have always been urban and historic places that used to be low-urban.
- This layer likely underrepresents some historic areas in the U.S. Virgin Islands compared to Puerto Rico because we were unable to incorporate historic places data from the USVI SHPO during the timeline of this Blueprint update. As a result, some sites on the National Register of Historic Places are not depicted in this indicator.
- OpenStreetMap is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the boundary of a historic site) or incorrect tags (e.g., labelling an area as a historic site that does not have historic value). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new historic sites to improve the accuracy and coverage of this indicator in the future.
- Because open water is considered a non-urban landcover for the purposes of this analysis, this indicator is likely overprioritizing some urbanized historic areas that are close to water, such as marinas and bridges.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

OpenStreetMap. Historic. Data extracted through Geofabrik downloads. Accessed March 14, 2023. [https://wiki.openstreetmap.org/wiki/Key:historic].

UNESCO (2012) Operational Guidelines for the Implementation of the World Heritage Convention [1]. UNESCO World Heritage Centre. Paris. Page 14. [https://whc.unesco.org/archive/opguide12-en.pdf].

Caribbean Reforestation Potential

This indicator prioritizes areas to increase tree cover in the U.S. Caribbean based on current land uses and potential benefits for local drinking water supplies. It includes opportunities to improve water quality and species habitat by transitioning sun-grown coffee production to shade-grown; enhancing the overstory of shade-grown coffee areas; and reforesting open space in developed areas, pasture, and agricultural lands. The highest scores represent coffee plantations in watersheds with reservoirs. This indicator uses LANDFIRE landcover and U.S. Geological Survey watershed boundaries.

This indicator was last updated in October 2023.

Reason for Selection

This indicator reflects the importance of restoring forests in the Caribbean to provide quality habitat for species and protect local drinking water supplies. Agriculture, timber harvesting, and development have greatly reduced the extent of Caribbean forests, resulting in habitat loss for many species and declines in water quality due to increased nutrient runoff and sedimentation (Newkirk II 2018, Brash 1987, Lyons and Gartner 2017). In watersheds that contain reservoirs used for local drinking water, reducing sedimentation is particularly important. Not only does sediment accumulation in reservoirs cause “loss in storage capacity of the dam, but it may also lead to many negative effects both within and downstream of reservoirs which may include high turbidity, loss of flood-carrying capacity and reduction in water quality” (Adeogun et al. 2018). These negative impacts can incur high financial and environmental costs (Kawashima 2007).

Protecting drinking water and improving the environmental sustainability of coffee production is a major focus of reforestation efforts in the Caribbean by partners like the Natural Resources Conservation Service (NRCS) (Frank Velazquez – Assistant State Conservationist for the NRCS Caribbean Area State Office, personal communication, 2-9-2023). The intent of this indicator is not to eliminate coffee production, but rather to support farmers in maintaining and enhancing forest cover in coffee plantations to “protect, enhance, and conserve soil, water, and wildlife habitat” (NRCS 2019).

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- U.S. Geological Survey Watershed Boundary Dataset (WBD), accessed 1-5-2023: HUC12s; download the data

Mapping Steps

- Export the HUC12 watersheds in the Caribbean from the WBD using states = 'VI' or states = 'PR'.
- Make a raster of coffee locations with LANDFIRE using the classes "Tropical agroforestry plantation" and "Caribbean bush fruit and berries". The Caribbean bush fruit and berries LANDFIRE class is intended to represent sun-grown coffee, while tropical agroforestry plantation is intended to capture shade-grown coffee (LANDFIRE help desk, personal communication, 2-22-23).
- Make a raster of potential reforestation areas with LANDFIRE using coffee areas from the step above, plus the Developed-Open Space class and agriculture classes (Pasture and Hay, Cultivated Crops and Irrigated Agriculture).
- From the WBD, identify HUC12 watersheds with reservoirs. A watershed is assumed to contain a reservoir if it contains the word "Dam" in the "name" attribute.
- Convert HUC12 watersheds with reservoirs from polygon to raster.
- Combine the above rasters and reclassify to produce the final indicator values seen below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = Highest reforestation potential (coffee production in watershed with reservoir)
3 = Very high reforestation potential (other coffee production)
2 = High reforestation potential (open space in developed area, pasture, or crops in watershed with reservoir)
1 = Medium reforestation potential (other open space in developed area, pasture, or crops)
0 = Low reforestation potential (already natural or developed)

Known Issues

- This indicator does not account for differences in canopy cover among existing coffee plantations. It assigns the same level of priority to both sun-grown and shade-grown coffee. While sun-grown coffee is often considered a higher priority for reforestation (Almodovar 2013, NRCS 2019), even plantations that are considered shade-grown can vary in their level of canopy cover and would benefit from management or restoration to maintain or enhance the overstory.
- The methods used to identify watersheds that contain reservoirs do not distinguish between reservoirs used for drinking water and reservoirs used for other purposes, such as recreation

or flood control. While drinking water quality is a major driver of this indicator, not all reservoirs identified in this indicator are necessarily used for drinking water.

- This indicator underprioritizes reforestation in some areas that are classified as natural but have significantly fewer trees than they should. Some ecosystems in the Caribbean (e.g., thorn scrub in eastern St. Croix), do not have many trees due to climate and geology. However, other areas classified as natural are lacking trees due to invasive species, grazing, or other human disturbance.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

Adeogun, Adeniyi Ganiyu and Bolaji Fatai Sule, Adebayo Wahab Salami. Cost effectiveness of sediment management strategies for mitigation of sedimentation at Jebba Hydropower reservoir, Nigeria. Journal of King Saud University - Engineering Sciences, Volume 30, Issue 2. 2018. Pages 141-149, ISSN 1018-3639. [https://doi.org/10.1016/j.jksues.2016.01.003].

Almodovar, Edwin. Natural Resources Conservation Service. Shade-Grown Coffee Protects Puerto Rico Bay Coral Reefs. April 8, 2013. Accessed July 26, 2023. [https://www.usda.gov/media/blog/2013/04/08/shade-grown-coffee-protects-puerto-rico-bay-coral-reefs].

Brash, Alexander R. The history of avian extinction and forest conversion on Puerto Rico. Biological Conservation, Volume 39, Issue 2, 1987, Pages 97-111, ISSN 0006-3207. [https://doi.org/10.1016/0006-3207(87)90028-0].

Kawashima, S. (2007), Conserving reservoir water storage: An economic appraisal, Water Resources Research, 43, W05417. [https://doi.org/10.1029/2006WR005090].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Lyons, Katie, and Todd Gartner. World Resources Institute. 3 Surprising Ways Water Depends on Healthy Forests. March 21, 2017. [https://www.wri.org/insights/3-surprising-ways-water-depends-healthy-forests].

Natural Resource Conservation Service (2019). Shade Coffee for the Caribbean. Accessed July 25, 2023. [https://www.nrcs.usda.gov/conservation-basics/conservation-by-state/caribbean-area/shade-coffee-for-the-caribbean].

Newkirk, Vann R. II. The 150-Year Mission to Reforest Puerto Rico. The Atlantic – Science. September 25, 2018. Accessed July 26, 2023. [https://www.theatlantic.com/science/archive/2018/09/the-150-year-mission-to-reforest-puerto-rico/571219/].

U.S. Geological Survey. National Hydrography Products – Watershed Boundary Dataset. Accessed January 5, 2023. [https://www.usgs.gov/national-hydrography/access-national-hydrography-products].

Caribbean Urban Park Size

This cultural resource indicator measures the size of parks and beaches in the urban environment in the U.S. Caribbean. It uses several protected areas and beach datasets (e.g., the Protected Areas Database of the United States, OpenStreetMap) and U.S. Census Bureau urban areas.

This indicator was last updated in October 2023.

Reason for Selection

Protected natural areas in urban environments provide urban residents a nearby place to connect with nature and offer refugia for some species. Because beaches in Puerto Rico and the U.S. Virgin Islands are open to the public, beaches also provide important outdoor recreation opportunities for urban residents, so we include beaches as parks in this indicator.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI) Coastal Relief Model, accessed 11-22-2022
- Protected Areas Database of the United States (PAD-US) 3.0: VI, PR, and Marine Combined Fee Easement
- Puerto Rico Protected Natural Areas 2018 (December 2018 update): Terrestrial and marine protected areas (PACAT2018_areas_protegidasPR_TERRESTRES_07052019.shp, PACAT2018_areas_protegidasPR_MARINAS_07052019.shp)
- 2020 Census Urban Areas from the U.S. Census Bureau's urban-rural classification; download the data, read more about how urban areas were redefined following the 2020 census
- OpenStreetMap data "multipolygons" layer, accessed 3-14-2023

  A polygon from this dataset is considered a park if the "leisure" tag attribute is either "park" or "nature_reserve" and considered a beach if the value in the "natural" tag attribute is "beach". OpenStreetMap describes leisure areas as "places people go in their spare time" and natural areas as "a wide variety of physical geography, geological and landcover features". Data were downloaded in .pbf format and translated to an ESRI shapefile using R code. OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the

OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- TNC Lands - Public Layer, accessed 3-8-2023
- U.S. Virgin Islands beaches layer (separate vector layers for St. Croix, St. Thomas, and St. John) provided by Joe Dwyer with Lynker/the NOAA Caribbean Climate Adaptation Program on 3-3-2023 (contact jdwyer@lynker.com for more information)

Mapping Steps

- Most mapping steps were completed using QGIS (v 3.22) Graphical Modeler.
- Fix geometry errors in the PAD-US PR data using Fix Geometry. This must be done before any analysis is possible.
- Merge the terrestrial PR and VI PAD-US layers.
- Use the NOAA coastal relief model to restrict marine parks (marine polygons from PAD-US and Puerto Rico Protected Natural Areas) to areas shallower than 10 m in depth. The deep offshore areas of marine parks do not meet the intent of this indicator to capture nearby opportunities for urban residents to connect with nature.
- Merge into one layer the resulting shallow marine parks from marine PAD-US and the Puerto Rico Protected Natural Areas along with the combined terrestrial PAD-US parks, OpenStreetMap, TNC Lands, and USVI beaches. Omit from the Puerto Rico Protected Areas layer the “Zona de Conservación del Carso”, which has some policy protections and conservation incentives but is not formally protected.
- Fix geometry errors in the resulting merged layer using Fix Geometry.
- Intersect the resulting fixed file with the Caribbean Blueprint subregion.
- Process all multipart polygons to single parts (referred to in Arc software as an “explode”). This helps the indicator capture, as much as possible, the discrete units of a protected area that serve urban residents.
- Clip the census urban area to the Caribbean Blueprint subregion.
- Select all polygons that intersect the census urban extent within 1.2 miles (1,931 m). The 1.2 mi threshold is consistent with the average walking trip on a summer day (U.S. DOT 2002) used to define the walking distance threshold used in the greenways and trails indicator. Note: this is further than the 0.5 mi distance used in the continental version of the indicator. We extended it to capture East Bay and Point Udall based on feedback from the local conservation community about the importance of the park for outdoor recreation.
- Dissolve all the park polygons that were selected in the previous step.
- Process all multipart polygons to single parts (“explode”) again.
- Add a unique ID to the selected parks. This value will be used to join the parks to their buffers.
- Create a 1.2 mi (1,931 m) buffer ring around each park using the multiring buffer plugin in QGIS. Ensure that “dissolve buffers” is disabled so that a single 1.2 mi buffer is created for each park.
- Assess the amount of overlap between the buffered park and the census urban area using overlap analysis. This step is necessary to identify parks that do not intersect the urban area,

but which lie within an urban matrix. This step creates a table that is joined back to the park polygons using the UniqueID.

- Remove parks that had ≤2% overlap with the urban areas when buffered. This excludes mostly non-urban parks that do not meet the intent of this indicator to capture parks that provide nearby access for urban residents. Note: In the continental version of this indicator, we used a threshold of 10%. In the Caribbean version, we lowered this to 2% in order to capture small parks that dropped out of the indicator when we extended the buffer distance to 1.2 miles.
- Calculate the GIS acres of each remaining park unit using the Add Geometry Attributes function.
- Join the buffer attribute table to the previously selected parks, retaining only the parks that exceeded the 2% urban area overlap threshold while buffered.
- Buffer the selected parks by 15 m. Buffering prevents very small parks and narrow beaches from being left out of the indicator when the polygons are converted to raster.
- Reclassify the polygons into 7 classes, seen in the final indicator values below. These thresholds were informed by park classification guidelines from the National Recreation and Park Association, which classify neighborhood parks as 5-10 acres, community parks as 30-50 acres, and large urban parks as optimally 75+ acres (Mertes and Hall 1995).
- Export the final vector file to a shapefile and import to ArcGIS Pro.
- Convert the resulting polygons to raster using the ArcPy Polygon to Raster function. Assign values to the pixels in the resulting raster based on the polygon class sizes of the contiguous park areas.
- Assign a value of 0 to all other pixels in the Southeast Blueprint 2023 Caribbean subregion not already identified as an urban park in the mapping steps above. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = 75+ acre urban park
5 = >50 to <75 acre urban park
4 = 30 to <50 acre urban park
3 = 10 to <30 acre urban park
2 = 5 to <10 acre urban park
1 = <5 acre urban park
0 = Not identified as an urban park

Known Issues

- This indicator does not include park amenities that influence how well the park serves people and should not be the only tool used for parks and recreation planning. Park standards should be determined at a local level to account for various community issues, values, needs, and available resources.
- This indicator includes some protected areas that are not open to the public and not typically thought of as "parks", like mitigation lands, private easements, and private golf courses. While we experimented with excluding them using the public access attribute in PAD, due to numerous inaccuracies, this inadvertently removed protected lands that are known to be publicly accessible. As a result, we erred on the side of including the non-publicly accessible lands.
- This indicator includes parks and beaches from OpenStreetMap, which is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the boundary of a park) or incorrect tags (e.g., labelling an area as a park that is not actually a park). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new parks to improve the accuracy and coverage of this indicator in the future.

Other Things to Keep in Mind

- This indicator calculates the area of each park using the park polygons from the source data. However, simply converting those park polygons to raster results in some small parks and narrow beaches being left out of the indicator. To capture those areas, we buffered parks and beaches by 15 m and applied the original area calculation to the larger buffered polygon, so as not to inflate the area by including the buffer. As a result, when the buffered polygons are rasterized, the final indicator has some areas of adjacent pixels that receive different scores. While these pixels may appear to be part of one contiguous park or suite of parks, they are scored differently because the park polygons themselves are not actually contiguous.
- The Caribbean version of this indicator uses a slightly different methodology than the continental Southeast version. It includes parks within a 1.2 mi distance from the census urban area, compared to 0.5 mi in the continental Southeast. We extended it to capture East Bay and Point Udall based on feedback from the local conservation community about the importance of the park for outdoor recreation. Similarly, this indicator uses a 2% threshold of overlap between buffered parks and the census urban areas, compared to a 10% threshold in the continental Southeast. This helped capture small parks that dropped out of the indicator when we extended the buffer distance to 1.2 miles. Finally, the Caribbean version does not use the impervious surface cutoff applied in the continental Southeast because the landcover data available in the Caribbean does not assess percent impervious in a comparable way.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

2020 Census Qualifying Urban Areas and Final Criteria Clarifications, 87 Fed. Reg. 80114 (December 29, 2022). Federal Register: The Daily Journal of the United States. Web. [https://www.federalregister.gov/documents/2022/12/29/2022-28286/2020-census-qualifying-urban-areas-and-final-criteria-clarifications].

Mertes, James D. and James R. Hall. 1995. Park, recreation, open space, and greenway guidelines. A project of the National Recreation and Park Association and American Academy for Park and Recreation Administration. Alexandria, VA: NPRA.

OpenStreetMap. Leisure. Accessed March 14, 2023. Data extracted through Geofabrik downloads. [https://wiki.openstreetmap.org/wiki/Key:leisure].

OpenStreetMap. Natural. Accessed March 14, 2023. Data extracted through Geofabrik downloads. [https://wiki.openstreetmap.org/wiki/Key:natural].

Protected Areas Conservation Action Team. 2018. Puerto Rico Protected Areas Database. December 2018 version. GIS data. San Juan, PR. [https://research.fs.usda.gov/iitf/products/dataandtools/puerto-rico-protected-natural-areas-2018].

U.S. Census Bureau, Geography Division, Spatial Data Collection and Products Branch. January 4, 2023. TIGER/Line Shapefile, Current, Nation, U.S., 2020 Census Urban Area. [https://www2.census.gov/geo/tiger/TIGER2020/UAC/tl_2020_us_uac20.zip].

U.S. Department of Health and Human Services. Physical Activity Guidelines for Americans, 2nd edition. Washington, DC: U.S. Department of Health and Human Services; 2018. [https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf].

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2022, Protected Areas Database of the United States (PAD-US) 3.0: U.S. Geological Survey data release. [https://doi.org/10.5066/P9Q9LQ4B].

Puerto Rico Potential Access to Parks

This cultural resource indicator prioritizes places to create new parks that would fill gaps in access to open space within socially vulnerable communities in urban areas in Puerto Rico. It identifies areas where residents currently lack access to parks within a 10-minute walk (accounting for walkable road networks and access barriers like highways and fences), then prioritizes based on park need using demographic and environmental metrics. This indicator originates from the Trust for Public Land's ParkServe park priority areas and the Center for Disease Control's Social Vulnerability Index.

This indicator was last updated in November 2025.

Reason for Selection

Protected natural areas help foster a conservation ethic by providing opportunities for people to connect with nature, and also support ecosystem services like offsetting heat island effects (Greene and Millward 2017, Simpson 1998), water filtration, stormwater retention, and more (Hoover and Hopton 2019). In addition, parks, greenspace, and greenways can help improve physical and psychological health in communities (Gies 2006).

Parks contribute to Puerto Rico's natural infrastructure and help increase resiliency to high-intensity storms and flooding. For example, urban forests can support soil stabilization, improve air and water quality, reduce urban heat effects, and improve stormwater management (Resetar et al. 2020). Similarly, parks along Puerto Rico's coastline that support coral reefs, seagrasses, and marine wetlands can help reduce storm energy and protect the coastline from flooding and erosion (Guannel et al. 2016). Parks also help connect Puerto Rico's local population and tourists to natural and cultural resources. Every Puerto Rican municipality has local and commonwealth parks that are significant to local residents and help support local tourism. Parks that are nearby and walkable to communities also offer recreation benefits and opportunities to connect with wildlife, in addition to localized ecosystem services.

However, parks are not equitably distributed within easy walking distance for everyone, especially outside of major municipalities like San Juan. This indicator prioritizes places where new parks can help to serve communities that lack nearby park access. It also complements the urban park size indicator by capturing the value of potential new parks.

Input Data

- The Trust for Public Land (TPL) ParkServe database, accessed 31-12-2024: Park priority areas (ParkServe_ParkPriorityAreas, ParkServe_10MWServiceAreas, ParkServe_Parks)

  From the TPL ParkServe documentation:

  The ParkServe database maintains an inventory of parks for every urban area in the U.S., including Puerto Rico. This includes all incorporated and Census-designated places that lie within any of the country's 3,000+ census-designated urban areas. All populated areas in a city that fall outside of a 10-minute walk service area are assigned a level of park priority, based on a comprehensive index of six equally weighted demographic and environmental metrics:

  - Population density
  - Density of low-income households – which are defined as households with income less than 75 percent of the urban area median household income
  - Density of people of color
  - Community health – a combined index based on the rate of poor mental health and low physical activity from the 2020 CDC PLACES census tract dataset
  - Urban heat islands – surface temperature at least 1.25o greater than city mean surface temperature from The Trust for Public Land, based on Landsat 8 satellite imagery

- Pollution burden - Air toxics respiratory hazard index from 2020 EPA EJScreen

The 10-minute walk

For each park, we create a 10-minute walkable service area using a nationwide walkable road network dataset provided by Esri. The analysis identifies physical barriers such as highways, train tracks, and rivers without bridges and chooses routes without barriers.

- Centers for Disease Control (CDC) Social Vulnerability Index 2022 (Puerto Rico): RPL_Themes

  Social vulnerability refers to the capacity for a person or group to "anticipate, cope with, resist and recover from the impact" of a natural or anthropogenic disaster such as extreme weather events, oil spills, earthquakes, and fires. Socially vulnerable populations are more likely to be disproportionately affected by emergencies (Wolkin et al. 2015).

  In this indicator, we use the "RPL_THEMES" attribute from the Social Vulnerability Index, described here. "The Geospatial Research, Analysis, and Services Program (GRASP) at Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry developed the Social Vulnerability Index (SVI). The SVI is a dataset intended to help state, local, and tribal disaster management officials identify where the most socially vulnerable populations occur (Agency for Toxic Substances and Disease Registry [ATSDR] 2018)" (Flanagan et al. 2018).

  "The SVI database is regularly updated and includes 15 census variables (ATSDR 2018). Each census variable was ranked from highest to lowest vulnerability across all census tracts in the nation with a nonzero population. A percentile rank was calculated for each census tract for each variable. The variables were then grouped among four themes.... A tract-level percentile rank was also calculated for each of the four themes. Finally, an overall percentile rank for each tract as the sum of all variable rankings was calculated. This process of percentile ranking was then repeated for the individual states" (Flanagan et al. 2018).

- Southeast Blueprint 2025 extent

Mapping Steps

- From the Puerto Rico ParkServe data, remove the 10-minute walk service areas and current parks from the park priority areas. This removes areas that are already being served by an existing park, which are not the focus of this indicator.
- Convert the resulting layer to raster using the ParkRank field. Note: The ParkRank scores are calculated using metrics classified relative to each city. Each city contains park rank values that range from 1-3. For the purposes of this indicator, we chose to target potential park areas to improve equity. Because the ParkRank scores are relative for each city, a high score in one city is not necessarily comparable to a high score from another city. In an effort to try to bring

more equity into this indicator, we also use the CDC Social Vulnerability Index to narrow down the results.

- Reclassify the ParkServe raster to make NoData values 0.
- Convert the SVI layer from vector to raster based on the "RPL_Themes" field. In the source geodatabase, the field name is "RPL_Themes" and the alias for that field is "Overall percentile ranking"
- To limit the ParkRank layer to areas with high SVI scores, first identify census tracts with an "RPL_Themes" field value >0.65. Make a new raster that assigns a value of 1 to census tracts that score >0.65, and a value of 0 to everything else. Take the resulting raster times the ParkRank layer.
- Reclassify this raster into the 4 classes seen in the final indicator below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2025.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

3 = Very high priority for a new park to serve people who lack nearby park access
2 = High priority for a new park to serve people who lack nearby park access
1 = Moderate priority for a new park to serve people who lack nearby park access
0 = Not identified as a priority for a new park to serve people who lack nearby park access (within urban areas)

Known Issues

- The ParkServe database has many missing and incomplete parks in Puerto Rico (e.g., Cabo Rojo National Wildlife Refuge, parts of Balneario Seven Seas in Fajardo). As a result, it overestimates park need in the surrounding areas. TPL regularly updates ParkServe to incorporate the best available park data. We encourage users to report missing parks or errors in the park boundaries or attributes. You can submit corrections through the ParkReviewer tool or by contacting TPL staff.
- Within a given area of high park need, the number of people served by the creation of a new park depends on its size and how centrally located it is. This indicator does not account for this variability. Similarly, while creating a new park just outside an area of high park need would create access for some people on the edge, the indicator does not capture the benefits of new parks immediately adjacent to high-need areas. For a more granular analysis of new park benefits, ParkServe's ParkEvaluator tool allows you to draw a new park, view its resulting 10-minute walk service area, and calculate who would benefit.
- Beyond considering distance to a park and whether it is open to the public, this indicator does not account for other factors that might limit park access, such as park amenities or public safety. The TPL analysis excludes private or exclusive parks that restrict access to only certain

individuals (e.g., parks in gated communities, fee-based sites). The TPL data includes a wide variety of parks, trails, and open space as long as there is no barrier to entry for any portion of the population.

- The indicator does not incorporate inequities in access to larger versus smaller parks. In predicting where new parks would benefit nearby people who currently lack access, this indicator treats all existing parks equally.
- This indicator identifies areas where parks are needed but does not consider whether a site is available to become a park. We included areas of low intensity development in order to capture vacant lots, which can serve as new park opportunities. However, as a result, this indicator also captures some areas that are already used for another purpose (e.g., houses, cemeteries, and businesses) and are unlikely to become parks. In future updates, we would like to use spatial data depicting vacant lots to identify more feasible park opportunities.
- This indicator underestimates places in rural areas where many people within a socially vulnerable census tract would benefit from a new park. ParkServe covers incorporated and census-designated places within census-designated urban areas, which leaves out many rural areas. We acknowledge that there are still highly socially vulnerable communities in rural areas that would benefit from the development of new parks. However, based on the source data, we were not able to capture those places in this version of the indicator.
- Some high-scoring areas may have access to nearby greenways and trails that can serve many of the roles of parks. This indicator does not account for access to nearby greenways and trails.

Other Things to Keep in Mind

- The zero values in this indicator contain three distinct types of areas that we were unable to distinguish between in the legend: 1) Areas that are not in a community analyzed by ParkServe (ParkServe covers incorporated and census-designated places within census-designated urban areas); 2) Areas in a community analyzed by ParkServe that were not identified as a priority; 3) Areas that ParkServe identifies as a priority but do not meet the SVI threshold used to represent areas in most need of improved equitable access.
- This indicator only includes park priority areas that fall within the 65th percentile or above from the Social Vulnerability Index. We did not perform outreach to community leaders or community-led organizations for feedback on this threshold.
- This indicator is intended to generally help identify potential parks that can increase equitable access but should not be solely used to inform the creation of new parks. As the social equity component relies on information summarized by census tract, it should only be used in conjunction with local knowledge and in discussion with local communities (NRPA 2021, Manuel-Navarete et al. 2004).

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Centers for Disease Control and Prevention/ Agency for Toxic Substances and Disease Registry/ Geospatial Research, Analysis, and Services Program. CDC/ATSDR Social Vulnerability Index [2022] Database [Puerto Rico]. [https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html].

Flanagan BE, Hallisey EJ, Adams E, Lavery A. Measuring Community Vulnerability to Natural and Anthropogenic Hazards: The Centers for Disease Control and Prevention's Social Vulnerability Index. J Environ Health. 2018;80(10):34-36. [https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7179070/].

Gies E. (2006). The health benefits of parks: how parks help keep Americans fit & healthy. The Trust for Public Land. Accessed February 16, 2022. [https://cloud.tpl.org/pubs/benefits_HealthBenefitsReport.pdf].

Greene, C. S., & Millward, A. A. (2017). Getting closure: The role of urban forest canopy density in moderating summer surface temperatures in a large city. Urban ecosystems, 20(1), 141-156. [https://link.springer.com/article/10.1007/s11252-016-0586-5].

Guannel, G., Arkema K., Ruggiero, P., Verutes, G. 2016. The Power of Three: Coral Reefs, Seagrasses and Mangroves Protect Coastal Regions and Increase Their Resilience. PLoS ONE, Vol. 11, No. 7, [https://doi.org/10.1371/journal.pone.0158094].

Hoover, F. A., & Hopton, M. E. (2019). Developing a framework for stormwater management: leveraging ancillary benefits from urban greenspace. Urban ecosystems, 22(6), 1139-1148. [https://link.springer.com/article/10.1007/s11252-019-00890-6].

Manuel-Navarrete D, Kay JJ, Dolderman D. Ecological integrity discourses: linking ecology with cultural transformation. Human Ecology Review 2004, 11:215–229. [https://www.jstor.org/stable/24707715].

National Recreation and Park Association. "NRPA Park Metrics." 2021. Accessed April 5, 2021. [https://www.nrpa.org/publications-research/research-papers/agency-performance-review/].

Resetar, S. A., Tingstad, A., Mendelsohn, J., Marlier, M. E., Lachman, B. E., Anania, K., Garber, C., & Adamson, D. M. 2020. Recovery planning for natural resources and parks in Puerto Rico: Natural and cultural resources sector report (Vol. 1). Homeland Security Operational Analysis Center, RAND Corporation. Accessed May 14, 2020. [https://recovery.preventionweb.net/publication/recovery-planning-natural-resources-and-parks-puerto-rico-natural-and-cultural].

Simpson, J. R. (1998). Urban forest impacts on regional cooling and heating energy use: Sacramento County case study. Journal of Arboriculture, 24, 201-214. [https://www.fs.usda.gov/research/treesearch/61731].

Wolkin A, Patterson JR, Harris S, et al. Reducing Public Health Risk During Disasters: Identifying Social Vulnerabilities. J Homel Secur Emerg Manag. 2015;12(4):809-822. [https://doi.org/10.1515/jhsem-2014-0104].

U.S. Virgin Islands Imperiled Terrestrial Species

This indicator identifies potential habitat to support terrestrial species across the U.S. Virgin Islands. It focuses on species of conservation interest like endangered species and Species of Greatest Conservation Need (SGCN), which are identified in each state or territory's Wildlife Action Plan as most in need of conservation action. The indicator evaluates overall species richness, but places special emphasis on three narrow endemics that are restricted to very small ranges, which are therefore more vulnerable to habitat loss or change. The data originates from WildMon and the U.S. Fish and Wildlife Service Environmental Conservation Online System.

This indicator was last updated in November 2025.

Reason for Selection

The Caribbean Islands are recognized as a global biodiversity hotspot, with one of the highest numbers of globally threatened species among all such designated areas (CEPF 2019). This exceptional biodiversity is supported by the region's unique geography and diverse habitats. The U.S. Virgin Islands (USVI) are home to hundreds of plant, fish, and bird species, including endangered species like the St. Croix ground lizard *(Pholidoscelis polops)* and other species that would benefit from conservation action like Wilson's plover *(Charadrius wilsonia)* and the Antillean nighthawk *(Chordeiles gundlachii*).

The species used in this indicator are identified as SGCN within the 2015 USVI Wildlife Action Plan and also include two endangered plant species. Each state and territorial fish and wildlife agency identifies a list of SGCN, which are considered species most in need of conservation action. Wildlife Action Plans focus on not only identifying SGCN but also important habitats, key threats, and needed conservation actions in order to proactively conserve these species. Understanding the distribution of native wildlife and plant species in the USVI—especially those that are listed as endangered or classified as SGCN—is essential for informing effective conservation strategies.

This indicator identifies potential habitat for one amphibian, one reptile, eighteen birds, and two plant species and subspecies on the USVI SGCN list. While all listed species and SGCN warrant consideration, this indicator emphasizes places that support potential habitat for range-restricted species and narrow endemics that only occur in very small and often singular places, making them more vulnerable to extinction. For example, Egger's century plant *(Agave eggersiana)* is a succulent plant, endemic to St. Croix, and was listed for protection in 2014 under the Endangered Species Act (USFWS 2022). Once widespread in the subtropical dry forests of St. Croix, habitat loss and longer dry spells have greatly reduced the population and it only persists in some small pockets on the island (USFWS 2022). In contrast, the royal tern *(Thalasseus maximus)* is a seabird that nests along coastlines

and has a much broader range. While it is considered a regional SGCN (Rice et al. 2019), the Royal Tern is found in 10 states within the Southeast region as well as along the Northeast coast.

Input Data

- WildMon occupancy models for mapping species distributions across the U.S. Virgin Islands, provided by Dr. Marconi Campos-Cerqueira on March 28, 2025 (available on request by emailing marconi@wildmon.ai)

  WildMon is a nonprofit specializing in ecoacoustics, camera trapping, environmental DNA (eDNA), and artificial intelligence to deliver a holistic understanding of biodiversity patterns and guide effective conservation actions. In partnership with the U.S. Virgin Islands Department of Planning and Natural Resources, WildMon conducted a comprehensive wildlife monitoring project aimed at mapping biodiversity patterns across the region. The project focused on identifying locations of Species of Greatest Conservation Need. Through surveys at 140 sites, the team detected 70 species, providing critical data on rare, common, and invasive wildlife.

  A full list of all species included appears in the Full Species List section below.

- 2012 National Oceanic and Atmospheric Administration (NOAA) Coastal Change Analysis. Program (C-CAP) land cover files for the U.S. Virgin Islands (St. Thomas, St. John, and St. Croix are provided as separate rasters) accessed 4-26-2022; learn more about C-CAP high resolution land cover and change products.
- 2020 (v2.2.0) LANDFIRE Existing Vegetation Type (EVT) for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- Caribbean island size: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Species range maps from U.S. Fish and Wildlife Service Environmental Conservation Online System (ECOS) for Eggars agave *(Agave eggersiana)*, St. Croix ground lizard *(Ameiva polops)*, *Catesbaea melanocarpa*, and *Peperomia wheeleri*
- Southeast Blueprint 2025 extent

Mapping Steps

- Sum the occupancy probabilities across all the WildMon occupancy models for SGCN, then resample to 30 m and snap to the Blueprint extent.
- Convert the four ECOS species range vectors to 30 m rasters snapped to the Blueprint extent. We include these species ranges because they're detailed enough to be used in the Blueprint and cover species not captured by the WildMon acoustic monitoring. Sum these values, then add them to the WildMon estimate of species richness to get a total species richness layer.

- Select pixels classified as “Developed” or “Agriculture” in the LANDFIRE EVT “Lifeform” attribute and assign them a value of 0 to create a developed and agriculture mask. Then use the mask to set developed and agriculture in the combined weighted richness layer to 0.
- Reclassify the richness layer to get the final indicator categories. Three of the species pulled from ECOS have very small ranges (St. Croix ground lizard, Eggar’s agave, and *Catesbaea melanocarpa*). Include any pixel with at least one of these species in the top indicator class.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = Very high species richness (top 10%) or presence of species with very small ranges
3 = High species richness (upper 10-30%)
2 = Medium species richness (above average)
1 = Low species richness (below average)
0 = Developed or agriculture

Known Issues

- The WildMon data only used acoustic sampling and cannot differentiate between species of coquí as well as some other species. Also, this sampling approach cannot detect species that are not vocal or were not vocal at the time of sampling.
- The WildMon data had a low number of detections for some rarer species. This resulted in higher uncertainty in predictions for those species—especially in unsampled areas.
- The indicator overprioritizes some altered areas that are difficult to separate with landcover data. For example, the landfill near Smalley Bay in St. Thomas is overprioritized.

Full Species List

Table 12. SGCN included in this indicator.

	Scientific name	**Common name**
1	*Agave eggersiana*	Eggers’ agave
2	*Ardea alba*	Great white egret
3	*Calidris pusilla*	Semipalmated sandpiper
4	*Catesbaea melanocarpa*	Tropical lily-thorn
5	*Charadrius vociferus*	Killdeer
6	*Charadrius wilsonia*	Wilson’s plover
7	*Chordeiles gundlachii*	Antillean nighthawk
8	*Egretta thula*	Snowy egret
9	*Eleutherodactylus lentus*	Virgin Islands yellow frog
10	*Geotrygon mystacea*	Bridled quail-dove

	Scientific name	Common name
11	*Numenius phaeopus*	Whimbrel
12	*Nycticorax nycticorax*	Black-crowned night-heron
13	*Orthorhyncus cristatus*	Antillean crested hummingbird
14	*Patagioenas leucocephala*	White-crowned pigeon
15	*Patagioenas squamosa*	Scaly-naped pigeon
16	*Pholidoscelis polops*	St. Croix ground lizard
17	*Progne dominicensis*	Caribbean martin
18	*Rallus crepitans*	Clapper rail
19	*Setophaga adelaidae*	Adelaide's warbler
20	*Sternula antillarum*	Least tern
21	*Tachybaptus dominicus*	Least grebe
22	*Thalasseus maximus*	Royal tern

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Critical Ecosystem Partnership Fund (CEPF). 2019. CEPF 2019 Impact Report. [https://www.cepf.net/sites/default/files/cepf-2019-impact_report-website.pdf].

Rice, T. M., Crisfield, E., Terwilliger, K. 2019. Regional species of greatest conservation need in the Southeastern United States. Terwilliger Consulting, Inc. Prepared for the Wildlife Diversity Committee, Southeast Association of Fish and Wildlife Agencies. [https://secassoutheast.org/pdf/SEAFWA_RSGCN_Final_Report_20190715.pdf].

U.S. Fish & Wildlife Service. 2022. Eggers' Century Plant (*Agave eggersiana*). U.S. Department of the Interior. Accessed May 16, 2025. [https://www.fws.gov/species/eggers-century-plant-agave-eggersiana].

Freshwater

Caribbean Imperiled Aquatic Species

This indicator measures the number of aquatic animal Species of Greatest Conservation Need (SGCN) observed within each 12-digit HUC subwatershed in Puerto Rico and the U.S. Virgin Islands, including fish, shrimp, crabs, and crayfish. This indicator originates from data compiled by the Southeast Aquatic Resources Partnership from university researchers. It applies to the floodplain predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood), derived from the Federal Emergency Management Agency's National Flood Hazard Layer, and buffered flowlines representing other streams.

This indicator was last updated in October 2026.

Reason for Selection

While Puerto Rico is widely known for its marine fisheries, the island's freshwater habitats also support many species of freshwater fish that are valuable for sport fishing and natural heritage. The health of these fish populations is affected by non-native species, aquatic barriers, changes to instream flow patterns, and water pollution (Kwak et al. 2007). In the U.S. Virgin Islands, steep slopes, small drainage basins, and low rainfall makes freshwater habitats scarce. As a result, nearly all streams are ephemeral, and no natural freshwater lakes occur across St. Croix, St. John, and St. Thomas (USGS 1996). This makes it all the more important for the Blueprint to capture areas of high freshwater aquatic biodiversity across both U.S. Caribbean territories.

Broadly, ten species of freshwater fish have been identified in Puerto Rico and also reported in the U.S. Virgin Islands (USFWS 2018b). In addition, eighteen species of native freshwater crustaceans (e.g., crabs, crayfish) have also been identified in Puerto Rico, many of which have also been reported in the U.S. Virgin Islands (USFWS 2018a). This indicator represents observations of species identified as Species of Greatest Conservation Need (SGCN) in the draft 2025 U.S. Virgin Islands and Puerto Rico Wildlife Action Plans. It identifies areas with abundant rare and endemic aquatic species that would benefit from conservation action. It captures patterns of rare and endemic species diversity not well-represented by other freshwater indicators.

The Southeast Aquatic Resources Partnership (SARP) tracks the number of aquatic animal species in different conservation categories, including SGCN, Regional SGCN (RSGCN), and threatened or endangered. Not all these categories are tracked across Puerto Rico and the U.S. Virgin Islands because of differences in data availability compared to the continental geography. To depict the aquatic priorities of state fish and wildlife agencies, this indicator uses SGCN, which are identified in each Wildlife Action Plan as most in need of conservation action. In contrast, the continental version of this indicator uses RSGCN.

Input Data

- Southeast Blueprint 2026 extent
- U.S. Geological Survey (USGS) Watershed Boundary Dataset (WBD), accessed 1-5-2023: HUC12s; download the data
- Southeast Aquatic Resources Partnership (SARP) SGCN HUC12 summaries for USVI

  SARP provided these summaries on March 3, 2026 as a spreadsheet containing the ID number for each HUC12 watershed and the species names of aquatic animal SGCN observed in that watershed. This dataset is based on observational data for fish, shrimp, and crabs collected by in the U.S. Virgin Islands by Jacob Laurain at Clemson University. It was last updated in March 2026. SECAS staff updated the SGCN status for observed species based on the list of SGCN in the draft 2025 U.S. Virgin Islands Wildlife Action Plan and recalculated the total number of aquatic animal SGCN observed in each watershed. A list of all aquatic animal SGCN used in the indicator in both USVI and Puerto Rico appears in the Full Species List section below.

- Fishery Population and Habitat Assessment in Puerto Rico Streams: Phase 1 and Phase 2 Final Reports (Kwak et al. 2007 and 2013); Phase 3 data provided by Dr. Bonnie Myers with the U.S. Geological Survey and Dr. Gus Engman with the University of Tennessee on April 6, 2026

  For Phase II data, SECAS staff exported into Excel tabular data from these final reports representing sampling locations and species of native fish, shrimp, crabs, and crayfish observed at each sampling site from 2005-2007 and 2008-2010. Phase III data was already delivered in Excel format. Using the sampling site coordinates, we converted the data to geospatial point locations and intersected them with HUC12 subwatersheds. We then calculated the number of aquatic animal SGCN observed in each watershed according to the list of SGCN in the draft 2025 U.S. Puerto Rico Wildlife Action Plan. These data have been shared with SARP for long-term stewardship. A list of all aquatic animal SGCN used in the indicator in both Puerto Rico and USVI appears in the Full Species List section below.

- Caribbean natural landcover in floodplains (Southeast Blueprint indicator): used to represent spatial footprint of expanded floodplains and define the spatial extent of this indicator. Input data used to define these floodplains are listed below.

  - Federal Emergency Management Agency (FEMA) National Flood Hazard Layer flood zones for Puerto Rico and the U.S. Virgin Islands, accessed 10-22-2022; to download the data, visit the FEMA Flood Map Service Center, search by jurisdiction (Puerto Rico or Virgin Islands), download all FIRM (Flood Insurance Rate Maps) panels, and locate the "S_FLD_HAZ_AR" shapefile in each download package.

    We used the "FLD_ZONE" attribute of the S_FLD_HAZ_AR shapefile to define an estimated floodplain depicting areas predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood). To create the estimated floodplain for

Puerto Rico and the U.S. Virgin Islands, we combined all areas with flood zone codes beginning with the letter "A". These zones represent the inland (non-coastal) portions of FEMA Special Flood Hazard Areas considered at high risk of flooding. This excludes coastal areas where the high risk of flooding stems from storm waves, areas of moderate-low flood risk, and areas with possible but undetermined flood hazards where no hazard analysis has been conducted. For more details on FEMA flood zones, read this FEMA blog or visit the FEMA glossary (detailed definitions are under "Z" for "zones").

- National Hydrography Dataset Plus High Resolution (NHDPlus HR) National Release catchments and flowlines, accessed 11-30-2022; download the data

Mapping Steps

- Join the SGCN count table to the HUC12 spatial data.
- Convert to a 30 m raster, where the values represent the number of SGCN.
- Reclassify the resulting raster so all counts of 8 or greater receive a value of 8. This is consistent with how counts are capped in the continental version of the indicator.
- Add 1 to the raster values to allow separation of areas in the floodplain with 0 observations and areas outside of the floodplain in a later step.
- Reclassify the Caribbean natural landcover in floodplains indicator, assigning a 1 to areas in the floodplain and a 0 to areas outside the floodplain. In this step, using the Caribbean natural landcover in floodplains indicator serves a shortcut for a more complex series of analysis steps:
    - Convert the FEMA floodplain polygons to a 30 m raster, giving floodplain areas a value of 1.
    - Extract the stream and river lines from the NHDPlus HR flowlines (ftype IN (460, 558)). Convert extracted stream and river lines to a 30 m raster. Use the ArcPy Spatial Analysis Expand function to "buffer" the streams by 1 cell. This is the method that SARP uses to create a total stream width of approximately 90 m.
    - Combine the FEMA floodplains and buffered flowlines using the Mosaic function to make an enhanced floodplain layer.
- Multiply the two rasters above so everything outside of the floodplain receives a value of 0 and areas inside the floodplain receive the count of species observations in a given watershed, plus 1, as shown in the indicator values below.
- As a final step, clip to the spatial extent of Southeast Blueprint 2026.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

9 = 8+ aquatic animal Species of Greatest Conservation Need (SGCN) observed
8 = 7 aquatic animal SGCN observed
7 = 6 aquatic animal SGCN observed
6 = 5 aquatic animal SGCN observed
5 = 4 aquatic animal SGCN observed
4 = 3 aquatic animal SGCN observed
3 = 2 aquatic animal SGCN observed
2 = 1 aquatic animal SGCN observed
1 = 0 aquatic animal SGCN observed
0 = Not identified as a floodplain in surveyed watersheds

Known Issues

- Some watersheds in Puerto Rico and USVI may be underprioritized in this indicator because survey data was not available. Unsurveyed watersheds receive a value of NoData in this indicator, while surveyed watersheds where no aquatic animal SGCN were found receive a value of 1 inside the floodplain and 0 outside the floodplain. In the continental version of this indicator, we do not have data on survey locations, so are unable to distinguish between areas where species were found to be absent and unsurveyed areas.
- This indicator may underprioritize areas important for aquatic plants. At this point SARP is only collecting information on aquatic animal species. We will explore ways to incorporate aquatic plant species in the future.
- This indicator may underprioritize the U.S. Virgin Islands because sampling data is incomplete. Data from an additional sampling season for the U.S. Virgin Islands should be available in the next indicator update.
- Sirajo goby *(Sicydium plumieri)* has been taxonomically split into four *Sicydium* species (*S. buscki, S. gilberti, S. plumieri,* and *S. punctatum*), but Kwak et al. combined them as *S. plumieri*. Both *S. plumieri* and *S. punctatum* are identified as unique SGCN in the Puerto Rico Wildlife Action Plan. As a result, watersheds where both *S. plumieri* and *S. punctatum* co-occur may be undervalued by 1 in the indicator because the source data treats them as one species.
- This indicator does not account for aquatic barriers, so there is a chance that the species reported are unable to use the full watershed they are assigned to in this indicator.

Full Species Lists

U.S. Virgin Islands

Table 13. List of freshwater aquatic animal Species of Greatest Conservation Need included in the indictor.

	Scientific Name	Common Name
1	*Albula vulpes*	Bonefish
2	*Anguilla rostrata*	American eel
3	*Atya scabra*	Fat shrimp/Camacuto shrimp
4	*Awaous banana*	River goby
5	*Cardisoma guanhumi*	Bluecrab/Blue land crab
6	*Epilobocera sinuatifrons*	Freshwater crab
7	*Lutjanus apodus*	Schoolmaster snapper
8	*Macrobrachium carcinus*	Big claw river shrimp/Kribishees
9	*Macrobrachium faustinum*	Caribbean long-arm shrimp

Puerto Rico

Table 14. List of freshwater aquatic animal Species of Greatest Conservation Need included in the indictor.

	Scientific Name	Common Name
1	*Agonostomus monticola*	Mountain mullet
2	*Anguilla rostrata*	American eel
3	*Awaous banana*	River goby
4	*Centropomus pectinatus*	Tarpon snook
5	*Dormitator maculatus*	Fat sleeper snook
6	*Epilobocera sinuatifrons*	Puerto Rican freshwater crab/Buruquena
7	*Gobiomorus dormitor*	Bigmouth sleeper
8	*Lutjanus griseus*	Gray snapper/mangrove snapper
9	*Lutjanus jocu*	Dog snapper
10	*Macrobrachium acanthurus*	Cinnamon river shrimp
11	*Macrobrachium carcinus*	Bigclaw river shrimp/Kribishees
12	*Macrobrachium crenulatum*	Big-claw river shrimp
13	*Macrobrachium faustinum*	Caribbean long-arm shrimp
14	*Macrobrachium heterochirus*	Cascade river prawn
15	*Microphis brachyurus*	Short-tailed pipefish
16	*Pomadasys crocro*	Burro grunt
17	*Sicydium plumieri*	Sirajo goby

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Southeast Aquatic Resources Partnership. Species Summaries by HUC12. Accessed March 2026.

Kwak, T. J., P. B. Cooney, and C. H. Brown. 2007. Fishery population and habitat assessment in Puerto Rico streams: phase 1 final report. Federal Aid in Sportfish Restoration Project F-50 Final Report. Submitted to Marine Resources Division. Puerto Rico Department of Natural and Environmental Resources. San Juan. [https://www.drna.pr.gov/wp-content/uploads/2015/04/Fishery-Population-and-Habitat-Assessment-in-Puerto-Rico-Streams.pdf].

Kwak, T. J., W. E. Smith, E. N. Buttermore, P. B. Cooney, and W. G. Cope. 2013. Fishery population and habitat assessment in Puerto Rico streams: phase 2 final report. Federal Aid in Sport Fish Restoration Project F-50 Final Report, Submitted to Marine Resources Division, Puerto Rico Department of Natural and Environmental Resources, San Juan. [https://www.govinfo.gov/content/pkg/GOVPUB-I49-PURL-gpo190937/pdf/GOVPUB-I49-PURL-gpo190937.pdf].

U.S. Fish and Wildlife Service. February 2018a. Caribbean Freshwater Crustaceans. Fish and Aquatic Conservation Initiative, Caribbean Ecological Services Field Office. [https://www.fws.gov/sites/default/files/documents/2024-05/caribbean-freshwater-crustaceans.pdf].

U.S. Fish and Wildlife Service. February 2018b. Caribbean Freshwater Fish. Fish and Aquatic Conservation Initiative, Caribbean Ecological Services Field Office. [https://www.fws.gov/sites/default/files/documents/2024-05/caribbean-freshwater-crustaceans.pdf].

U.S. Geological Survey. 1996. National Water Summary: U.S. Virgin Islands Wetland Resources, pages 369-270. [https://www.fws.gov/sites/default/files/documents/National-Water-Summary-Wetland-Resources-US-Virgin-Islands.pdf].

Caribbean Natural Landcover in Floodplains

This indicator measures the amount of natural landcover in the estimated floodplain of rivers and streams within each catchment in the U.S. Caribbean. It assesses the stream channel and its surrounding riparian buffer, measuring the percent of unaltered habitat like forests, wetlands, or open water (rather than agriculture or development) in the floodplain. This indicator originates from LANDFIRE landcover. It applies to the floodplain predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood), derived from the Federal Emergency Management Agency's National Flood Hazard Layer, and buffered flowlines representing other streams.

This indicator was last updated in October 2023.

Reason for Selection

Habitat near rivers and streams is strongly linked to water quality and instream flow (Naiman 1997). Intact vegetated buffers within the floodplain of rivers and streams provide aquatic habitat, improve water quality, reduce erosion and flooding, recharge groundwater, and more (WeConservePA 2014). Natural floodplain landcover provides a "front line defense" for aquatic systems.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- National Hydrography Dataset Plus High Resolution (NHDPlus HR) National Release catchments and flowlines, accessed 11-30-2022; download the data
- Federal Emergency Management Agency (FEMA) National Flood Hazard Layer flood zones for Puerto Rico and the U.S. Virgin Islands, accessed 10-22-2022; to download the data, visit the FEMA Flood Map Service Center, search by jurisdiction (Puerto Rico or Virgin Islands), download all FIRM (Flood Insurance Rate Maps) panels, and locate the "S_FLD_HAZ_AR" shapefile in each download package.

  We used the "FLD_ZONE" attribute of the S_FLD_HAZ_AR shapefile to define an estimated floodplain depicting areas predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood). To create the estimated floodplain for Puerto Rico and the U.S. Virgin Islands, we combined all areas with flood zone codes beginning with the letter "A". These zones represent the inland (non-coastal) portions of FEMA Special Flood Hazard Areas considered at high risk of flooding. This excludes coastal areas where the high risk of flooding stems from storm waves, areas of moderate-low flood risk, and areas with possible but undetermined flood hazards where no hazard analysis has been conducted. For more details on FEMA flood zones, read this FEMA blog or visit the FEMA glossary (detailed definitions are under "Z" for "zones").

Catchments

A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.

To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

Mapping Steps

- Convert the FEMA floodplain polygons to a 30 m raster, giving floodplain areas a value of 1.
- Extract the stream and river lines from the NHDPlus HR flowlines (ftype IN (460, 558)). Convert extracted stream and river lines to a 30 m raster. Use the ArcPy Spatial Analysis Expand function to "buffer" the streams by 1 cell. This is the method that SARP uses to create a total stream width of approximately 90 m.
- Combine the FEMA floodplains and buffered flowlines using the Mosaic function to make an enhanced floodplain layer.
- Clip the 2020 LANDFIRE EVT to the enhanced floodplain layer. This limits the indicator values to the floodplain areas, where they are most relevant.
- Reclassify the clipped 2020 LANDFIRE EVT to identify natural landcover. The following classes were considered not natural: Quarries-Strip Mines-Gravel Pits-Well and Wind Pads, Developed-Low Intensity, Developed-Medium Intensity, Developed-High Intensity, Developed-Roads, Developed-Open Space, Agriculture-Pasture and Hay, Agriculture-Cultivated, Crops and Irrigated Agriculture, Caribbean Bush fruit and berries. All other classes were considered natural.
- The original NHDPlus HR catchment data was missing coverage of a small area on the west coast of Puerto Rico (just east of Parcelas Aguas Claras). Create an additional catchment polygon for this missing area so that the indicator covers the entire island of Puerto Rico.
    - The missing area is essentially outlined by extremely thin catchment polygons. To fill the gap, make a new rectangular feature class covering the missing area, then union it together with the original NHDPlus HR catchments. From that output, select the newly created polygon that fills in the hole.
    - The resulting polygon is a multipart feature, so use the explode tool to separate out just the missing catchment. Export it as a shapefile.
    - Union together the missing catchment with the other NHDPlus HR catchments and use that combined output as the catchment layer for the rest of the mapping steps.
- Calculate the percent of riparian natural landcover inside each NHDPlus catchment using ArcPy Spatial Analyst Zonal Statistics "MEAN" function.
- Reclassify the above raster into the 1-5 classes seen in the final indicator values below.
- Clip the resulting raster back to the enhanced floodplain layer. It is necessary to do this again since the Zonal Statistics function outputs pixel values for the entire catchment. During this step, assign a value of 0 to areas outside the enhanced floodplain. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = >90% natural landcover within the estimated floodplain, by catchment
4 = >80-90%
3 = >70-80%
2 = >60-70%
1 = ≤60% natural landcover within the estimated floodplain, by catchment
0 = Not identified as a floodplain

Known Issues

- This indicator does not account for the accumulated impacts of upstream riparian buffers. Buffers at the headwaters are treated the same as those downstream.
- This indicator does account for river or stream size in relation to the estimated floodplain. Aquatic habitat needs may differ based on the river size class. For example, smaller headwater streams may need more natural landcover than larger rivers to maintain aquatic health. It also does not account for variation in buffer quality within the floodplain at a scale below the catchment. This means that within the estimated floodplain, loss of natural habitat adjacent to the river is treated the same as loss farther away.
- The NHDPlus flowlines in the headwaters could represent intermittent or ephemeral streams. They were not excluded, so the indicator could be overprioritizing headwater areas relative to second-, third-, or fourth-order streams.
- The National Hydrography Dataset digitizes surface water systems at the 1:24,000 resolution. It does not digitize every small, ephemeral stream. As a result, some stream channels that contribute excess sediment to downstream streams or the ocean are not included in this indicator.
- NHDPlus HR contains multiple catchments that are very small. The reduced size of these catchments may result in exaggerating their values in the indicator.

Other Things to Keep in Mind

- Headwater streams are important to freshwater systems but are not always well-captured in the FEMA floodplain dataset. To better represent headwater streams in this indicator, we chose to buffer flowlines and include them as additional floodplain areas to give those areas the opportunity to be included in the Blueprint.
- You may notice stream sections in this indicator that appear to be unconnected to other stream networks. These occur because the National Hydrography Dataset digitizes surface water streams that are not always connected to downstream stream segments. We assume that in these cases, the surface water streams sink into karst areas (e.g., in northwest Puerto Rico around Quebrada). We still included these stream segments in this indicator because they could provide freshwater water habitat for aquatic species.

Disclaimer: Comparing with Older Indicator Versions

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Literature Cited

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

Moore, R.B., McKay, L.D., Rea, A.H., Bondelid, T.R., Price, C.V., Dewald, T.G., and Johnston, C.M., 2019, User's guide for the national hydrography dataset plus (NHDPlus) high resolution: U.S. Geological Survey Open-File Report 2019– 1096, 66 p. [https://doi.org/10.3133/ofr20191096].

Naiman, Robert J., and Henri Decamps. "The Ecology of Interfaces: Riparian Zones." Annual Review of Ecology and Systematics 28 (1997): 621–58. [https://www.jstor.org/stable/2952507].

U.S. Geological Survey (USGS). National Hydrography Dataset Plus High Resolution (NHDPlus HR). Accessed November 30, 2022. [https://www.usgs.gov/national-hydrography/nhdplus-high-resolution].

WeConservePA. 2014. ConservationTools.org: The Science Behind the Need for Riparian Buffer Protection. [https://conservationtools.org/guides/131-the-science-behind-the-need-for-riparian-buffer-protection].

Caribbean Network Complexity

This indicator depicts the number of connected stream size classes in a river network between dams or waterfalls in the U.S. Caribbean. It originates from the Southeast Aquatic Resources Partnership. It applies to the estimated floodplain, which spatially defines areas predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood), based on the Federal Emergency Management Agency's National Flood Hazard Layer, and buffered flowlines representing other streams.

This indicator was last updated in October 2023.

Reason for Selection

River networks with a variety of connected stream size classes are more likely to have a wide range of available habitat to support a greater number of species. This will help retain aquatic biodiversity in a changing climate by allowing species to access climate refugia and move between habitats (Morelli et al. 2016).

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- Southeast Aquatic Resources Partnership's Network Complexity metric

The Southeast Aquatic Resources Partnership (SARP) developed metrics for their Southeast Aquatic Barrier Prioritization Tool. In February 2023, Brendan Ward with Astute Spruce (software developer working on behalf of SARP) shared high resolution NHDPlus flowlines with attributes depicting the network complexity attribute for each functional network (see definition of "functional network" below). The network complexity attribute calculates the total number of different stream size classes within each functional network. SARP assigned stream and river reaches to size classes based on total drainage area:

- 1a: Headwaters (<3.861 sq mi)
- 1b: Creeks (≥3.861 and <8.61 sq mi)
- 2: Small Rivers (≥38.61 and <200 sq mi)
- 3a: Medium Tributary Rivers (≥200 and <1,000 sq mi)
- 3b: Medium Mainstem Rivers (≥1,000 and <3,861 sq mi)
- 4: Large Rivers (≥3,861 and <9,653 sq mi)
- 5: Great Rivers (≥9,653 sq mi)

Functional network

SARP compiles the Southeast Aquatic Barrier Inventory from national, regional, state, and local partner databases across the Southeast region. These include the National Inventory of Dams (2018), National Anthropogenic Barrier Dataset (2012), databases from state dam safety programs and other state agencies, information from local partners, and dam locations estimated by SARP. Waterfalls are compiled from national datasets and local partners. Dams and waterfalls are snapped to hydrologic networks extracted from the National Hydrography Dataset (NHD) - High Resolution Beta version. All dams and waterfalls are treated as "hard" barriers for network connectivity analysis. Aquatic networks are cut at the location of each barrier. All network "loops" (non-primary flowlines) are omitted from the analysis.

An upstream functional network is constructed by traversing upstream from each barrier through all tributaries to the upstream-most origination point or upstream barrier, whichever comes first. Additional functional networks are defined from downstream-most non-barrier termination points, such as marine areas or other downstream termination points.

The total length of all network segments within a functional network is summed to calculate the total network length of each functional network. Each flowline segment within the NHD is assigned to a size class based on total drainage area. This was used to calculate the number of unique size classes per functional network.

- Federal Emergency Management Agency (FEMA) National Flood Hazard Layer flood zones for Puerto Rico and the U.S. Virgin Islands, accessed 10-22-2022; to download the data, visit the FEMA Flood Map Service Center, search by jurisdiction (Puerto Rico or Virgin Islands), download all FIRM (Flood Insurance Rate Maps) panels, and locate the "S_FLD_HAZ_AR" shapefile in each download package.

  We used the "FLD_ZONE" attribute of the S_FLD_HAZ_AR shapefile to define an estimated floodplain depicting areas predicted to be inundated by a 100-year flood (also known as the 1% annual chance flood). To create the estimated floodplain for Puerto Rico and the U.S. Virgin Islands, we combined all areas with flood zone codes beginning with the letter "A". These zones represent the inland (non-coastal) portions of FEMA Special Flood Hazard Areas considered at high risk of flooding. This excludes coastal areas where the high risk of flooding stems from storm waves, areas of moderate-low flood risk, and areas with possible but undetermined flood hazards where no hazard analysis has been conducted. For more details on FEMA flood zones, read this FEMA blog or visit the FEMA glossary (detailed definitions are under "Z" for "zones").

- National Hydrography Dataset Plus High Resolution (NHDPlus HR) National Release catchments and flowlines, accessed 11-30-2022; download the data

  #### Catchments

  A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics. To learn more about catchments and how they're defined, check out these resources:

  - An article from USGS explaining the differences between various NHD products
  - The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

Mapping Steps

- Convert the SARP network complexity values from the NHDPlus HR flowlines to a 30 m raster.
- The original NHDPlus HR catchment data was missing coverage of a small area on the west coast of Puerto Rico (just east of Parcelas Aguas Claras). Create an additional catchment polygon for this missing area so that the indicator covers the entire island of Puerto Rico.
  - The missing area is essentially outlined by extremely thin catchment polygons. To fill the gap, make a new rectangular feature class covering the missing area, then union it together with the original NHDPlus HR catchments. From that output, select the newly created polygon that fills in the hole.
  - The resulting polygon is a multipart feature, so use the explode tool to separate out just the missing catchment. Export it as a shapefile.

    - Union together the missing catchment with the other NHDPlus HR catchments and use that combined output as the catchment layer for the rest of the mapping steps.
- Apply the network complexity values to the NHDPlus HR catchments using the ArcPy Zonal Statistics “MAJORITY” function. This results in a raster where each catchment is given the majority network complexity value that intersects the catchment. Most catchments have only one intersecting line, but for catchments with interior dams, the analysis uses the majority network complexity value. This creates a raster with network complexity value assigned to catchments.
- Convert the FEMA floodplain polygons to a 30 m raster, giving floodplain areas a value of 1.
- Extract the stream and river lines from the NHDPlus HR flowlines (ftype IN (460, 558)). Convert extracted stream and river lines to a 30 m raster. Use the ArcPy Spatial Analysis Expand function to “buffer” the streams by 1 cell. This is the method that SARP uses to create a total stream width of approximately 90 m.
- Combine the FEMA floodplains and buffered flowlines using the Mosaic function to make an enhanced floodplain layer.
- Clip the raster with network complexity values assigned to catchments to the enhanced floodplain layer. This limits the indicator values to the floodplain areas, where they are most relevant.
- Some areas of the floodplain are not scored in the resulting layer because they are missing SARP network complexity values. This is due to the fact that some small reaches, such as braids and loops in the stream network, are not assigned a network complexity value. SARP has to remove loops and braided streams in order to calculate network complexity because the analysis can only accommodate a one-way flow of water. Identify these holes in the floodplain and fill them in by looking at the network complexity value of the surrounding pixels and assigning the maximum value to the missing catchments in the floodplain. Note: This explanation simplifies a complex series of analysis steps. For more specifics, please consult the code.
- Assign zero values to all areas that are covered by the NHDPlus HR catchments, but that are outside the enhanced floodplain layer. Zero values are intended to help users better understand the extent of this indicator and make it perform better in online tools.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = 4 connected stream classes
3 = 3 connected stream classes
2 = 2 connected stream classes
1 = 1 connected stream class
0 = Not identified as floodplain

Known Issues

- This indicator does not include other smaller scale attributes of complexity (e.g., sinuosity, mixtures of riffles/pools/runs) that influence the habitat quality of the connections.
- This indicator likely overestimates the number of connected stream classes in some areas due to missing barriers in the inventory, such as smaller dams or road-stream crossings. It could also underestimate the number of connected stream classes, given the extensive ongoing restoration work to improve aquatic connectivity across the SECAS geography. If you identify a missing barrier or a removed barrier, please let SARP know by emailing Kat Hoenke at kat@southeastaquatics.net. You can learn more about the current inventory of dams and road-stream crossings by visiting https://connectivity.sarpdata.com/.
- SARP did a lot of work to snap the dam locations to the line network, but there are likely still dams (including some large ones) that didn't get snapped correctly due to the large distance between the centerpoint of the dam and the nearest flowline. If you see any of these cases when reviewing the data, please let SARP know (the giveaway is networks that look longer than they should on a map).
- The NHDPlus flowlines in the headwaters could represent intermittent or ephemeral streams. They were not excluded, so the indicator could be overprioritizing headwater areas relative to second-, third-, or fourth-order streams.
- The National Hydrography Dataset digitizes surface water streams that are not always connected to downstream stream segments. We assume that in these cases, the surface water streams sink into karst areas (e.g., in northwest Puerto Rico around Quebrada). We still included these stream segments in this indicator because they could provide freshwater water habitat for aquatic species.
- The FEMA Floodplain layer sometimes misses the small, linear connections made by artificial canals, especially when they go through areas that wouldn't naturally be part of the floodplain. As a result, some areas (like lakes) that are connected via canals may appear to be disconnected but still receive high scores.
- The National Hydrography Dataset digitizes some surface water streams that are not always connected to downstream stream segments. We assume that some of these areas have surface water streams but sink into karst areas. An example of this is in the northwest portion of Puerto Rico around Quebrada. We still included these stream segments in this indicator because they could provide freshwater water habitat for aquatic species.
- NHDPlus HR contains multiple catchments that are very small. The reduced size of these catchments may result in exaggerating their values in the indicator.

Other Things to Keep in Mind

- You may notice stream sections in this indicator that appear to be unconnected to other stream networks. These occur because the National Hydrography Dataset digitizes surface water streams that are not always connected to downstream stream segments. We assume that in these cases, the surface water streams sink into karst areas (e.g., in northwest Puerto Rico around Quebrada). We still included these stream segments in this indicator because they could provide freshwater water habitat for aquatic species.
- Headwater streams are important to freshwater systems but are not always well-captured in the FEMA floodplain dataset. To better represent headwater streams in this indicator, we chose to buffer flowlines and include them as additional floodplain areas to give those areas the opportunity to be included in the Blueprint.
- We considered using different indicator values that would account for varying island sizes and the natural complexity of freshwater systems on each island. By using consistent indicator values across all islands, regardless of their maximum aquatic connectivity potential, this indicator essentially assigns lower scores to smaller islands, despite the fact that they will never reach the same level of network complexity as larger islands. Ultimately, we decided to use the same classes across all islands because more complex and connected aquatic systems on larger islands do provide more climate refugia and available habitat to aquatic species, and therefore should have higher scores.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Martin, E. H, Hoenke, K., Granstaff, E., Barnett, A., Kauffman, J., Robinson, S. and Apse, C.D. 2014. SEACAP: Southeast Aquatic Connectivity Assessment Project: Assessing the ecological impact of dams on Southeastern rivers. The Nature Conservancy, Eastern Division Conservation Science, Southeast Aquatic Resources Partnership. [https://secassoutheast.org/pdf/SEACAP_Report.pdf].

Moore, R.B., McKay, L.D., Rea, A.H., Bondelid, T.R., Price, C.V., Dewald, T.G., and Johnston, C.M., 2019, User's guide for the national hydrography dataset plus (NHDPlus) high resolution: U.S. Geological Survey Open-File Report 2019– 1096, 66 p. [https://doi.org/10.3133/ofr20191096].

Morelli TL, Daly C, Dobrowski SZ, Dulen DM, Ebersole JL, Jackson ST, et al. (2016) Managing Climate Change Refugia for Climate Adaptation. PLoS ONE 11(8): e0159909. [https://doi.org/10.1371/journal.pone.0159909].

Southeast Aquatic Barrier Prioritization Tool. Southeast Aquatic Resources Partnership (SARP). [https://connectivity.sarpdata.com].

U.S. Geological Survey (USGS). National Hydrography Dataset Plus High Resolution (NHDPlus HR). Accessed November 30, 2022. [https://www.usgs.gov/national-hydrography/nhdplus-high-resolution].

Caribbean Permeable Surface

This indicator measures the average percent of non-impervious cover within each catchment in the U.S. Caribbean. High levels of impervious surface degrade water quality and alter freshwater flow, impacting both aquatic species communities and ecosystem services for people, like the availability of clean drinking water. It originates from the National Oceanic and Atmospheric Administration's Coastal Change Analysis Program landcover.
This indicator was last updated in October 2023.

Reason for Selection

Impervious cover is strongly linked to water quality, estuary condition, eutrophication, and freshwater inflow. Impervious surface affects not only aquatic habitats and biodiversity, but also human communities. High levels of impervious surface cause more frequent flooding by increasing the volume of stormwater runoff, reduce the amount of available drinking water by preventing groundwater recharge, and pollute waterways where people swim and fish (Chesapeake 2023, USGS 2018, EPA 2018).

The 90% permeable surface threshold (i.e., 10% impervious) is a well-documented signal of major, negative changes to aquatic ecosystems (Schueler et al. 2009). The 95% permeable surface threshold (i.e., 5% impervious) has been documented to impact Piedmont fish [tricolor shiner *(Cyprinella trichroistia)*, bronze darter *(Percina palmaris)*, Etowah darter *(Etheostoma etowahae)*] (Wenger et al. 2008) and estuarine species [blue crab *(Callinectes sapidus)*, white perch *(Morone americana)*, striped bass *(M. Saxatilis)* and spot *(Leiostomus xanthurus)*] (Uphoff Jr. et al. 2011).

While most of these species do not occur in Puerto Rico and the U.S. Virgin Islands, we kept these thresholds in the Caribbean for consistency with the continental version of the indicator.

Input Data

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent
- 2012 National Oceanic and Atmospheric Administration (NOAA) Coastal Change Analysis Program (C-CAP) land cover files for the U.S. Virgin Islands (St. Thomas, St. John, and St. Croix are provided as separate rasters) accessed 11-10-2022; learn more about C-CAP high resolution land cover and change products
- 2010 NOAA C-CAP land cover files for Puerto Rico, accessed 11-10-2022; learn more about C-CAP high resolution land cover and change products
- National Hydrography Dataset Plus High Resolution (NHDPlus HR) National Release catchments, accessed 11-30-2022; download the data

Catchments

A catchment is the local drainage area of a specific stream segment based on the surrounding elevation. Catchments are defined based on surface water features, watershed boundaries, and elevation data. It can be difficult to conceptualize the size of a catchment because they vary significantly in size based on the length of a particular stream segment and its surrounding topography—as well as the level of detail used to map those characteristics.

To learn more about catchments and how they're defined, check out these resources:

- An article from USGS explaining the differences between various NHD products
- The glossary at the bottom of this tutorial for an EPA water resources viewer, which defines some key terms

- NOAA Continually Updated Shoreline Product (CUSP), accessed 1-11-2023; read a 1-page factsheet about CUSP; view and download CUSP data in the NOAA Shoreline Data Explorer (to download, select "Download CUSP by Region" and select Southeast Caribbean)

Mapping Steps

- NHDPlus HR catchments are currently only available for the islands of Puerto Rico, Vieques, Culebra, St. Croix, St. John, and St. Thomas. Because the catchments don't cover many of the smaller islands, use CUSP to add islands larger than 900 m^2 (the area of a 30 m pixel). Start by converting CUSP shoreline lines to polygons.
- Dissolve interior waterbodies on islands to represent each island with only one polygon.
- To eliminate alignment issues between the CUSP and catchment polygons, remove most island areas that overlap with or are near (<10 m from) the NHDPlus HR catchments, ensuring that all of Culebra is retained.
- The original NHDPlus HR catchment data was missing coverage of a small area on the west coast of Puerto Rico (just east of Parcelas Aguas Claras). Create an additional catchment polygon for this missing area so that the indicator covers the entire island of Puerto Rico.
    - The missing area is essentially outlined by extremely thin catchment polygons. To fill the gap, make a new rectangular feature class covering the missing area, then union it together with the original NHDPlus HR catchments. From that output, select the newly created polygon that fills in the hole.
    - The resulting polygon is a multipart feature, so use the explode tool to separate out just the missing catchment. Export it as a shapefile.
    - Union together the missing catchment with the other NHDPlus HR catchments and use that combined output as the catchment layer for the rest of the mapping steps.
- Remove islands created from the CUSP dataset that are less than 900 m^2.
- Merge the remaining CUSP islands with the NHDPlus catchments to create a single set of polygons in which to calculate average permeable surface.
- Convert the C-CAP landcover rasters for Puerto Rico (2 m resolution) and the U.S. Virgin Islands (separate downloads for St. Thomas, St. John, and St. Croix with 2.4 m resolution) from .img format to .tif using the Copy Raster function.

- For each individual C-CAP layer, use the ArcPy Conditional function to make a binary raster assigning the impervious class a value of 100 (representing fully impervious) and all other classes a value of 0 (representing fully permeable). This mimics the data format of the 2019 National Land Cover Database used in the continental Southeast permeable surface indicator, which provides a continuous impervious surface value ranging from 0 to 100.
- Using the ArcPy Mosaic to New Raster function, mosaic all 4 rasters into 1 raster. Reproject to match the Blueprint projection and the 2 m cell size of the original Puerto Rico C-CAP data.
- Calculate the average percent of impervious surface for each NHDPlus catchment or CUSP island using the ArcPy Spatial Analyst Zonal Statistics "MEAN" function, assigning the average impervious surface value to each catchment or island.
- Convert percent impervious to percent permeable using the formula [percent permeable = 100 - percent impervious] to maintain consistent scoring across Southeast Blueprint indicators (where high values indicate better ecological condition).
- Reclassify the above raster into 4 classes, seen in the final indicator values below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = >95% of catchment or small island permeable (likely high water quality and supporting most sensitive aquatic species)
3 = >90-95% of catchment or small island permeable (likely declining water quality and supporting most aquatic species)
2 = >70-90% of catchment or small island permeable (likely degraded water quality and not supporting many aquatic species)
1 = ≤70% of catchment or small island permeable (likely degraded instream flow, water quality, and aquatic species communities)

Known Issues

- This indicator may not account for differences in permeability between different types of soils and land uses.
- The C-CAP impervious layer used in this indicator contains classification inaccuracies that may cause this indicator to overestimate or underestimate the amount of permeable surface in some catchments.
- C-CAP dates from 2010 for Puerto Rico and 2012 for the U.S. Virgin Islands. As a result, this indicator likely overestimates permeable surface values in areas that have been developed since the data was collected.

- C-CAP landcover is not available for some islands over 900 m^2. While these islands exceeded the size threshold for inclusion in this indicator, they are therefore scored as NoData. This indicator only covers areas where C-CAP landcover is present, and either NHDPlus HR catchments or islands over 900 m^2 that were generated using CUSP data are also present.
- NHDPlus HR contains multiple catchments that are very small. The reduced size of these catchments may result in exaggerating their values in the indicator.

Other Things to Keep in Mind

- The impervious surface in the C-CAP data has impervious surface as one class in the landcover, which differs from the 2019 NLCD percent developed impervious layer used in the continental Southeast version of the permeable surface indicator. NLCD 2019 is served up as a continuous raster ranging from 0-100% impervious.
- We used the Caribbean island size layer for this indicator and not others because landcover data was available for small islands that were not covered by catchments, which otherwise would have been excluded. This was not the case for other indicators. For example, while we use catchments in natural landcover in floodplains, the floodplains and flowlines did not occur on small islands, anyway, so we did not leave any data out by using the catchments only and not supplementing with the islands layer.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Chesapeake Bay Program. 2023. Stormwater Runoff. Accessed September 7, 2023. [https://www.chesapeakebay.net/issues/threats-to-the-bay/stormwater-runoff].

Environmental Protection Agency. EnviroAtlas. Data Fact Sheet. January 2018. Percent of Stream and Shoreline with 15% or More Impervious Cover within 30 Meters. Accessed September 7, 2023. [https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/ESN/Percstreamw15percentimperviousin30meters.pdf].

Moore, R.B., McKay, L.D., Rea, A.H., Bondelid, T.R., Price, C.V., Dewald, T.G., and Johnston, C.M., 2019, User's guide for the national hydrography dataset plus (NHDPlus) high resolution: U.S. Geological Survey Open-File Report 2019– 1096, 66 p. [https://doi.org/10.3133/ofr20191096].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, National Geodetic Survey. NOAA Continually Updated Shoreline Product (CUSP): Southeast Caribbean. [https://shoreline.noaa.gov/cusp.html].

National Oceanic and Atmospheric Administration, Office for Coastal Management. “C-CAP Land Cover Files for Puerto Rico and US Virgin Islands”. Coastal Change Analysis Program (C-CAP) High-Resolution Land Cover. Charleston, SC: NOAA Office for Coastal Management. Accessed November 2022. [https://www.coast.noaa.gov/htdata/raster1/landcover/bulkdownload/hires/].

Schueler, T., Fraley-McNeal, L., and Cappiella, K. 2009. “Is Impervious Cover Still Important? Review of Recent Research.” J. Hydrol. Eng. 14, SPECIAL ISSUE: Impervious Surfaces in Hydrologic Modeling and Monitoring, 309–315. [https://ascelibrary.org/doi/10.1061/%28ASCE%291084-0699%282009%2914%3A4%28309%29].

Uphoff Jr. JH, McGinty M, Lukacovic R, Mowrer J, Pyle B. 2011. Impervious surface, summer dissolved oxygen, and fish distribution in Chesapeake Bay subestuaries: linking watershed development, habitat conditions, and fisheries management. North American Journal of Fisheries Management 31:554-566. [https://dnr.maryland.gov/fisheries/Documents/2011_Impervious_Surface.pdf].

U.S. Geological Survey (USGS). Water Science School. Impervious Surfaces and Flooding. June 5, 2018. Accessed September 7, 2023. [https://www.usgs.gov/special-topics/water-science-school/science/impervious-surfaces-and-flooding].

U.S. Geological Survey (USGS). National Hydrography Dataset Plus High Resolution (NHDPlus HR). Accessed November 30, 2022. [https://www.usgs.gov/national-hydrography/nhdplus-high-resolution].

Wenger, S. J., J. T. Peterson, M. C. Freeman, B. J. Freeman, D. D. Homans. 2008. Stream fish occurrence in response to impervious cover, historic land use and hydrogeomorphic factors. Canadian Journal of Fisheries and Aquatic Sciences 65, 1250-1264. [https://cdnsciencepub.com/doi/10.1139/F08-046].

Coastal & Marine

Caribbean Beach Habitat

This indicator evaluates beach habitat for six species of birds and sea turtles that nest on beaches in the U.S. Caribbean (Wilson's plover, American oystercatcher, and hawksbill, leatherback, green, and loggerhead sea turtles). It includes beach locations, sea turtle nest observations, and predicted suitable habitat for birds and sea turtles. This indicator combines multiple datasets from the Gap Analysis Project, State of the World's Sea Turtles, OpenStreetMap, and more.

This indicator was last updated in October 2023.

Reason for Selection

Beaches in Puerto Rico and the U.S. Virgin Islands support a wide array of shorebirds, colonial seabirds, and sea turtles (ACJV et al. 2015, USFWS 2022). However, their limited spatial extent makes beaches an ecosystem of special concern for conservation (ACJV et al. 2015). In addition, coastal dunes and beaches are some of the Caribbean ecosystems facing the greatest threat from disturbance and development (ACJV et al. 2015). This indicator focuses on a suite of bird and sea turtle species that nest on beaches, though it includes beach habitat used for other activities like foraging and breeding, in addition to nesting.

Input Data

- Puerto Rico Gap Analysis Project predicted vertebrate species distributions: data provided by Dr. Bill Gould with the Caribbean Climate Hub on 4-4-2022 (contact william.a.gould@usda.gov for more information); read the final report

  In Puerto Rico, we used the following GAP species models:

  - Wilson's plover

- Puerto Rico Gap Analysis Project landcover; download the data; read the final report
- U.S. Virgin Islands Gap Analysis Project predicted vertebrate species distributions and landcover; data and report appendices provided by Dr. Bill Gould with the Caribbean Climate Hub on 2-6-2023 (contact william.a.gould@usda.gov for more information); read the final report

  In the USVI, we used the following GAP species models:

  - Wilson's plover
  - American oystercatcher
  - Hawksbill sea turtle
  - Leatherback sea turtle
  - Green sea turtle

- State of the World's Sea Turtles (SWOT) nest locations for hawksbill, leatherback, green, and loggerhead sea turtles; download the data using the download icon on the left side of the SWOT mapping application. Note: loggerhead was only observed in USVI, while the other species were observed in both PR and USVI.
- OpenStreetMap data, accessed 6-28-2023

  A polygon from this dataset is considered a beach if the value in the "nature" attribute is beach. OpenStreetMap describes natural areas as "a wide variety of physical geography, geological and landcover features". Data were downloaded in .pbf format using R code. OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- U.S. Virgin Islands beaches layer (separate vector layers for St. Croix, St. Thomas, and St. John) provided by Joe Dwyer with Lynker/the NOAA Caribbean Climate Adaptation Program on 3-3-2023 (contact jdwyer@lynker.com for more information)
- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- Buffer by 15 m the beach polygons from OpenStreetMap and the USVI beaches layer. This is consistent with the methodology used in the urban park size indicator to avoid the loss of narrow beaches when converting to raster. Project and convert to raster.
- Extract from GAP landcover the relevant beach classes and resample to a 30 m pixel. In Puerto Rico, the beaches include two classes: "Gravel beaches and stony shoreline" and "Fine to coarse sandy beaches, mixed sand and gravel beaches". In the U.S. Virgin Islands, the beaches include three classes: "Fine to Medium Grained Sandy Beaches," "Gravel Beaches" and "Mixed Sand and Gravel Beaches".
- Extract the predicted habitat class from the GAP predicted habitat rasters for species that nest on beaches. These include Wilson's plover, American oystercatcher, and hawksbill, leatherback, and green sea turtles. Note: Only Wilson's plover was predicted in Puerto Rico. Project and resample the rasters to 30 m.
- Extract from the SWOT data nest all point locations for hawksbill, leatherback, green and loggerhead sea turtles and convert them to 30 m rasters.
- Merge together the SWOT and GAP predicted habitat rasters for each species and identify each pixel that contains at least one species. Then clip the resulting raster to the beach extent.
- To define individual beaches, run a region group on beach extent.
- Run the Zonal Statistics "MAXIMUM" function to apply species presence to the entirety of each beach.
- Reclassify to 0 the beach extent layer created above.
- To create the final indicator values seen below, mosaic together three rasters: the beaches containing at least one species, the beach extent, and the Caribbean Blueprint 2023 extent.

This adds back in a 0 value for areas not identified as beaches and a 1 value for beaches that did not contain any species predictions or observations. Zero values better represent the extent of the source data and make the indicator perform better in online tools.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

2 = Beach with 1+ nesting species predicted or observed
1 = Other beach
0 = Not identified as a beach

Known Issues

- This indicator likely underprioritizes beaches in Puerto Rico due to disparities in both beach extent and species data coverage between Puerto Rico and the U.S. Virgin Islands. USVI has a comprehensive hand-digitized beach layer that is not available in Puerto Rico. GAP models only one beach-nesting species in Puerto Rico, compared to five in the U.S. Virgin Islands (though the additional species are known to occur in Puerto Rico as well). In addition, the SWOT sea turtle observations better aligned with the beach polygons in USVI. We will explore additional datasets and methods for addressing these disparities in future revisions. To help mitigate this issue for this year, we set the maximum species richness as 1+ rather than using the full range of species richness values, since Puerto Rico had a maximum species richness of 3 in the available data, compared to a maximum value of 6 in USVI.
- This indicator includes beaches from OpenStreetMap, which is a crowdsourced dataset. While members of the OpenStreetMap community often verify map features to check for accuracy and completeness, there is the potential for spatial errors (e.g., misrepresenting the boundary of a beach) or incorrect tags (e.g., labelling an area as a beach that is not actually a beach). However, using a crowdsourced dataset gives on-the-ground experts, Blueprint users, and community members the power to fix errors and add new parks to improve the accuracy and coverage of this indicator in the future.
- This indicator may exclude some small beaches that aren't captured in the source data. We encourage interested partners and citizens to add any missing beaches to OpenStreetMap so we can better capture them in future updates.
- This indicator does not account for other factors that influence the quality of beach habitat, such as distance to roads, light pollution, and vulnerability to erosion and sea level rise.

Other Things to Keep in Mind

- The species chosen for this indicator are birds and sea turtles that nest on beaches. However, the indicator also includes beach habitat used for activities other than nesting, like foraging and breeding.
- This indicator does not always align with Caribbean coastal shoreline condition. Some areas identified as important beach habitat in this indicator, especially those coming from the GAP Wilson's plover model, are scored as armored in coastal shoreline condition (e.g., the Hyatt Regency Grand Hotel in Río Grande, Puerto Rico). This often occurs where riprap is present along narrow beaches, or occasionally near bulkheads. There is often a section of beach present behind the riprap or bulkhead that could still provide habitat, or the riprap is sporadically placed on a long stretch of beach to protect inland structures. In these cases, the mismatch reflects the different intent of these complementary indicators. In some cases, hardened structures may be actually misclassified as beach. Inconsistencies in alignment and classification likely result from the older age and coarser resolution of the GAP data (10 m raster based on 2001 landcover) compared to the more recent and fine-scale CUSP shorelines (vectors dating primarily from 2014-2021) and challenges in distinguishing the unique remote sensing signature of beach vs. riprap and other hardened structures. Because of the 30 m resolution of the Blueprint and underlying data, a single pixel may contain a mix of beach habitat and hardened structures and be reflected differently in each of these two indicators due to their different functions.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Atlantic Coast Joint Venture, Caribbean Landscape Conservation Cooperative, and U.S. Fish and Wildlife Service. Avian Conservation Planning Priorities for Puerto Rico and the U.S. Virgin Islands (BCR 69). February 2015. [https://acjv.org/documents/PRUSVI_plan.pdf].

Gould, William A.; Alarcón, Caryl; Fevold, Brick; Jiménez, Michael E.; Martinuzzi, Sebastián; Potts, Gary; Quiñones, Maya; Solórzano, Mariano; Ventosa, Eduardo. 2008. The Puerto Rico Gap Analysis Project. Volume 1: Land cover, vertebrate species distributions, and land stewardship. Gen. Tech. Rep. IITF-GTR-39. Río Piedras, PR: U.S. Department of Agriculture, Forest Service, International Institute of Tropical Forestry. 165 p. [https://data.fs.usda.gov/research/pubs/iitf/iitf_gtr39.pdf].

Gould WA, Solórzano MC, Potts GS, Quiñones M, Castro-Prieto J, Yntema LD. 2013. U.S. Virgin Islands Gap Analysis Project – Final Report. USGS, Moscow ID and the USDA FS International Institute of Tropical Forestry, Río Piedras, PR. 163 pages and 5 appendices. [http://www.thinkamap.com/share/IndividualGISdata/PDFs/USVI_FINAL_REPORT.pdf].

Halpin, P.N., A.J. Read, E. Fujioka, B.D. Best, B. Donnelly, L.J. Hazen, C. Kot, K. Urian, E. LaBrecque, A. Dimatteo, J. Cleary, C. Good, L.B. Crowder, and K.D. Hyrenbach. 2009. OBIS-SEAMAP: The world data center for marine mammal, sea bird, and sea turtle distributions. Oceanography 22(2):104-115. [https://tos.org/oceanography/assets/docs/22-2_halpin.pdf].

Kot, C.Y., E. Fujioka, A.D. DiMatteo, B.P. Wallace, B.J. Hutchinson, J. Cleary, P.N. Halpin and R.B. Mast. 2021. The State of the World's Sea Turtles Online Database: Data provided by the SWOT Team and hosted on OBIS-SEAMAP. Oceanic Society, Conservation International, IUCN Marine Turtle Specialist Group (MTSG), and Marine Geospatial Ecology Lab, Duke University. [https://seamap.env.duke.edu/swot].

OpenStreetMap. Natural. Accessed 6-28-2023. Data extracted through Geofabrik downloads. [https://wiki.openstreetmap.org/wiki/Key:natural].

Puerto Rico GAP Analysis Project. 2006. PRGAP Landcover. USDA Forest Service, International Institute of Tropical Forestry.

U.S. Fish and Wildlife Service Caribbean Ecological Services Field Office. July 2022. Conservation Measures for sea turtles in Puerto Rico and the U.S. Virgin Islands. [https://www.fws.gov/sites/default/files/documents/Conservation%20Measures%20for%20sea%20turtles%20in%20Puerto%20Rico%20and%20the%20U.S.%20Virgin%20Islands.pdf].

Caribbean Coastal Shoreline Condition

This indicator assesses shoreline alteration based on the presence of hardened structures like seawalls, groins, and riprap at the dynamic interface between land and water along the U.S. Caribbean coast. It originates from the National Oceanic and Atmospheric Administration's Continually Updated Shoreline Product.

This indicator was last updated in October 2023.

Reason for Selection

Armoring along shorelines, such as jetties, groins, seawalls, revetments, and other structures, provide a measure of habitat alteration at the dynamic interface between land and water along the coast. Human infrastructure along shorelines generally stabilizes the coastline, impeding natural beach migration processes. Groins, seawalls, jetties, and revetments have resulted in narrowing of beaches, or greater beach loss, compared to unstructured beaches (Dugan et al. 2008, Hall and Pilkey 1991, Mohanty et al. 2012, Pilkey and Wright III 1988). Jetties also alter sand transport and may result in downdrift erosion (Bruun 1995). Hardened structures landward of tidal wetlands can cause "coastal squeeze" by accelerating erosion during storms and preventing inland migration in response to sea level rise (Gittman 2015).

In Puerto Rico, shoreline stabilization intended to stop shoreline erosion has made beaches narrower and steeper, resulting in "dramatic loss of beach recreational quality and ease of beach access" (Bush et al. 2009).

In addition, hardened shorelines, particularly seawalls, generally support lower levels of biodiversity (Gittman et al. 2015, 2016). Studies funded by the National Oceanic and Atmospheric Administration's (NOAA) National Centers for Coastal Ocean Science found that shoreline hardening has a negative impact on amounts of submerged aquatic vegetation and on fish density and egg-laying (NOAA NCOS 2015, 2013).

Input Data

- National Oceanic and Atmospheric Administration (NOAA) Continually Updated Shoreline Product (CUSP), accessed 1-11-2023; read a 1-page factsheet about CUSP; view and download CUSP data in the NOAA Shoreline Data Explorer (to download, select "Download CUSP by Region" and select Southeast Caribbean)
- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- If the "Attribute" field in the CUSP dataset contained the word "natural", assign the feature a value of 1. Assign a value of 0 to all other features.
- Convert the entire CUSP linework to a 30 m raster.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

1 = Natural
0 = Armored

Known Issues

- This indicator overestimates shoreline condition in areas near hardened structures or urban development. It only assigns a low value to places that are currently hardened and not to nearby areas that can also be impacted by that hardening.
- This indicator overestimates shoreline condition in areas with active beach renourishment. Beach renourishment negatively affects some beach and dune species but is not captured by this indicator.

Other Things to Keep in Mind

- This indicator uses a simplified approach compared to its counterpart for the continental Southeast. The continental indicator gives a slightly higher score to partially armored and natural shorelines that fall within the Coastal Barrier Resource System (CBRS). These areas are not eligible for federal flood insurance and can therefore be considered "harder to develop". We dropped the CBRS from the Caribbean indicator because these areas can still be developed through non-federal investments and will not necessarily remain in a more natural condition. In addition, using just the CBRS does not give a score bump to shorelines in protected conservation lands that are much less likely to develop. We considered giving a higher score to protected shorelines to fix that issue but determined that making already protected shorelines a higher priority for conservation action in the final Blueprint was inconsistent with the intent of the Blueprint and the approach used for other similar indicators. We plan to update the continental version to match in the next Blueprint update.
- This indicator also uses a different data source compared to the continental version. The Caribbean version uses CUSP shoreline data, while the continental uses ESI. Since CUSP is more recent, we plan to update the continental version to match in the next Blueprint update.
- This indicator does not always align with Caribbean beach habitat. Some areas identified as armored in this indicator are scored as important habitat beach habitat, especially in areas where the beach habitat data comes from the GAP Wilson's plover model (e.g., the Hyatt Regency Grand Hotel in Río Grande, Puerto Rico). This often occurs where riprap is present along narrow beaches, or occasionally near bulkheads. There is often a section of beach present behind the riprap or bulkhead that could still provide habitat, or the riprap is sporadically placed on a long stretch of beach to protect inland structures. In these cases, the mismatch reflects the different intent of these complementary indicators. In some cases, hardened structures may be actually misclassified as beach. Inconsistencies in alignment and classification likely result from the older age and coarser resolution of the GAP data (10 m raster based on 2001 landcover) compared to the more recent and fine-scale CUSP shorelines (vectors dating primarily from 2014-2021) and challenges in distinguishing the unique remote sensing signature of beach vs. riprap and other hardened structures. Because of the 30 m resolution of the Blueprint and underlying data, a single pixel may contain a mix of beach habitat and hardened structures and be reflected differently in each of these two indicators due to their different functions.

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Literature Cited

Bruun, P., 1995. The development of downdrift erosion. J. Coast. Res., 1242-1257. [https://www.jstor.org/stable/4298427].

David M. Bush, William J. Neal, Chester W. Jackson, 2009. “Summary of Puerto Rico’s vulnerability to coastal hazards: Risk, mitigation, and management with examples”, America’s Most Vulnerable Coastal Communities, Joseph T. Kelley, Orrin H. Pilkey, J. Andrew, G. Cooper. [https://caribbeanclimatehub.org/wp-content/uploads/2019/08/SummaryofPuertoRicosVulnerabilitytoCoastalHazard-RiskMitigationandManagement_2009.pdf].

Dugan, J.E., Hubbard, D.M., Rodil, I.F., Revell, D.L., Schroeter, S., 2008. Ecological effects of coastal armoring on sandy beaches. Marine Ecology 29, 160-170. [https://onlinelibrary.wiley.com/doi/full/10.1111/j.1439-0485.2008.00231.x].

Gittman, R.K., Fodrie, F.J., Popowich, A.M., Keller, D.A., Bruno, J.F., Currin, C.A., Peterson, C.H. and Piehler, M.F. (2015), Engineering away our natural defenses: an analysis of shoreline hardening in the US. Frontiers in Ecology and the Environment, 13: 301-307. [https://doi.org/10.1890/150065].

Gittman, Rachel K., Steven B. Scyphers, Carter S. Smith, Isabelle P. Neylan, Jonathan H. Grabowski, Ecological Consequences of Shoreline Hardening: A Meta-Analysis, BioScience, Volume 66, Issue 9, 01 September 2016, Pages 763-773. [https://doi.org/10.1093/biosci/biw091].

Hall, M.J., Pilkey, O.H., 1991. Effects of hard stabilization on dry beach width for New Jersey. J. Coast. Res., 771-785. [https://journals.flvc.org/jcr/article/view/78532/75937].

Mohanty, P.K., Patra, S.K., Bramha, S., Seth, B., Pradhan, U., Behera, B., Mishra, P., Panda, U.S., 2012. Impact of groins on beach morphology: a case study near Gopalpur Port, east coast of India. J. Coast. Res. 28, 132-142. [https://bioone.org/journals/journal-of-coastal-research/volume-28/issue-1/JCOASTRES-D-10-00045.1/Impact-of-Groins-on-Beach-Morphology--A-Case-Study/10.2112/JCOASTRES-D-10-00045.1.full].

National Oceanic and Atmospheric Administration National Centers for Coastal Ocean Science. August 10, 2015. Hardened Shorelines Make it Hard for Submerged Aquatic Vegetation. [https://coastalscience.noaa.gov/news/hardened-shorelines-make-hard-submerged-aquatic-vegetation/].

National Oceanic and Atmospheric Administration National Centers for Coastal Ocean Science. January 23, 2013. Project Finds Fish Prefer Natural Shorelines. [https://coastalscience.noaa.gov/news/riprap-project-finds-fish-prefer-natural-shorelines-human-response-to-sea-level-rise-at-issue/].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, National Geodetic Survey. NOAA Continually Updated Shoreline Product (CUSP): Southeast Caribbean. [https://shoreline.noaa.gov/cusp.html].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, Office of Response and Restoration, Hazardous Materials Response Division, Seattle, Washington. June 2001. Puerto Rico ESI/RSI: HYDRO. [https://response.restoration.noaa.gov/esi_download].

Pilkey, O.H., Wright III, H.L., 1988. Seawalls versus beaches. J. Coast. Res., 41-64. [https://www.jstor.org/stable/25735351].

Rice, T.M., 2012a. Inventory of Habitat Modifications to Tidal Inlets in the Continental U.S. Coastal Migration and Wintering Range of the Piping Plover *(Charadrius melodus)*. Appendix 1b, in Comprehensive Conservation Strategy for the Piping Plover *(Charadrius melodus)* in its Coastal Migration and Wintering Range in the Continental United States. U.S. Fish and Wildlife Service, East Lansing, Michigan.

Caribbean Fish Hotspots

This indicator represents areas of high predicted fish density and diversity in the U.S. Caribbean based on the presence of mangroves, seagrass, and coral in close proximity to one another. It draws from research in Puerto Rico that examines fish density and the number of fish species present at different distances from various habitat types (Pittman et al. 2007). It uses benthic habitat data from The Nature Conservancy and landcover from the National Oceanic and Atmospheric Administration's Coastal Change Analysis Program.

This indicator was last updated in October 2023.

Reason for Selection

Nearshore waters where mangroves, seagrass, and coral are all present and within close proximity to one another are likely to support higher densities and diversity of fish species (Pittman et al. 2007). The co-occurrence of these three habitats supports healthy coastal ecosystems and connected seascapes (Gillis et al. 2017). While the movement and dispersal patterns for different fish species can vary widely, mangroves, seagrass, and coral provide key ecological services and functions for many species. For example, mangroves and seagrass beds serve as important nursery habitats, especially for fish species that, as adults, also depend on coral reefs (Nagelkerken et. al 2001). Many fish species, like mangrove snapper and yellowtail snapper, move through all three habitat types within their home ranges (Pittman et al. 2007).

The 300 m and 600 m distance thresholds used in this indicator draw on personal communication with Dr. Simon Pittman (1-25-2023) and several studies examining seascape structure and the number and diversity of fish species present at different distances from various habitat types. Research in southwest Puerto Rico shows that the positive impact of co-occurring mangrove, seagrass, and coral reef habitat on fish abundance is species-specific and strongest at 100 m but ranges between 50 and 600 m (Pittman et al. 2007). In a decision support framework developed for the U.S. Virgin Islands, "coral reefs were deemed strongly connected where they existed within 300 m of seagrasses, mangroves and other reefs" (Pittman et al. 2018). These findings align with research in Australia and

the western Pacific that considered habitats within 250-500 m of one another to be highly connected (Olds et al. 2012; Martin et al. 205), and a study in the United Arab Emirates that used a 500 m buffer to prioritize relationships between mangrove, seagrass, and reefs (Pittman et al. 2022).

Input Data

- The Nature Conservancy's (TNC) Caribbean benthic habitat maps; read a press release about the data; read a scientific journal article about the data; request to download the data
- 2012 National Oceanic and Atmospheric Administration (NOAA) Coastal Change Analysis Program (C-CAP) land cover files for the U.S. Virgin Islands (St. Thomas, St. John, and St. Croix are provided as separate rasters) accessed 4-26-2022; learn more about C-CAP high resolution land cover and change products
- 2010 NOAA C-CAP land cover files for Puerto Rico, accessed 4-26-2022; learn more about C-CAP high resolution land cover and change products
- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- Mosaic the benthic data for Puerto Rico and the U.S. Virgin Islands.
- Mosaic the C-CAP landcover data for Puerto Rico and the U.S. Virgin Islands.
- Reproject and do a majority resample of the TNC benthic data to 30 m pixels.
- Reproject and do a majority resample of the C-CAP data to 30 m pixels.
- Create a seagrass raster by reclassifying the TNC benthic data so that "dense seagrass" is 1, all other data is 0, and NoData is 1,000. The NoData value helps later in the analysis to remove land and deal with differences in NoData between the C-CAP and TNC benthic data.
- Create a coral raster by reclassifying the TNC benthic data so that "Reef Crest", "Fore Reef", "Back Reef", "Coral/Algae", and "Spur and Groove Reef" are 1, all other classes are 0, and NoData is 0.
- Create a mangrove raster by reclassifying the C-CAP data so that "Estuarine forested wetland" is 1, all other classes are 0, and NoData is 0.
- Combine the mangrove and seagrass data to make a water mask to remove land (including mangroves) from the final indicator.
- 600 m analysis: For each habitat raster (mangrove, seagrass, and coral), use a 20 cell radius circle and the Focal Statistics "MAXIMUM" function to identify areas with at least one pixel of those habitats. Sum these rasters together and multiply by the presence of mangrove (0/1) to get habitat diversity. This step calculates how many of the distinct habitat types occur within a 600 m radius. It also removes the actual mangrove pixels, as this indicator targets the estuarine and marine habitats near mangroves.
- 300 m analysis: Repeat the same steps from the 600 m analysis but use a 10 cell radius.
- Combine the 600 m analysis and 300 m analysis to get final indicator classes.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

4 = Highest predicted fish density/diversity (mangrove, coral, and dense seagrass all present within 300 m)
3 = Very high predicted fish density/diversity (either mangrove and coral, mangrove and dense seagrass, or coral and dense seagrass present within 300 m)
2 = High predicted fish density/diversity (mangrove, coral, and dense seagrass all present within 600 m)
1 = Medium predicted fish density/diversity (either mangrove and coral, mangrove and dense seagrass, or coral and dense seagrass present within 600 m)
0 = Low predicted fish density/diversity (no coral, mangrove, or dense seagrass present within 600 m of one other)

Known Issues

- For some pixels at the edge of the Caribbean subregion, less than half of the 30 m pixel is covered by the finer resolution TNC benthic data. These cells are classified as NoData in the indicator.
- The distances used in this indicator are primarily based on a study conducted in southwest Puerto Rico (Pittman et al. 2007), a decision support framework developed for the U.S. Virgin Islands, and personal communication with the principal investigator of those projects, Dr. Simon Pittman (1-25-2023). While other similar studies in Australia, the western Pacific, and the United Arab Emirates support the distance thresholds chosen for this analysis (Olds et al. 2012, Martin et al. 2015, Pittman et al. 2022), different distances may be more appropriate for other parts of the U.S. Caribbean.
- This indicator may overestimate the fish habitat value of some terrestrial areas that were evaluated by the TNC benthic habitat dataset (e.g., areas near Limetree Bay Refinery in the southern part of St. Croix, USVI).

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Literature Cited

Gillis, L. G., Jones, C. G., Ziegler, A. D., van der Wal, D., Breckwoldt, A., and Bouma, T. J. 2017. Opportunities for protecting and restoring tropical coastal ecosystems by utilizing a physical connectivity approach. Front. Marine Sci. 4:374. [https://www.frontiersin.org/articles/10.3389/fmars.2017.00374/full].

Martin TSH, Olds AD, Pitt KA, Johnston AB, Butler IR, Maxwell PS, Connolly RM. 2015. Effective protection of fish on inshore coral reefs depends on the scale of mangrove-reef connectivity. Mar Ecol Prog Ser 527:157-165. [https://doi.org/10.3354/meps11295].

Nagelkerken, Ivan & Kleijnen, Sarah & Klop, T & Brand, RACJ & Morinière, EC & Van der Velde, Gerard. (2001). Dependence of Caribbean reef fishes on mangroves and seagrass beds as nursery habitats: A comparison of fish faunas between bays with and without mangroves/seagrass beds. Marine Ecology-progress Series - MAR ECOL-PROGR SER. 214. 225-235. 10.3354/meps214225. [https://www.int-res.com/articles/meps/214/m214p225.pdf].

Olds, AD, Connolly, RM, Pitt, KA., Maxwell, PS. (2012). Habitat connectivity improves reserve performance. Conservation Letters 5: 56–63. [https://conbio.onlinelibrary.wiley.com/doi/10.1111/j.1755-263X.2011.00204.x].

Pittman SJ, Caldow C, Hile SD, Monaco ME. 2007. Using seascape types to explain the spatial patterns of fish in the mangroves of SW Puerto Rico. Marine Ecology Progress Series. Vol. 348: 273–284. [https://doi.org/10.3354/meps07052].

Pittman, S.J., Poti, M., Jeffrey, C.F., Kracker, L.M. and Mabrouk, A., 2018. Decision support framework for the prioritization of coral reefs in the US Virgin Islands. Ecological Informatics, 47, pp.26-34. [https://www.sciencedirect.com/science/article/abs/pii/S1574954117300614].

Pittman, S.J. et al 2022. Rapid site selection to prioritize coastal seascapes for nature-based solutions with multiple benefits. Frontiers in Marine Science, 9, p.571. [https://www.frontiersin.org/articles/10.3389/fmars.2022.832480/full].

Schill SR, McNulty VP, Pollock FJ, Lüthje F, Li J, Knapp DE, Kington JD, McDonald T, Raber GT, Escovar-Fadul X, Asner GP. Regional High-Resolution Benthic Habitat Data from Planet Dove Imagery for Conservation Decision-Making and Marine Planning. Remote Sensing. 2021; 13(21):4215. [https://doi.org/10.3390/rs13214215].

Caribbean Fish Nursery Habitat

This indicator represents nursery and spawning habitat or other concentration areas for fish in the U.S. Caribbean. It captures places like mangrove lagoons, bays, estuaries, and some coral reefs. These areas serve as important nursery habitat for many fish species including snook, tarpon, snapper, great barracuda, grunt, mojarra, mullet, jack, bonefish, and more. This data originates from the National Oceanic and Atmospheric Administration's Environmental Sensitivity Index.

Reason for Selection

Areas that provide nursery habitat for juvenile fish, as well as areas where fish spawn and otherwise congregate, are important for ensuring the persistence of abundant and diverse fisheries. Many fish use these habitats during multiple life stages and throughout all seasons. For example, many coastal

fish lay their eggs in estuarine habitats because of the combination of food availability for juveniles and protection from predators (Sheaves 2015).

This indicator was last updated in October 2023.

Input Data

- National Oceanic and Atmospheric Administration (NOAA) Environmental Sensitivity Index (ESI) data for Puerto Rico and Virgin Islands; download ESI maps and data (PuertoRico_2000_GDB and VirginIsl_2000_GDB, last modified October 10, 2012); read NOAA's definitions of the biological resource classes

  The ESI biological resources data is intended to represent species that are particularly vulnerable to the impacts of oil spills. NOAA classifies these species into seven general categories, referred to as "elements": birds, fish, invertebrates, habitats, marine mammals, terrestrial mammals, and reptiles. NOAA further divides the elements into sub-elements by grouping species with similar characteristics (e.g., taxonomy, morphology, life history, and/or oil spill sensitivity). NOAA compiled the biological resources data with the assistance of biologists from the U.S. Fish and Wildlife Service and other organizations.

  In this indicator, we use polygons associated with the fish element and two sub-elements: estuarine nursery fish and estuarine resident fish. ESI defines the estuarine nursery fish class as "spawning, nursery, or other concentration areas". ESI defines the estuarine resident fish polygons as "spawning or other concentration areas; locations of threatened, endangered, or rare species". The estuarine nursery fish polygons represent habitat for the following estuarine nursery fish: tarpon (*Megalops atlanticus*), mutton snapper (*Lutjanus analis*), yellowtail snapper (*Ocyurus chrysurus*), great barracuda (*Sphyraena barracuda*), common snook (*Centropomus undecimalis*), bonefish (*Albula vulpes*), mojarras (*Diapterus* spp.), mullet (*Mugil* spp.), jacks, grunts, other snappers, and other species generally classified as "nursery fish". The estuarine resident fish polygons represent habitat for mangrove molly (*Poecilia orri*).

  Many of these areas also provide habitat for other species of diadromous, freshwater, marine benthic, and marine pelagic fish. ESI also classifies the e_nursery sub-element fish polygons used in this indicator as associated with the following diadromous fish: sirajo goby (*Sicydium plumieri*), hog-nosed mullet (J*oturus pichardi*), mountain mullet (*Agonostomus monticola*), and other species generally classified as "native stream fish"; the following freshwater fish: largemouth bass (*Micropterus salmoides*), redear sunfish (*Lepomis microlophus*), channel catfish (*Ictalurus punctatus*), peacock bass (*Cichla ocellaris*), and tilapia (*Tilapia* spp.); the following marine benthic fish: red hind (*Epinephelus guttatus*), permit (*Trachinotus falcatus*), butterfly fish (*Chaetodon* spp.), damselfish (*Chromis* spp.), groupers, parrotfish, wrasses, and other species generally classified as "reef fish"; and the following pelagic fish: species generally classified as "pelagic fish".

- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- From the ESI geodatabase for Puerto Rico and the U.S. Virgin Islands, select the "fish_polygon" vector layer and the associated "biofile" table. Join the biofile table to the fish polygons using the attribute "RARNUM". Note: be sure to use a one-to-many join because many of the fish polygons are coded to multiple sub-elements.
- Merge the Puerto Rico and USVI fish polygon data into a single layer.
- From the merged fish polygon layer, select by attributes where SUBELEMENT = 'e_nursery' or SUBELEMENT = 'e_resident' and extract the selected polygons.
- Convert the extracted nursery polygons from vector to raster.
- Using the LANDFIRE EVT data, remove all pixels with a value in the EVT_NAME field of "Developed-High Intensity", "Developed-Medium Intensity", "Developed-Low Intensity", "Developed-Roads", "Developed-Open Space" or "Quarries-Strip Mines-Gravel Pits-Well and Wind Pads". This removes areas from the fish nursery polygons that have been urbanized since the ESI data was created.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

1 = Fish spawning, nursery, or other concentration area
0 = Not identified as a fish spawning, nursery, or other concentration area

Known Issues

- The indicator source data was published in 2001. As a result, this indicator may overestimate fish nursery habitat value in areas that have been altered by storms, urbanized, or converted to agriculture since the data was collected. We attempted to remove areas that have been developed using 2020 LANDFIRE EVT landcover. We left in agricultural areas because they may still provide some fish nursery habitat value if inundated.
- The 2020 LANDFIRE EVT landcover misclassifies some parking lots as wet areas due to aerial imagery. These areas may be overprioritized in the indicator.

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Literature Cited

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, Office of Response and Restoration, Hazardous Materials Response Division, Seattle, Washington. Published June 2001. Puerto Rico ESI/RSI: FISH (Fish Polygons). Second Edition. [https://response.restoration.noaa.gov/oil-and-chemical-spills/oil-spills/download-esi-maps-and-gis-data#PuertoRico].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, Office of Response and Restoration, Hazardous Materials Response Division, Seattle, Washington. Published August 2001. U.S. Virgin Islands and British Virgin Islands ESI: FISH (Fish Polygons), First Edition. [https://response.restoration.noaa.gov/esi_download#VirginIslands].

Sheaves, M., R. Baker, I. Nagelkerken, and R. M. Connolly. 2015. True value of estuarine and coastal nurseries for fish: Incorporating complexity and dynamics. Estuaries and Coasts, 38: 401-414. DOI: 10.1007/s12237-014-9846-x. [https://link.springer.com/article/10.1007/s12237-014-9846-x].

Caribbean Mangrove Migration

This indicator represents current and potential future locations of mangroves in the U.S. Caribbean based on landcover and barriers to mangrove migration like slope, elevation, and development. This indicator adapts methods from the U.S. Virgin Islands Climate Change Ecosystem-Based Adaptation guidance document and expands the analysis to the full U.S. Caribbean. It uses landcover from the National Oceanic and Atmospheric Administration's Coastal Change Analysis Program as well as slope, elevation, roads, and landcover from LANDFIRE.

This indicator was last updated in November 2025.

Reason for Selection

Mangroves are an iconic Caribbean coastal habitat that provide a wide diversity of natural and cultural resource benefits. On a global scale, mangroves in the Caribbean region make up about 12% of the worldwide population (TNC). Unfortunately, these systems have experienced significant declines globally, losing an estimated 2.1% of their area from 2000-2016 (Goldberg 2020), and as much as 20% over the last 40 years (Smith 2024).

Mangroves help protect human and wildlife populations by enhancing coastal resilience. They buffer coastal communities from storm surge, which can reduce inland flooding. In addition, the aerial root

systems provide natural protection in coastal environments by increasing sediment deposition, which reduces erosion rates and nutrient losses (Akram 2023). Mangroves also serve as "one of the world's most effective natural carbon capture systems" (TNC), storing as much as five times as much organic carbon as tropical upland forests (Donato et al. 2011).

Mangroves provide additional ecological benefits by serving as important habitat for resident and migratory bird communities (Acevendo 2008) as well as "hundreds of endemic species of fish" (TNC). The International Union for the Conservation of Nature (IUCN) describes mangroves as "nurseries for the world's seafood supply", recognizing the important they play in commercial and recreational fisheries. According to The Nature Conservancy, "with the high land-to-water ratio in the Caribbean, it is not surprising that the presence of mangroves heavily influences the health of the shorelines of multiple countries, as well as their local economies and communities. Without healthy mangroves, adjacent reefs and seagrasses are not well protected, posing challenges to species that the fishing industry heavily relies on like the Queen Conch, spiny lobster and Atlantic blue crab."

Input Data

- Southeast Blueprint 2025 extent
- 2012 National Oceanic and Atmospheric Administration (NOAA) Coastal Change Analysis Program (C-CAP) land cover files for the U.S. Virgin Islands (St. Thomas, St. John, and St. Croix are provided as separate rasters) accessed 4-26-2022; learn more about C-CAP high resolution land cover and change products
- 2010 NOAA C-CAP land cover files for Puerto Rico, accessed 4-26-2022; learn more about C-CAP high resolution land cover and change products
- 2020 (v2.2.0) LANDFIRE Existing Vegetation Type (EVT), Operational Roads (Roads), Slope Percent Rise (SlpP), and Elevation for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data

Mapping Steps

- Combine the C-CAP data for Puerto Rico and the U.S. Virgin Islands, reproject to Albers Equal Area, and resample to 30 m to match the resolution of the Southeast Blueprint.
- Create current mangrove layer by extracting the Estuarine Forested Wetland class from C-CAP.
- Create a barrier layer using LANDFIRE EVT, roads, slope, and elevation based on thresholds used by Schill et al. 2014. Include all developed classes from the EVT layer, all roads, slopes ≥10%, and elevation >2 m. Combine them all into a single barrier layer, assigning a value of 1 if any of those barriers to mangrove migration are present.
- Use the ArcGIS Distance Accumulation tool to identify migration space not blocked by barriers.
- Make a mask to remove areas of open water and ocean using areas of NoData (-9999) and open water (7292) in LANDFIRE EVT.
- Use the resulting mask to remove open water and ocean. We use the mask after estimating migration space in order to allow mangroves to use open water and ocean to move to new

sites that don't currently have mangroves, but could in the future. This is different from the methods originally used by Schill et. al 2014.
- Use the ArcGIS Region Group tool to remove isolated pixels and very small patches, retaining only areas roughly 2 acres or more in size (9 or more 30 m pixels)
- Reclassify the resulting layer into the indicator categories below.
- As a final step, clip to the extent of the Southeast Blueprint.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

4 = Current mangrove
3 = Potential future mangrove <0.5 km from current mangrove
2 = Potential future mangrove 0.5-1 km from current mangrove
1 = Potential future mangrove >1 km from current mangrove
0 = Not likely to be current or future mangrove

Known Issues

- The indicator underestimates mangrove in some areas due to classification issues with C-CAP and the age of the data missing some newly established mangroves.
- The indicator overestimates mangrove extent in some areas where mangrove was cleared after 2010 (Puerto Rico) or 2012 (Virgin Islands). This is based on the age of the landcover data.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Acevedo, Miguel A.; Aide, Miguel and T. Mitchell. "Bird Community Dynamics and Habitat Associations in Karst, Mangrove and *Pterocarpus* Forest Fragments in an Urban Zone in Puerto Rico," *Caribbean Journal of Science* 44(3), 402-416, (1 December 2008). [https://doi.org/10.18475/cjos.v44i3.a15].

Akram H, Hussain S, Mazumdar P, Chua KO, Butt TE, Harikrishna JA. Mangrove Health: A Review of Functions, Threats, and Challenges Associated with Mangrove Management Practices. *Forests*. 2023; 14(9):1698. [https://doi.org/10.3390/f14091698].

Bimrah, K.; Dasgupta, R.; Hashimoto, S.; Saizen, I.; Dhyani, S. Ecosystem Services of Mangroves: A Systematic Review and Synthesis of Contemporary Scientific Literature. 2022. Sustainability, 14, 12051. [https://doi.org/10.3390/su141912051].

Donato, D. C., Kauffman, J. B., Murdiyarso, D., Kurnianto, S., Stidham, M., and Kanninen, M. (2011). Mangroves among the most carbon-rich forests in the tropics. Nat. Geosci. 4, 293–297. doi: 10.1038/ngeo1123. [https://www.nature.com/articles/ngeo1123].

Goldberg L, Lagomasino D, Thomas N, Fatoyinbo T. Global declines in human-driven mangrove loss. Glob Chang Biol. 2020 Oct;26(10):5844-5855. [https://doi.org/10.1111/gcb.15275].

International Union for Conservation of Nature and Natural Resources (IUCN). August 24, 2017. Mangroves: nurseries for the world's seafood supply. [https://iucn.org/news/forests/201708/mangroves-nurseries-world%E2%80%99s-seafood-supply].

Smith, Renée. Caribbean Biodiversity Fund. February 2, 2024. Unveiling the Marvels of Mangrove Forests: The Mighty Coastal Wetlands. [https://caribbeanbiodiversityfund.org/blog/unveiling-the-marvels-of-mangrove-forests-the-mighty-coastal-wetlands/].

Rull, Valentí. 2023. Rise and fall of Caribbean mangroves. Science of The Total Environment, Volume 885. Accessed November 25, 2024. [https://www.sciencedirect.com/science/article/pii/S0048969723024725].

The Nature Conservancy. Mangroves of the Caribbean: Vital Protectors of Reefs, Seagrass, and Coastal Economies. Accessed May 13, 2025. [https://www.nature.org/en-us/about-us/where-we-work/caribbean/stories-in-caribbean/exploring-caribbean-mangroves/].

Caribbean Seagrass

This indicator represents the presence and density of seagrass at various depths in the U.S. Caribbean. It uses benthic habitat data from The Nature Conservancy and bathymetry data from the National Oceanic and Atmospheric Administration.

This indicator was last updated in October 2023.

Reason for Selection

Seagrass provides food and habitat for a range of marine and estuarine wildlife, including fish, sea turtles, octopuses, and more (NWF 2021). Seagrass in shallow water (<13 m) and very shallow water (<5 m) is particularly important for many species like manatee and queen conch (Marshak et al. 2006, Drew et al. 2012). Most manatee grazing occurs in waters <5 m deep (Drew et al. 2012). Seagrass also produces oxygen, filters water, controls erosion, and buffers storms. It serves as an important indicator of the overall health of coastal ecosystems because it is sensitive to water quality and requires sufficiently clear water for sunlight to penetrate (NPS 2021, NWF 2021). In addition, seagrass meadows are being increasingly recognized for their value as a carbon sink in the face of climate change (McKenzie et al. 2020).

Input Data

- The Nature Conservancy's (TNC) Caribbean benthic habitat maps, accessed 12-21-2022; read a press release about the data; read a scientific journal article about the data; request to download the data
- National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI) Coastal Relief Model (Vol.9 - Puerto Rico), accessed 11-22-2022

  This layer was preprocessed to create a Caribbean Blueprint bathymetry raster layer. The coastal relief data is in NetCDF format. The NetCDF layer was imported into ArcPro using the Make NetCDF Raster Layer function in the Multidimensions Toolbox, then saved as a .tif. The horizontal datum was converted to EPSG 5070 and the vertical datum is in meters representing depth.

- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- Mosaic together TNC's benthic data for Puerto Rico and the Virgin Islands.
- Reclassify the Caribbean bathymetry layer to identify land (0 m and above), very shallow waters (down to 5 m), shallow waters (5-13 m), and deeper waters (below 13 m).
- Reproject and do a majority resample of the benthic data to 30 m pixels.
- Reproject and do a majority resample of the bathymetric data to 30 m pixels.
- Reclassify the benthic data to assign "Dense Seagrass" a value of 2, "Sparse Seagrass" a value of 1, and all other classes a value of 0.
- Combine the benthic and bathymetric data and reclassify into the final indicator values below. The bathymetric data are coarser than 30 m, which results in some nearshore areas being incorrectly classified as land based on the finer resolution benthic data. In these cases, include benthic data classified as "land" as part of shallow water.
- Clip to the Blueprint 2023 Caribbean subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

6 = Very shallow and dense seagrass
5 = Very shallow and sparse seagrass
4 = Shallow and dense seagrass
3 = Shallow and sparse seagrass
2 = Deep and dense seagrass
1 = Deep and sparse seagrass
0 = Not seagrass

Known Issues

- For some pixels at the edge of the Caribbean subregion, less than half of the 30 m pixel is covered by the finer resolution benthic data. These cells are classified as NoData.
- Seagrass depth classes are over and underestimated in some areas—especially nearshore and in the very shallow depth class. The bathymetry data is coarser than 30 m resolution, which results in more coarse estimates of depth.
- Seagrass deeper than 13 m can be hard to differentiate from algae with the Planet Dove imagery used in the TNC benthic map source data. It's likely some areas classified as deep seagrass are actually algae or a mix of algae and seagrass.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Drew CA, Alexander-Vaughn LB, Collazo JA. Science summary in support of Manatee Protection Area (MPA) design in Puerto Rico. US Fish and Wildlife Service; 2012. [https://doi.org/10.3996/CSS.101.2013].

Marshak AR, Appeldoorn RS, Jiménez NI. Utilization of GIS mapping in the measurement of the spatial distribution of queen conch (*Strombus gigas*) in Puerto Rico. 57th Gulf and Caribbean Fisheries Institute. 2006. [https://aquadocs.org/bitstream/handle/1834/29739/gcfi_57-3.pdf?sequence=1].

McKenzie, Len J. and Lina M. Nordlund, Benjamin L. Jones, Leanne C. Cullen-Unsworth, Chris Roelfsema, and Richard K. F. Unsworth. The global distribution of seagrass meadows. 2020. Environmental Research Letters 15 074041. [https://doi.org/10.1088/1748-9326/ab7d06].

National Geophysical Data Center, 2005. U.S. Coastal Relief Model - Puerto Rico. National Geophysical Data Center, NOAA. Accessed November 22, 2022. [https://www.ncei.noaa.gov/products/coastal-relief-model].

National Park Service. Seagrass Meadows. Accessed December 22, 2021. [https://www.nps.gov/subjects/oceans/seagrass-meadows.htm].

NOAA Office for Coastal Management, 2021: Seagrasses. NOAA National Centers for Environmental Information. [https://www.fisheries.noaa.gov/inport/item/56960].

Pittman SJ, Caldow C, Hile SD, Monaco ME. Using seascape types to explain the spatial patterns of fish in the mangroves of SW Puerto Rico. Marine Ecology Progress Series. 2007; Vol. 348: 273–284. [https://doi.org/10.3354/meps07052].

Schill SR, McNulty VP, Pollock FJ, Lüthje F, Li J, Knapp DE, Kington JD, McDonald T, Raber GT, Escovar-Fadul X, Asner GP. Regional High-Resolution Benthic Habitat Data from Planet Dove Imagery for Conservation Decision-Making and Marine Planning. Remote Sensing. 2021; 13(21):4215. [https://doi.org/10.3390/rs13214215].

The National Wildlife Federation. Seagrasses. Accessed December 22, 2021. [https://www.nwf.org/Educational-Resources/Wildlife-Guide/Plants-and-Fungi/Seagrasses].

Caribbean Shallow Hardbottom & Coral

This indicator measures the presence of hardbottom habitat and coral in the U.S. Caribbean. It also predicts the ability of coral to survive the impacts of climate change based on reef locations, past and future thermal conditions, hurricane impacts, and coral larval connectivity. This indicator combines benthic habitat and coral climate refugia data from The Nature Conservancy.

This indicator was last updated in October 2023.

Reason for Selection

Coral reefs and hardbottom serve as important habitat for much of the marine biodiversity in the Caribbean. They also provide important economic and cultural goods and services that benefit nearby coastal communities, such as supporting fisheries, filtering seawater, and buffering the impacts of storms (NOAA 2020). In addition, coral reefs can serve as environmental indicators of water quality because of their narrow temperature, salinity, and water clarity requirements (EPA 2005).

While climate change will affect all habitats in the Caribbean, coral reefs are likely to face the largest impacts. According to The Nature Conservancy, "...in recent decades, coral cover throughout the region has declined by almost 60 percent. Remaining reefs are threatened by warmer waters, harsher storms, disease, overfishing and pollution" (2019).

In this indicator, we score hardbottom with sparse algae higher than hardbottom with dense algae because algae negatively impact coral colonization and resilience. According to Bonaldo et al., "macroalgae can suppress the settlement, growth and survivorship of corals, thereby limiting the capacity of coral populations to recover following disturbances" (2017).

Input Data

- The Nature Conservancy's (TNC) Caribbean benthic habitat maps, accessed 12-21-2022; read a press release about the data; read a scientific journal article about the data; request to download the data
- TNC coral climate refugia; read a journal article about the data
- Southeast Blueprint 2023 subregions: Caribbean
- Southeast Blueprint 2023 extent

Mapping Steps

- Convert the coral refugia data from vector to raster.
- Reclassify the coral refugia ranks to above and below average values using the field "rank_PRU". Lower values in the rank_PRU class correspond to higher resilience. Values ranging from 1-775 are considered above average, and values ranging from 776-1552 are considered below average. The rank of 0 is not assessed in the Puerto Rico/U.S. Virgin Islands portion of the coral refugia analysis, so treat it as NoData.
- Mosaic together The Nature Conservancy's benthic data for Puerto Rico and the Virgin Islands.
- Reproject and do a majority resample of the benthic data to 30 m pixels.
- Reclassify the benthic data to assign "Reef Crest" (1), "Fore Reef" (2), "Back Reef" (3), "Coral/Algae" (4) and "Spur and Groove Reef" (5) a value of 3; "Hardbottom Sparse Algae" (6) a value of 2, "Hardbottom Dense Algae" (7) a value of 1; and all other classes a value of 0. While conceptually "Boulders and Rocks" (11) would score more highly than the hardbottom sparse and dense algae classes, because algae have a negative impact on coral settlement, no boulders and rocks pixels occur in the TNC benthic data in Puerto Rico and the U.S. Virgin Islands.
- Combine the above rasters and reclassify into the final indicator values seen below.
- Clip to the Caribbean Blueprint 2023 subregion.
- As a final step, clip to the spatial extent of Southeast Blueprint 2023.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Final indicator values

Indicator values are assigned as follows:

5 = Coral with above average climate resilience
4 = Coral with unknown climate resilience
3 = Coral with below average climate resilience
2 = Hardbottom with sparse algae
1 = Hardbottom with dense algae
0 = Not coral or hardbottom

Known Issues

- While the benthic data was at a finer resolution than 30 m, the coral climate refugia analysis was at a 1 km^2 resolution. This likely resulted in this indicator undervaluing or overvaluing a number of 30 m pixels at the edge of the 1 km^2 squares.
- For some pixels at the edge of the Caribbean Blueprint 2023 subregion, less than half of each 30 m pixel is covered by the finer resolution benthic data. These cells are classified as NoData in the final indicator.

Disclaimer: Comparing with Older Indicator Versions

There are numerous problems with using Southeast Blueprint indicators for change analysis. Please consult Blueprint staff if you would like to do this (email hilary_morris@fws.gov).

Literature Cited

Bonaldo RM, Pires MM, Guimarães PR Junior, Hoey AS, Hay ME (2017) Small Marine Protected Areas in Fiji Provide Refuge for Reef Fish Assemblages, Feeding Groups, and Corals. PLoS ONE 12(1): e0170638. [https://doi.org/10.1371/journal.pone.0170638].

Chollett I, Escovar-Fadul X, Schill, SR, Croquer A, Dixon AM, Beger M, Shaver E, McNulty VP, Wolff NH, Planning for resilience: Incorporating scenario and model uncertainty and trade-offs when prioritizing management of climate refugia. Global Change Biology. 2022; 28, 4054– 4068. [https://doi.org/10.1111/gcb.16167].

Environmental Protection Agency. Coral Reefs Fact Sheet. October 2005. [https://archive.epa.gov/water/test/web/pdf/2007_6_28_oceans_coral_documents_coralreeffactsheet.pdf].

Hoegh-Guldberg O, Pendleton L, Kaup A. People and the changing nature of coral reefs. Regional Studies in Marine Science. 2019; Volume 30,100699. [https://doi.org/10.1016/j.rsma.2019.100699].

Kleypas J, Allemand D, Anthony K, Baker AC, Beck MW, Hale LZ, Hilmi N, Hoegh-Guldberg O, Hughes T, Kaufman L, Kayanne H, Magnan AK, Mcleod E, Mumby P, Palumbi S, Richmond RH, Rinkevich B, Steneck RS, Voolstra CR, Wachenfeld D, Gattuso JP. Designing a blueprint for coral reef survival. Biological Conservation. 2021; Volume 257,109107. [https://doi.org/10.1016/j.biocon.2021.109107].

National Oceanic and Atmospheric Administration Coral Reef Conservation Program. 2020. Coral reef condition: A status report for Puerto Rico. [https://icriforum.org/wp-content/uploads/2020/05/Puerto-Rico-English.pdf].

Schill SR, McNulty VP, Pollock FJ, Lüthje F, Li J, Knapp DE, Kington JD, McDonald T, Raber GT, Escovar-Fadul X, Asner GP. Regional High-Resolution Benthic Habitat Data from Planet Dove Imagery for Conservation Decision-Making and Marine Planning. Remote Sensing. 2021; 13(21):4215. [https://doi.org/10.3390/rs13214215].

The Nature Conservancy. July 17, 2019. Stories in the Caribbean: New Hope for Coral Reefs. [https://www.nature.org/en-us/about-us/where-we-work/caribbean/stories-in-caribbean/caribbean-a-revolution-in-coral-conservation/].

Running Zonation

Defining Subregions

We break down the Southeast Blueprint into subregions to reduce processing time and help with balancing weights across indicators in such a large and varied region. The subregions are aggregated into larger "zones" that define the extent of each Zonation run. The individual subregions then serve as inputs to the Zonation run for their zone to help balance priorities across the zone. The Southeast Blueprint 2026 subregions are available on the Blueprint page of the SECAS Atlas.

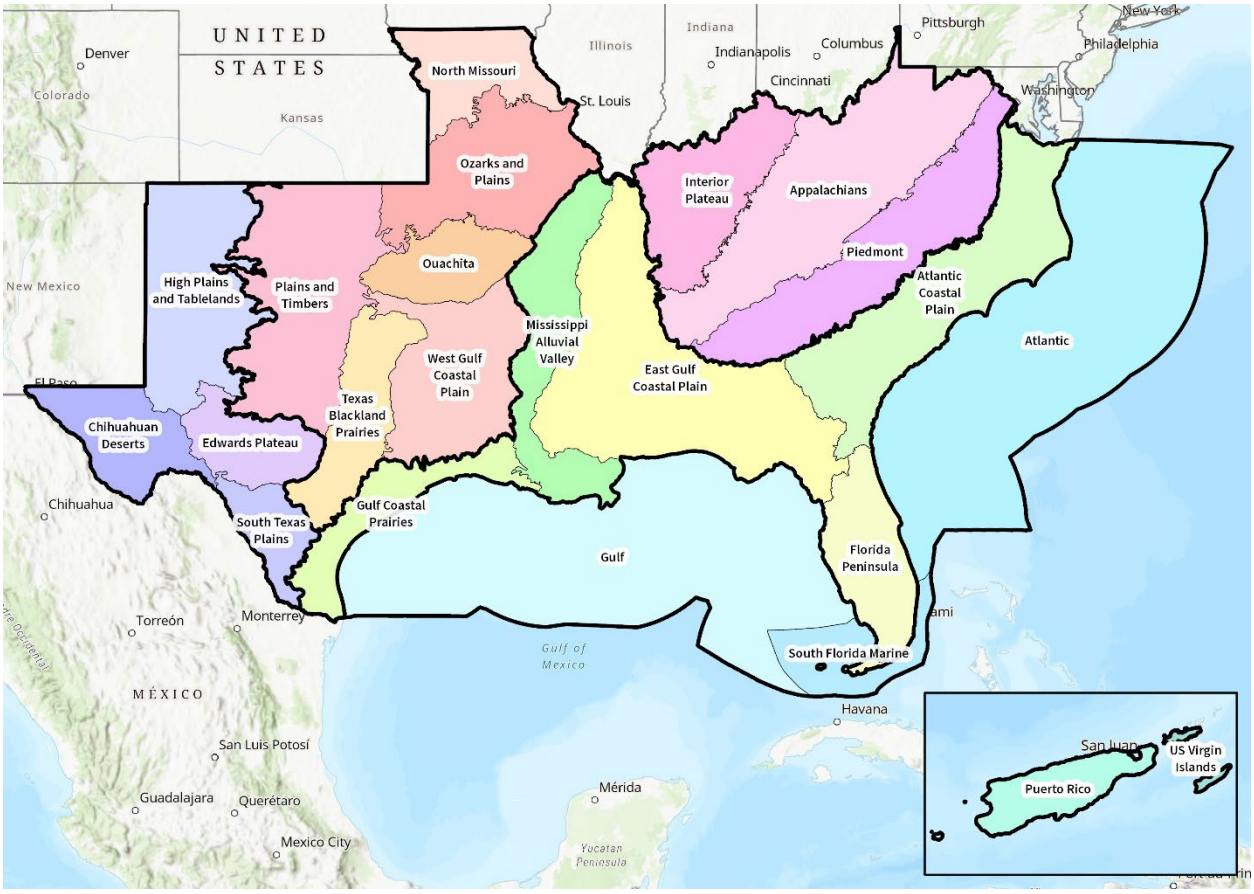


Figure 4. Subregions (colored polygons) and Zonation zones (black lines) used in Southeast Blueprint 2026. The purple subregions in west Texas define the Arid West Zone; the orange and red subregions from east Texas to Missouri define the Central West Zone; the green and yellow subregions along the Gulf and Atlantic coast define the Coastal Zone; the pink subregions near the Appalachians define the Greater Appalachians Zone; the blue offshore marine subregions define the Continental Marine Zone, and the turquoise subregions within the inset map define the Puerto Rico and U.S. Virgin Islands Zones

Continental Subregions

The continental portion of the Southeast Blueprint is composed of 15 Southeast states and the surrounding marine environment to the end of U.S. waters in the Atlantic Ocean and Gulf of America. We further subdivide the continental area into subregions as described below.

Input Data

- U.S. Environmental Protection Agency (EPA) Level IV Ecoregions shapefile with state boundaries (EPA L4), accessed 12-8-2021
- U.S. Geological Survey (USGS) Watershed Boundary Dataset (WBD), accessed 12-8-2021: HUC4s; download the data
- 2021 Census TIGER/Line State boundary, accessed 12-8-2021
- National Oceanic and Atmospheric Administration (NOAA) U.S. Maritime Limits and Boundaries, accessed 9-20-2023
- Marine Ecoregions Level III from the Commission for Environmental Cooperation North American Environmental Atlas, accessed 12-8-2021
- Southeast Blueprint extent
- Southeast Blueprint 2023 subregions

Mapping Steps

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code. Generally, we used the following mapping steps.

Inland Continental

- Do some initial grouping of the EPA ecoregions to make larger ecoregions. Usually, we grouped by Level III ecoregions; sometimes we went down to the Level IV ecoregions to make smoother, larger ecoregions.
- The EPA ecoregions include some “islands” where part of an ecoregion is not contiguous with the rest. Because we need contiguous subregions to use in running Zonation, identify those “islands” and assign them to the contiguous ecoregion that surrounds them. In addition, some ecoregions have long, finger-like, “peninsulas” that protrude from the ecoregion. Sometimes we assigned these to the surrounding ecoregion to make smoother edges.
- Union together the modified EPA ecoregions and HUC4 watershed layers and reclassify to assign subregions to unique combinations of EPA ecoregions and HUC4 watersheds.
- For the “islands” and “peninsulas” mentioned above, we made exceptions to the above rules and assigned those polygons to the subregion that surrounds them.
- Use the census TIGER/Line state boundaries to remove HUC4 areas that extend beyond the Southeast Blueprint extent (i.e., states not part of SECAS). This removes some coastal water areas in the Northeast United States that are outside the Southeast Blueprint area.

Marine Continental

- From the NOAA maritime limits and boundaries layer, select the lines from the U.S. Exclusive Economic Zone (EEZ) that define the outer limit of the U.S. offshore marine waters in the Atlantic Ocean and Gulf of America.
- Intersect the EEZ lines with the inland continental subregions to create a large marine polygon bounded by the coastal edge of the inland subregion and the EEZ. In the vast majority of the coastline, the coastal boundaries of the inland subregions are defined by HUC4s. In a tiny area of south Texas, the coastal boundary is instead defined by EPA Level III Ecoregions, and minor hand-editing is required to make the EEZ intersect with the subregional boundary there.
- From the marine ecoregions layer, pull out the South Florida/Bahamian Atlantic Level 1 marine ecoregion.
- Use that selected South Florida ecoregion to split the bigger marine subregion created above into three distinct subregions: Gulf of America, Atlantic, and South Florida Marine.

Continental Subregion Descriptions

Table 15. Definitions of all inland and marine continental subregions used in the Blueprint. The subregions were aggregated into Zonation zones that define the extent of each Zonation run. The subregions are also used as inputs to those Zonation runs to help balance priorities within zones.

Zonation Zone	Subregion	Extent
Arid West Zone	Chihuahuan Deserts	Defined based on the TX state boundary and the following EPA ecoregion: • EPA L3: Chihuahuan Desert-Code 24 • It also contains a “peninsula” of EPA L3 Arizona/New Mexico Mountains-Code 23.
	Edwards Plateau	Defined based on the following EPA ecoregion: • EPA L3: Edwards Plateau-Code 30
	High Plains and Tablelands	Defined based on the TX state boundary and the following EPA ecoregions and HUC4s: • EPA L3: High Plains-Code 25, Southwestern Tablelands-Code 26 • HUC4: Upper Cimarron-1104, Lower Cimarron-1105, Arksansas-Keystone-1106, Lower Canadian-1109, Red Headwaters-1112, Red-Washita-1113, North Canadian-1110, Brazos Headwaters-1205, Middle Brazos-1206, Upper Colorado-1208, Lower Colorado-San Bernard Coastal-1209, Lower Pecos-1307 • It also contains an “island” of EPA L3 Edwards Plateau-Code 30.

Zonation Zone	Subregion	Extent
	South Texas Plains	Defined based on the following EPA ecoregion: • EPA L3: Southern Texas Plains-Code 31
Central West Zone	North Missouri	Defined based on the MO state boundary and the following EPA ecoregions and HUC4s: • EPA L3s: Central Irregular Plains-Code 40, Western Corn Belt Code 47, Interior River Valleys and Hills-Code 72 • HUC4s: Des Moines-0710, Upper Mississippi-Salt-0711, Missouri-Nishnabotna-1024, Kansas-1027, Chariton-Grand-1028, Gasconade-Osage-1029, Lower Missouri-1030
	Ouachita	Defined based on the following EPA ecoregions: • EPA L3s: South Ouachita Mountains-Code 36, Arkansas Valley-Code 37, Boston Mountains-Code 38
	Ozarks and Plains	Defined based on the OK and MO state boundaries and the following EPA ecoregions and HUC4s: • EPA L3s: Ozark Highlands-Code 39, Central Irregular Plains-Code 40, Interior River Valleys and Hills-Code 72 • HUC4s: Upper Mississippi-Salt-0711, Upper Mississippi-Kaskaskia-Meramec-0714, Lower Mississippi-St. Francis-0802, Chariton-Grand-1028, Gasconade-Osage-1029, Lower Misssouri-1030, Upper White-1101, Neosho-Verdigris-1107, Lower Canadian-1109, North Canadian-1110, Lower Arkansas-1111
	Plains and Timbers	Defined based on the TX state boundary and the following EPA ecoregions and HUC4s: • EPA L3s: Central Great Plains-Code 27, Cross Timbers-Code 29 • HUC4: Upper Cimarron-1104, Lower Cimarron-1105, Arksansas-Keystone-1106, Neosho-Verdigris-1107, Lower Canadian-1109, North Canadian-1110, Lower Arkansas-1111, Red Headwaters-1112, Red-Washita-1113, Red Sulphur-1114, Trinity-1203, Brazos Headwaters-1205, Middle Brazos-1206, Lower Brazos-1207, Upper Colorado-1208, Lower Colorado-San Bernard Coastal-1209 • It also contains a “peninsula” of EPA L3 Flint Hills-Code 28.

Zonation Zone	Subregion	Extent
Central West Zone	Texas Blackland Prairies	Defined based on the following EPA ecoregions: • EPA L3s: Texas Blackland Prairies-Code 32, East Central Texas Plains-Code 33
	West Gulf Coastal Plain	Defined based on the following EPA ecoregion: • EPA L3: South Central Plains-Code 35
Coastal Zone	Atlantic Coastal Plain	Defined based on the VA state boundary and the following EPA ecoregions and HUC4s: • EPA L3s: Middle Atlantic Coastal Plain-Code 63, Southeastern Plains-Code 65, Southern Coastal Plain-Code 75 • HUC4s: Delaware-Mid Atlantic Coastal-0204, Potomac-0207, Lower Chesapeake-0208, Chowan-Roanoke-0301, Neuse-Pamlico-0302, Cape Fear-0303, Pee Dee-0304, Edisto-Santee-0305, Ogeechee-Savannah-0306, Altamaha-St. Marys-0307 • It also includes a small "peninsula" of EPA L4 Northern Outer Piedmont-Code 45f.
	East Gulf Coastal Plain	Defined based on the following EPA ecoregions and HUC4s: • EPA L3s: Southeastern Plains-Code 65, Mississippi Valley Loess Plains-Code 74, Southern Coastal Plain-Code 75 • HUC4s: Mobile-Tombigbee-0316, Pascagoula-0317, Pearl-0318, Middle Tennessee-Elk-0603, Lower Tennesee-0604, Lower Mississippi-Hatchie-0801, Lower Mississippi-Yazoo-0803, Lower Mississippi-Big Black-0806, Lower Mississippi-Lake Maurepas-0807, Lower Mississippi-0809, Choctawhatchee-Escambia-0314, Alabama-0315, Suwannee-0311, Ochlockonee-0312, Apalachicola-0313
	Florida Peninsula	Defined based on the following EPA ecoregions and HUC4s • EPA L3s: Southern Coastal Plain-Code 75, Southern Florida Coastal Plain-Code 76 • HUC4s: St. Johns-0308, Southern Florida-0309, Peace-Tampa Bay-0310
	Gulf Coastal Prairies	Defined based on the following EPA ecoregion: • EPA L3: Western Gulf Coastal Plain-Code 34 • It also contains an "island" of EPA L3 Mississippi Alluvial Plain-Code 73.

Zonation Zone	Subregion	Extent
Coastal Zone	Mississippi Alluvial Valley	Defined based on the MO state boundary and the following EPA ecoregion: • EPA L3: Mississippi Alluvial Plain-Code 73 • It also includes a few "islands" of EPA L3 South Central Plains-Code 35 and Mississippi Valley Loess Plains-Code 74, as well as a "peninsula" of EPA L3 Interior River Valleys and Hills-Code 72.
Greater Appalachians Zone	Appalachians	Defined based on the state boundaries of VA, WV, and KY, as well as the following EPA ecoregions and HUC4s: • EPA L3s: Blue Ridge-Code 66, Ridge and Valley-Code 67, Southwestern Appalachians-Code 68, Central Appalachians-Code 69, Western Allegheny Plateau-Code 70 • HUC4s: Potomac-0207, Lower Chesapeake-0208, Chowan-Roanoke-0301, Pee Dee-0304, Monongahela-0502, Upper Ohio-0503, Kanawha-0505, Scioto-0506, Big Sandy-Guyandotte-0507, Middle Ohio-0509, Kentucky-Licking-0510, Edisto-Santee-0305, Ogeechee-Savannah-0306, Apalachicola-0313, Alabama-0315, Mobile-Tombigbee-0316, Cumberland-0513, Upper Tennessee-0601, Middle Tennessee-Hiwassee-0602, Middle Tennessee-Elk-0603
	Piedmont	Defined based on the VA state boundary and the following EPA ecoregions and HUC4s: • EPA L3s: Piedmont-Code 45, Northern Piedmont-Code 64 • HUC4s: Potomac-0207, Lower Chesapeake-0208, Chowan-Roanoke-0301, Neuse-Pamlico-0302, Cape Fear-0303, Pee Dee-0304, Edisto-Santee-0305, Ogeechee-Savannah-0306, Altamaha-St. Marys-0307, Apalachicola-0313, Alabama-0315 • It also includes several small "islands" and "peninsulas" of EPA L3 Blue Ridge-Code 66.
	Interior Plateau	Defined based on the state boundary of KY and the following EPA ecoregions: • EPA L3s: Interior Plateau-Code 71, Interior River Valleys and Hills-Code 72 • It also includes a few small "islands" and "peninsulas" of EPA L3 Southwestern Appalachians-Code 68.

Zonation Zone	Subregion	Extent
Continental Marine Zone	Atlantic	Defined based on: • Northern extent: hand-edited line connecting the northern boundary of Virginia to the U.S. EEZ along a line of latitude • Offshore extent: U.S. EEZ • Southern extent: South Florida marine subregion, described below • Inland extent: inland subregion boundaries (HUC4s), described above
	South Florida Marine	Defined based on the following marine ecoregion: • Level 1: South Florida/Bahamian Atlantic
	Gulf	Defined based on: • Western extent: South Florida marine subregion, described above • Offshore extent: U.S. EEZ • Inland/eastern extent: inland subregion boundaries (HUC4s or EPA L3 ecoregions), described above

Literature Cited

Commission for Environmental Cooperation. 2009. Marine Ecoregions of North America, 2008. Vector digital data. Montréal, Québec, Canada. [https://www.cec.org/north-american-environmental-atlas/marine-ecoregions/].

Flanders Marine Institute (2019). Maritime Boundaries Geodatabase: Maritime Boundaries and Exclusive Economic Zones (200NM), version 11. Available online at [https://www.marineregions.org/]. [https://doi.org/10.14284/386].

Moilanen, A., I. Kohonen, P. Lehtinen, I. Kivistö, J. Jalkanen, E. Virtanen, H. Kujala. March 2024. Zonation 5 v2.0 User Manual. [https://github.com/zonationteam/Zonation5/releases/download/v2.0/manual_and_example_setups.zip].

Southeast Conservation Adaptation Strategy (SECAS). 2021. South Atlantic Extent and Subregions 2021. [https://www.sciencebase.gov/catalog/item/607f1b76d34e8564d6809f2f].

U.S. Department of Commerce, U.S. Census Bureau, Geography Division, Spatial Data Collection and Products Branch. TIGER/Line Shapefile, 2021, U.S. Current State and Equivalent National. 2021. [https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html].

U.S. Environmental Protection Agency. Level IV Ecoregions of the Conterminous United States. U.S. EPA Office of Research & Development (ORD) - National Health and Environmental Effects Research Laboratory (NHEERL). Corvallis, OR. Published April 16, 2013. [https://www.epa.gov/eco-research/level-iii-and-iv-ecoregions-continental-united-states].

U.S. Geological Survey (USGS). National Watershed Boundary Dataset in FileGDB 10.1 format. Published July 28, 2020. [https://www.usgs.gov/national-hydrography/watershed-boundary-dataset].

Caribbean Subregions

The Caribbean portion of the Southeast Blueprint is composed of Puerto Rico, the U.S. Virgin Islands, and a subset of the surrounding marine environment. It was intended to cover as broad a swath as possible of the terrestrial, freshwater, and marine environments of Puerto Rico and the U.S. Virgin Islands given the data available.

It is composed of two distinct subregions, Puerto Rico and the U.S. Virgin Islands, each of which serves as its own Zonation zone.

Input Data

- The Nature Conservancy's (TNC) Caribbean benthic habitat maps, accessed 12-21-2022: read a press release about the data; read a scientific journal article about the data; request to download the data
- National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI) Coastal Relief Model (Vol.9 - Puerto Rico), accessed 11-22-2022
- NOAA U.S. Maritime Limits and Boundaries, accessed 11-22-2022
- NOAA Continually Updated Shoreline Product (CUSP), accessed 1-18-2023; read a 1-page factsheet about CUSP; view and download CUSP data in the NOAA Shoreline Data Explorer (to download, select "Download CUSP by Region" and select Southeast Caribbean)
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data

Mapping Steps

- Processing steps were performed in the QGIS v3.22 graphical modeler.
- Prepare the TNC benthic data by merging together the U.S. Virgin Islands and Puerto Rico layers into one dataset using the DGDAL Merge utility. Using the Reclassify utility, reclassify the TNC benthic classes into a single class representing the extent of the TNC benthic analysis.
- Prepare the coastal relief data by importing the NetCDF file using the Import NetCDF function. Reclassify the coastal relief data to assign a value of 0 to elevations ranging from -10,000 to 0 and a value of 1 to elevations ranging from 0.0001 to 1,500. This separates out all land areas, which have an elevation >0.
- Prepare the U.S. maritime limits and boundaries data by selecting the marine boundaries for Puerto Rico and the U.S. Virgin Islands. Select all lines with a REGION field value of "Puerto

Rico and U.S. Virgin Islands". Hand-edit the marine boundary lines to create marine boundary polygons.

- Fill in the holes in the marine boundary polygons, which represent land areas, using the Delete Holes utility. This creates a layer representing the combined land and sea jurisdiction of Puerto Rico and the U.S. Virgin Islands.
- Clip the processed coastal relief data to the Puerto Rico and the U.S. Virgin Islands land and sea jurisdiction layer using the GDAL Clip utility. This step removes data in the British Virgin Islands, creating an elevation layer that distinguishes land and sea areas within Puerto Rico and the U.S. Virgin Islands.
- Combine the U.S. Caribbean elevation layer with the TNC benthic data to create a preliminary extent using the Raster Boolean OR utility. The resulting layer includes all land pixels with an elevation >0 in NOAA's coastal relief model and all sea pixels in TNC's benthic habitat layer.
- Convert that layer to vector using the GRASS Raster to Vector utility and extract the desired TNC benthic and land areas from the background values using Select by Attribute.
- The resulting layer has gaps in coverage between the land and TNC benthic ocean values. Remove holes <1,000,000 m^2 (247 acres) using the Delete Holes utility.
- Hand-digitize to further modify the extent to include areas captured in the CUSP shoreline dataset. Use the CUSP line data to create a polygon covering Isla Desecheo, which was not included in the coastal relief model.
- Cricket Rock, Dutchcap Key, Cockroach Island, Mona Island and Isla Monita were all excluded from the coastal relief model but did have a marine extent included in the benthic habitat data. Guided by the CUSP shoreline data, hand-edit each of the polygons for these small islands to fill in the land areas.
- Convert the final vector dataset to raster using the GDAL Vector to Raster utility in QGIS and snap to the extent and cell size of the LANDFIRE 2020 EVT raster.
- In 2025, we added several new Caribbean indicators that apply to only the U.S. Virgin Islands or to Puerto Rico, but not both. To make these work in Zonation without inadvertently penalizing the areas of the Caribbean not covered by these new indicators, we ran Zonation separately in Puerto Rico and USVI, instead of together in one Caribbean subregion. To facilitate this, we added an extra mapping step to create unique subregions for Puerto Rico and the U.S. Virgin Islands. To do this, in ArcPro, bring in the 2023 raster extent created above and convert it to a polygon (these data use the spatial projection EPSG 5070). Then, for the Caribbean area, use the "x" coordinate of each polygon to assign it to either the U.S. Virgin Islands (x>3326000) or Puerto Rico (x>3000000).

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Literature Cited

Flanders Marine Institute (2019). Maritime Boundaries Geodatabase: Maritime Boundaries and Exclusive Economic Zones (200NM), version 11. Available online at https://www.marineregions.org/. [https://doi.org/10.14284/386].

LANDFIRE, Earth Resources Observation and Science Center (EROS), U.S. Geological Survey. Published August 1, 2022. LANDFIRE 2020 Existing Vegetation Type (EVT) Puerto Rico US Virgin Islands. LF 2020, raster digital data. Sioux Falls, SD. [https://www.landfire.gov].

National Geophysical Data Center, 2005. U.S. Coastal Relief Model - Puerto Rico. National Geophysical Data Center, NOAA. Accessed November 22, 2022. [https://www.ncei.noaa.gov/products/coastal-relief-model].

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, National Geodetic Survey. NOAA Continually Updated Shoreline Product (CUSP): Southeast Caribbean. [https://coast.noaa.gov/digitalcoast/data/cusp.html].

Schill SR, McNulty VP, Pollock FJ, Lüthje F, Li J, Knapp DE, Kington JD, McDonald T, Raber GT, Escovar-Fadul X, Asner GP. Regional High-Resolution Benthic Habitat Data from Planet Dove Imagery for Conservation Decision-Making and Marine Planning. Remote Sensing. 2021; 13(21):4215. [https://doi.org/10.3390/rs13214215].

Combining Subregions

Input Data

- Continental inland subregions (created above)
- Continental marine subregions (created above)
- Caribbean subregions (Puerto Rico and U.S. Virgin Islands, created above)

Mapping Steps

- Merge the continental inland, continental marine, and Caribbean subregions into one subregion file.
- Note: In 2024 , 2025, and 2026, we decided to use the same Blueprint extent as version 2023. This extent leaves out some estuarine areas in Virginia that are included in the TIGER/Lines state boundaries. In the future, we may want to extend the Blueprint to those areas, but in order to do that, we would first have to extend all our indicators to cover the new areas. Our approach to creating the 2025 subregions will allow us to do that in future iterations. However, to avoid confusion, for the Blueprint 2026 release, we are limiting the subregions back to the 2023 extent. Consequently, we do not include the larger subregion layer in the Blueprint 2026 download package, but rather save it in a project archive for future reference.
- To limit the 2026 subregions back to the 2023 Blueprint extent, convert the 2023 Southeast Blueprint extent raster to a polygon and union it with the subregions made above. Calculate

an attribute identifying which areas were in the 2023 Southeast Blueprint extent, and export that as the final 2026 subregions layer.
- Add a new zone attribute and group the 2026 subregions into the zones that will be used in Zonation. The continental zone assignments are described in Table 15 above. The Puerto Rico and U.S. Virgin Islands subregions in the Caribbean each serve as their own zones.
- Export the subregion vector file, the full Blueprint extent raster file (including the Caribbean), and the continental-only Blueprint raster extent files used in the 2026 Southeast Blueprint.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Defining Extent

The exterior boundary of the collection of subregions described above represents the Southeast Blueprint 2026 extent (which is exactly the same as the previous Blueprint 2023, 2024, and 2025 extents).

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code. The final step of the code used to create the subregions exports the extent layer. The Southeast Blueprint extent is available on the Blueprint page of the SECAS Atlas.

Overall Zonation Approach

The Southeast Blueprint indicators were modeled at a 30 m resolution, tested and reviewed, and used as inputs to identify priority areas for a connected network of lands and waters. The indicators then served as inputs into Zonation, a conservation planning framework and software program that produces a hierarchical prioritization of the landscape. In addition to the indicators, we also use the subregions as inputs to Zonation in the continental inland and marine geographies (but not the Caribbean) to better balance priorities across subregions.

We used version 5 of Zonation for the entire Southeast Blueprint. We ran Zonation within seven zones (see Figure 4 above). Zonation ranks the pixels in each zone according to their indicator scores, using a modeling approach that tries to conserve high-value representations of all indicators collectively. Pixels that rank higher in Zonation become higher priority in the Blueprint.

Inland Continental

Zonation Inputs

To create the inputs for each inland continental Zonation run, clip the inland continental indicators to the 4 inland continental zones. Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

The zones serve as the boundaries of each Zonation run. Each Zonation run in the inland continental geography includes:

- All indicators that occur in a given zone, clipped to that zone
- All subregions that occur in a given zone

Zonation Run Settings

Determining Indicator Weights

We use Zonation to identify a network of priorities that includes the areas most important for all indicators collectively. Since the goal is to create a balanced portfolio where all indicators are represented, in a perfect world, we would weight each indicator equally. However, in some cases, we have to downweight indicators that would otherwise have an outsized impact on the Blueprint priorities.

The way Zonation ranks pixels on the landscape can be influenced by factors like an indicator's spatial rarity within a given zone and the distribution of high and low values (i.e., is the indicator's distribution top-heavy or bottom-heavy). In addition, some indicators are based on coarse-scale data. Other indicators have a wide spread of high to low values, but the data provider only intends the highest values to be considered good candidates for conservation action. In these situations, we use weighting to limit the disproportionate influence of certain indicators.

We developed four standard indicator weighting rules to ensure that:

- Indicators with coarse scale or information have less of an influence
- Spatially limited (i.e., rare) indicators were not overprioritized
- Subregional indicators that don't cover most of their target ecosystem were not overprioritized
- Indicators with limited overlap of below-average values in a given zone were not overprioritized

In a handful of edge cases, we developed additional exceptions to those four rules. The section below describes the application of the standard rules and additional exceptions.

Indicators with coarse scale or information

One indicator (estuarine coastal condition) has a much coarser spatial scale than the other indicators in the Blueprint. Estuarine coastal condition interpolates data from a relatively small number of sampling points into broad condition scores. Four other indicators (imperiled mammals, imperiled amphibians and reptiles, grassland and savanna extent, and grassland and savanna restoration) are based on relatively coarse information. For the imperiled species indicators, they use very broad definitions of what is potential species habitat and what areas are within a species range. For the grassland indicators, estimating grassland condition is particularly challenging, which results in large areas with coarse estimates of their current or potential condition. All of these indicators oversimplify more site-specific variation in ecosystem health and habitat value, so we reduce their weight by 0.7 across all zones to limit their influence on the Blueprint priorities. This allows other indicators with more detailed information to play a stronger role in teasing out key places within broad areas of importance for these coarser indicators.

Spatially limited indicators

With equal weights, the entire area of many indicators that cover a limited part of a zone (e.g., South Atlantic beach birds, greenways and trails) is identified as a high priority. To address this issue, we weighted indicators based on the proportion of their total area with values ≥1 in a given zone.

For indicators where the proportion of ≥1 values was between 0.09 and 0.5, we set their weights equal to that proportion. For example, we set the imperiled aquatic species indicator weight to approximately 0.26 in the Coastal Zone because that was the proportion of indicator pixels analyzed by Zonation in that zone with a value ≥1. For indicators where the proportion of ≥1 values was ≤0.09, we set their weights equal to that proportion multiplied by 3. We found that indicators in this range began to be underprioritized when only using the unadjusted proportion.

For indicators where the proportion of ≥1 values was 0.5 and higher, we set their weight to 1.0 unless they were covered by the exceptions discussed below.

Subregional indicators that don't cover most of their target ecosystem

Nine indicators use source datasets that only cover specific subregions and do not apply to the majority of their target ecosystem (Mississippi Alluvial Valley forest birds - protection, Mississippi Alluvial Valley forest birds - reforestation, West Coastal Plain and Ouachitas forested wetland birds, West Coastal Plain and Ouachitas open pine birds, West Gulf Coast mottled duck nesting, South Atlantic forest birds, South Atlantic beach birds, South Atlantic low-urban historic landscapes, South Atlantic maritime forest). For example, South Atlantic forest birds does not capture important forest bird habitat that occurs elsewhere in the Southeast. While these indicators help the Blueprint identify important areas within these subregions, they can also unfairly penalize areas that aren't covered by the models that would otherwise score highly due to their habitat or ecosystem value. This can cause Zonation to overprioritize the subregions that happen to be covered by these indicators. To reduce these unintended negative impacts, we reduce the weight of these indicators by 0.5.

Indicators with limited overlap of below-average values in a given zone

In a few cases, an indicator that targets a specific ecosystem will occur primarily in one zone, but spill over the boundary slightly into a neighboring zone. When this handful of overlapping pixels scores relatively low for that indicator, Zonation will often overprioritize these low-scoring areas. This occurs because Zonation can only see the range of indicator values that occur in a given zone—so while they score low overall, they are the highest values in that zone, which makes these areas seem more important than they actually are. To address this, we multiply the weight of the four indicators that have a small amount of overlapping below-average values in a given zone by 0.1. This rule applies to East Coastal Plain open pine birds and resilient coastal sites in the Greater Appalachians, Gulf coral and hardbottom in the Central West, and imperiled aquatic species in the Marine.

Other exceptions

Intact habitat cores in the Arid West

The Arid West Zone is composed of much drier ecosystems that are climatically distinct from the rest of the SECAS geography. Intact habitat cores does not perform as well in this zone for distinguishing natural, unfragmented areas from more altered areas. As a result, the standard “spatially limited indicators” weighting rule caused Zonation to overprioritize the Arid West. To mitigate this, we reduced the weight of this indicator to 0.7 in the Arid West only.

Lakes & reservoirs

The open water areas of lakes and reservoirs are particularly tricky. None of the indicators provide highly detailed information about the relative importance of pixels in these areas. This can result in computational challenges for Zonation, particularly when the weight for this indicator is too high. To address this, we used a modified version of the weight rule described under “spatially limited indicators”. Just like the spatially limited indicator rule, we set the weight equal to the proportion of ≥1 values. Unlike the spatially limited indicator rule, we did not multiply that proportion by 3 when the proportion was ≤0.09. This kept the weight low enough to avoid computational issues in Zonation.

Spatially limited Indicators on the edge of weighting thresholds

In some cases, indicators that are right on the edge of weighting thresholds can cause issues. Intact habitat cores in the Coastal Zone is just below the 0.5 “spatially limited indicators” threshold, and applying that weighing rule caused this indicator to be underrepresented. Falling on the other side of the threshold would have resulted in a weight of 1.0 instead of <0.5. To address this, we set the weight to 1.0.

Fire frequency had the opposite problem where some subregions barely crossed over the 0.5 weight threshold, resulting in a weight of 1.0. That increase in weight led areas even with low indicator values to become high priority only because of that increased weight. For fire frequency values over the 0.5 threshold, we left the weight as-is instead of increasing it to 1.0.

Marginal Loss Rule

CAZMAX: This setting is consistent with basic core-area Zonation (commonly CAZ) in Zonation 4, which was used in previous versions of the Blueprint. It tries to always cover high-value locations for all indicators, even if this comes with the cost of achieving lower average coverage across all indicators.

Additional Connectivity Methods

To increase the overall aggregation of the priorities, we included each indicator twice in each zone. The first version of the indicator in the input file is just the indicator layer as-is. For the second version, we have Zonation do a connectivity transformation. This approach of using both regular and connectivity transformed versions of indicators is described in the Zonation manual. It helps balance locally important places and nearby connectivity.

For the connectivity transformation, we used the negative exponential dispersal kernel. This is commonly used in modeling animal movements (Moilanen et al. 2024). We used a parameter of 1, which aligns with a mean dispersal distance of 1 pixel. For the cut parameter, which sets a distance to stop the connectivity transformation, we used a value of 5 pixels. Beyond that distance, the distribution already has only a minimal impact on values anyway.

Zonation Weights for Each Zone

To see the weights used for each indicator in each zone, see the Zonation feature list files available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Marine Continental

Zonation Inputs

To create the inputs for the marine continental Zonation run, clip the marine continental indicators to the Continental Marine Zone. Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

The Continental Marine Zone serves as the boundaries of the Zonation run. This Zonation run includes:

- All indicators that occur in the Continental Marine Zone, clipped to that zone
- All marine continental subregions

Zonation Run Settings

Determining Indicator Weights

We developed indicator weighting rules to ensure that:

- Spatially limited indicators were not overprioritized
- Priorities were relatively balanced between Gulf and Atlantic areas

Spatially limited indicators

With equal weights, the entire area of many indicators that cover a limited part of the Continental Marine Zone (e.g., estuarine coastal condition, Atlantic estuarine fish habitat) is identified as a high priority. To address this issue, we weighted indicators based on the proportion of their total area with values ≥1.

For indicators where the proportion of ≥1 values was between 0.1 and 0.5, we set their weights equal to that proportion. For example, we set the Gulf coral and hardbottom indicator weight to approximately 0.217 because that was the proportion of pixels analyzed by Zonation in the Continental Marine Zone with a value ≥1. For indicators where the proportion of ≥1 values was ≤0.1, we set their weights equal to that proportion multiplied by 3. We found that indicators in this range began to be underprioritized when only using the unadjusted proportion.

For indicators where the proportion of ≥1 values was 0.5 and higher, we set their weight to 1.0 unless they were covered by the "balancing Atlantic and Gulf" rule below.

Balancing Atlantic & Gulf priorities

The Continental Marine Zone includes many indicators that only cover either the Atlantic or the Gulf. In order to better balance priorities across those areas, we made three modifications to the "spatially limited indicators" weighting rule. The first modification was only weighting bird, mammal, and turtle indicators when the proportion of ≥1 values was less than 0.3. The second change was increasing the weight of two indicators that were getting significantly under-represented in the Blueprint priorities (Atlantic coral and hardbottom, Gulf deep-sea coral richness) from 0.3 to 0.6. The last change was not increasing the weight of the Gulf marine subregion to 1.0 even though it was slightly over the weight threshold (0.51). That improved consistency with the Atlantic subregion.

Marginal Loss Rule

CAZMAX: This setting is consistent with basic core-area Zonation (commonly CAZ) in Zonation 4, which was used in previous versions of the Blueprint. It tries to always cover high-value locations for all indicators, even if this comes with the cost of achieving lower average coverage across all indicators.

Additional Connectivity Methods

To increase the overall aggregation of the priorities, we included each indicator twice in each zone. The first version of the indicator in the input file is just the indicator layer as-is. For the second version, we have Zonation do a connectivity transformation. This approach of using both regular and connectivity transformed versions of indicators is described in the Zonation manual. It helps balance locally important places and nearby connectivity.

For the connectivity transformation, we used the negative exponential dispersal kernal. This is commonly used in modeling animal movements (Moilanen et al. 2024). We used a parameter of 1, which aligns with a mean dispersal distance of 1 pixel. For the cut parameter, which sets a distance to stop the connectivity transformation, we used a value of 5 pixels. Beyond that distance, the distribution already has only a minimal impact on values anyway.

Zonation Weights for Each Zone

To see the weights used for each indicator in each zone, see the Zonation feature list files available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Caribbean

Zonation Inputs

To create the inputs for the two Caribbean Zonation runs, clip the Caribbean indicators to the two Caribbean zones (Puerto Rico and U.S. Virgin Islands). Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

The Puerto Rico and U.S. Virgin Islands Zones serve as the boundaries of each Zonation run. Each Zonation run in the Caribbean geography includes all indicators that occur in a given zone, clipped to that zone. Subregions are not used as inputs in the Caribbean because the Caribbean zones are not subdivided into multiple subregions.

Zonation Run Settings

Determining Indicator Weights

We developed indicator weighting rules to ensure that:

- Spatially limited indicators were not overprioritized

Spatially limited indicators

With equal weights, the entire area of many indicators that cover a limited part of either the Puerto Rico or U.S. Virgin Island Zone (e.g., Caribbean beach habitat, Caribbean greenways and trails) is identified as a high priority. To address this issue, we weighted indicators based on the proportion of their total area with values ≥1.

For indicators where the proportion of ≥1 values was between 0.1 and 0.5, we set their weights equal to that proportion. For example, in the Puerto Rico Zone, we set the Caribbean karst habitat indicator weight to approximately 0.228 because that was the proportion of pixels analyzed by Zonation in the Puerto Rico Zone with a value ≥1. For indicators where the proportion of ≥1 values was ≤0.1, we set

their weights equal to that proportion multiplied by 3. We found that indicators in this range began to be underprioritized when only using the unadjusted proportion.

For indicators where the proportion of ≥1 values was 0.5 and higher, we set their weight to 1.0.

Other exceptions

Spatially limited Indicators on the edge of weighting thresholds

In the Puerto Rico Zone, Caribbean habitat patch size (large Islands) was just over the 0.5 weighting threshold. It was being overprioritized, so we kept the weight equal to the proportion of ≥1 values and did not increase the weight 1. The U.S. Virgin Islands imperiled terrestrial species indicator was being underprioritized and was close to a 0.3 threshold we used in previous versions of the Blueprint, so we doubled its weight in the U.S. Virgin Islands Zone to compensate.

Marginal Loss Rule

CAZMAX: This setting is consistent with basic core-area Zonation (commonly CAZ) in Zonation 4, which was used in previous versions of the Blueprint. It tries to always cover high-value locations for all indicators, even if this comes with the cost of achieving lower average coverage across all indicators.

Zonation Weights for Each Zone

To see the weights used for each indicator in each zone, see the Zonation feature list files available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Combining Caribbean & Continental Results

Next, we mosaiced together the Zonation results for all the inland continental zones, the Continental Marine Zone, and the two Caribbean zones. In this mosaiced Zonation results layer, each pixel in the Southeast Blueprint geography has a continuous value ranging from 0 to 100 according to its rank by the Zonation prioritization. The output of this step, the combined Zonation results for the Southeast Blueprint, is available on the Blueprint page of the SECAS Atlas. The continuous Zonation results are also available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip), but are provided as separate files for the continental and Caribbean geographies to reduce file size.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Literature Cited

Moilanen, A., I. Kohonen, P. Lehtinen, I. Kivistö, J. Jalkanen, E. Virtanen, H. Kujala. March 2024. Zonation 5 v2.0 User Manual. [https://github.com/zonationteam/Zonation5/releases/download/v2.0/manual_and_example_setups.zip].

Moilanen A, Lehtinen P, Kohonen I, Virtanen E, Jalkanen J, Kujala H. 2022. Novel methods for spatial prioritization with applications in conservation, land use planning and ecological impact avoidance. Methods in Ecology and Evolution. [https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13819].

Connectivity

The Southeast Blueprint uses a least-cost path connectivity analysis to identify connections between priority areas. A program called Linkage Mapper defines corridors that link hubs across the shortest distance possible, while also routing through as much Blueprint priority as possible. The hubs are defined as large patches of highest priority Blueprint areas and/or protected areas.

Because the continental and Caribbean portions of the Southeast Blueprint are non-contiguous, we run a separate connectivity analysis for each area. We use the same overall methods in both places; the primary difference is a smaller size threshold for hubs in the Caribbean (~125 acres vs. ~5,000 acres) to reflect the relative maximum size potential compared to the continental Southeast. Other minor differences between the continental and Caribbean methods—like the use of wildlife road crossing data and state connectivity priorities where available, and running the analysis in a buffer area around the Blueprint extent to align with neighboring plans—are detailed in the subsequent mapping steps for each analysis.

The results of the combined connectivity analysis, the Southeast Blueprint 2026 hubs and corridors, are available on the Blueprint page of the SECAS Atlas.

Continental

Continental Resistance Raster & Hubs

This is the resistance raster or cost surface used in the Linkage Mapper-based connectivity analysis for the continental portion of Blueprint 2026.

Input Data

- 2026 Southeast Blueprint subregions
- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)
- 2026 Midwest Conservation Blueprint subregions, resistance raster, and hubs
- The Nature Conservancy's (TNC) Terrestrial Resilience - Local Connectedness (Anderson et al. 2016)
- Florida Wildlife Road crossings (previously maintained by the FL Fish and Wildlife Conservation Commission, but now maintained by the FL Department of Transportation), last updated 12-7-2020
- Select North Carolina wildlife road crossings

  Eastern NC crossings were provided by Gary Jordan at the Raleigh, NC U.S. Fish and Wildlife Service Ecological Services field office on 7-22-2021; Gary received these data from Travis

Wilson at the North Carolina Wildlife Resources Commission. Other wildlife road crossings in NC were provided by Alex Vanko from the Wildlands Network and Steve Goodman from the National Parks Conservation Association. Some of these are dedicated crossing structures and some are bridge replacements that were specifically widened to accommodate wildlife; this is not a census of all widened bridge replacements.

- OpenStreetMap data "roads" layer, accessed 2-23-2022; used to estimate the fencing that is built around the wildlife road crossings

  OpenStreetMap® is open data, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). Additional credit to OSM contributors. Read more on the OSM copyright page.

- 2025 National Land Cover Database (NLCD): Land cover
- Protected Areas Database of the United States (PAD-US): PAD-US 4.1 national geodatabase - Combined Proclamation Marine Fee Designation Easement
- Southeast Blueprint reservoirs mask: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Select state-led connectivity priorities used to encourage consistency of the Blueprint corridors with existing plans
  - Virginia Wildlife Corridor Action Plan, accessed 7-24-25
  - Florida Ecological Greenways Network (FEGN): download 2021 FEGN Priorities 1-5 Raster
  - Georgia Conservation Opportunity Areas (COAs) from 2025 State Wildlife Action Plan (SWAP): access COA data on the Wildlife Resources Division Open Data Hub
  - North Carolina COAs from 2025 SWAP (analysis provided to NC Wildlife Resources Commission SWAP Coordinator Cindy Simpson on 4-7-2025), available on request (amy_keister@fws.gov)
  - South Carolina COAs from the 2025 SWAP (provided by Joe Lemeris with the SC Dept. of Natural Resources on 10-17-2024), available on request (amy_keister@fws.gov)
- Marine mammal and turtle movement areas used to ensure the marine/estuarine corridors reflect connectivity for marine mammals and sea turtles: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed
- Elevation-based mask used to separate the marine environment from terrestrial areas: available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip); see Appendix II of the Southeast Blueprint 2026 Development Process for details on how this layer was developed

Mapping Steps

Define Extent of the Continental Connectivity Analysis

- Dissolve the continental Southeast subregions into a single polygon covering the continental Blueprint extent.
- Buffer the continental extent by 26.1 km. The distance of 26.1 km is based on dispersal distances of subadult black bears (White et al. 2000), a species that disperses from within the Southeast Blueprint geography into neighboring areas.
- Convert the buffered continental extent from vector to raster.
- Dissolve the Midwest subregions and convert to raster.
- Using the vector file, erase the Southeast Blueprint area from the buffered Southeast Blueprint area. This defines only the buffer area around the Southeast Blueprint.
- Erase the Midwest Blueprint area from the Southeast Blueprint buffer area. This defines only the portion of the Southeast Blueprint buffer area that is not covered by the Midwest Blueprint. In this area, the source of potential hubs and resistance will be the TNC local connectedness data.
- Convert the resulting layer to a raster.

Begin Creation of the Resistance Raster

- The resistance raster is primarily derived from three data sources: the Southeast Blueprint combined Zonation results, the Midwest Blueprint potential hubs, and TNC local connectedness. Wildlife road crossings and protected areas will be integrated in a later step.
- Zonation outputs a continuous floating point decimal raster that ranges from 0-1. Take the raw Zonation results times 100 and convert to an integer.
- Flip the Southeast Blueprint continental combined Zonation results so that areas that ranked higher in Zonation are easier to move through.
- Start with the Midwest Blueprint resistance raster. They use a value of 500 for urban areas, and in the Southeast we have switched to using 150. Change the 500 values to 150 to match ours. Clip the MLI resistance raster to the area where it applies, which is the portion of the Southeast Blueprint buffer covered by the Midwest Blueprint.
- Rescale and flip the local connectedness layer from TNC's Resilient Land project so it has similar values to the Zonation results. Clip these data to the buffer area around the Southeast Blueprint. Using the ArcPy Rescale "Linear" function, rescale the local connectedness layer to assign values ranging from 1 to 100. Also, flip the high and low values so that areas with high local connectedness are easier to move through (i.e., lower resistance) than areas of low local connectedness. Convert these results to integer. This will be used in the buffer area around the continental Southeast Blueprint in places that aren't covered by the Midwest Blueprint.
- Mosaic together the resulting Zonation outputs and the TNC local connectedness layer, using the Southeast Zonation results anywhere they occur, the Midwest Zonation results in places where there are no Southeast Zonation results, and the local connectedness in the remaining buffer area.

Make Urban Areas Harder to Move Through

- To get the Linkage Mapper connectivity analysis to run successfully across the whole area, we have to resample the resistance raster from 30 m to 90 m. During this process, developed areas often get resampled out. To make sure that the connectivity analysis has sufficient information to move around developed areas, select the high and medium developed classes from the NLCD, resample them to 90 m, and burn them into the resistance raster with a high resistance value (150).

Make Known Wildlife Road Crossings Easier to Move Through

We obtained a small sample of known wildlife road crossings for North Carolina and Florida. This served as a test to route corridors through these crossings. We would like to get more spatial data for wildlife road crossings in the future.

- Buffer known wildlife road crossings by 210 m and convert them to raster with a low resistance value (1) that makes it easy for corridors to route through.
- Select roads surrounding known wildlife road crossings from the OpenStreetMap data. We do not want corridors to go across these roads except at the designated crossings, so we needed to make the resistance very high in these areas. Buffer the roads by 180 m and convert to raster with a very high resistance value (150).
- In testing, the methods above weren't enough to encourage corridors to route through the crossings, so we buffered the crossings again, this time by 900 m, erased the road areas from the larger buffered crossings (so the roads remained a high value) and converted to a raster with value that is easier to move through (10).
- Combine the roads with the combined Zonation results, using a maximum value so that these roads are very hard to cross.
- Combine the above raster with the two buffered road crossing layers, using a minimum value so that the areas with wildlife road crossings get the lower value from the buffered road crossing layers or Zonation results.

Make Protected Areas Easier to Move Through

- Protected conservation lands are an important component of regional connectivity. Many are captured as hubs in a later step, but not all conserved areas meet the size threshold to qualify as a hub. To better capture their value, regardless of size, burn protected areas into the resistance raster to make them easier to move through. To do this, first prepare the protected areas data, starting with the PAD-US 4.1 combined layer. To exclude areas that do not meet the intent of this step to capture the value of protected conservation lands for connectivity:
    - Remove areas with location designations of 'School Trust Land', 'School Lands', 'School Land', 'State Land Board', or '3201'. These extensive lands are leased out and are not open to the public.
    - Remove areas with the designation type of "Proclamation'. The proclamation category represents the approved boundary of public lands, within which land

protection is authorized to occur, but not all lands within the proclamation boundary are necessarily currently in a conserved status.
    - Remove areas with the owner name of 'BOEM'. These Outer Continental Shelf lease blocks do not represent actively protected marine parks, but serve as the "legal definition for BOEM offshore boundary coordinates...for leasing and administrative purposes" (BOEM).
    - Remove areas with a category of 'Proclamation' (see explanation above).
- Convert the selected polygons from PAD-US 4.1 to raster and reclassify to assign a value of 0 to unprotected areas and a value of 10 to protected areas.
- Subtract the protected area raster from the resistance raster. Lowering the resistance value makes it easier for corridors to route through protected areas.
- The above subtraction created some negative values, which Linkage Mapper does not accept. To correct for that, use a conditional statement to change all values <1 to 1.

Make Additional Corridor Priority Areas Easier to Move Through

- Make a copy of the Virgina Wildlife Corridor Action Plan corridors, add a field, and convert it to raster, assigning corridor pixels a value of 5.
- Reclassify the Florida Ecological Greenways Network raster to assign a value of 5 to tier 1, 2, and 3 pixels. These top three tiers correspond to the Florida Wildlife Corridor.
- Make a copy of the Georgia Conservation Opportunity Areas, which have a strong connectivity component. Add a field and convert to a raster, assigning corridor pixels a value of 5.
- Reclassify the South Carolina Conservation Opportunity Areas, which have a strong connectivity component. Assign a value of 5 to tier 1, 2, and 3 pixels.
- Reclassify the North Carolina Conservation Opportunity Areas, which have a strong connectivity component. Assign a value of 5 to Tier 1, 2, 3, and 4 pixels.
- In previous versions of the Blueprint, we created a marine mammal and turtle movement areas mask using spatial density models. Remove terrestrial areas from these marine mammal and turtle movement areas using the elevation-based marine/terrestrial mask. Assign a value of 5 to pixels in these marine mammal and turtle movement areas.
- Combine the six additional corridor priority area rasters listed above, using the ArcPy Cell Statistics "MAXIMUM" function.
- Subtract the combined additional corridor priority areas raster from the resistance raster. Lowering the resistance value makes it easier for corridors to route through pixels that were already identified as a corridor priority through other methods.
- The above subtraction created some negative values, which Linkage Mapper does not accept. To correct for that, use a conditional statement to change all values <1 to 1.

Finalize the Resistance Raster

- Using the ArcPy Resample "Bilinear" function, resample the resistance layer from 30 m to 90 m to allow it to run successfully in Linkage Mapper. The result is the final continental resistance raster.

Create the Hubs

- Hubs are derived from four different data sources: the Southeast Blueprint combined Zonation results, the merged protected areas layer (created above) the Midwest Blueprint potential hubs, and TNC local connectedness.
- Start with the Southeast combined Zonation results. Identify as potential hubs the highest scoring 10% of the of the combined Zonation results.
- Use the protected areas raster (derived from PAD-US 4.1 as described in the "Make protected areas easier to move through" section above) as potential hubs.
- Modify the protected areas raster to limit the influence of long, thin protected areas such as parkways, which do not align with the intent of the hubs. Shrink the protected areas raster by 3 pixels using the ArcPy Shrink function, then grow that result by 3 pixels using the ArcPy Expand function. This process breaks up very long thin potential hubs that are derived from PAD-US.
- Next, remove reservoirs from the modified protected areas raster. Reservoirs are still eligible to become hubs if they are included in the top 10% of the Zonation results, but not just because of their protected status, given that they are innately altered systems.
- Outside the Southeast Blueprint area, use the potential hubs from the Midwest Blueprint. Convert these to raster.
- Outside the Southeast Blueprint area, also use the local connectedness from TNC's Resilient Land project by identifying as potential hubs areas ≥1 standard deviation above average.
- Combine the potential hubs rasters from the four layers above using the ArcPy Cell Statistics "MAXIMUM" function.
- Convert the potential hubs raster to a cell size of 90 m using the default Esri method (nearest neighbor). Since we have to run Linkage Mapper at 90 m to overcome computational limitations, when Linkage Mapper converts the 30 m hubs to a 90 m raster, it breaks up long linear features into small pieces, creating many small hubs. This produces errors that prevented Linkage Mapper from running successfully. Converting the hubs to 90 m before running Linkage Mapper removes any long, narrow protuberances from the cores.
- Keep as hubs contiguous patches greater than 2,000 ha (5,000 acres) in size. Identify these patches using the ArcGIS Spatial Analyst Region Group function (8-neighbor). This size threshold from Hoctor et. al (2000) is also used in the Florida Conservation Blueprint for connectivity purposes.
- Convert the raster patches to polygons. Select polygons that intersect the buffered continental Southeast Blueprint extent. These will be used as hubs in the Linkage Mapper analysis.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Prepare Continental Hubs & Resistance Raster for Linkage Mapper

Due to the large size of the continental area, we must break it into 15 hexagons to run the continental connectivity analysis in Linkage Mapper. This requires breaking both the continental hubs and resistance raster into hexagons.

Input Data

- Continental resistance raster (created above)
- Continental hubs (created above)

Mapping Steps

- Use the Generate Tessellation tool to create hexagons with an area of 500,000 km^2. This size was the largest we could feasibly run in Linkage Mapper.
- To eliminate edge effects, buffer each hexagon by 200 km.
- For each buffered hexagon, clip the continental resistance raster to the buffered hexagon and select all continental hub polygons that intersect each buffered hexagon.
- Use these layers as the inputs to the Linkage Mapper run for each buffered hexagon.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

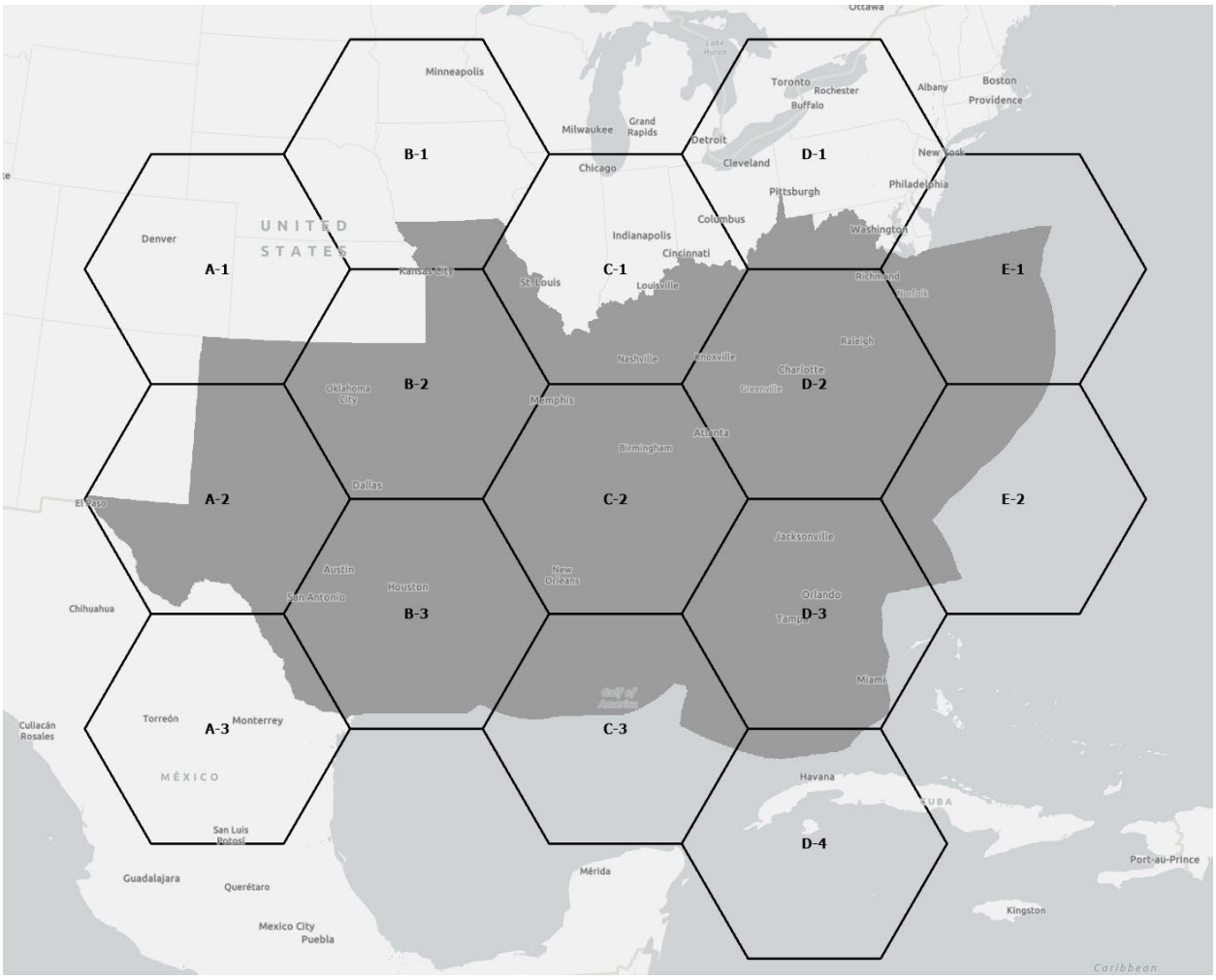


Figure 5. Illustration of hexagons used for the continental Linkage Mapper connectivity analysis.

Run Continental Corridor Analysis

Running the continental corridor analysis creates a continuous corridor surface for each buffered hexagon.

Input Data

- Continental resistance raster within each buffered hexagon (created above)
- Continental hubs that intersect each buffered hexagon (created above)

Mapping Steps

- Perform a Linkage Mapper corridor analysis for each buffered hexagon in ArcGIS Pro V 3.7.1 using the settings seen in this example log file from one of the buffered hexagon runs:

  LINKAGE MAPPER LOG FILE
  Start time: 14:05 2025-08-09

Version: 3.1.0
Tool used: Linkage Mapper

Version: 3.1.0
Tool used: Linkage Mapper

SYSTEM AND DATA INFORMATION
Operating system: Windows-11-10.0.26200-SP0
Processor type: Intel64 Family 6 Model 151 Stepping 2, GenuineIntel
Total & available RAM: 63.6 GB & 36.8 GB

ARCGIS INFORMATION
Product name & version: ArcGISPro 3.7.1
Build number: 1901
Product license: ArcInfo
Spatial Analyst status: Available

MODEL INPUTS
Parameter Value
--- -----
PROJ_DIR D:\b2
CORE_LYR D:\b2\Hubs_b2.shp
CORE_FLD UID
RESIS_LYR D:\b2\Resist_b2.tif
STEP1 true
STEP2 true
S2_ADJ_METHOD Cost-Weighted & Euclidean
S2_EUCDIST_FILE D:\e1\Hubs_Classic2.shp_dists.txt
STEP3 true
S3_DROP_LCCS true
STEP4 true
S4_MAX_NN 4
S4_NN_UNIT Cost-Weighted
S4_CONNECT true
STEP5 true
S5_TRUNC_RAST true
S5_CWD_THRESH 200000
CUS_BUF_DIST 200000
CUS_MAX_CWD #
CUS_MAX_EUCDIST #
CUS_OUTPUT_MB #
CUS_SET_FILE #

Custom Settings
CALCNONNORMLCCS False
MINCOSTDIST None
MINEUCDIST None
SAVENORMLCCS True
SIMPLIFY_CORES True

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Merge Outputs of Corridor Analysis

The corridor analysis outputs 15 buffered Linkage Mapper runs. This step merges them together into a seamless continuous corridor surface for the continental area.

Input Data

- Outputs of all 15 buffered hexagon Linkage Mapper runs (created above)
- Hexagons used to subset continental area (created above)
- Continental resistance raster (90 m version)

Mapping Steps

- After running all 15 buffered hexagon Linkage Mapper runs, combine the outputs to create a corridor surface covering the whole continental Southeast Blueprint area.
- Clip each Linkage Mapper output raster back down to its original hexagon and mosaic them all together. This ended up leaving a one cell empty space in some areas, so we went back and buffered the hexagons by just 90 m and used that to clip down instead so there were no cells "lost" on the line between two hexagons.
- Snap to the 90 m version of the continental resistance raster to avoid a cell shift.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Make Final Continental Corridors

This step extracts the final corridors for the continental area from the combined continuous corridor surface. It creates the corridor raster used in the continental portion of Southeast Blueprint 2026.

Input Data

- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is

available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)

- Continental hubs (created above)
- Full continental corridor continuous surface (created above)

Mapping Steps

- The above steps produce a corridor raster with a continuous surface covering the entire continental Southeast Blueprint area.
- To turn the continuous corridor output into corridors, we have to find a cutoff value in the output. We do this by combining it with the Zonation results, with the goal of making the corridor class cover 5% of the continental area.
- Zonation outputs a continuous floating point decimal raster that ranges from 0-1. Take the raw Zonation results times 100.
- We tested different cutoffs and found that keeping all pixels <81,100created corridors that cover approximately 5% of the area of the continental Southeast Blueprint area (that were not already covered by highest, high, and medium priority pixels). The result is the full corridor layer for the continental portion of Southeast Blueprint. Continental corridor pixels not already identified as highest, high or medium priority were incorporated into the priority connections class of the Southeast Blueprint. We chose the 81,100cutoff because it resulted in a priority connections area of approximately 5% of the continental region (not already covered by highest, high, or medium priority pixels).

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip) under > 6_Code.

Caribbean

Caribbean Resistance Raster

This is the resistance raster or cost surface used in the Linkage Mapper-based connectivity analysis for the Caribbean portion of Southeast Blueprint 2026.

Input Data

- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)
- 2020 LANDFIRE Existing Vegetation Type (EVT) [LF 2.2.0] for Puerto Rico and the U.S. Virgin Islands (U.S. Insular Areas); access the data
- Protected Areas Database of the United States (PAD-US): PAD-US 4.1 national geodatabase - Combined Proclamation Marine Fee Designation Easement
- Southeast Blueprint 2026 extent (Caribbean only)
- The Nature Conservancy Insular Caribbean Protected Areas 2025
- U.S. Virgin Islands Territorial Parks: spatial data provided by Kenneth Douyon, NOAA Fellow with the USVI Division of Territorial Parks and Protected Areas, on 8-27-2026

Mapping Steps

Begin Creation of the Resistance Raster

- Zonation outputs a continuous floating point decimal raster that ranges from 0-1. Take the raw Zonation results times 100 and convert to an integer.
- Flip the Zonation results so that areas that rank higher in Zonation are easier to move through.

Make Urban Areas Harder to Move Through

- Burn the LANDFIRE urban classes into the resistance raster, giving urban pixels the highest resistance value (150), making them harder for corridors to move through. Pixels with an EVT_NAME value of 'Developed-Medium Intensity', 'Developed-High Intensity', or 'Developed-Roads' were considered urban.

Make Protected Areas Easier to Move Through

- Protected conservation lands are an important component of regional connectivity. Many are captured as hubs in a later step, but not all conserved areas meet the size threshold to qualify as a hub. To better capture their value, regardless of size, burn protected areas into the resistance raster to make them easier to move through. We use 3 sources of protected areas in the U.S. Caribbean.

- First, prepare the protected areas data, starting with the PAD-US 4.1 combined layer. To exclude areas that do not meet the intent of this step to capture the value of protected conservation lands for connectivity:
    - Remove areas with location designations of 'School Trust Land', 'School Lands', 'School Land', 'State Land Board', or '3201'. These extensive lands are leased out and are not open to the public.
    - Remove areas with the designation type of 'Military' or "Proclamation'. Military lands are not primarily managed for conservation. The proclamation category represents the approved boundary of public lands, within which land protection is authorized to occur, but not all lands within the proclamation boundary are necessarily currently in a conserved status.
    - Remove areas with the owner name of 'BOEM'. These Outer Continental Shelf lease blocks do not represent actively protected marine parks but serve as the "legal definition for BOEM offshore boundary coordinates...for leasing and administrative purposes" (BOEM).
    - Remove areas with a category of 'Proclamation' (see explanation above).
    - Remove marine protected areas by removing areas with a category of 'Marine'. In the U.S. Caribbean, the number and size of the marine protected areas relative to the islands themselves had a much more significant impact on the connectivity analysis than they did in the continental United States, and produced undesirable corridor results.
- Next, from The Nature Conservancy's Insular Caribbean Protected Areas 2025, select only polygons where the Agency is The Nature Conservancy. This adds in TNC preserves that are missing from PAD-US.
- Use all polygons provided for U.S. Virgin Islands Territorial Parks.
- Combine the 3 sources of protected areas using the Union function, then convert to a raster. Reclassify to assign a value of 0 to unprotected areas and a value of 10 to protected areas.
- Subtract the protected area raster from the resistance raster. Lowering the resistance value makes it easier for corridors to route through protected areas.
- The above subtraction created some negative values, which Linkage Mapper does not accept. To correct for that, use a conditional statement to change all values <1 to 1.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Caribbean Hubs

These are the hubs used in the Linkage Mapper-based connectivity analysis for the Caribbean portion of Southeast Blueprint 2026.

Input Data

- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)
- Southeast Blueprint 2026 extent (Caribbean only)
- Protected Areas Database of the United States (PAD-US): PAD-US 4.1 national geodatabase - Combined Proclamation Marine Fee Designation Easement
- The Nature Conservancy Insular Caribbean Protected Areas 2025
- U.S. Virgin Islands Territorial Parks: spatial data provided by Kenneth Douyon, NOAA Fellow with the USVI Division of Territorial Parks and Protected Areas, on 8-27-2026

Mapping Steps

- Use a conditional statement to pull out the top 10% (value >89) from the combined Zonation results and clip to the Caribbean extent. Reclassify to turn values of NoData to 0.
- Use the protected areas raster (derived from 3 sources described in the "Make protected areas easier to move through" section above) as potential hubs.
- Combine the top 10% of the Caribbean Zonation results with the protected areas using the ArcPy Cell Statistics "MAXIMUM" function.
- Keep as hubs contiguous patches greater than 50 ha (124 acres) in size. Identify these patches using the ArcPy Spatial Analyst Region Group function (8-neighbor). This size threshold is ~2.5% of the threshold used for hubs in the continental part of the Blueprint. When compared to the continental part of the Blueprint, many key species in the Caribbean have adapted to use the significantly smaller areas of habitat. Compared to other thresholds tested, this smaller hub size allowed for many important areas, especially in the U.S. Virgin Islands, to be potentially identified as hubs in the analysis, and produced corridor results that better aligned with user feedback and input from Caribbean workshops.
- Convert the raster patches to polygons.
- Use all resulting polygons as hubs in the Caribbean Linkage Mapper analysis.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Run Caribbean Corridor Analysis

Running the Caribbean corridor analysis creates a continuous corridor surface for the Caribbean subregion of Southeast Blueprint 2026.

Input Data

- Caribbean resistance raster (created above)
- Caribbean hubs (created above)

Mapping Steps

- Perform a Linkage Mapper corridor analysis ArcGIS Pro V 3.3.6 using the settings seen in this log file:

  LINKAGE MAPPER LOG FILE
  Start time: 16:50 2025-08-01
  Version: 3.1.0
  Tool used: Linkage Mapper

  SYSTEM AND DATA INFORMATION
  Operating system: Windows-10-10.0.22631-SP0
  Processor type: Intel64 Family 6 Model 85 Stepping 7, GenuineIntel
  Total & available RAM: 63.6 GB & 53.3 GB

  ARCGIS INFORMATION
  Product name & version: ArcGISPro 3.3.6
  Build number: 52636
  Product license: ArcInfo
  Spatial Analyst status: Available

  MODEL INPUTS
  Parameter Value
  --- -----
  PROJ_DIR C:\LV1
  CORE_LYR C:\LV1\VIPR_LM_Input\VIPRHubs50ha.shp
  CORE_FLD UID
  RESIS_LYR C:\LV1\VIPR_LM_Input\VIPRResistance.tif
  STEP1 false
  STEP2 false
  S2_ADJ_METHOD Cost-Weighted & Euclidean
  S2_EUCDIST_FILE C:\LV1\VIPRHubs50ha.shp_dists.txt
  STEP3 false
  S3_DROP_LCCS false
  STEP4 false
  S4_MAXIMUM_NN 4
  S4_NN_UNIT Cost-Weighted
  S4_CONNECT true
  STEP5 true
  S5_TRUNC_RAST true
  S5_CWD_THRESH 200000
  CUS_BUF_DIST #
  CUS_MAXIMUM_CWD #

CUS_MAXIMUM_EUCDIST #
CUS_OUTPUT_MB #
CUS_SET_FILE #

Custom Settings
CALCNONNORMLCCS False
MINCOSTDIST None
MINEUCDIST None
SAVENORMLCCS True
SIMPLIFY_CORES True

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Make Final Caribbean Corridors

This step extracts the final corridors for Caribbean area from the continuous corridors surface. It creates the corridor raster used in the Caribbean portion of Southeast Blueprint 2026.

Input Data

- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)
- Caribbean hubs (created above)
- Full Caribbean corridor continuous surface (created above)
- Southeast Blueprint 2026 extent (Caribbean only)

Mapping Steps

- The above steps produce a corridor raster with a continuous surface covering the entire Caribbean Southeast Blueprint area.
- To turn the continuous corridor output into corridors, we have to find a cutoff value in the output. We do this by combining it with the Zonation results, with the goal of making the corridor class cover 5% of the Caribbean portion of the Southeast Blueprint.
- Zonation outputs a continuous floating point decimal raster that ranges from 0-1. Take the raw Zonation results times 100.
- We tested different cutoffs and found that keeping all pixels <13,170 created corridors that cover approximately 5% of the Caribbean region of the Southeast Blueprint (that were not already covered by highest, high, and medium).
- This result is the full Caribbean corridor layer for Southeast Blueprint 2023. Caribbean corridor pixels not already identified as highest, high or medium priority were incorporated into the

priority connections class of the Southeast Blueprint. We chose the 13,170 cutoff because it resulted in a priority connections area of approximately 5% of the Caribbean region (not already covered by highest, high, or medium priority pixels).

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Known Issues

- To remove land areas from the marine mammal and turtle movement areas, we used a version of an elevation-based mask that is typically used in terrestrial indicators to remove marine areas from land, which errs on the side of including more land areas to preserve barrier islands. We intended to use a version of an elevation-based mask that is typically used in marine indicators to remove land areas from the marine, which errs on the side of including more marine areas to avoid cutting off values in the nearshore marine environment. As a result, we may have excluded some marine mammal and turtle movement areas from the nearshore marine environment. We will use the marine version of the mask in the future.
- In the U.S. Caribbean, some military lands are underrepresented in the hubs layer. Unlike in the continental Southeast, military lands were not included in the protected lands layer that we use to supplement the top 10% Zonation areas and define potential hubs.
- The corridor layer may not cover enough area to maintain connectivity for key species. The extent of the corridors layer was chosen so that the “priority connections” class in the Blueprint covers 5% of the landscape, ensuring the total Blueprint covers 50% of the SECAS geography. It was not chosen for the purpose of achieving any particular goal for structural or functional connectivity.

Other Things to Keep in Mind

- On its own, the Blueprint Zonation prioritization—which largely tends to identify areas that are already in “good” condition—does not always produce a connected network of priorities. The corridor analysis is our attempt to identify areas that would enhance structural connectivity, if conservation actions were taken in those places. We then use this analysis to create the “priority connections” class in the Blueprint.
- The approach used to create the Blueprint corridors is a structural rather than a functional approach to connectivity. In other words, the corridors identify places with the potential to create physical habitat linkages that could serve a variety of purposes, from species movement to marsh migration to outdoor recreation. This contrasts with a functional approach to connectivity, which optimizes corridors for a particular purpose, such as the dispersal of a species. Even though we are not optimizing the corridors for any particular set of species, over time, we have added information to encourage terrestrial corridors to route through wildlife road crossings, and to encourage marine corridors to route through marine mammal and turtle movement areas.

- The methods used to create these corridors were developed for the purpose of creating the "priority connections" class in the final Blueprint. The priority connection class occurs only in places not already identified as highest, high, or medium priority in the Zonation results. The extent of the corridors layer was chosen to ensure the "priority connections" Blueprint class covers 5% of the landscape. The hubs and corridors are interim data layers necessary to create the "priority connections" class in the Blueprint. We provide these layers as a part of our commitment to transparency in methods. We also see value in the hubs and corridor layer because it shows where areas underneath the highest/high/medium pixels of the Blueprint may also potentially contribute to regional connectivity.
- The hubs and corridors identified in this analysis vary from year to year. This can pose a barrier to implementation (i.e., it's difficult to conserve a corridor's entire length over the long-term if the route keeps changing). This variability results in part from our use of the Zonation results to help identify hubs and define the resistance raster, which makes the corridors sensitive to even minor changes in the Zonation output between Blueprint versions. Ultimately, we would like the corridors in the Blueprint to strike a balance between stability and responsiveness to landscape change. Corridor stability has increased incrementally since the first precursor to the Southeast Blueprint was released in 2014—both because the suite of indicators used in the Blueprint has matured, painting a more complete and consistent picture of the region's key resources, and as a result of corridor methods specifically introduced to improve stability (e.g., encouraging corridors to route through areas prioritized in state corridor plans). However, with our current methods, the stability of corridors has not increased as fast as we hoped. We are exploring new methods to limit the shifting of corridors that result from minor changes in the Blueprint Zonation results. Currently, we recommend thinking of the hubs and corridors as potential areas to work to enhance connectivity, rather than a durable roadmap for parcel-level corridor decisions. These data will be most helpful when used in combination with the rest of the Blueprint priorities, and—as with all Blueprint data—with local data and knowledge.
- There are inconsistencies in the availability and use of state-level data in the Blueprint connectivity analysis. State corridor priorities are only used to influence corridor routing in VA, NC, SC, GA, and FL. We would like to incorporate priorities from AL's State Wildlife Action Plan, but data was not available in time for Blueprint 2026. So far, we have only incorporated wildlife road crossing data in NC and FL.

Literature Cited

Anderson, M.G., A. Barnett, M. Clark, C. Ferree, A. Olivero Sheldon, J. Prince. 2016. Resilient Sites for Terrestrial Conservation in Eastern North America. The Nature Conservancy, Eastern Conservation Science. [https://easterndivision.s3.amazonaws.com/Resilient_Sites_for_Terrestrial_Conservation.pdf].

Hoctor, T. S., M. H. Carr, P. D. Zwick. 2000. Identifying a linked reserve system using a regional landscape approach: the Florida ecological network. Conservation Biology 14:984-1000. [https://www.jstor.org/stable/2641997].

Krueck, N. C., Legrand, C., Ahmadia, G. N., Estradivari, E., Green, A., Jones, G. P., Riginos, C., Treml, E. A. and Mumby, P. J. 2017. Reserve Sizes Needed to Protect Coral Reef Fishes. Conservation Letters. Accepted Author Manuscript. [https://doi.org/10.1111/conl.12415].

McRae, B.H. and D.M. Kavanagh. 2011. Linkage Mapper Connectivity Analysis Software. The Nature Conservancy, Seattle WA. [https://linkagemapper.org/].

OpenStreetMap. Download OpenStreetMap data for this region: North America. Accessed February 23, 2022. [https://download.geofabrik.de/north-america.html].

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2024, Protected Areas Database of the United States (PAD-US) 4.1: U.S. Geological Survey data release. [https://doi.org/10.5066/P96WBCHS].

U.S. Geological Survey, 2024, Annual NLCD Collection 1 Science Products (ver. 1.2, June 2026): U.S. Geological Survey data release. [https://doi.org/10.5066/P94UXNTS].

White, T.H., JR., J.L. Bowman, B.D. Leopold, H.A. Jacobson, W.P. Smith, and F.J. Vilella. 2000. Influence of Mississippi alluvial valley rivers on black bear movements and dispersal: Implications for Louisiana black bear recovery. Biological Conservation 95:323–331. [https://www.sciencedirect.com/science/article/abs/pii/S0006320700000240].

Creating Southeast Blueprint 2026

The Southeast Conservation Blueprint is a living, spatial plan to achieve the SECAS vision of a connected network of lands and waters across the Southeast and Caribbean. It identifies priority areas based on a suite of natural and cultural resource indicators representing terrestrial, freshwater, and marine ecosystems. A connectivity analysis identifies corridors that link coastal and inland areas and span climate gradients. So far, more than 2,500 people from over 650 different organizations have actively participated in developing the Blueprint.

Priority Categories

The Southeast Blueprint covers 50% of the SECAS geography, as described below.

Priorities for a Connected Network of Lands & Waters

Highest priority: Areas where conservation action would make the biggest impact, based on a suite of natural and cultural resource indicators. This class covers roughly 10% of the Southeast Blueprint geography.

High priority: Areas where conservation action would make a big impact, based on a suite of natural and cultural resource indicators. This class covers roughly 15% of the Southeast Blueprint geography.

Medium priority: Areas where conservation action would make an above-average impact, based on a suite of natural and cultural resource indicators. This class covers roughly 20% of the Southeast Blueprint geography.

Priority connections: Connections between priority areas that cover the shortest distance possible while routing through as much Blueprint priority as possible. This class covers roughly 5% of the Southeast Blueprint geography.

Combining Zonation Results with Corridors to Create the Southeast Blueprint

Input Data

- 2026 Southeast Blueprint combined Zonation results (decimal version); the decimal version is too large to share, but the integer version produced in the following mapping steps is available on the SECAS Atlas and in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip)
- Southeast Blueprint 2026 hubs and corridors
- Southeast Blueprint 2026 subregions

Mapping Steps

Creating the Continental Blueprint

- Start with the mosaiced Zonation results for all continental subregions. This layer has been taken times 100 but has not been converted to an integer. In this layer, each pixel in the continental Southeast Blueprint geography has a continuous value ranging from 0 to 100 according to its rank by Zonation prioritization.
- Pixels with values ≥90 are in the highest tier of indicator value. Select all pixels with values ≥90 and classify them as “highest priority for a connected network of lands and waters”.
- Pixels with values ≥75 that aren’t already classified as highest priority are in the second-highest tier of indicator value. Select all pixels ≥75 and <90 and classify them as “high priority for a connected network of lands and waters”.
- Creating the “medium priority for a connected network of lands and waters” is a bit more complicated. We need to add hubs that aren’t captured in the highest, high, or medium classes to the medium class. We must adjust the bottom threshold of the medium class to account for these non-captured hubs. This year we found that using a cut off value of 55.57 in the Zonation results allowed us to add in the non-covered hubs while maintaining the correct percentage of the landscape covered by each class. Select all pixels >55.57 and <75 and classify them as “medium priority for a connected network of lands and waters”. This makes up the first portion of the medium priority class.
- Add to the medium priority class any hubs used in the connectivity analysis that that were not already classified as highest, high, or medium priority in the steps above. This ensures that the

large patches of protected lands used as hubs in the connectivity analysis can score no lower than medium priority in the Blueprint. This adds an additional 0.57% of total area to the medium priority class.

- Use the continental corridors to fill in the priority connections class. Classify as “priority connections” any pixel identified as a corridor in the inland corridor analysis that is not already assigned to the highest, high or medium priority categories in the steps above. This contributes an additional 5% to the total Blueprint area, ensuring the final Blueprint ultimately covers 50% of the Southeast Blueprint landscape.

Creating the Caribbean Blueprint

- Start with the mosaiced Zonation scores for the Caribbean subregions. This layer has been taken times 100 but has not been converted to an integer. In this layer, each pixel in the Caribbean Blueprint geography has a continuous value ranging from 0 to 100 according to its rank by Zonation prioritization.
- Pixels with values ≥90 are in the highest tier of indicator value. Select all pixels with values ≥90 and classify them as “highest priority for a connected network of lands and waters”.
- Pixels with values ≥75 that aren’t already classified as highest priority are in the second-highest tier of indicator value. Select all pixels ≥75 and <90 and classify them as “high priority for a connected network of lands and waters”.
- Creating the “medium priority for a connected network of lands and waters” is a bit more complicated. We need to add hubs that aren’t captured in the highest, high, or medium classes to the medium class. We must adjust the bottom threshold of the medium class to account for these non-captured hubs. This year we found that using a cut off value of 55.27 in the Zonation results allowed us to add in the non-covered hubs while maintaining the correct percentage of the landscape covered by each class. Select all pixels >55.27 and <75 and classify them as “medium priority for a connected network of lands and waters”. This makes up the first portion of the medium priority class.
- Add to the medium priority class any hubs used in the Caribbean connectivity analysis that that were not already classified as highest, high, or medium priority in the steps above. This ensures that the large patches of protected lands used as hubs in the connectivity analysis can score no lower than medium priority in the Blueprint. This adds an additional 0.27% of total area to the medium priority class.
- Use the Caribbean corridors to fill in the priority connections class. Classify as “priority connections” any pixel identified as a corridor in the corridor analysis that is not already assigned to the highest, high or medium priority categories in the steps above. This contributes an additional 5% to the total Blueprint area, ensuring the final Blueprint ultimately covers 50% of the Southeast Blueprint landscape.

Combining the Continental & Caribbean Components into Southeast Blueprint 2026

- As a final step, combine the continental and Caribbean results into a single raster representing final Southeast Blueprint 2026. Do this using the ArcPy Cell Statistics “MAXIMUM” function.

Note: For more details on the mapping steps, code used to create this layer is available in the data download package on ServCat (Southeast_Blueprint_2026_Data_Download_YYYY-MM-DD.zip or Southeast_Blueprint_2026_Data_Download_Caribbean_YYYY-MM-DD.zip) under > 6_Code.

Known Issues

Continental

Terrestrial

Uplands

- Some managed private grasslands are underprioritized. Examples include Prairie Wildlife grasslands west of Vinton, MS; East Foundation lands in South TX; grasslands of special significance southeast of Starkville, MS and northwest of Egypt, MS; and a grassland restoration area northeast of Starkville, MS. Improvements to the fire frequency and grassland indicators could fix this in the future.
- Some managed public grasslands are underprioritized. Examples include multiple prairies in MO (McGee Family Conservation Area, Jerry Smith Park, and Stilwell Prairie). These will likely be fixed in the next version of the Blueprint.
- Parts of some important ecological corridors are underprioritized. Examples include parts of the corridor between Ocmulgee Mounds National Historic Park and Bond Swamp National Wildlife Refuge in GA; the South Fork of the Forked Deer River in TN; parts of the corridor between Fort Campbell, Land Between the Lakes, and Clarks River National Wildlife Refuge in KY and TN; some of the areas from Alligator River National Wildlife Refuge to Pocosin Lakes National Wildlife Refuge in NC; multiple corridors coming out of Okefenokee Swamp in GA; and the Osceola to Ocala corridor in FL. Improvements to prioritization methods and indicators will likely fix these in the future.
- Some patches of open pine with good local conditions are underprioritized. Examples include important gopher tortoise habitat in an area just east of Mauk, GA; a pitcher plant flat south of Rowlands, MS; the Farmer's Home tract managed by Mississippi Sandhill Crane National Wildlife Refuge; wet pine savanna mitigation bank areas east of Pearl River Wildlife Management Area, MS; longleaf south of Pine Hill, NC; savanna northwest of Mississippi Sandhill Crane National Wildlife Refuge in MS; and a shortleaf pine site north of Marshes Siding in Daniel Boone National Forest in KY. Ongoing updates to the grasslands and savannas and fire frequency indicators could continue to improve this issue in future updates.
- Some parts of small, low-elevation islands are underprioritized. The exact boundaries of these highly dynamic islands can be hard to predict. The boundaries used in the islands indicator and areas used for critical habitat of key island species don't always align perfectly—especially in the most dynamic parts of the island. A potential improvement to address this is under investigation. Examples include Tybee Bar in GA, Crab Bank Seabird Sanctuary in SC, Lanark Reef in FL, and the Chandeleur Islands off of LA.

- Some recently developed areas are overprioritized (e.g., a solar field near Wedgefield, FL; a limestone barren west of Lime, TN; and the Moncure Megasite in NC). Updated landcover and indicator updates based on newer landcover should fix this issue.
- Some new conservation areas where restoration has only started recently are underprioritized. Examples include Wolfe Creek Forest in FL and the Wolf River corridor in MS. Updated landcover and indicator updates based on newer landcover should fix this issue.
- Some important urban natural areas are underprioritized. Examples include Kapok Park in Clearwater, FL; the West Atlanta Watershed Alliance education hub in Atlanta, GA; Lost Corner Preserve in Sandy Springs, GA; and Blucher Park in Corpus Christi, TX. Improvements in indicators related to urban natural areas could fix this in the future.
- Parts of some larger natural areas, especially forested areas, near urban centers are underprioritized (e.g., the Western Highland Rim in Nashville, TN; the ridge south of Elkins, WV; and Rocky Face Ridge in Dalton, GA). Improving how the indicators address the role these areas play in connecting people to nature and providing climate change solutions to urban areas, as well as updated corridor methods, could help address this in the future.
- Some ridges in the Ridge and Valley ecoregion are underprioritized. Future indicator improvements should fit this.
- Culturally important historical areas are underprioritized through the Blueprint. This is particularly true in areas outside of the Piedmont, Atlantic Coastal Plain, and East Gulf Coastal Plain subregions. Even within these subregions, where there is an indicator for these areas, there are significant gaps. Some low-urban historic areas in these subregions are underprioritized because 1) they are not yet part of the National Register of Historic Places (e.g., Lost Island Farm on Roanoke Island, the likely landing site for the Lost Colony at the mouth of the Chowan River, and Native American sites on the Dan River near the NC/VA border), 2) because their location isn't publicly shared (e.g., sensitive archeological sites), or because 3) the GIS depiction of their spatial boundaries has significant errors (e.g., sites in GA and AL).
- Some areas with important ecological communities are underprioritized in TX and OK (e.g., shinnery oak scrub).
- Some upland areas in the Upper Coastal Plain of GA are overprioritized. Improvements in prioritization methods could improve this in the future.
- Some sandy ridges with significant longleaf restoration potential and importance as movement corridors for longleaf-associated species are underprioritized. Examples include the Trail Ridge area east of Okefenokee Swamp in GA and a sand ridge off of Hwy 421 northwest of Wilmington, NC. Improvements in indicators and prioritization methods could improve this in the future.
- Some small or newer quarries are overprioritized (e.g., American Stone Quarry near Chapel Hill, NC). While most quarries are classified correctly as developed, smaller or newer ones don't have large enough areas of barren landcover in the 2019 National Land Cover Database to be filtered out in the resilient terrestrial sites indicator. If not identified as non-natural, quarries tend to score very highly on landscape diversity given all the elevation change that

happens within them. This issue could be fixed in a future update to the resilient terrestrial sites indicator.

- Some oil platforms north of Monahans, TX are overprioritized due to errors in the natural landcover in floodplains indicator that come from the EPA floodplain map. Indicator improvements in the near future could fix this.
- Some urban areas are overprioritized due to issues with the potential access to parks indicator (e.g. areas east of Sarasota, FL). This indicator identifies areas where parks are needed but does not consider whether a site is available to become a park. It includes areas of low intensity development in order to capture vacant lots, which can serve as new park opportunities. However, as a result, this indicator also captures some areas that are already used for another purpose (e.g., houses, cemeteries, and businesses) and are unlikely to become parks.
- Some dirt roads through forests are overprioritized based on the grasslands and savannas indicator (e.g., a road northwest of Hawthorne, FL). The mix of sparse tree cover and sandy soil in the pixel can look like a savanna. This can also cause speckling of priorities—especially in northern FL and southern GA and AL.
- Some narrow powerline rights-of-way and roadsides important for sun-loving plants of high conservation concern are underprioritized. Improved methods for the grasslands and savannas indicator could fix this in the future.
- Some mowed areas on military installations are overprioritized based on the grasslands and savannas indicator (e.g., parts of Camp Robinson in AR and Milan Army Ammunition Plant in TN).
- Some greenways and trails are overprioritized in areas where those trails degrade important habitat (e.g., trails in Bastrop State Park, TX and their impacts on Houston toad habitat).
- Greenways and trails missing from OpenStreetMap are underprioritized (e.g., Bayport Berm in Seabrook, TX). This will likely be fixed with future indicator updates.
- There are some overprioritized speckles in highly developed areas due to issues with the greenways and trails indicator (e.g., North Raleigh, NC and NorthPark Center mall in Dallas, TX). Future indicator improvements could fix this.
- At least one private easement southwest of Frankfort, KY is overprioritized due to its inclusion in the urban park size indicator. We will investigate ways to better exclude private easements from this indicator in the next update.

Wetlands

- Some important wetlands are underpriorized. Examples include the east and west sides of Felsenthal National Wildlife Refuge in AR; Ten Mile Pond Conservation Area in MO; the southeast part of Big Oak Tree State Park in MO; bottomland hardwood and xerohydric flatwoods in Clarks River National Wildlife Refuge in KY; white fringeless orchid habitat in Daniel Boone National Forest in KY; depressional wetlands southeast of Raeford, NC; the Stony Run wetlands in northeast Dunn, NC; and marshes in a TNC conservation easement off of Port Bay in TX. Improvements to indicators and prioritization methods could fix these in the future.

- Some ephemeral wetlands are underprioritized. New prioritization methods under investigation for next year could address this.
- A small set of bottomland forest areas in areas flooded by dams are underprioritized (e.g., the area between Summerfield and Faircloth, LA and the Little River wetlands in Millwood Recreation Area in AR). These areas are misidentified as open water reservoirs in the current reservoir mask used by the landscape condition indicator. Improvements to that mask for future Blueprints are under investigation.
- Important Carolina bays are often included in large patches of medium priority, but the bays and nearby areas should be higher priority. Different methods for resolving this issue are under investigation.
- Some important areas of tidal freshwater marsh are underprioritized (e.g., Mackay National Wildlife Refuge). Indicators that better represent waterfowl habitat needs could improve this in the future.
- Some areas along the shores of small lakes that are surrounded by development or agriculture are underprioritized (e.g., Lake Lawne in northern Orlando, FL; a historical wood stork colony northeast of Lake Poinsett in FL; and a sandhill crane roost east of Palm Bay, FL). These areas can provide important habitat for waterbirds and corridors for species like bobcat.
- Some coastal wetlands in the coverage gap between resilient coastal sites and resilient terrestrial sites are underprioritized (e.g., Jamestown, VA).

Species-Specific

- Some important caves for Ozark big-eared bat, Northern long-eared bat, and Indiana bat are underprioritized. A potential indicator to address this is under investigation.
- Some coastal marsh areas important for diamondback terrapin are underprioritized (e.g., Cedar Point marsh north of Dauphin Island, AL).
- Some islands particularly important for species that don't yet have critical habitat spatially mapped (e.g., red knot) or are not federally listed species (e.g., seabirds, heron rookeries) are underprioritized (e.g., Ogeechee Bar in GA and Walker and Robinson Islands in AL).
- Some parts of Cape Sable seaside sparrow critical habitat are being underprioritized. All critical habitat for this species is prioritized in the Blueprint, but it is currently a mix of high and medium priority. Multiple indicator improvements under investigation could fix this in the future.
- Important areas for beach birds that are not on islands are underprioritized in the FL Peninsula and all other Gulf coast subregions.
- Some areas that are important for mottled duck nesting in the edge of the Texas Blackland Prairie and West Gulf Coastal Plain subregions are underprioritized.
- Some coastal habitat important for aboriginal prickly pear in Florida is underprioritized. Future indicator updates could improve this.
- Some habitat for cave and karst species is being underprioritized (e.g., Ozark cave fish habitat in Republic, MO and a watershed north of Key Cave in AL).
- Some habitat for range-restricted plants is underprioritized (e.g., an area southwest of Manning, GA).

Freshwater

- Some river sections important for aquatic diversity are underprioritized. Underprioritized areas include some sections of Shoe Heel Creek in NC that are important for broadtail madtoms and other endemics; Neuse River waterdog habitat in the Trent River near Croatan National Forest in NC; the VA section of the Nottoway River; the lower section of the Little River in NC; some parts of the lower Neuse River in NC; the headwaters of the Flint River; and sections of the Flint River north and south of the Atlanta airport. Ongoing improvements in the imperiled aquatic species indicators and prioritization methods should improve these issues.
- Some aquatic areas, particularly smaller rivers and streams, are overprioritized. The imperiled aquatic species indicator is at a subwatershed (HUC12) scale while the species hotspots it seeks to depict are often only a part of that subwatershed.
- Some important areas for migratory fish in Gulf of America drainages are underprioritized. The source data for the Gulf migratory fish connectivity indicator didn't include migratory fish species that are important for the Mississippi Alluvial Valley and all Gulf coast watersheds west of that subregion.
- Some open water areas of artificial waterbodies are underprioritized. In some cases, parts of these waterbodies can provide important species habitat (e.g., American crocodile habitat in cooling canals in South FL and waterfowl habitat in parts of some reservoirs).
- Some river areas are underprioritized in South FL (e.g., the north fork of the St. Lucie River and Ten Mile Creek). Potential aquatic indicator improvements under investigation could fix this in the future.
- Some canals are overprioritized in South FL. Potential aquatic indicator improvements under investigation could fix this in the future.
- Some streams are overprioritized due to lack of information on local hydrologic alterations (e.g., a stream just south of Bastrop State Park in TX).
- Some sections of small streams are underprioritized where the estimated floodplain map doesn't include them (e.g., parts of Upper Crabtree Creek in Durham, NC). Improvements to freshwater indicators to better capture small streams should fix this in the future.

Marine

- Mouths of many priority rivers are underprioritized where they transition into the estuarine ecosystem. Improved estuarine indicators should improve this issue in the future.
- Some marine Blueprint priorities are at a coarser resolution due to the marine birds, sea turtles, and marine mammals indicators. The coarser data results in what looks like parts of large squares or hexagons and unnatural edges in priority. Improvements in indicator resolution should fix this in the future.
- Some marine areas in the far eastern part of the Blueprint, particularly beyond the Blake Plateau, may be underprioritized due to a lack of survey data for marine birds and mammals in that region.
- Some areas along the Atlantic shelf break are overprioritized due to overprediction in deep-sea coral richness models (e.g., select areas east of GA). While the models predict high coral

richness in these areas, surveys show that they are almost entirely sand. Improvements in the models could fix this in the future.
- Some important areas of the Charleston Gyre upwelling are underprioritized. While the exact location of the upwelling and its high concentration of nutrients and fish larvae isn't fixed, there are some areas where it commonly occurs that are underprioritized.
- Some parts of the deeper waters off Onslow Bay, NC, which are important for reef fish and many tropical/subtropical species, are likely underprioritized.
- Some areas important for corals in the Gulf are underprioritized (e.g., an area east of the West Florida Escarpment that's being considered for a Coral Habitat Area of Particular Concern). Future coral indicator improvements will likely fix this.
- Some important areas for Rice's whale may be underprioritized. The Blueprint currently covers habitat predicted by the GoMMAPPS model for Rice's whale, but NOAA core distribution polygons cover a much larger area.
- DeSoto Canyon—an important area for corals, fish and nutrient upwellings south of Pensacola, FL—is underprioritized. Future improvements in coral, fish, or marine bird models could improve this in the future.
- Parts of important species movement corridors are underprioritized (e.g., a Kemp's ridley sea turtle corridor near the TX border). Improved corridor methods could fix this in the future.
- Some important hardbottom and coral areas are underprioritized in the Atlantic (e.g., parts of Gray's Reef in GA, part of the Savannah Banks off of GA, and some corals in Southeast FL). Future indicator updates could fix this.
- Some paleo channels are underprioritized. These areas that were historically freshwater rivers have significant archeological value. No indicator currently targets paleo channel locations.

Caribbean

Terrestrial

Uplands

- Parts of some protected areas in the east part of St. Croix in USVI (e.g, Point Udall, Jack and Issac Bay) are underprioritized. Indicator and/or method improvements could fix this in the future.
- Some important areas related to Taíno history are underprioritized (e.g., sections of Salt River Bay National Historical Park and Ecological Preserve in St. Croix, USVI). Work to better integrate this information into cultural indicators is ongoing.
- Parts of the coastal hills in eastern PR, important for their unique vegetation composition, are underprioritized.
- Some upland military areas that would have otherwise received a score of medium priority were underprioritized in the Blueprint. When we supplemented the top 10% Zonation areas with protected lands to define hubs in the Blueprint connectivity analysis, we inadvertently omitted military areas from the protected lands layer in the U.S. Caribbean (but not the continental Southeast).

Wetlands

- The Altona Lagoon area of St. Croix, USVI—an important area for birds, sportfish, and the potential reintroduction of the St. Croix ground lizard—is underprioritized.
- Some salt ponds, which provide important habitat for birds and other species, are underprioritized.
- Some upland military areas that would have otherwise received a score of medium priority were underprioritized in the Blueprint. When we supplemented the top 10% Zonation areas with protected lands to define hubs in the Blueprint connectivity analysis, we inadvertently omitted military areas from the protected lands layer in the U.S. Caribbean (but not the continental Southeast).

Species-Specific

- Some areas important for future Puerto Rican parrot habitat are underprioritized (e.g., northeast of El Yunque in PR; east of Lares, PR). Improvements to species-specific indicators could fix this in the future.
- Some areas important for rare plants are underprioritized (e.g., east of Lares, PR).
- Some areas that are important for coquí species in PR are underprioritized (e.g., Puerto Rican rock frog *[Eleutherodcactylus cooki]* habitat near water in Southeast PR). Work on a coquí-specific indicator is ongoing.
- Habitat important for the Virgin Islands boa on the east side of St. Thomas is underprioritized. Improvements in prioritization methods should fix this soon.

Marine

- Some important areas for coral reef restoration are underprioritized (e.g., north of Teague Bay in St. Croix, USVI; north of Christiansted in St. Croix, USVI). Better incorporation of restoration in coral indicators could fix this in the future.
- Some marine areas strongly impacted by point source pollution may be overprioritized (e.g., area of south St. Croix, USVI impacted by Cruzan rum distillery discharge).
- Some marine corridor routes do not sufficiently account for variation in habitat quality between hubs (e.g., corridors south of Culebra, PR; corridors southwest of the big island of PR).

Appendices

Appendix I: Additional information on SARP aquatic animal SGCN HUC12 summaries used in imperiled aquatic species indicator

Species Summaries by HUC12 Methodology*

All species data was obtained by contacting the appropriate representatives from State Natural Heritage Programs. The request made was:

Data Request: I am the GIS coordinator for the Southeast Aquatic Resources Partnership, and I am contacting you to obtain GIS (lat/long) data for occurrences of rare aquatic species: **fishes, mussels, snails, crayfishes, amphibians and reptiles.** If this is not possible, the ultimate product that we will be creating are species genus and species names by HUC 12, so we can calculate the number of T&E, State Listed, and SGCN species in a HUC 12 where particular dams are located. Please see below for project description.

What is SARP: The Southeast Aquatic Resources Partnership (SARP) is a regional collaboration of natural resource and science agencies, conservation organizations and private interests developed to strengthen the management and conservation of aquatic resources in the southeastern United States. SARP is a joint party committee created by the SEAFWA Directors to complete the mission of conserving aquatic habitats across political boundaries (more under SEAFWA - https://seafwa.org/committee/sarp) We are state governed by a steering Committee with representatives from every state with advisory roles for our federal, corporate, and NGO partners. You can learn more at https://southeastaquatics.net/.

How will the data be used: SARP has an Aquatic Connectivity Program where we are working to inventory and prioritize dams and road related barriers for removal based on ecological metrics. One additional piece of information that we have and we will use your data for, is to determine the number of threatened and endangered species, number of state listed species, and number of species of greatest conservation need within a HUC12. You can see our tool here: https://connectivity.sarpdata.com

Who will view the data: This is a public facing tool, so the HUC12 summaries will be viewed by whoever frequents the tool, mostly aquatic resource managers that we work with, but also anyone who happens upon it. For the public facing tool, it DOES NOT show species names, only number of TE, state listed, and SGCN in a HUC12.

Internally, we use lat/long information to work with partners to look at how far upstream or downstream a species of concern is from potential projects. So if raw lat/long information is provided, only I will see this information.

Methodology:

1) Project all species data feature classes into one geodatabase and then add to map.
2) Select out only aquatic occurrences if necessary. Many states did not have a method to determine which amphibians and reptiles were aquatic. SARP together with USFWS created a list of amphibians and reptiles and eliminated those that were largely terrestrial. It is possible that some non-aquatic amphibians and reptiles remain and will be refined for next year's update.
3) Merge blank schema containing fields for scientific name, common name, federal status, state status, SGCN Listing, historic status.
4) Spatial join (points) species dataset to HUC12 and populate HUC12 name and HUC12 code fields. If dataset is a polygon Intersect polygon occurrences by HUC 12 so that features split by HUC12 and contain only one HUC12 ID. Export to file geodatabase.
5) Remove all occurrences listed as extirpated, and calculate a 'Yes' in the historic status for those older than 1980, or listed as historical. Note: for those with no historical status, separate out the year of the date last observed using this code in field calculator: DatePart("yyyy", [DATEFIELD]).
6) Remove historical records.
7) Calculate scientific, common, and historic fields using scripts.
8) Delete unwanted fields.
9) Join on SGCN species look up table for each state (https://www.sciencebase.gov/catalog/item/655288d7d34ee4b6e05c390a) and calculate SGCN List field, calculate yes or no based on field.
10) Join on Regional SCGN species look up table for the Southeast (https://www.nwf.org/Our-Work/Wildlife-Conservation/southeast-regional-priority-species) and calculate yes or no based on the look up results.
11) Export to geodatabase tables.
12) Merge into one geodatabase table that contains species names and status and HUC12ID for each.
13) Send to Astute Spruce to calculate numbers of TE, State, SGCN, and Regional SGCN species by HUC12 and join to dams/crossings. Astute Spruce cleans up any spelling errors.

State Status

Table 16. Date imperiled aquatic species data was last updated for each state.

State	Data in tool
TN	Feb 2024
SC	January 2024
MO	March 2024
AR	Nov 2022
VA	March 2024
LA	March 2024
KY	March 2024
FL	March 2024
NC	March 2024
AL	March 2024
GA	March 2024
MS	March 2024 for crayfishes; Dec 2022 for fishes and mussels
OK	March 2024
TX	March 2024
WV	Feb 2023

*Information provided by Kat Hoenke with the Southeast Aquatic Resources Partnership (SARP) in May 2024 accompanying the aquatic animal SGCN HUC12 summary data. Please direct any questions to kat@southeastaquatics.net.

Appendix II: Ancillary Blueprint Data

Caribbean Island Size

This layer represents the size of islands in the U.S. Caribbean. It was created primarily as an input to several indicators and the Southeast Blueprint 2023 Caribbean subregion to make sure that the boundaries of the Blueprint and underlying data are consistently aligned to a shared depiction of shoreline contours and island shape. It was also used to stratify indicator thresholds according to island size.

Input Data

- NOAA Continually Updated Shoreline Product (CUSP), accessed 1-11-2023; read a 1-page factsheet about CUSP; view and download CUSP data in the NOAA Shoreline Data Explorer (to download, select "Download CUSP by Region" and select Southeast Caribbean)
- Southeast Blueprint 2023 subregions: Caribbean

Mapping Steps

- Make a copy of the Southeast Caribbean CUSP feature line dataset and reproject it to ESPG 5070.
- For the big island of Puerto Rico, special steps were required to deal with CUSP shorelines that did not connect across large rivers.
    - Add and calculate a field to use to dissolve the lines.
    - Dissolve the lines using the Dissolve function, which reveals where there are gaps in the shoreline.
    - Use the integrate tool to snap together nearby nodes, using a tolerance of 8 m. This connects the disconnected lines on the big island of Puerto Rico.
    - Convert these modified shorelines to a polygon.
    - Add and calculate a dissolve field, then dissolve using the dissolve tool. This is necessary because interior waterbodies on the big island of Puerto Rico also have shorelines in the CUSP data. This step produces a layer where inland waterbodies are included as a part of the island where they occur.
    - From the resulting layer, select the big island of Puerto Rico and create a separate polygon feature layer from it. This extracts a modified shoreline boundary for the big island of Puerto Rico only. We don't want to use the modified shorelines created above for other islands that didn't have an issue of disconnected shoreline segments near large rivers.
- Go back to the original Caribbean CUSP lines and convert them to polygons.
- Add a dissolve field and dissolve using the dissolve tool. This produces a layer where all inland waterbodies are included as a part of the island where they occur.
- From the island boundaries derived from the original CUSP data, remove the polygons that overlap with the big island of Puerto Rico derived from the modified CUSP data. This produces a layer representing all U.S. Caribbean islands except the big island of Puerto Rico.

- Merge the modified big island of Puerto Rico layer with the layer for all other islands.
- Create and populate a field that has unique IDs for all islands.
- Convert the island polygon to a raster using the ArcPy Feature to Raster function. This makes a raster that correctly represents the interior of the islands. However, because the Feature to Raster function for polygons works differently than the Line to Raster function, the shoreline doesn't perfectly match the result we get when we convert the CUSP lines to a raster.
- Because the Caribbean coastal shoreline condition indicator is created from the CUSP lines, we need the shorelines to match exactly. To reconcile this, go back to the original Caribbean CUSP line data and use the Feature to Raster function again, this time converting the lines to a raster.
- Use the ArcPy Cell Statistics "MAXIMUM" function to combine the two rasters above (one created from the CUSP lines and one created from the CUSP-derived polygons).
- Export the raster that represents the extent of Caribbean islands.
- Use the Region Group function to give unique values to each island.
- Reclassify to make 3 island size classes. The big island of Puerto Rico is the only island in the highest class. The medium island class contains the following islands: Isla Mona, Isla de Vieques, Isla de Culebra, St. Thomas, St. John, and St. Croix. All other islands were put in the smaller class. All other non-island pixels in the Caribbean were given a value of marine.

Note: For more details on the mapping steps, code used to create this layer is available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip).

Literature Cited

National Oceanic and Atmospheric Administration (NOAA), National Ocean Service, National Geodetic Survey. NOAA Continually Updated Shoreline Product (CUSP): Southeast Caribbean. [https://coast.noaa.gov/digitalcoast/data/cusp.html].

Elevation-Based Marine/Terrestrial Mask

This detailed elevation-based marine mask is used to separate the marine environment from terrestrial/freshwater areas in some indicators (e.g., imperiled mammals) and Blueprint processes (e.g., connectivity analysis). In this mask, terrestrial/freshwater areas have a value of 1, and marine areas have a value of NoData.

Input Data

- 2023 NOAA coastal relief model: volumes 1 (Northeast Atlantic), 2 (Southeast Atlantic), 3 (Florida and East Gulf of Mexico), 4 (Central Gulf of Mexico), and 5 (Western Gulf of Mexico), accessed 6-17-2025
- USGS current 1/3 arc-second Digital Elevation Models (DEMs) from the 3D Elevation Program, downloaded using uGET and USGS The National Map download application v.2, accessed 3-03-2025
- Southeast Blueprint 2025 extent (continental only)

Mapping Steps

- Convert the NOAA Coastal Relief models from netCDF format to .tif using the gdal Translate function.
- Add the resulting .tif files to a Mosaic file in ArcPro.
- Use a conditional statement to pull out all pixels ≥-1 from the mosaic file. Assign a value of 1 to all pixels ≥-1 and assign everything else a value of NOATA. Save the output as a .tif.
- To match the 30 m resolution and projection used for the Blueprint, use the ArcPy Resample "MAJORITY" function on the resulting .tif file.
- Use the ArcPy Extract by Mask function to clip the result above to the 2025 continental Southeast Blueprint extent.
- Because the NOAA Coastal relief models do not cover the interior of the continental Southeast Blueprint area, use USGS current 1/3 arc-second DEMs to fill in the interior areas. We had already downloaded and mosaiced the 1/3 arc second DEMs for the Southeast Blueprint geography, so we used those. Trying to use the methods above to perform a conditional statement on the mosaic and output as a .tif file kept throwing errors, so we exported the mosaic file to .tif format manually by right clicking on the mosaic file in the contents window of an ArcGIS Pro project, choosing "data", then choosing "export raster", and saving as a .tif file.
- Use a conditional statement on the 1/3 arc second mosaic export to pull out all areas >0. Assign all pixels >0 a value of 1, assign all other pixels a value of NoData, and save the output as a .tif.
- Convert the resulting .tif file to the 30 m resolution and projection used for the Blueprint using the ArcPy Resample "MAJORITY" function.
- Combine the raster derived from the NOAA coastal relief models with the raster derived from the 1/3 arc-second DEM data using the ArcPy Cell Statistics "MAXIMUM" function with the ignore data option of "DATA".

- The steps up to this point have produced a raster that only contains values of 1 or NoData. Areas with a value of 1 are largely terrestrial, with an elevation >-1 m in the NOAA coastal relief model (above 1 m below sea-level). Areas with a value of NoData are largely marine, with an elevation ≤-1 m in the NOAA coastal relief model (at or below 1 m below sea level). This raster has interior holes of NoData within the 1 values representing low-lying coastal areas that are disconnected from the ocean, such as quarries and coastal lakes. As this layer is intended to provide a relatively clean separation between terrestrial and marine areas, we wanted to backfill those interior holes.
- Reclassify the raster to change the NoData pixels to 0.
- Use the ArcGIS Spatial Analyst Region Group function (4-neighbor) to create unique clumps of contiguous values.
- Use the result of the region group to find large interior clumps of 0 pixels (low-lying coastal areas surrounded by higher elevations) in order to convert those values to 1 and essentially remove the holes. We used Lake Mattamuskeet in eastern NC as a reference to determine the size of the clumps to change to 1, since that is a freshwater lake that we wanted to include as part of the terrestrial (i.e. non-marine) environment. After testing multiple clump size thresholds, we ultimately chose a threshold of 50,000 pixels, representing interior holes roughly 4,500 ha in size.
- Use a conditional statement with "where" clause to fill in the clumps of 0 values.
- Use the ArcPy Resample "MAJORITY" function on the resulting .tif file to convert to the 30 m resolution and projection used for the Blueprint.

Note: For more details on the mapping steps, code used to create this layer is available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip).

Known Issues

- This layer is intended to distinguish between terrestrial and marine ecosystems, where freshwater is included as part of the terrestrial environment. However, in aquatic systems, it is difficult to determine where freshwater ends and saltwater begins, as that transition is highly dynamic and influenced by many factors including topography, instream flow, tides, storms, and more. By including all areas greater than -1 m in elevation (i.e., above 1 m below sea level) and backfilling to capture low-lying coastal areas, we intentionally erred on the side of a more generous depiction of the terrestrial environment, and a more narrow depiction of the marine environment. We wanted the terrestrial portion to include small islands and coastal freshwater lakes (like Lake Mattamuskeet in NC). This is an imperfect methodology but produced better results than some of the other alternatives we explored.

Literature Cited

NOAA National Centers for Environmental Information. (2023). Coastal Relief Models (CRMs) [Data set]. NOAA National Centers for Environmental Information. [doi: 10.25921/5ZN5-KN44].

United States Geological Survey. United States Geological Survey 3D Elevation Program 1/3 arc-second Digital Elevation Model. Accessed March 3, 2025. [https://apps.nationalmap.gov/downloader/].

Marine Mammal & Turtle Movement Areas Mask

To ensure the corridors in the marine and estuarine environment reflect connectivity for marine mammals and sea turtles, we created a marine mammal and turtle movement areas mask using spatial density models.

Input Data

- Habitat-based Marine Mammal Density Models for the U.S. Atlantic (last updated 6-20-2022), accessed 12-12-2023

  These models use aggregated survey information, distance-sampling, and oceanographic variables to predict cetacean density throughout the region.

- Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) - GoMMAPPS marine mammal and sea turtle spatial density model outputs (version 2.2), accessed 12-14-2022

  Based on ship-based and aerial line-transect surveys conducted in the U.S. waters of the Gulf of Mexico between 2003 and 2019, the National Oceanic and Atmospheric Administration (NOAA) Southeast Fisheries Science Center developed spatial density models (SDMs) for cetacean and sea turtle species for the entire Gulf of Mexico. SDMs were developed using a generalized additive modeling framework to determine the relationship between species abundance and environmental variables (monthly averaged oceanographic conditions during 2015-2019).

Mapping Steps

- We selected 3 species for the Atlantic (North Atlantic right whale, humpback whale, and pilot whale) and 4 species for the Gulf (Rice's whale, pilot whale, Kemp's ridley sea turtle, and loggerhead sea turtle). The movement areas for these species also cover most of the major movement areas used by other marine mammals and sea turtle species.
- For each of the marine mammal and sea turtle species, sum the monthly abundance rasters across all months. The only exception to this is pilot whales in the Atlantic, where the data were yearly and did not need to be summed.
- Use the following thresholds for summed species data to ensure movement areas are connected and represent a wide range of species.
    - For Rice's whale, pilot whale (Atlantic), North Atlantic right whale, and Kemp's ridley sea turtle, use the top 70% of values based on a 10-class quantile classification.
    - For more cohesive and selective areas for pilot whale (Gulf), use the top 60% of values based on a 10-class quantile classification.
    - For more cohesive and selective areas for loggerhead sea turtles, use the top 50% of values based on a 10-class quantile classification.

    - For more cohesive and selective areas for humpback whales, use a manually identified threshold (0.04). For context, this threshold is slightly less than the top 85% of values based on a 10-class quantile classification.
- Combine all species movement areas into a single layer identifying whether a pixel is in at least one movement area.

Note: For more details on the mapping steps, code used to create this layer is available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip).

Literature Cited

Litz J, Aichinger Dias L, Rappucci G, Martinez A, Soldevilla M, Garrison L, Mullin K, Barry K, Foster M. 2022. Cetacean and sea turtle spatial density model outputs from visual observations using line-transect survey methods aboard NOAA vessel and aircraft platforms in the Gulf of Mexico from 2003-06-12 to 2019-07-31 (NCEI Accession 0256800). NOAA National Centers for Environmental Information. Dataset. [https://doi.org/10.25921/efv4-9z56].

Roberts, J., Best, B., Mannocci, L. et al. Habitat-based cetacean density models for the U.S. Atlantic and Gulf of Mexico. Sci Rep 6, 22615 (2016). [https://doi.org/10.1038/srep22615].

Roberts JJ, Yack TM, Halpin PN (2023) Marine mammal density models for the U.S. Navy Atlantic Fleet Training and Testing (AFTT) study area for the Phase IV Navy Marine Species Density Database (NMSDD). Document version 1.3. Report prepared for Naval Facilities Engineering Systems Command, Atlantic by the Duke University Marine Geospatial Ecology Lab, Durham, NC. [https://seamap.env.duke.edu/seamap-models-files/Duke/Reports/AFTT_Marine_Mammal_Density_Models_2022_v1.3.pdf].

Reservoirs

Though reservoirs are highly altered systems, they still have conservation value. In earlier versions of the Blueprint that lacked the lakes and reservoirs indicator, the indicators did not do a good job of capturing important parts of reservoirs or distinguishing the relative value of different reservoirs. As a result, we used to remove the open water areas of reservoirs from the zones used to define the boundaries of each Zonation run so that were not eligible to be prioritized in the Blueprint. (Note, however, that we did always include terrestrial areas surrounding reservoirs, which were represented by the indicators, and are also where most conservation actions occur to improve reservoir condition.)

While we no longer mask out the open water parts of reservoirs in Zonation in Southeast Blueprint 2055 and 2026, we still use the previous Blueprint reservoir mask as an input dataset in indicators (e.g., landscape condition) and the connectivity analysis.

Input Data

- USGS National Hydrography Database (High Resolution) in FileGDB 10.1 format (published 08-30-2021) - NHDWaterbody and NHDFlowlines, accessed 10-14-2021
- National Inventory of Dams (NID), accessed 10-15-2021; download the data
- 2019 National Land Cover Database (NLCD): Land cover
- Southeast Floodplain Inundation Frequency mosaic
- Base Blueprint 2022 extent

Mapping Steps

- Make copies of the NHDWaterbody and NHDFlowlines layers for editing.
- Extract features identified as either "LakePond" or "Reservoir" from the NHDWaterbody layer. Most reservoirs in the Southeast region are coded as "LakePond" in this dataset.
- Make a geospatial layer of the National Inventory of Dams (NID) from the source .csv file.
- Select NHD waterbodies that are within 200 m of NID locations.
- Add to the selection all NHD waterbodies that are within 5 m of the selection generated in the previous step to ensure that all parts of a single waterbody are selected.
- Select NHD flowlines that are within 50 m of NID locations.
- Select NHD waterbodies that are within 50 m of the selection generated in the previous step.
- At this stage, we did some hand-editing to add in obvious large reservoirs (especially in Texas) that were omitted from the above selections because the NID did not capture the dam locations. We used Inundation Frequency and the 2019 NLCD to assist in this step. In addition, the NHD contains some misclassified reservoirs (e.g., reservoirs classified as swamp/marsh or stream/river) that we manually added in. Note: The NID is also missing many dam locations associated with small farm ponds, which are too numerous to add by hand.
- Convert to raster using the ArcPy Polygon to Raster function and clip to the spatial extent of Base Blueprint 2022.
- Create a mask to use for Zonation runs by creating a raster where everything in the SECAS extent is 1 except for the reservoirs, which are NoData. For rebalancing later, we also create a raster where everything in the SECAS extent is 1 and the reservoirs have a value of 0.

Note: For more details on the mapping steps, code used to create this layer is available in the ancillary data download package on ServCat (Southeast_Blueprint_2026_Ancillary_Data_Download_YYYY-MM-DD.zip).

Literature Cited

Allen, Y. 2016. Landscape Scale Assessment of Floodplain Inundation Frequency Using Landsat Imagery. River Research and Applications 32:1609–1620. [https://doi.org/10.1002/rra.2987].

U.S. Army Corps of Engineers. National Inventory of Dams. Accessed October 15, 2021. [https://nid.sec.usace.army.mil/nid/].

U.S. Geological Survey. National Hydrography in FileGDB 10.1 format (08-30-2021) – NHDWaterbody and NHDFlowlines. Accessed 14 October 2021. [https://www.usgs.gov/national-hydrography/access-national-hydrography-products].

U.S. Geological Survey (USGS). Published June 2021. National Land Cover Database (NLCD) 2019 Land Cover Conterminous United States. Sioux Falls, SD. [https://doi.org/10.5066/P9KZCM54].

Yang, L., Jin, S., Danielson, P., Homer, C., Gass, L., Case, A., Costello, C., Dewitz, J., Fry, J., Funk, M., Grannemann, B., Rigge, M. and G. Xian. 2018. A New Generation of the United States National Land Cover Database: Requirements, Research Priorities, Design, and Implementation Strategies, ISPRS Journal of Photogrammetry and Remote Sensing, 146, pp.108-123. [https://doi.org/10.1016/j.isprsjprs.2018.09.006].